

North-Eastern Neo-Aramaic in the Context of the Languages of Western Asia

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Preface

This volume is one of the outcomes of the ERC project *Echoes of Vanishing Voices in the Mountains: A Linguistic History of Minorities in the Near East*, which was hosted by the University of Cambridge from 2021 to 2025 (Principal Investigator: Geoffrey Khan; Grant Agreement ID: 101021183). The aim of the project was to investigate the endangered subgroup of Neo-Aramaic dialects known as North-Eastern Neo-Aramaic and the ways in which these dialects have converged with other languages in contact with them in Western Asia. This work involved in-depth studies and documentation of Neo-Aramaic dialects and also of several of the dialects of the contact languages, many of which had not been adequately described.

The volume has been written by members of the project research team and by several other researchers who participated in conferences of the project.

Geoffrey Khan, the Principal Investigator, who coordinated and edited the volume, wrote the sections on Neo-Aramaic in the chapters on phonology, demonstratives, the morphosyntax of nominal modification, the copula, perfective and perfect, future, habitual, progressive, and narrative verbal forms, including the discussion sections in these chapters. He also wrote the concluding remarks and parts of the introduction.

Paul Noorlander wrote the sections on Neo-Aramaic and contributed to the sections on Arabic and Iranian in the chapters on case morphology and adpositional marking, the introduction to verbal morphology and its inflectional categories, the morphology of verbal person marking, valency changing constructions, alignment typology, word order in verbal clauses, and experiential predicates and predicative possessors, including the discussion sections in these chapters. He has also helped with the editing of these chapters of the volume and is responsible for the creation and design of the maps for Neo-Aramaic dialectology.

Katherine Hodgson wrote the sections on Armenian in the various chapters and in the introduction. She also helped with the editing of parts of the volume.

Masoud Mohammadirad wrote the sections on Iranian in the chapters on morphology of verbal person marking, valency changing constructions, alignment

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typology, syntax of case and verbal agreement, word order in verbal clauses, experiential predicates and predicative possessors, and the copula. He was the main contributor to the phonology chapters on interdental, bilabial, and palatalisation of dorsals. He contributed to the Iranian section in the introduction chapter, nominal modification, and different chapters on syntax of verbs.

Ergin Öpengin wrote the sections on Iranian in the chapters on emphatics, pharyngeals, unaspirated stops and affricates, future, uvular stops, and was the main contributor to the Iranian section in chapters on demonstratives, the expression of habitual, perfective and perfect, verbal forms used in the narrative, and the morphosyntax of nominal modification. He also contributed to the rest of the phonology chapters.

Shuan Karim wrote the sections on Iranian in the chapters on case morphology and adpositional marking, and the introduction to verbal morphology. He contributed to the phonology chapters, and to the chapters on the expression of progressive, the expression of habitual, and the morphosyntax of nominal modification. Shuan also helped with the technical aspects of preparing the volume in LaTeX.

A number of other scholars made contributions to the volume. Eleanor Coghill contributed to the sections on Arabic dialects. Mahîr Dogan contributed to the sections on Zazaki. Murad Suleymanov contributed the sections on Turkic. Dorota Molin contributed to the Neo-Aramaic sections on case morphology and adpositional marking and discussions of data from the Jewish Neo-Aramaic dialect of Dohok throughout the volume.

Geoffrey Haig was a visitor in Cambridge for several months and attended our workshops. He greatly enhanced our work on the project by many discussions and by helping create a questionnaire for Kurdish. Parvin Mahmoudveysi was also a visitor in Cambridge for a short period of time and attended our workshops. She contributed important data on the dialects of Gorani.

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Geoffrey Khan, January 2026

Preface

Abbreviations

1PL	first-person plural	CONJ	conjunction
1SG	first-person singular	COP	copula
2PL	second-person plural	CSTR	construct
2SG	second-person singular	<i>cvb</i>	converb
3PL	third-person plural	DAT	dative
3SG	third-person singular	DEF	definite article
A	the agent argument of a transitive clause	DEIC	deictic
ABL	ablative	DEM	demonstrative
ABS	absolutive	DEM1	proximal demonstrative
ANT	?	DEM2	medial demonstrative
ACC	accusative	DEM3	distal demonstrative
ACT	active	DOM	differential object marker
ADD	additive	DTRZ	detransitiviser
ADP	adposition	DIR	direct case
ALL	allative	DIST	distal
ANAPH	anaphoric	DO	direct object
AOR	aorist	DRCT	directional particle
APRT	active participle	DU	dual
Arab.	Arabic	ERG	ergative
ART	article	<i>evid</i>	evidential
ART1	proximal/first-person article	<i>evt</i>	eventual (Turkic)
ART2	medial/second-person article	<i>excl</i>	exclamation
ART3	distal/third-person article	EXIST	existential
ATTR	attributive	EZ	ezafe/construct case
AUG	augment	F	feminine
C	common gender	F.PTCP	future participle
CA	Classical Armenian	FUT	future tense
CAUS	causative	G.	Gorani
CK	Central Kurdish	GEN	genitive
CIRC	circumposition	HAB	habitual
CLF	classifier	IMP	imperative
CMPD	compound	IND	indicative
COM	comitative	INDF	indefinite
COMP	complementizer	INF	infinitive
COMPL	completive	INS	instrumental
COND	conditional	IO	indirect object
		IPF	imperfect
		IPFV	imperfective

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I.PTCP	imperfective participle	PRO	pronoun
IRR	irrealis	PROG	progressive
<i>jus</i>	jussive	PROH	prohibitive
Kurd.	Kurdish	PRON	pronoun
LNK	linker	PRP	preposition
LOC	locative	PRS	present tense
M	masculine	PROX	proximal
MAL	malifactive?	PST	past (stem/marker)
MED	middle voice	PTCL	particle
NEG	negative	PTCP	participle
NK	Northern Kurdish	PVB	preverb
NNK	Northern Northern Kurdish	QNTF	quantifier
NOM	nominative	R	recipient-like argument
N.PTCP	negative participle	REFL	reflexive
OBJ	direct object	REL	relative
OBL	oblique (case/clitic/L-suffix)	RES.PTCP	resultative participle
OM	object marker	S	single argument of an intransitive verb
<i>opt</i>	optative	SBJV	subjunctive
<i>ost</i>	ostensive	SBRD	subordinator
P	the patient argument of a transitive clause	<i>sc</i>	suffix conjugation (Semitic)
PAM	person agreement marker	SCK	southern Central Kurdish
PAR	particle	SEA	Standard Eastern Armenian
PASS	passive	SG	singular
PC	prefix conjugation (Arabic)	SK	Southern Kurdish
Pers.	Persian	S.PTCP	subject participle
PFV	perfective	<i>stat.</i>	stative
PIr	Proto-Iranian	SWA	Standard Western Armenian
PL	plural	T	theme argument of a ditransitive verb
PN	proper noun	TR	transitive
POS	Possessive	TRZR	transitiviser
POSS	possessive suffix	VIS	visible
POSTP	postposition	VOC	vocative
PPRT	passive participle	Z.	Zazaki
PST.PTCP	past participle		
PRF	perfect		

In transcriptions of NENA dialects, an acute accent marks word stress, a grave accent marks the nuclear stress of the intonation group and raise [|] symbol marks the end of an intonation group, e.g.,

- (1) C. Barwar (Khan 2008a: 599)
b-šàly-ən[|] *b-tápq-ən* *bìy-e.[|]*
FUT-pray.PRS-1SG.M FUT-meet.PRS-1SG.M on-him
'I shall go down, I shall meet him.'

Part I

Introduction

1 Introduction

1.1 The aims and scope of the volume

This volume aims to examine the development of Neo-Aramaic dialects in the context of the languages they have been in contact with in Western Asia. The region we refer to as “Western Asia” is not clearly demarcated, either politically or topographically. It can be defined as the northern part of Mesopotamia, comprising the upper catchment regions of the Tigris and Euphrates rivers. This is a region of a high degree of linguistic diversity characterised by the intersection of several large genetic groupings. Four distinct language families are represented (Turkic, Semitic, Indo-European, Kartvelian), and within Indo-European, three branches (Armenian, Iranian, and Hellenic). The Semitic language Aramaic and the Indo-European language Armenian have the deepest historical roots in the region, followed by Iranian, Arabic and Turkic. These languages are diverse both in their genetic heritage and in their typology. They have, however, undergone various degrees of convergence due to contact and so one may describe the region as a linguistic area or Sprachbund.

An overview of the languages of Western Asia and discussions of their areal convergence were presented in the volume *The Languages and Linguistics of Western Asia: An Areal Perspective* (Haig & Khan 2019). The aim of the present volume is to focus in greater depth on Neo-Aramaic, and specifically the subgroup of Neo-Aramaic known as North-Eastern Neo-Aramaic (henceforth NENA), and study in detail a range of linguistic developments it has undergone that have correlations with other languages of the region and, therefore, are likely to have been induced by this contact. The major contact languages of the region that are relevant are Kurdish, Gorani, Armenian and vernacular Arabic, and more peripherally Turkic and Zazaki. This establishes to what extent the NENA dialects in the form they have in modern times have converged with their linguistic environment. It also casts light on the social and migration history of NENA-speaking communities in the region. The study describes diverse structures across the NENA dialects in different geographical areas of the region and shows how these generally correlate with structures in the contact languages of these specific areas. In some cases, however, these geographical correlations are skewed. In such cases, the

1 Introduction

main reasons for this skewing are migrations of NENA speakers from one area of the region to another or shifts in the languages in contact with NENA (for further details, see §??).

In addition to the volume *The Languages and Linguistics of Western Asia* (ed. Haig & Khan 2019), there have been a number of other previous studies of the contact of NENA with languages in the region. These include studies that focus on a selection of features, sometimes taken from various dialects of NENA. Articles of this nature include Chyet (1995), Matras (2002), Khan (2007a; 2020b; 2022c; 2022b), Noorlander (2014), Borghero (2015), Haig (2015), Noorlander & Stilo (2015), Stilo & Noorlander (2015), Coghill (2025a), Cohen (2015), Gutman (2018), Molin (2025). (Khan & Mohammadirad 2024a) have published a study of contact-induced change across all levels of grammar and lexicon in the Neo-Aramaic dialect spoken by the Jews of Sanandaj. The volume edited by Noorlander (2026) presents comparative studies of processes of passivisation in NENA and various languages of the region. The present volume aims to present a more comprehensive coverage of contact-induced change in a range of features in NENA across the diverse dialects in the entire NENA area with in-depth examinations of the relevant features in the contact languages in the region. The contact languages that are examined in greatest depth are Kurdish, Gorani and Armenian, with shorter treatments of parallels in Arabic, Zazaki and Turkic. Our target audience is primarily that of specialists in Neo-Aramaic and the other languages of Western Asia. It is hoped, however, that this case study will be of interest to the wider community of linguists working in the field of language contact studies.

Two or more languages spoken in the same region may exhibit structural similarities through (i) mere chance, (ii) inheritance from a common ancestor, (iii) contact influence, or (iv) a combination of (i)–(iii). We are primarily interested in (iii), but assigning similarities to contact influence requires caution. It should be taken into account, for example, that parallel structures in two languages could be cross-linguistically common typological developments (e.g., locative progressive constructions) or the elimination of features could be due to their being typologically marked (e.g., the elimination of interdental stops in sound systems). Moreover, many developments are partial convergence and not full replication. Crucial support for the identification of features in NENA that are shared with languages in contact as induced by contact is their geographical restriction to Neo-Aramaic of the northern Mesopotamian region compared to Neo-Aramaic of other regions. This applies to the vast majority of features that are considered as the result of contact-induced change in the ensuing chapters.

For practical reasons the scope of the volume has been limited in various ways, which leaves various desiderata for future research.

1.1 *The aims and scope of the volume*

As remarked, the focus is specifically on the NENA subgroup of Neo-Aramaic. It does not include an examination of contact-induced change in all Neo-Aramaic subgroups (see §1.2.1). Such a broader coverage of Neo-Aramaic would require separate studies and take into account a broader range of contact languages.

The languages in contact with NENA that have been incorporated into the study are vernacular dialects of the region, many of which are spoken by minority groups and are endangered. These are the contact languages that have had the greatest impact on NENA. The official state languages of the region, i.e., the standard versions of Arabic, Persian and Turkish, are, in principle, not engaged with. This is because they have generally not been sources of linguistic influence on NENA.

The purpose of the volume is not to trace contact-induced changes throughout the attested history of Aramaic in the region. Rather its purpose is to document correlations between Neo-Aramaic dialects of one particular subgroup spoken in modern times and contemporary vernacular languages in the region and to identify cases and processes of contact-induced change in the Neo-Aramaic dialects in the light of these correlations. Aramaic has deep historical roots in Western Asia and can be traced back in written records to the beginning of the first millennium BCE. The NENA dialects spoken in modern times are descendants of vernacular forms of Aramaic that existed in antiquity. There was a continuous interaction of Aramaic with languages in contact throughout its history in the region. Some of the results of such linguistic contact in antiquity are indeed still visible in NENA dialects, such as loanwords from Akkadian, one of the major languages of Mesopotamia in the first millennium BCE (Krotkoff 1985; Khan 2017; Mutzafi 2018). Documenting correlations between NENA dialects and contemporary vernacular languages, of course, has historical implications, since it reflects contact-induced changes that in many cases may have a considerable time depth. There is some discussion of attested earlier forms of Aramaic and of the various contact languages in the chapters of the volume, but there is no systematic treatment of these.

The ensuing chapters focus on selected features of phonology, morpho-syntax and syntax. The volume does not include a detailed examination of the impact of contact languages on the lexicon of NENA. The features that have been selected as topics for the chapters all exhibit influence from contact languages and so were judged to be valuable case studies of contact-induced change in NENA dialects.

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1.2 The languages of the region

1.2.1 Neo-Aramaic

The Neo-Aramaic dialects are modern vernacular forms of Aramaic, a Semitic language, which has a documented history in the Middle East of over 3,000 years, the earliest inscriptions being datable to approximately 1,000 BCE.

Spoken vernacular dialects of Aramaic, generally known as Neo-Aramaic dialects, have survived down to modern times in four subgroups:

1. Central Neo-Aramaic
2. Neo-Mandaic
3. Western Neo-Aramaic
4. North-Eastern Neo-Aramaic

The dialect geography of Neo-Aramaic has undergone radical changes over the last one hundred years, as a result of which a large number of the speakers of the dialects have been displaced from the places where they have lived for many centuries. The following geographical description, therefore, relates to the situation that existed at the beginning of the twentieth century, before these major upheavals.

1.2.1.1 Central Neo-Aramaic

The subgroup that is generally termed Central Neo-Aramaic was spoken by Christian communities in south-eastern Turkey in the region of Țūr ‘Abdīn, which extends from the town of Mardin in the west up to the boundary of the Tigris river in the east and north. The main component of this sub-group is the cluster of dialects of the Neo-Aramaic variety generally known in the academic literature as Țuroyo. Native speakers of the language generally refer to it by the term Țurāyt. The main dialect split in Țuroyo is between the dialect of the town of Midyat and the dialects of the surrounding villages. The differences between the Țuroyo dialects are small, and they are mutually comprehensible (Jastrow 1985; Ritter 1990; Waltisberg 2016). In addition to the Țuroyo cluster, one other dialect is known to have existed in the Central Neo-Aramaic sub-group. This is the dialect of the Christians of the village of Mlaḥso (now Yünlüce), situated near Lice in northern Diyarbakir province, which is related to Țuroyo but exhibits a number of significant differences (Jastrow 1994a, 2011).

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1.2.1.2 Neo-Mandaic

Neo-Mandaic is spoken by Mandaean in south-western Iran. This exists in two known varieties originally spoken in the towns of Ahvāz and Khorramshahr, respectively (Macuch 1965, 1993; Häberl 2009, 2011; Mutzafi 2014a). The native name of this spoken language is *raṭnā*, derived from a verbal root meaning ‘to whisper or mutter’.

1.2.1.3 Western Neo-Aramaic

Western Neo-Aramaic is spoken in three mountain villages in the Anti-Lebanon of Syria, north of Damascus, namely Maʿlula, Baxʿa and Jubbʿadin. The population of Maʿlula is Christian with a small minority of Muslims. The inhabitants of Baxʿa (official name Ṣarxa) and Jubbʿadin are all Muslim (Arnold 1989-1991, 2011). There are no significant differences in the dialect of Maʿlula between the speech of Christians and Muslims. The native name is *siryōn* or *arōmay*.

1.2.1.4 North-Eastern Neo-Aramaic

North-Eastern Neo-Aramaic (NENA), which is the subgroup of Neo-Aramaic that is the focus of this volume, is highly diverse, consisting of over 150 dialects spoken by Christians and Jews originating from towns and villages east of the Tigris river in northern Iraq, south-eastern Turkey and western Iran.

The NENA dialects are not direct descendants of any of the earlier literary forms of Aramaic, although they exhibit close affinities to Syriac and Jewish Babylonian Aramaic. The dialects rather have their roots in a vernacular form of Aramaic that existed in antiquity in the region of northern Mesopotamia, which differed from the vernacular underlying the literary languages of Syriac to the west and Jewish Babylonian Aramaic to the south. This is shown by the fact that, although exhibiting numerous innovations, they are more conservative than Syriac and Jewish Babylonian Aramaic in some features (Khan 2007b; Fox 2008). Some of the dialects, moreover, have, as remarked already, preserved lexical items of apparently Akkadian origin that do not appear in dictionaries of the earlier forms of literary Aramaic.¹

Within NENA itself, one may identify a number of subgroups on the basis of linguistic structure and lexicon.

There is a fundamental split between the dialects spoken by the Christians and those spoken by the Jews. This applies even to cases where Jewish and Christian

¹For Christian dialects see Khan (2002a: 515; 2017: 262), and for Jewish dialects Sabar (2002: 12).

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communities lived in the same town, such as Koy Sanjak, Sulemaniyya (both in northern Iraq), Urmi (north-western Iran) and Sanandaj (western Iran). In these towns, the dialect of the Christians differed radically from the dialect of the Jews in all levels of grammar (phonology, morphology, syntax) and lexicon.

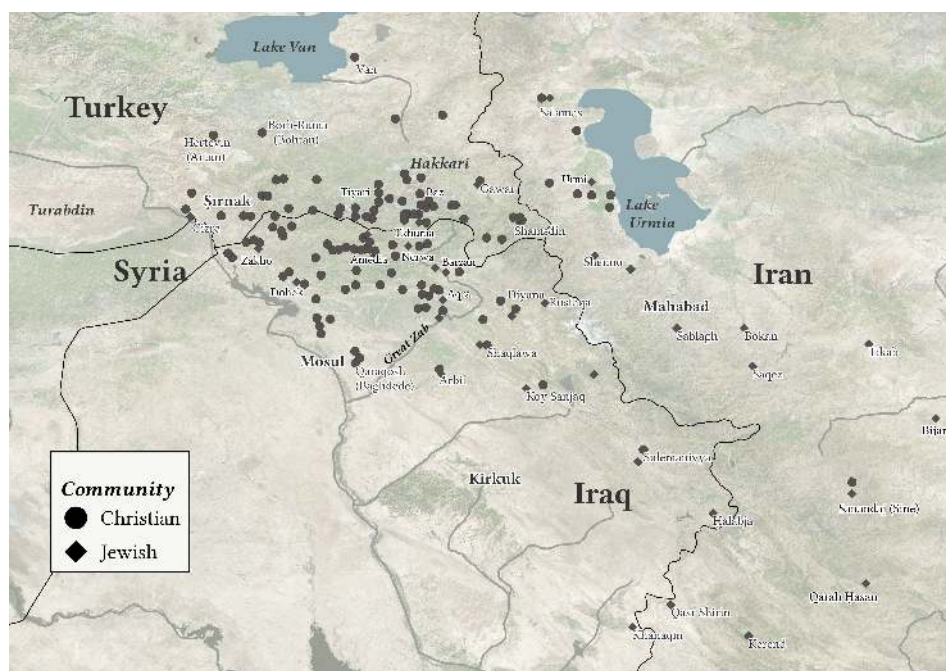


Figure 1.1: The NENA dialect area

Within Jewish NENA dialects, two main subgroups are clearly identifiable:

One of these was spoken in the north-west of Iraq mainly in Dohuk province in locations to the west of the Great Zab river, such as Zakho (Avinery 1988; Sabar 2002; Cohen 2012), Dohuk, Amedia (Greenblatt 2011), Betanure (Mutzafi 2008a), also across the Zab in Iraq near the Turkish border in villages such as Nerwa and in small communities in what is now south-eastern Turkey in, for example, Challa (Fassberg 2010) in Hakkâri province and Cizre (Nakano 1969, 1973) in Şırnak province. This subgroup is generally referred to as *lišana deni* ('our language'), the native term used by speakers of the dialects, which contains the form of the 1PL genitive pronoun that is distinctive of the group.

The dialects of the other Jewish subgroup were spoken in locations east of the Great Zab river in Iraq, north-western Iran and western Iran. This subgroup is generally referred to as trans-Zab (following Mutzafi 2008c). In Iraq, this included

1.2 The languages of the region



Figure 1.2: Detail of NENA-speaking communities

the dialects of locations in the Arbel (Erbil/Hawler) and Sulemaniyya provinces, e.g., Rustaqa (Khan 2002b), Ruwanduz, Koy Sanjak (Mutzafi 2004a), villages of the plain of Arbel (Khan 1999a), the village of Dobe, which is on the western bank of the Great Zab, Ḥalabja and Sulemaniyya to the east (Khan 2004), and as far south as Khanaqin on the Iranian border. In north-western Iran, it includes the Jewish dialects of the towns of Urmi (also known as Urmia), Šəno (official name Ushnuye), Solduz (official name Naqadeh) and Sablagh (now Mahabad) (Garbell 1965b; Khan 2008b), the district of Salamas north of the Urmi plain (Duval 1883; Mutzafi 2015) and in adjacent towns that are now situated in the east of Turkey, such as Başkale and Gawar (official name Yüksekova). In western Iran, the trans-Zab subgroup includes a cluster of dialects spoken by Jewish communities in various localities in the Kordestan and Kermanshah provinces in an area that includes Sainqala, Bokon, Saqqez on its northern border, Sanandaj in the centre, Bijar on the eastern border, and in the south, Kerend and Qasr-e Širin (Hopkins 1999a; Khan 2009; Yisraeli 1998). The Jewish Trans-Zab dialects of the Kordestan and Kermanshah provinces of western Iran and the adjacent area of Sulemaniyya in north-eastern Iraq exhibit very little diversity. Most of the data on this variety

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of NENA for this chapter are taken from the Jewish dialect of Sanandaj (henceforth referred to as J. Sanandaj), which has been described in detail in [Khan \(2009\)](#). Studies on other Jewish dialects of this cluster include those by [Yisraeli \(1998\)](#) on the dialect of Sāqəz and by [Hopkins \(2002\)](#) on Kerend. Their relationship to the Jewish trans-Zab cluster of dialects of the West Azerbaijan province of Iran in the Urmi region is more distant to the extent that speakers of dialects from the western Iran cluster had difficulties communicating with Jews from Urmi.

Various native names of the language are used by the Trans-Zab Jewish communities, e.g., *Lišanət Nošan* (north-eastern Iraq), *Lišana Nošan* (western Iran), *Lišana Didan* (north-west Iran), all of which mean ‘our language’, also *Hulaula* (western Iran), which is an abstract noun meaning ‘Jewishness/Judaism’ (< **hūdāyūtā*).

In addition, there was a small cluster of Jewish dialects in the region of Barzan, located in Iraq between the areas of the *Lišana Deni* and Trans-Zab dialects, which exhibit a linguistic profile that is transitional between the two main subgroups ([Mutzafi 2002a; 2004b](#)). The native term for the language in this cluster is *Lišān Dideni* ‘our language’.

The divisions among the Christian NENA dialects on structural and lexical grounds are not so clear-cut. One may, nevertheless, identify clusters of dialects with distinctive features.

In Iraq, the following clusters of Christian dialects are found:

1. The dialects of the Mosul plain in Nineveh province. e.g., Qaraqosh ([Khan 2002a](#)), Alqosh ([Coghill 2003](#)), Telkepe ([Coghill 2018](#)), Barəṭle ([Mole 2015; al-Saka 2024](#)), Karəmləsh ([Borghero 2008](#)).
2. The dialects in the far north close to the Turkish border in Dohok province, mainly west of the Zab, e.g., Araḏin ([Krotkoff 1982](#)), Barwar [= Barwari Bala] ([Khan 2008c](#)), Nerwa ([Talay 2001](#)), Derigne.
3. The dialects east of the Zab in Arbil and Sulemaniyya provinces, e.g., Ankawa ([Borghero 2015](#)), Shaqlawa ([Khan 2022b](#)), Bədyal, Koy Sanjak ([Mutzafi 2004a](#)), Sulemaniyya.

As can be seen, the second two clusters correspond broadly to the Jewish *Lišana Deni* and trans-Zab subgroups, respectively.

The Christian NENA dialects of south-eastern Turkey and Iran may be classified into several clusters. These include the following ([Talay 2008a](#), 47–48; [Hopkins 1999a](#)):

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1. The Bohtan cluster, spoken in villages in the area that is now the Şırnak and Siirt provinces of Turkey, such as Hertevin (Jastrow 1988a), Ruma, Shwata, Borb (Fox 2009a), Umra (Hobrack 2000), Jənnēt (Jastrow 1994b; Talay 2008a, 44–45).
2. The Cudi cluster, spoken in villages in the area of the Cudi mountain (Cudi Daği) that is now in the Şırnak province of Turkey (Sinha 2000).
3. The Țyare cluster, divided into Upper Țyare and Lower Țyare, spoken in villages on the western side of what is now the Hakkâri province of Turkey (Talay 2008a; 2008b; Borghero 2005).
4. The Txuma cluster, spoken in villages lying to the east of the Lower Țyare area (Jacobi 1973; Talay 2008a).
5. The Hakkari cluster, spoken in a variety of villages in the Hakkari mountains east and north-east of Țyare, including villages of the Jilu (Fox 1997) and Baz tribes (Mutzafi 2000; Talay 2008a), Sat (Mutzafi 2008b), and the villages in the area of Lake Van, and Salamas (Mutzafi 2015) in north-western Iran.
6. The cluster of dialects spoken in the far east of Turkey in the areas of Şamməsdin, Gawar (Talay 2008a) and, over the border in north-western Iran, in the mountains of Tergawar (Khan 2016a).
7. The Urmi cluster, which includes varieties of a dialect spoken by Christians in villages situated on the plain of Urmi in north-western Iran and within the town of Urmi (Khan 2016a).
8. The dialect of Sanandaj in western Iran (Panoussi 1990; 1991; Heinrichs 2002).

Native names of the language among Christian NENA-speakers include those designating the religious community, such as *surəṭ* (northern Iraq), *lišanət suraya* (Urmi), and names such as *ḥadiṭan* ‘our speech’ (Qaraqosh, Iraq).

1.2.1.5 History of the NENA speech communities

The Christian and Jewish NENA-speaking communities have deep historical roots in the region. This is reflected by the linguistic background of the NENA

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dialects, which, as remarked above, are shown by a number of features that originate in vernacular Aramaic in antiquity when Akkadian was still a living language. The Christians are descendants of communities that converted to Christianity in the first half of the 1st millennium C.E. (Fiey 1965). Until the end of the 1st millennium CE, Aramaic was spoken more widely by Christians and Jews throughout Iraq. By around the 10th century, the communities in the centre and south of Iraq had largely become Arabic-speaking. In the course of the Middle Ages, many of the Christians of central Iraq migrated to the north of the country to avoid forced conversion with its consequent Arabisation. We know, for example, that around the eleventh century a large proportion of the Christian population of Tikrit migrated northwards and settled in the Mosul region, many in the town of Qaraqosh (Khan 2002a: 2-3). In the Timurid invasions of the fourteenth century, the Christian communities of central and southern Iraq were decimated, which would have hastened the end of any vestiges of Aramaic in this region. However, the Christian communities in the mountains in the north of the country and in villages on the nearby plain who fled to the mountains were largely spared the ravages of these invaders.

There are only scant historical records concerning the Jews of the region before the modern period. Mann (Mann 1931: 216) published letters datable to the beginning of the sixteenth century that mention villages in the highlands of northern Iraq that had Jewish communities. Some of these had been abandoned by their Jewish population by the twentieth century or the number of Jewish inhabitants was considerably reduced, reflecting the fact that the Jewish population of the region declined after the sixteenth century. One reason for this reduction is likely to have been the forcible conversion of some Jews to Islam, especially in the nineteenth century in some areas (Soane 1912: 186).

The isolated Christian dialect spoken in Sanandaj is likely to have been the result of migration from Iraq. It is closely related to the Christian dialect of Sulemaniyya, and there were family relationships between these two communities in the living memory of speakers. It is likely, therefore, that the Christians of Sanandaj were originally migrants from the region of Sulemaniyya in Iraq. The Christians of Sanandaj belong to the Chaldean Church. In the 19th century, several Christian families moved to Qazvīn, where their speech developed the distinctive trait of the realisation of /w/ as /v/ under the influence of Persian (Heinrichs 2002: 238). In the middle of the twentieth century, the Chaldean diocese of Sanandaj moved to Tehran, and the Christian Neo-Aramaic speakers moved with it.

The lack of diversity in the cluster of Jewish Trans-Zab dialects in the Kordistan and Kermanshah provinces of western Iran suggests that the communi-

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ties that spoke them migrated in a single wave into the region in relatively recent centuries. The ancient heartland of NENA must have been in what is now northern Iraq and south-eastern Turkey, where there is great dialectal diversity. The close relationship of the Jewish dialects of western Iran with the Jewish Trans-Zab dialect of Sulemaniyya points to north-eastern Iraq as the origin of the migration. We know that some of the Jewish communities that settled in the towns of the region originally lived in surrounding villages. The Jews of Sanandaj, for example, moved into the town after its foundation in the 17th century from a village known as Qal'at Ḥasan'ābād (Khan 2009: 1).

The geographical division between the Central Neo-Aramaic subgroup and the NENA subgroup coincides with the ancient border between the Romans and Parthians, and later between the Byzantines and the Sassanians (Kim 2008). The linguistic boundary between the two subgroups also coincides with an early Christian ecclesiastical division between the Jacobite (Syrian Orthodox) denomination of the communities west of the Tigris and the Nestorian (Church of the East) denomination of the communities to the east of the Tigris. The Christianity of the NENA-speaking communities has become more diverse in recent centuries, especially after the formation of the Chaldean Church, which is in communion with Rome, and the activities of Protestant missionaries in the 19th century. There are only sparse historical records relating to the various Neo-Aramaic-speaking settlements of the region. Some linguistic aspects of the dialects, however, give us insights into the history of the communities. The diversity of the NENA dialects, for example, can be interpreted as a reflection of the antiquity of settlement of the communities. Another factor that is likely to have had a bearing on this diversity is the fact that many of the NENA-speaking communities had the status of semi-independent tribes (*'aširatte*). The Jewish NENA dialects of the region exhibit major differences in their structure from the neighbouring Christian NENA dialects. This reflects not only distinct developments motivated by communal identity but also differing migration histories of the communities. The Jews of Urmi, for example, had settled in the town at an early period, whereas the Christians of the area were almost exclusively agriculturalists living in the surrounding villages and only began to settle in the town in large numbers in the late 19th century.

The Christian NENA-speaking communities, as well as the Turoyo-speaking Christians, underwent a major upheaval during the First World War in 1915, when they suffered massacres and mass displacement from their homes in an Ottoman-led campaign in south-eastern Turkey. No accurate statistics are available for the total death toll, but it is estimated that as much as half of the Neo-Aramaic-speaking population perished, either through violence, disease or starvation, pos-

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sibly amounting to around 250,000. Some of the survivors of the Christian communities subsequently returned to their homes in the Tūr ‘Abdīn area and the villages in the vicinity of the Cudi mountain. The NENA-speaking communities in the remainder of south-eastern Turkey, however, became permanently displaced. The majority settled initially in refugee camps in Iraq, then subsequently in Iraqi towns, in particular Baghdad and Kirkuk. Some of the Christians from the Bohtan region fled northwards and found safety in the Russian empire, eventually settling in the town of Gardabani in Georgia or in Krasnodar. From 1933 to 1935, about ten thousand refugees from south-eastern Turkey were settled in refugee camps in north-eastern Syria, then subsequently in villages on both sides of the Khabur River. During the Kurdish uprisings in the second half of the twentieth century, there were further upheavals. The villages in the Cudi region of south-eastern Turkey were destroyed in the 1980s and early 1990s, and the NENA-speaking population settled in Europe, mainly in France and Belgium.

The foundation of the State of Israel in 1948 was the decisive event for the Aramaic-speaking Jews of the region, since shortly thereafter the Jewish population of the region emigrated to the new state, except for a few isolated individuals, most of whom converted to Islam. The majority of the Jewish NENA-speaking communities migrated to Israel in 1952. Many Jews in western Iran, however, did not migrate until after the Iranian Revolution in 1979.

The NENA-speaking Christians belong to various church denominations, including the Assyrian Church of the East, Ancient Church of the East, Chaldean, Syriac Orthodox and Syriac Catholic.

Before the disturbances of the First World War, the majority of Christians were agriculturalists living in villages. After the settlement of refugees in the towns, many adopted various urban professions.

Most of the NENA-speaking Jews were town-dwellers, who were small traders, goldsmiths, tailors, weavers and dyers. Some of the traders were shopkeepers, while others were who hawked their wares around the surrounding countryside. Some of the Jews who remained in villages down to the twentieth century, such as the communities of the villages of Betanure, Shukho and Sandu, were agriculturalists (Mutzafi 2008a, 2014b).

1.2.1.6 Current status of the NENA-speech communities

As remarked, the NENA subgroup is extremely diverse, and the size of the speech community of individual dialects varies considerably. Some dialects are now reduced to a handful of final speakers. This applies in particular to the Jewish di-

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alects, all of which are now highly endangered and will not survive much beyond the next two decades.

Several Jewish dialects have recently become extinct, e.g., Nerwa in 2012, Sandu in 2010, Shahe in 2000, Bədyal in 1998 ([Mutzafi 2014b](#)), Challa in 2007 ([Fassberg 2010](#)). Some dialects of small Christian communities are also highly endangered, e.g., the Christian dialect of Bədyal. In general, however, Christian dialects are generally less endangered than Jewish ones. It is not possible to give precise statistics for individual dialects, but some of the larger ones have several thousand speakers, taking into account speakers in migrant communities.

Among Christian Aramaic-speakers from Iraq, a particularly widely spoken dialect is a koine that developed in the towns after the merging of various refugee communities after the First World War ([Odishe 1988](#): 19-38).

There are now no NENA-speaking communities in south-eastern Turkey. Those who had remained in the area of the Cudi mountain finally left in the 1980s when their villages were destroyed in the conflict between the Turks and the Kurds. There are still some Christians living in the Urmi region, almost exclusively now in the town of Urmi rather than the villages. A large proportion of the Christian community, however, has left the region. Some have settled in the large Iranian cities, mainly Tehran. In the 19th and early 20th centuries, many Christian speakers of the dialect of Urmi and other dialects of north-western Iran and the adjacent region moved to the Caucasus. These dialects, especially the Urmi dialect, are still spoken by communities in Georgia and Armenia, and elsewhere in the former Soviet Union ([Tsereteli 1970](#)). A large proportion of speakers have now settled in the USA, in particular in California, and Australia.

In recent times, the communities that settled in the Khabur region have been displaced due to the recent military conflict in Syria.

There are currently very few surviving speakers of the Urmi cluster of dialects who have a good knowledge of the language. In the 19th century, many families from north-western Iran, in particular from the region of Salamas, moved to Tbilisi in Georgia, and subsequently, in 1950, were settled by Stalin in Almaty in Kazakhstan, where several speakers of the dialect can be found today.

After the Christian NENA-speaking community of Sanandaj moved en masse from Sanandaj to Tehran in the middle of the twentieth century, they gradually left Iran and settled abroad. The majority of the migrants have settled in the Los Angeles area of the USA.

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1.2.1.7 Sociolinguistic situation

Most speakers of NENA across the NENA region were bilingual and in many cases trilingual. There are reports, however, that some Christian communities in isolated villages in the Tyare mountains spoke only NENA.

The Christians of the Mosul plain region speak Arabic in addition to their local Aramaic dialects. They have been exposed to both local Arabic vernaculars and to the literary Arabic of the Iraqi education system. In many cases, Arabic has had a major impact on their Aramaic speech, particularly in peripheral dialects such as Qaraqosh, where code-switching between Aramaic and Arabic is common. NENA speakers elsewhere in Iraq are generally fluent in Kurdish, the language of the majority of the surrounding Muslim population, and those who have been through the Iraqi education system know Arabic. In some areas of Iraq, NENA speakers spoke Turkoman, especially Turkoman converts to Christianity.

In the NENA-speaking area of south-eastern Turkey at the beginning of the twentieth century, before the upheavals, there was general bilingualism in Neo-Aramaic and Kurdish. There were also Armenian-speaking communities in the region. There was little exposure to Arabic, though Arabic loanwords within the dialects suggest that the contact with Arabic was greater at an earlier historical period. The communities from south-eastern Turkey who settled in the Khabur area of Syria in the 1930s speak Arabic in addition to their native NENA dialects. The communities of the Khabur generally preserved their individual dialects and did not develop a koine dialect as was the case among the refugee communities in Iraq (Talay 2008a).

In north-western Iran, speakers of the NENA dialects had contact with Kurdish and Azeri Turkish. Nowadays, most Christian speakers of NENA who still live in the area also speak Azeri, the vernacular of the Muslim population in this area, and also Persian, the official language of Iran, but not Kurdish. There are, however, numerous Kurdish loanwords in the NENA dialects of this area, and their morphology indicates that they form an older historical layer of the lexicon than the many Azeri words. This indicates that there must have been a more widespread knowledge of Kurdish in the NENA communities at an earlier period (Khan 2016b: vol. 3, 1-3).

The NENA speakers who live in the Caucasus also speak Russian, Armenian or Georgian. In Armenia, the Christian NENA speakers live in villages together with Armenians, and there are many intermarriages. Many of the native Armenian speakers in these villages also speak NENA.

In western Iran, NENA speakers were bilingual in Central Kurdish and Persian, which is the official language in schools and government administration in the

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region. There is evidence, however, that NENA speakers in this area spoke Gorani at an earlier period (Khan & Mohammadirad 2024a).

The surviving Jewish speakers of NENA from north-western Iran who settled in Israel are fluent in Modern Hebrew, which is now their primary language. The communities that have survived in Almaty now generally prefer to speak in Russian, especially the young generations.

Although the spoken NENA dialects have their roots in antiquity, they were not committed to writing in any systematic way until the seventeenth century, from which period onwards there are extant texts written by Christians (Mengozi 2002; Mengozzi & Braida 2011) and Jews (Sabar 1976). The literary form of Christian NENA that emerged in the seventeenth century was used by Chaldean Christians in the region of Mosul down to modern times (Macuch 1976). Another type of Christian literary NENA emerged in the nineteenth century in Urmi, in north-western Iran. This was originally developed by Western missionaries to that region in order to propagate Bible translations in the vernacular, but subsequently gave rise to a very variegated literature (Murre-van den Berg 1999). This form of written NENA was based on the spoken Christian dialect of the Urmi region, but came to be used widely by Christians in Iraq. Within the last few decades, there has been an increase in the production of literature written in this form of Christian literary NENA. Furthermore, education through the medium of the literary language has, in recent years, undergone considerable development in northern Iraq. All forms of literary NENA written by Christians are influenced by the classical Syriac language, in its orthography, grammatical structure and lexicon.

The written form of the Jewish NENA dialects, by contrast, is not influenced by a classical form of Jewish Aramaic. It has continued to be produced down to modern times, especially to commit to writing traditions of oral literature (Aloni 2014; 2022). No systematic form of education system in Jewish NENA was developed in the communities. Within Israel, in recent decades, however, the communities have organized various cultural events in an attempt to keep their spoken dialects alive. These include theatre plays, poetry readings and other types of entertainment.

1.2.2 Iranian

1.2.2.1 Kurdish

The Kurdish language family, a branch of North-Western Iranian, is traditionally divided into Northern Kurdish (Kurmanji), Central Kurdish (Sorani), and Southern Kurdish. While genealogically related, these three groupings have followed

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distinct historical, sociolinguistic, and contact-induced trajectories, producing considerable internal differentiation. The Northern, Central, and Southern designations reflect the three most genetically affiliated of the five major groups whose speakers identify as Kurds. The other two major groups are Gorani and Zazaki (discussed below). Additionally, Lekî (also spelled Laki), is debated as to whether it is part of Kurdish (see [Fattah 2000](#), [Belelli 2022](#), 2019). The varieties closest geographically to Southern Kurdish share enough similarity to Southern Kurdish to be included in [Fattah \(2000\)](#) as the Lekî-Kirmanşay group. The question of its relationship with Southern Kurdish is not relevant here, both thematically and geographically, as Lekî is spoken well south of the NENA-speaking areas. See Figure 1.3 for the distribution of Northern, Central, and Southern Kurdish varieties.

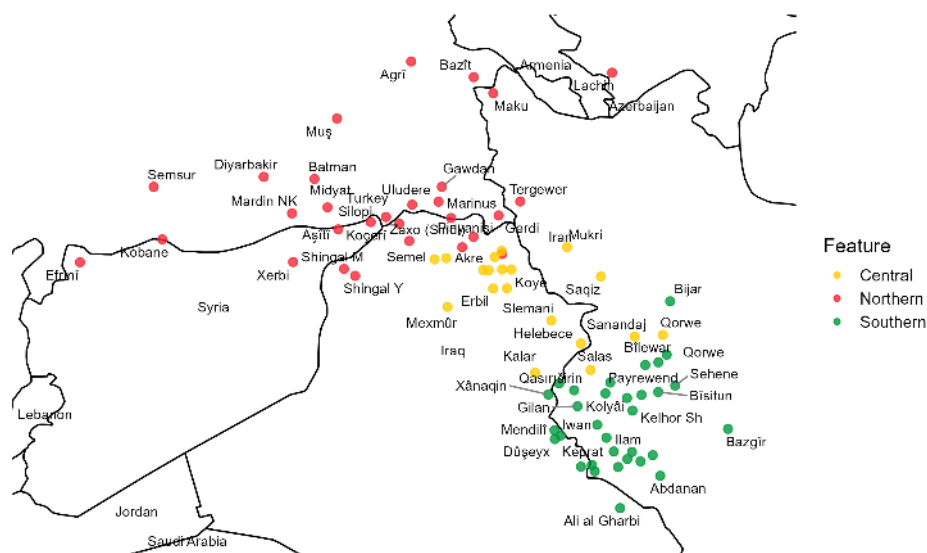


Figure 1.3: Northern, Central, and Southern Kurdish

1.2.2.1.1 Northern Kurdish

Northern Kurdish, also known as Kurmanji (Kurmançî), is the largest of the Kurdish dialects. It is internally diverse, and until recently lacked a systematic dialectological classification. The most significant modern contribution is [Haig & Öpengin \(2018\)](#), who identify three regional dialect zones: South-eastern Kurmanji (SEK), Southern Kurmanji (SK), and Western Kurmanji (WK).

1.2 The languages of the region

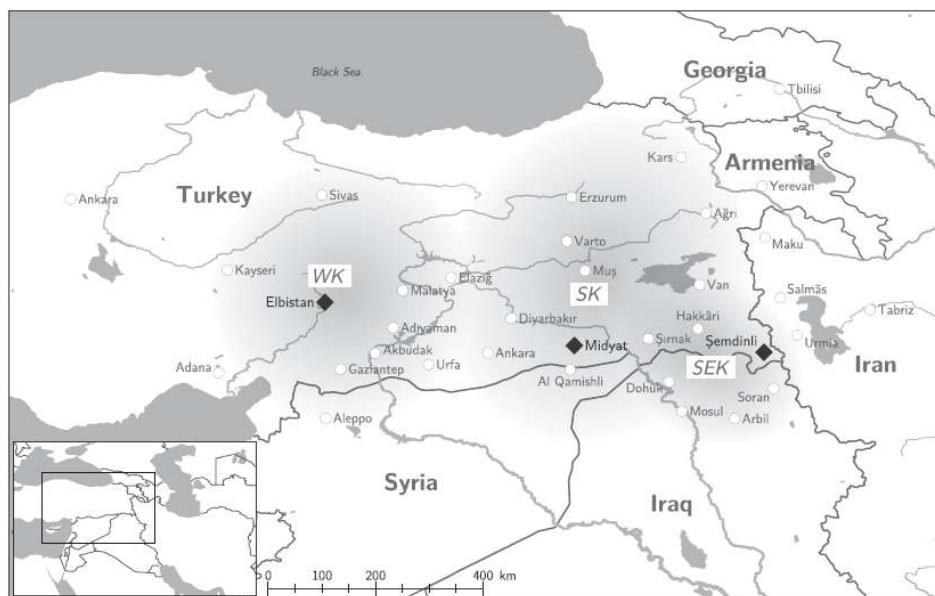


Figure 1.4: NK dialects (Haig 2019a: 133)

The traditional Kurmanji speech zone stretches west to east from Sivas and Erzincan to the Van–Hakkari region and into northern Iraq and north-west Iran, and north to south from Armenia into the Aleppo–Haseke region and the Sinjar area. Additional enclaves exist in Central Anatolia, in northern Iran, and in several former Soviet republics. Kurmanji dialects form a geographic continuum, with peripheral varieties such as SEK (Behdîni) and WK showing the strongest divergence, a pattern confirmed through lexical distance metrics and phonological correspondences. Significant diaspora communities, especially in Germany, France, Sweden, and the Netherlands, contribute to the global presence of Kurmanji. Estimates of speaker numbers vary, in part due to the political sensitivity of census data, but the figure is generally placed between 15 and 20 million. The majority reside in Turkey, where Kurmanji represents the dominant Kurdish variety.

Northern Kurdish is spoken in areas including the following cities, towns, and villages: Mosul, Imranli, Tatvan, Çukurca, Uludere, Muradiye, Pertek, Kahta, Nusaybin, Dolunay, Qamishli, Sabahiya, Duhok, Kobane, Zakho, Basselhaya, Akre, Sheran, Chiay Syan, Kanyder, Bismil, Kulp, Karilova, Adilcevaz, Kurtalan, Özalp, Kağızman, Kozluk, Siverek, Digor, Kiziltepe, Ergani, Doğubeyazit, Karakoçan, Suruç, Şemdinli, Muş, Varto, Erzincan, Bingöl, Sunak Köyü, Eleşkirt,

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Doda, Ad Darbasiyah, Raqqa, Qesirdib, Xirbe Cihwa, Jindires, Rajo, Elbistan, Bahçesaray, Sersink, Beroj, Tepkê, Kelhê.

Historically, Kurmanji preserves archaic North-West Iranian features including grammatical gender and a direct–oblique case system. Kurmanji also lacks pronominal clitics, an innovation shared with Zazaki and plausibly arising through historical contact with Armenian and Caucasian languages. Early attestations include a fifteenth-century prayer translation in Armenian script. Literary Kurmanji gained prominence in the sixteenth and seventeenth centuries through the works of Mela Ahmedê Cizîrî and Feqîyê Teyran.

Kurmanji evolved in a complex contact environment involving Armenian, Neo-Aramaic, Arabic, and Turkish. Armenian influence is particularly evident in phonology. Turkish influence, especially since 1923, has become pervasive in western Anatolia. Patterns of lexical, phonological, and morphosyntactic variation across the five regional zones include distinctive lexical items in SEK, archaic vocabulary in WK, and vowel or consonant shifts unique to particular regions. Notable correspondences include the shift of /a/ to a mid-back rounded vowel /ɔ/ in WK, fronting of /û/ to /ü/ or /i/ in SEK, and lenition of intervocalic /b/ in SWK and WK.

Morphosyntactically, SEK is particularly conservative, retaining final consonants in third singular present markers, a richer plural system, and additional TAM distinctions. NWK exhibits an unusual generalisation of the reflexive pronoun *xwe* as a third-person possessive. Kurmanji preserves a two-stem verbal system, a productive causative suffix *-and*, and the most elaborate *ezafe* system among Kurdish varieties. It maintains ergativity in the past tense, though alignment varies by TAM category.

Northern Kurdish usually appears in print in the “Hawar” Latin script. This was created by Celadet Bedirxan to offer a one-to-one grapheme-to-phoneme correspondence. He eventually launched the alphabet with his Kurdish literary magazine *Hawar* in the 1930s. In addition to Hawar, there is a history of publication in Cyrillic, Armenian, and Arabic scripts.

Sociolinguistically, Kurmanji faces strong pressures of language shift in Turkey due to historical state repression, lack of institutional support, and pervasive Turkish–Kurdish bilingualism. In Iraq and Syria, maintenance remains somewhat stronger, although displacement and Arabic shift affect younger populations.

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Table 1.2: Selected Northern Kurdish Isoglosses (Haig & Öpengin 2018: 183)

Feature	Şemz./SEK	SK	WK
suffixal singular masculine oblique	+	-	-
singular masculine oblique by vowel raising	-	+	+
neutralisation of gender with indefinite nouns	-/+	-	+
plural ezafe forms	-êd/-êt	-ê	-ê
proximal marker on nouns (SG/PL)	-e/-ene	-	-a/-ana
distinct 1PL verbal agreement suffix	+	-	-
3SG verbal agreement suffixes	-(î)t	-ê	-e
future tense particle	dê	=ê	=ê
future tense form of verb lacks <i>bi-</i> prefix	+	+(-)	-
plural indefiniteness suffix	-	+	-
aspectual verbal morpheme <i>-ewe</i>	+	-	-
mood-aspect particle <i>da</i>	+	-	-
incorporation of preverbs	+	+	-
ezafe used for progressive aspect	+	-	-
ezafe/gender marking of subjects in present	-	-	+
prepositional marking of goal/recipient	+	-	-
circumpositional marking of goal/recipient	-	+	-
postpositional marking of goal/recipient	-	-	+
directional particle <i>-e</i>	+	+/-	+/-
definiteness suffix <i>-eke</i>	+	-	-
heavy verb stems (for <i>kirin</i> , <i>birin</i> , <i>dan</i>)	+	-	-
possessive reflexive pronoun without a same-clause subject antecedent	-	-	+
particle for ‘intention’ mood	-	-	+

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1.2.2.1.2 Central Kurdish (Sorani)

Central Kurdish, conventionally referred to as Soranî, constitutes the second major Kurdish variety. It is spoken primarily in Iraqi Kurdistan, particularly in and around Slêmanî (Sulemaniyya), Hewlêr (Arbil), Kerkuk, and Helebce, as well as in western Iran, including the provinces of Kurdistan (Sanandaj), Kermansha, and parts of West Azerbaijan. It is estimated to have around 7.25 million speakers across Iraq, Iran, and the diaspora. The term “Sorani” is well-established, despite occasional objections regarding its historical origins; Garzoni already recorded the ethnonym “Soran” in 1787.² Dialectal variation within Sorani is less pronounced than in Kurmanji, though notable regional varieties include: Mukri, Hewleri, Sulemaniyya (Baban/Sharazur), Erdelani, and Germiyani.

Central Kurdish is spoken in areas including the following cities, towns, and villages: Erbil, Mehabad, Urmia, Oshnaviyeh, Marivan, Qoshachay-Miyandoab, Qalat Dizah, Khalakan, Naqadeh, Kamyaran, Sulemaniyya (Slêmanî), Kalar, Kirkuk, Khalifan, Baneh, Choman, Halabja, Sangaw, Saqqez, Mawat.

Central Kurdish is usually written in the Kurdo-Perso-Arabic script created by Sa’id Kaban Sedqi. Like the “Hawar” script used most commonly for Northern Kurdish and Zazaki, the Kurdish Arabic script is a true alphabet that features a one-to-one phoneme-to-grapheme correspondence. This stands in contrast to most other languages that employ an Arabic script, which does not represent all vowels. In scholarly articles and texts from the early 20th century, the “orientalists” created a transcription script based on the historical pronunciation of Persian to write all Iranic languages, which is often used to write Kurdish examples. The orientalist script is not employed here, both out of respect for Kurds and to promote comparability between varieties.

Sorani is distinguished by a high degree of standardisation, literacy, and institutional use and has been an official language of Iraq since 2005. Standardisation accelerated in the twentieth century through the growth of Kurdish media, education, and administration in the Kurdistan Region of Iraq.

Historically, Sorani developed in close contact with Gorani. Evidence for a Gorani substrate includes the aspectual suffix *-ewe*, the definiteness suffix *-eke*, and the preservation of pronominal clitics. These features distinguish Sorani sharply from Kurmanji.

²Note that the term Soranî originally referred to the variety of the Soran governorate *nawçey Soran*, centered around Rewanduz, Iraq. This is situated in the heart of the Central Kurdish area. However, the *de facto* standard Central Kurdish is the variety of Slêmanî, which gained prominence in the 19th century as the home of literary figures such as the poet Nalî. Slêmanî lies in the Baban region, making the proper dialect name Babanî.

1.2 The languages of the region

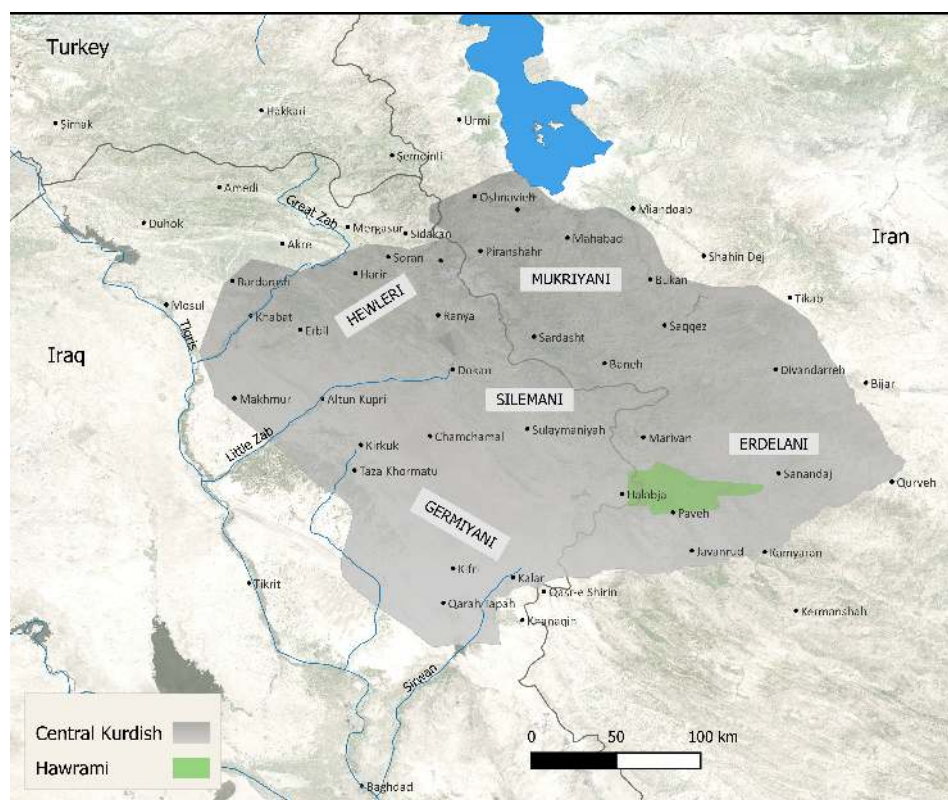


Figure 1.5: Varieties of Central Kurdish

The Erdelanî variety has been found to reflect a Gorani/Hewramî substrate. [Mohammadirad \(2025b\)](#) shows that many features that divide the southern varieties of CK from the central and northern varieties can be understood by reference to a Gorani substrate in the former group. Likewise, [Mohammadirad \(2024a\)](#) lists features in the word order profile of Erdelanî Kurdish that reflect a Gorani substrate. For example, while addressees of the verb ‘to say’ are predominantly post-verbal in Hewramî and Erdelanî, in the Mukri zone, they are overwhelmingly realised in the pre-verbal position.

Diagnostic isoglosses include the shift of velarised *l* to *r* in Hewleri; the third-singular present marker -î in Hewleri versus -ê elsewhere; the imperfective prefixes *de-* (Mukri, Hewleri) versus *e-* (Sulemaniyya, Erdelani, Germiyani); the preservation of an oblique suffix in Mukri and Hewleri; and different participial suffixes across dialects. Hewleri in particular forms a transitional zone with Northern Kurdish, sharing some lexical and morphological patterns; see Table

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1.3:

Table 1.3: Isogloss distribution and sub-dialect patterning in Central Kurdish (Öpengin n.d.)

Feature	Hewleri	Mukri	Sul.	Germiyani	Erdelani
l - r sound change	r	l	l	l	l
3SG agreement suffix	-î	-ê	-ê	-ê	-ê
Imperfective prefix	de-	de-	-e	-e	-e
Oblique suffix	+	+	-	-	-
2PL pronoun	engo	engo	êwe	êwe	êwe
Participle suffix	-î	-û	-û	-û/-ig	-ig

Structurally, Sorani lacks grammatical gender, has lost or reduced the oblique case in most dialects, maintains a rich system of pronominal clitics, and relies heavily on analytic TAM morphology. Unlike Kurmanji, it lacks phonemic aspiration. The *ezafe* system remains productive but less complex than in Kurmanji.

Sociolinguistically, Sorani speakers in Iran are typically bilingual in Persian, though language shift is reported in major urban centers such as Sanandaj and Tehran. In Iraq, younger generations increasingly acquire English due to English-medium education, resulting in concerns about balanced Kurdish literacy.

1.2.2.1.3 Southern Kurdish

Southern Kurdish is spoken in the southernmost Kurdish areas to the south of the Zagros mountains in Iran and Iraq (see Figure 1.6), primarily in the Kermanshah and Ilam provinces in Iran and in the Khanaqin–Mandali region of Iraq. It is often referred to with a number of epithets mostly drawn from place names, e.g., Kirmaşanî, Îlamî, Paşewanî, or tribe names, e.g., Kelhurî, Feylî. In local academic circles, the label *kurdî xwarîn* is also common. The number of speakers is estimated to be around 3-4 million. However, this figure is only tentative, as language shift has been accelerating, especially in large urban centres.

Southern Kurdish is spoken in areas including the following cities, towns, and villages: Bîcar, Kordelî Abdanan, Dînewar, Bîlewar, Sehena, Kolyay, Qorwe, Beyrey, Xanaqîn, Melikşay, Melikşay Xemîs, Melikşay Resulwan, Myexas, Kordelî Bazgîr, Îlam, Mihran, Wermizyar, Zurbatiya, Kelhor (Şahabad), Xalêsa, Çemçemal, Harasam, Kirmanşa, Qesri Şirin, Sencewî, Arkwaî, Duşeyx, Îwan, Keprat, Mendilî, Serpol, Şerwan Zûrî, Şerwan Kileway.

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The dialectology of Southern Kurdish is not well understood, partly due to socio-linguistic factors, contact, and the history of resettlement, but also due to lack of detailed grammatical descriptions of particular dialect groups. [Fattah \(2000\)](#) classifies Southern Kurdish into seven ‘important branches’, roughly corresponding to the urban localities in which the varieties are spoken:

1. Bîcarî (BIJ)
2. Kolyay (KOL)
3. Lekî-Kurdi (LAK-K)
4. Kelhorî-Sencewî-Zengene (KSZ)
5. Melikşay (MAL)
6. Beyrey (BAD)
7. Kordelî (KOR)

Disregarding the problems with the classifications of Southern Kurdish, a look at [Fattah’s \(2000\)](#) isoglosses suggests a two-way split between Southern Kurdish dialects; a first set of isoglosses separate the northern varieties from the southern varieties (e.g., the northern varieties have the inherited definite plural suffix *-ekan*, the more southern varieties have recruited the collective suffix *-(g)el* in place of the inherited plural *-an*, yielding the definite plural suffix *-(g)el-ek*. There is also a set of features that distinguishes the dialects in the south-west (and less so in the south) from those to the east, e.g., the completive particle *awa* has preverbal placement in the west but post-verbal placement in the east. In both of these isoglosses, the variety of Bijar aligns with opposing areas, pointing to the origin of speakers from the south-western Southern Kurdish speech zone (see [Mohammadirad & Karimi-Doostan in press](#) for details).

The northern varieties of Southern Kurdish have been the most in contact with the vernaculars of NENA. This includes localities such as Bijar and Kerend, though the latter has historically been a Gorani-speaking town, shifting to Southern Kurdish in the recent past.

Southern Kurdish lacks grammatical gender and oblique case, like Sorani, but retains pronominal clitics and exhibits analytic TAM morphology. Its lexicon includes forms absent from both Northern Kurdish and Central Kurdish, and its phonology and syntax bear the imprint of long-standing contact with Persian and Gorani. In Iraq, Arabic influence has also been significant. Structurally, Southern Kurdish occupies an intermediate space between Sorani and Gorani, though it remains genealogically closer to the other Kurdish varieties.

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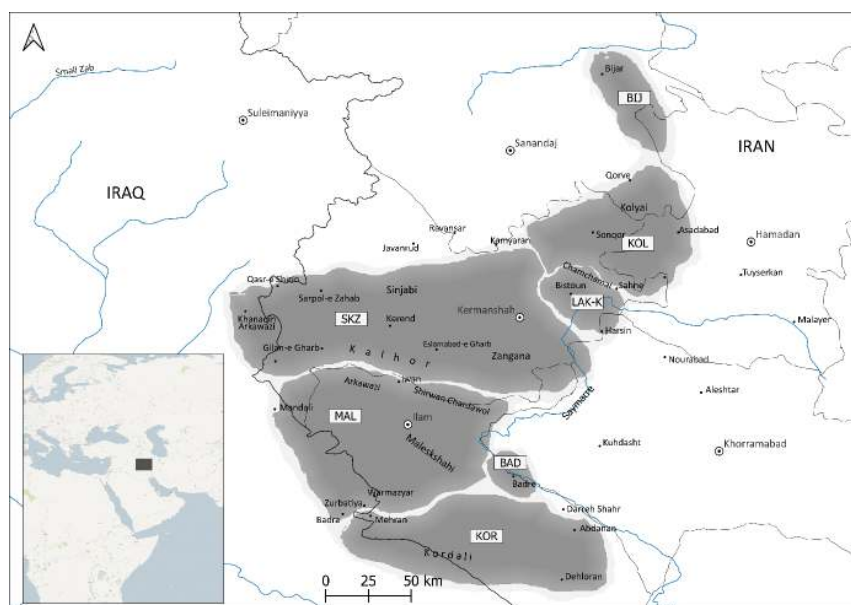


Figure 1.6: Approximate distribution of SK dialect groups (Mohammadi-rad & Karimi-Doostan in press based on Fattah 2000 and Belelli 2019: 80)

1.2.2.1.4 Comparison between Kurdish varieties

The Kurdish varieties exhibit both shared inheritance and substantial divergence arising from historical separation, sociopolitical context, and contact environments. Northern Kurdish is structurally the most conservative, preserving grammatical gender, oblique case, and a highly elaborated *ezafe* system. Central Kurdish aligns closely with Southern Kurdish in loss of gender and case, developing analytic TAM systems, and preserving clitics, reflecting shared innovations likely linked to historical contact with Gorani.

Contact patterns differ markedly: Kurmanji exhibits some influence from Armenian, Neo-Aramaic, and Turkish; Sorani reflects Persian, Gorani, and Arabic influences; and Southern Kurdish displays pervasive influence from contact with Persian and Gorani. These histories shaped divergent phonological and morphosyntactic outcomes.

Dialect continua characterise all three branches. Kurmanji exhibits the strongest internal diversity, with SEK and NWK as divergent poles. Sorani's dialects form a five-way chain with smooth transitions, and Southern Kurdish dialects mirror regional settlement histories. Sociolinguistically, Sorani enjoys the

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strongest institutional support. Kurmanji faces significant pressures of language shift in Turkey. Southern Kurdish remains understudied and less standardised.

Taken together, Northern, Central, and Southern Kurdish form a complex but coherent linguistic family, shaped by geography, migration, political structures, and contact.

1.2.2.2 Gorani

Gorani is a branch of the Iranian languages (Indo-European > Iranian > Central Iranian > north-western Iranian > (Adharic >) Gorani). Contemporary varieties of Gorani are spoken in pockets throughout Iraqi and Iranian Kurdistan. The Gorani vernaculars are generally divided into conservative Hewramî varieties, Text (Takht), Luhon, and Bizlâne, on the one hand, and the peripheral dialects comprising the rest of the dialects in Figure 1.7. As can be seen, Gorani dialects are scattered within the heart of Kurdish areas of Iran and Iraq. In Iran, there are Gorani-speaking communities south of the Hawraman region. The Gorani languages in Iraq are mostly situated to the west of the Hawraman region, from where they are found as far west as the Mosul plain, where they are the vernacular of the communities known as Kaka'î, Šabak, Sarlî, or Bâjaŋânî.

Gorani dialects used to be spoken across a much larger region. MacKenzie (1961b) claimed that the differences between Central Kurdish and Northern Kurdish is due to a Gorani substrate in the former group. The result was that the northern variety of Kurdish preserved its “purity”, while the variety that converged with Gorani, i.e., Central Kurdish, became less “conservative.” The resulting Central Kurdish variety later supplanted Gorani almost entirely, leaving only small pockets of Gorani dialects within a sea of Kurdish. This view is based on the idea that Kurdish as a whole was a continuum of dialects. The substrate account was rejected in Leezenberg (1992) and subsequent scholarship, where it has been argued that the linguistic contact between Gorani and Kurdish at earlier times was in the form of “prestige borrowing” rather than a “substratum hypothesis”. More recent studies, such as Mohammadirad (2025b, *in press*, 2024b), point to a Gorani substrate in the southern Central Kurdish speech zone in recent times since the middle of the 19th century. This hypothesis of a more recent substrate scenario is also supported by accounts of language shift in the region. Mahmoudveysi (2016: 3) reports that when Mann and Hadank Mann & Hadank (1930) conducted fieldwork among speakers in the localities of Bêwânîjî, Rijâbî and Gâhwârâi around Kerend (in western Iran), they were speaking Gorani, but now the speakers have shifted to the vernaculars of Southern Kurdish.

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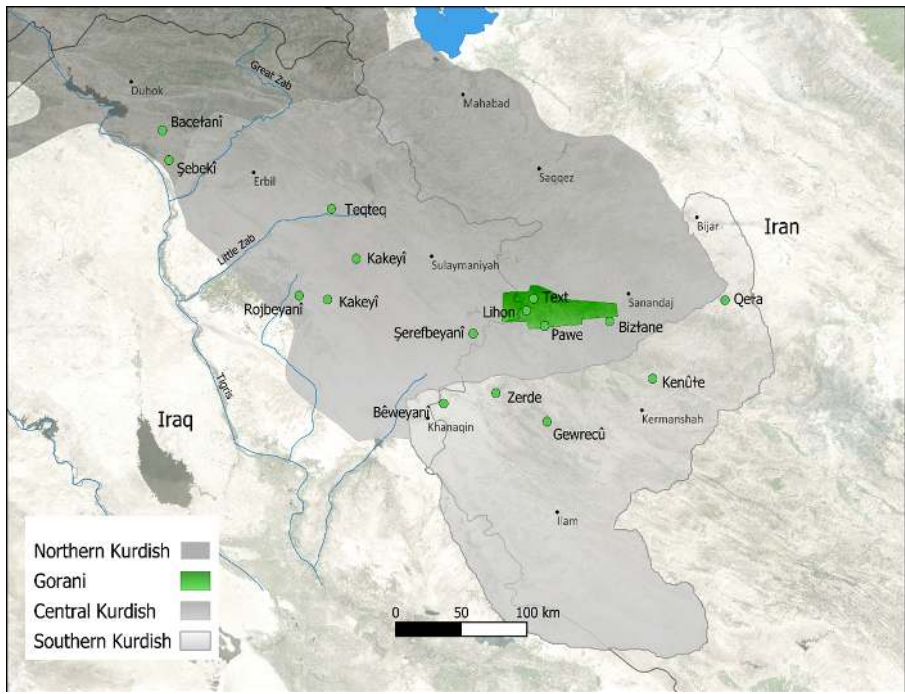


Figure 1.7: Current distribution of Gorani dialects

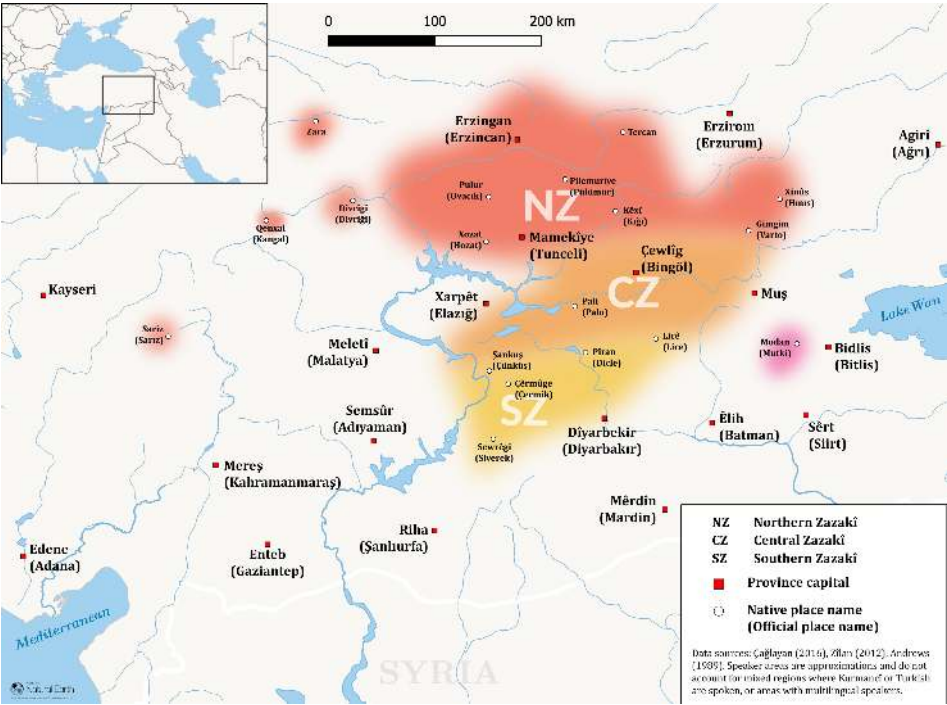
1.2.2.3 Zazaki

Zazaki, also referred to as Dimilî, Kirmanckî, or Kird(k)î, is an Iranian language spoken in the north-western area of Eastern Turkey. The traditional area of settlement stretches from the western districts of Sivas eastwards to Varto, and from Erzincan in the north southwards to Siverek. Today, Zazaki speakers form a majority in the provinces of Dêrsim (Tunceli) and Çewlîg (Bingöl), with additional communities in Elazığ, Palu, Siverek, and Çermik. There are also ‘speech islands’ of Zazaki outside the main territory, most notably in Central Anatolia (e.g., Ak-saray), and in the triangle around Bitlis, Siirt, and Batman.

The total number of speakers is uncertain, with estimates ranging from 1.5 to 3 million, though active speakers are fewer due to a significant language shift. Zazaki is classified as a north-west Iranian language, often grouped under Kurdish together with other languages such as Northern Kurdish (Kurmanji, Kurmançî), Central Kurdish (Sorani) and Gorani, reflecting the fact that most speakers identify themselves as Kurds. Traditionally, three main dialect groups are distinguished: Northern Zazaki (NZ), Central Zazaki (CZ), and Southern Za-

1.2 The languages of the region

zaki (SZ) (see the map in Figure 1.8). Recent research suggests a possible fourth group in the eastern peripheries around Modan (Mutki) and Hezo (Kozluk).



1.2.2.3.5 Contact with NENA

Unlike Northern Kurdish, there is very little direct documentation of contact between Zazaki and NENA dialects. The main geographical zones of overlap historically were in the south-eastern fringes of Zazaki, in areas such as Mardin and the broader Tur ‘Abdîn region, which lie close to the southern edge of the Zazaki-speaking area. It is plausible that in such transitional zones there was at least some degree of multilingualism involving (Central and Southern) Zazaki, Northern Kurdish, Aramaic, Arabic, and/or Armenian. However, direct evidence of contact between Zazaki and NENA is lacking. Given the absence of early documentation, it remains unclear what contact situations at earlier periods may have looked like.

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1.2.2.3.6 Current Status

Today, Zazaki is under intense pressure. In many regions, the language is not passed onto the younger generation, especially in urban settings. Due to migration, many Zazaki speakers now live in western Turkey and in large diaspora communities in Germany, Sweden, and France. Those who remain in the homeland face a gradual language shift towards Turkish. The decline in language use is not uniform across all regions. There is a stark urban-rural as well as age-related disparity. In general, the language decline is more advanced in the northern dialect regions than in the south.

Zazaki speakers are often bilingual, and also have competence in speaking Turkish or Northern Kurdish, depending on region and community. Historical multilingualism in eastern Anatolia likely included Aramaic, Arabic, and Armenian-speaking communities. However, systematic evidence of Aramaic contact is lacking.

1.2.3 Armenian

1.2.3.1 The origin of Armenian

Armenian is an Indo-European language occupying a separate branch of this language family. It was once spoken over a wide area of eastern Anatolia, northern Iran, and the southern Caucasus, as well as in parts of Mesopotamia. The earliest written attestations date from the 5th century CE, but it is known that Armenian populations have been in the area since a much earlier date. It is likely that Indo-European populations associated with the Catacomb culture, a later development from the Yamnaya culture that also shows similarities in material culture with Mycenaean Greece (see [Anthony 2007: 369](#)), crossed the Caucasus from the steppe through the more accessible passes on the eastern edge around 2000 BCE (see e.g., the Lagodekhi chariot burial in Eastern Georgia, [Aslanov 2018](#), [Makharadze 2015](#)). A common origin in the Catacomb culture of the early Bronze Age could explain the proposed (though debated) relationship of Armenian with Greek and Phrygian, as well as its links with Indo-Iranian, and possibly Tocharian. The Lchashen wagons discovered by Lake Sevan in North-West Armenia, produced around 2000-1500 BCE, show similarities in decoration to those of the Catacomb culture north of the Caucasus ([Shishlina et al. 2014](#)), and steppe ancestry appears in the population of Armenia in the same period ([Lazaridis, Alpaslan-Roodenberg, et al. 2022](#)). Pottery styles that were present in Eastern Georgia around 2000 BCE and have been linked to Indo-European settlement spread through Armenia and Erzurum and reached the area of Elazığ around 1100

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BCE, but did not spread further west into Anatolia (Sevin 1991). Despite a tradition going back to ancient authors such as Strabo linking the Armenians with the Phrygians, who are thought to have spread into Anatolia from the Balkans, there is no clear evidence for genetic or cultural links with the Balkans or Western Anatolia (Hovhannisyan et al. 2025, Lazaridis, Alpaslan-Roodenberg, et al. 2022, Sevin 1991). This suggests that the Armenian language probably evolved in the Early Bronze Age in the Southern Caucasus, possibly associated with the Lchashen cultural area, and may have spread to parts of Eastern Anatolia after the collapse of the Hittite Empire around 1150 BCE.

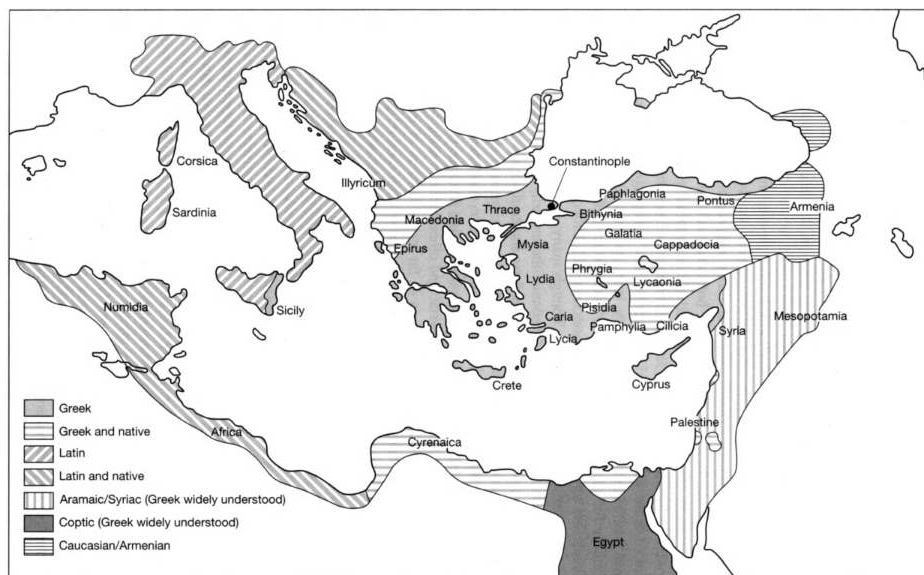
Armenian has a wide variety of dialects, not all of which are mutually intelligible. These are divided into Eastern and Western according to the form of the present tense (see section ??). Many of the Western dialects, which were mainly spoken in the territory of the Ottoman Empire, have become extinct since 1915. Others are only spoken outside their areas of origin. The Eastern dialects are spoken in territories that once belonged to Persia and Russia. Armenian has two standard languages, Standard Eastern Armenian (SEA), based on dialects spoken in the area of Yerevan, and used in Armenia, other areas of the former Soviet Union, in Iran, and in communities originating in these areas, and Standard Western Armenian (SWA), which is considered to have developed from Middle Armenian, which originated as the administrative language of the Kingdom of Cilicia, and was used from the 12th-16th century. SWA also exhibits some influence from the spoken language of Istanbul, but it, like SEA, differs from any form of colloquial spoken language. It is used by communities resident in and originating from the territory of the Ottoman Empire.

1.2.3.2 Armenian and Aramaic

Armenian has been in contact with Aramaic at least since the time of the Achaemenid Empire, in which Aramaic was the main written language. Armenian has some commonly used early Aramaic loanwords, including *berd* ‘fortress’, *xmor* ‘dough’, *kac’in* ‘axe’, *jet* ‘oil’, *mašk* ‘skin’, *c’ec* ‘clothes moth’ (Abajyan et al. 2017: 188). Abajyan et al. (2017: 188) give 26 “early Aramaic” loanwords in Classical Armenian, as well as 92 Syriac loanwords, many of which are Christian religious terms, but which also include some fairly basic vocabulary items such as *tlay* ‘child, boy’, *ar’net* ‘rat’, *xanut* ‘shop’, etc. Greppin (1997: 248) proposes that these words passed into Armenian as a result of the fact that at least part of the Armenian population was bilingual in Aramaic/Syriac. The map in Figure 1.9 showing languages spoken in the Byzantine Empire c.560 CE indicates that Armenian and Aramaic/Syriac were in contact all along

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the southern border of the Armenian-speaking area, before Arabic and Turkic languages became dominant. There were also Armenian communities in many urban areas of Mesopotamia, such as Urfa/Edessa, which had existed alongside an Aramaic-speaking majority at least since Hellenistic times.



Map 2 Linguistic map of the Byzantine empire c.560

Figure 1.9: Horrocks (2010: 208)

There is an Armenian tradition, reported by the 5th-century historian Movses Khorenatsi, that the Armenians of Sasun were of Assyrian ancestry, and while the genetic study of [Hovhannisyan et al. \(2025\)](#) does not indicate that the population of Sasun has closer links with the Assyrians than other Armenian populations, it does show that Armenians and Assyrian Christians are genetically extremely close, as seen in Figure 1.10. It is known that in various areas intermarriage took place between Armenians and Assyrians, and cases have been reported of language shift from Aramaic to Armenian among Assyrian communities in Anatolia, as discussed in the following sections on the individual dialects. Given that the Jacobite Assyrian Church is in communion with the Armenian Apostolic Church, this language shift and intermarriage could have led to integration of Assyrian populations into the Armenian community, especially in Armenian-majority areas such as Bitlis. It is also possible that the

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reverse development could have taken place in Assyrian-majority areas.

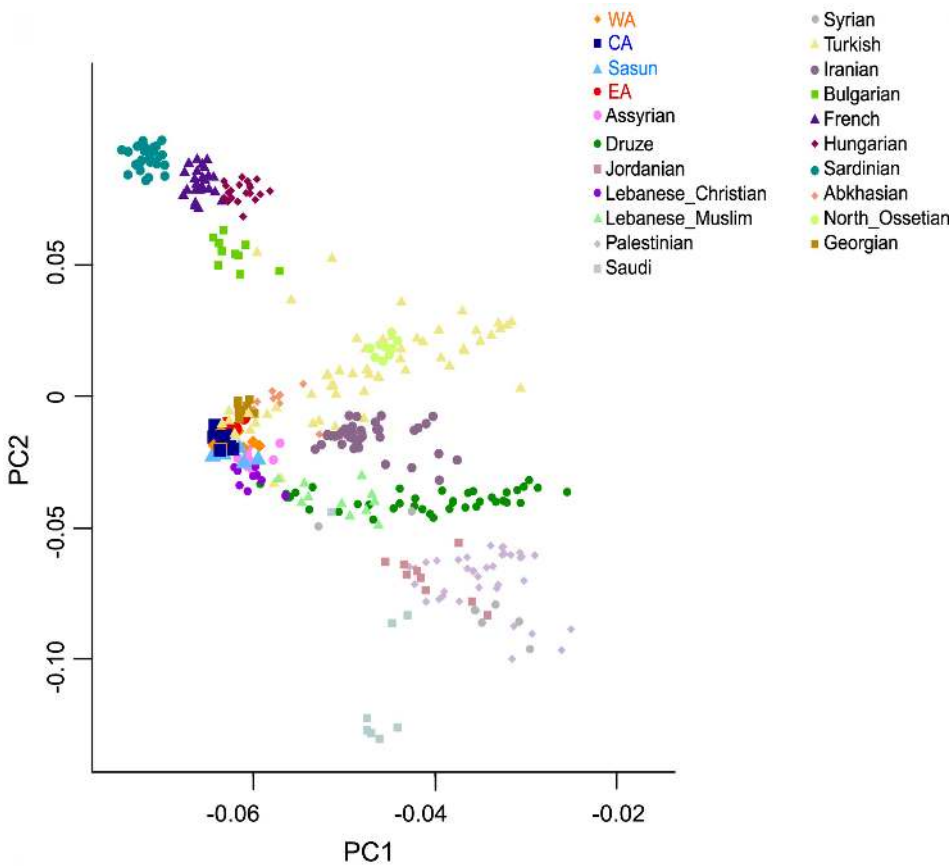


Figure 1.10: PCA analysis based on haplotype sharing showing Western Armenian (WA), Central Armenian (CA), Eastern Armenian (EA), Sasun Armenian, and other populations, including Assyrians (Hovhannisyan et al. 2025: 17)

Dialectological studies carried out in the early 20th century show that Armenian was once spoken as far south of Lake Van as Shatakh (Çatak) and Hertevin (Armenian Ardvan), as well as all along the north and west shores of Lake Urmia, as seen in Figure 1.11. There were Armenian communities further south still, for example, in Țur ‘Abdin and in parts of Hakkâri, as seen in Figure 1.12, a map showing the percentage of Armenians and other Christians in various regions of Eastern and Central Anatolia in 1896, but by the early 20th

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century, these communities spoke only Kurdish or Arabic (Acharyan 1952: 12). There are Armenian communities in the area of Zakho, notably the villages of Avzrog and Havresk, but these only date from the early 20th century, and were settled largely from the Van and Urmia areas (Tadros et al. 2025: 33).

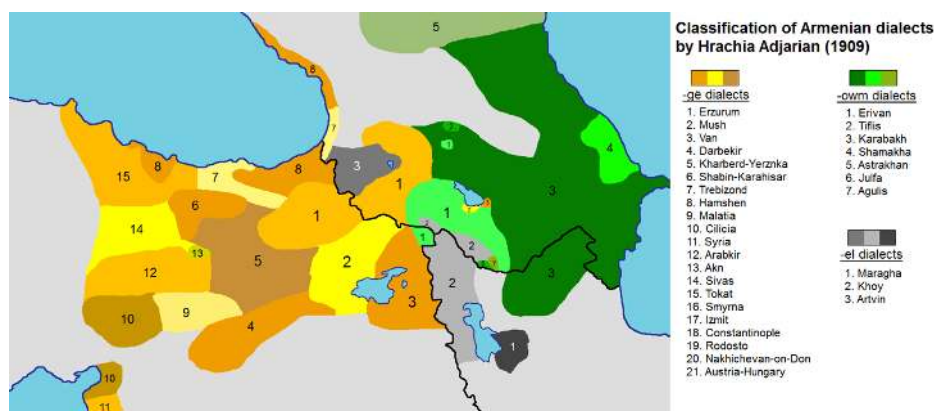


Figure 1.11

Historically, the dialects that were in contact with NENA were those of the Urmia, Van, and Mush groups. The NENA-speaking communities in Armenia, which have been present since around 1828, are also in contact with SEA, although some are in areas where the Armenian population mainly speaks Urmia dialect, having migrated from Iran to what was then Russian territory at the same time as the NENA-speakers following the treaty of Turkmencha in 1828.

1.2.3.2.7 Urmia dialect

The dialect here referred to as Urmia dialect is also referred to as Khoy dialect (as in Figure 1.11); the grammar of Asatryan (1962) is entitled Urmia (Khoy) Dialect, as the author establishes that essentially the same dialect was spoken by Armenian communities from Khoy to Urmia, with the dialect of Salmast belonging to the same group, although possessing some distinctive features, as discussed by Vaux (2022). It is known that there was, and still is, intermarriage between NENA-speaking Christians and Armenians in the Urmia area, and bilingualism between Urmia Armenian and Christian Urmi NENA (Rashel Pakbaz p.c.). Such bilingualism also exists in NENA-speaking villages in Armenia, where speakers of Christian Urmi NENA and Urmia Armenian settled in many of the same areas. Urmia dialect is widely spoken in Armenia today, mainly in rural communi-

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Figure 1.12: Map showing percentage of Armenian and other Christian populations in Anatolia (1896)

ties, but the younger generation may show less proficiency. There are also some speakers still resident in Iran.

This dialect is classed as an Eastern Armenian dialect based on the form of the present tense (see §??), but in terms of phonology, it is similar to the neighbouring Van dialect (Asatryan 1962), with devoicing of initial voiced stops and aspirates (see section 4), Acharyan's law (see §7), $h > x$, and extensive palatalisation of dorsals (see section 5). Its verbal system is unique within Armenian, making use of a past-converting particle identical to the 3sg past copula, as in many NENA dialects, including Christian Urmi (see §??). It is also one of the few Armenian dialects to possess object clitics, a feature it shares with some other dialects spoken in Iran (see §??).

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1.2.3.2.8 Van dialect

Van dialect was originally spoken around the eastern part of Lake Van, including the city of Van. [Jahukyan \(1972\)](#) considers Shatakh and Moks to be subdialects of the Van group, although [Muradyan \(1962\)](#) details the differences between Shatakh dialect and that of Van itself. Today, Van dialect is hardly spoken, although it is said to have survived in a small number of communities in Armenia, including Lernanist village and Atarbekyan, now part of the town of Hrazdan, in Kotayk province.

Van is considered a Western Armenian dialect based on the form of the present tense, but it differs considerably from SWA, sharing some features with the neighbouring Eastern dialects, for example, voiceless unaspirated consonants, and a mobile copula. Phonologically, it is very similar to Urmia dialect, with devoicing of initial voiced stops and aspirates (see §4), Acharyan's law (see §7), $h > x$, and extensive palatalisation of dorsals (see §5). However, the verbal system lacks the innovations characteristic of Urmia dialect (past particle, object clitics), as well as differing in the form of the present tense.

1.2.3.2.9 Mush dialect

Mush dialect was spoken in and around the town of Mush, and in a large area around the western part of Lake Van. The Wikipedia page for Hertevin mentions (without giving a source) that Mush dialect was spoken by the Armenian community of the area, which is consistent with the distribution shown in Figure 1.11. It is known that Armenians and Jacobite Assyrians were in contact in Bitlis, where [Lynch \(1901: 150\)](#) reports intermarriage, and language switch of the Assyrian community to Armenian, presumably to the local dialect, which is considered to form part of the Mush group ([Jahukyan 1972](#)). Also part of the wider Mush group according to [Jahukyan \(1972\)](#), but showing considerable differences from the dialect of Mush itself, are the dialects of Sasun ([Petoyan 1954](#)) and Tigranakert (Diyarbakir) ([Haneyan 1978](#)).

Mush dialect, like Van, is considered a Western Armenian dialect based on the form of the present tense, but shares some features typically associated with Eastern Armenian, such as the presence of voiceless unaspirated consonants and a mobile copula. Mush and Van also share the property of making more extensive use of prepositions than most modern Armenian dialects. However, Mush and Van dialects are separated by Acharyan's law (absent in Mush) and Weitenberg's

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isoglosses (Weitenberg 1996, 1999), implying that they developed separately in the early stages of the formation of the modern Armenian dialects.

1.2.4 Arabic dialects of the NENA region

Historically the Semitic language Arabic, like NENA, was a minority language in the NENA-speaking region. Where it was spoken, it was primarily used by religious minorities, namely communities of Jews, Christians and Yezidis. It should be noted that it is not the only language of these minorities: other Jewish and Christian communities in the region speak NENA, while other Yezidi communities speak Northern Kurdish. In and around the city of Mosul, located on the edge of the NENA-speaking region, Arabic is also spoken by Muslims.

During the late 20th century, the Baath regime conducted a policy of Arabicisation, which led to the relocation of large numbers of Muslim Arabs to northern Iraq. The most prominent target was, however, a city just outside the NENA-speaking region, namely Kirkuk, but many of the Arabs left the region after 2003 (Procházka 2019: 244). Further Arab settlement has occurred as a result of internal displacement from north-western Iraqi cities during their occupation by ISIS (2014–2017). These include refugees from Mosul who settled in Erbil/Hawler (Zupruk et al. 2018: 6).

The settlers and refugees of these two waves, coming from diverse regions, brought with them a wide variety of Arabic dialects. Despite their presence, however, competence in Arabic has overall declined in Iraqi Kurdistan since it gained autonomy in 1991 (Walker 2013; Procházka 2019). In this work, only the original Arabic dialects of the region will be covered, these having had more time potentially to exert influence on the NENA varieties which are the focus.³

The variety of Arabic spoken by long-standing communities in the NENA-speaking region is *Qeltu* Arabic (also known as Northern Mesopotamian Arabic). It is distinguished from the other type of Mesopotamian Arabic, *Gilit* Arabic, which is spoken by both settled and nomadic communities in central and southern Iraq and by nomads (and those of nomadic origin) elsewhere (Talay 2011: 909–910)). There seem historically not to have been any significant numbers of Arab nomads in the NENA-speaking region, thus it appears that only *Qeltu* Arabic is represented here. The dialects of Arabic in Iraq were first categorised into these two groups, *Gilit* and *Qeltu* dialects, by Blanc (1964: 5–8). They were named according to the shibboleth of the verbal form for ‘I said’ (Old Arabic **qultu*), but this is just one of many features distinguishing the two groups. The distribution

³The Jewish dialects were no longer spoken in Iraq, when these two waves of Arab settlement occurred, so could not have been affected by them.

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of the two dialect types is conditioned by geographical, social and communal (religious) factors. The *Qeltu* Arabic dialects are spoken in a much larger region than NENA is: not only in northern Iraq but also in adjacent regions of Turkey and Syria. In these regions *Qeltu* Arabic speakers include Muslims, Jews and Christians. In central-southern Iraq, by contrast, *Qeltu* Arabic is exclusively spoken by Jews and Christians, the Muslims in central-southern Iraq speaking *Gilit* dialects.

Most of the *Qeltu* dialects mentioned in Jastrow's groundbreaking (1978) survey or described in subsequent grammatical descriptions are not spoken in close proximity to NENA. As this study focuses on language contact, such dialects will not be considered here. The *Qeltu* Arabic dialects included here are those spoken within the outer boundaries of the NENA region—even if not in immediate proximity to any specific NENA variety—as well as those located just beyond these borders, where historical contact with NENA is plausible. Several additional dialects that meet these criteria are known to exist or have existed but remain undescribed and, therefore, cannot be analysed. These include the Jewish dialects of Kufri, Šoš and Cizre (Jastrow 1990b: 201), the extinct Christian dialect of Cizre (Jastrow 1978: 17), and the *Qeltu* variety (or varieties) spoken by Yezidis, Muslims and Christians in Ba'šīqa (Jastrow 1978: 24).

The dialects which meet the criteria and for which information exists, which will, therefore, be referenced in this study, are listed below.

1.2.4.1 Inside the borders of the NENA region

- Mosul (Muslims, Christians, Jews). Minor dialectal differences have been identified between the communities (Jastrow 1990b: 141-142). Data is available in Jastrow's (1978) survey and in short articles, consisting mostly of transcribed texts: mainly from Christians in Socin (1878, 1882, 1883), from Muslims in Jastrow (1979) and from Jews in Jastrow (1989, 1991a, 2001, 2005). An overview of the communal distinctions is provided in Jastrow (2004). Blanc (1964) also includes some information on the dialects of Mosul. Procházka (2019) treats mostly the (Muslim/Christian) dialect(s) of Mosul and covers some syntax, but is also brief.
- Bəḥzāni (Christians, Yezidis). Sparse data is available from Jastrow's (1978) survey, a lexicographical article (Jastrow 1982) and some notes in Hassan (2023). Jastrow's published texts in the dialect (Jastrow 1981: 372-417) are all recorded from Christians (Jastrow 1981: 374), but he makes no mention of any communal distinction in the dialect.

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- Mērgi (Christians) (Hassan 2023).
- Səndōr (Jews) (Jastrow 1991b, 1993).
- Arbīl and ‘Aqra (Jews) (Jastrow 1990a).
- Khānaqīn (Jews) (Bar-Moshe 2024).

1.2.4.2 On the periphery of the NENA region

- Āzəx (Christians, modern name İdil) (Wittrich 2001).
- Siirt town (only Muslims, before the First World War Christians too) (Jastrow 1978: 15-16 and various sections of the book); Biṭunā (2016a, 2016b).
- Siirt villages (only Muslims) (Jastrow 1978, 15–17 and various sections of the book), represented here by Tillo (Muslims) (Lahdo 2009).
- Kirkūk (Jews) (Bar-Moshe 2024).⁴
- Tūz Khurmātu (Jews) (Bar-Moshe 2024).

Information on Jewish dialects of Iraq is also found in Jastrow (1990b).

1.2.5 Turkic languages of the NENA region

1.2.5.1 Origin of Oghuz Turkic

The spread of Oghuz Turkic in Western Asia dates back to the eleventh century, when the Seljuq Turks conquered the region and established a long tradition of Turkic-speaking elite rule, which, with only brief interruptions, persisted for nine centuries and ended in the 1920s with the fall of the Ottoman dynasty in Turkey and the Qajar dynasty in Persia. During the Early Modern Period, the spread of Oghuz Turkic accelerated across the region. In Anatolia and the area north-west of the Iranian Plateau, historically known as Azerbaijan, and its adjacent areas, Turkic speakers achieved dominance not only politically but also demographically, through both migration and the assimilation of the local population. However, due to constant population movements, motivated by administrative or military objectives, Turkic-speaking communities became established

⁴Christian speakers of NENA live in Kirkuk but these families originated from elsewhere, e.g., Alqosh and ‘Ankawa, having arrived mostly since around 1920 to work for the oil companies. The original Christians of Kirkuk spoke and still speak Turkmen (Kawther Askar, p.c.).

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far beyond these core areas, extending into other parts of present-day Iran, as well as present-day Iraq and the Levant, and ranging from large, concentrated populations to smaller, isolated pockets.

Old Anatolian Turkic is the earliest written attestation of Oghuz and the earliest Turkic language of diwan-style poetry, dating to the thirteenth century. Despite its name, its area spanned the entire Seljuq realm, reaching as far east as Khorasan. By the fourteenth century, its westernmost dialectal area formed the basis of Ottoman Turkish, while much of the Old Anatolian Turkic literary tradition evolved in the fifteenth century into Middle Azeri, also known as Ajem Turkic, the literary Turkic language of Early Modern Persia (Bellér-Hann 1995: 33; Stein 2014a).

1.2.5.2 Classification

The Oghuz varieties spoken in Iran can be divided into West Oghuz, represented primarily by Azeri, and East Oghuz, represented by Turkmen, as well as Khorasani Turkic, a variety that may be described as transitional but leaning toward West Oghuz (Knüppel 2000). For the purposes of examining contact phenomena involving NENA, only West Oghuz varieties will be considered in this volume.

The classification of modern spoken West Oghuz varieties has itself been the subject of debate, due in part to their far-from-exhaustive description and documentation. While the distinction between West and East Oghuz is largely clear because of their discontinuous geographical distribution and more distinct grammatical features, identifying and labelling clear-cut dialect clusters within West Oghuz is problematic. This is due to the fact that Balkan, Anatolian, Levantine, Iraqi, Caucasian, and Iranian varieties of Oghuz Turkic are spoken across a continuum characterised by centuries of high population mobility, shifting alliances and a shared Islamicate Turco-Iranian cultural heritage (Bulut 2014; Anonby et al. 2020). It is generally agreed, however, that major isoglosses—such as the maintenance vs. drop of word-initial *y*, Kipchak influence on morphophonology, the form of singular and first-person plural copulae, the form and function of indirective TAM categories, lexical and grammatical borrowings from Chaghatay, complement clause constructions, and ubiquitous Iranian substrate influence—lie across the historical borders of the majority-Sunni Ottoman Empire and the majority-Shia Safavid Empire. This border, however, shifted repeatedly until the mid-eighteenth century, e.g., Eastern Anatolia, most of which passed from Safavid to Ottoman control by the mid-sixteenth century, or the Caucasus, which was briefly under Ottoman control in the sixteenth and again in the eighteenth

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century. The varieties are nowadays divided into those that can be loosely described as the “Anatolian” and “Azeri” types, respectively.

Some scholars have argued for a more fine-grained clustering of West Oghuz, proposing distinctions such as West (Anatolian), Central (Caucasus and Northern Iran) and South (Central and South-western Iran, Iraq, as well as Afghanistan, where some descendants of migrants from Western Iran from the first half of the eighteenth century continue to speak a West Oghuz variety) sub-branches of Oghuz (Doerfer 2006) or grouping the Turkic varieties of the Caucasus and Northern Iran with Anatolian Turkish into a single sub-branch distinct from South Oghuz (Bulut 2021). The criteria for these classifications are not always clear. For instance, Bulut (2021: 290) ascribes Turkish and Azeri to West Oghuz while classifying “South-east” Anatolian varieties (i.e., those located in the transitional zone between Turkish and Azeri) as South Oghuz. Interestingly, in an earlier work, the same author concludes that there are no criteria that would allow for a positive delimitation of a South Oghuz sub-branch, given that phenomena seen as characteristic of its supposed area almost always occur outside of it, including in regions located quite far away (Bulut 2014).

1.2.5.3 Naming and Geography

The terms “Azeri” and “Azerbaijani” first appeared in scientific literature in the mid-nineteenth century to designate the Oghuz Turkic varieties of the Caucasus and Iranian Azerbaijan, previously referred to as Ajem Turkic or more ambiguously as Turco-Tatar, and locally as *türki* ‘Turkic’. They gained popularity in the twentieth century when *Azərbaycan dili* ‘Azerbaijani language’ became the official name of the language in the Republic of Azerbaijan and came to be widely adopted by speakers in the post-Soviet realm (and communities tracing their origin to it) to refer to their language.

In Iran, Turkic varieties continue to be commonly referred to as *türki* ‘Turkic’, with *azəri* ‘Azeri’ used only occasionally and primarily by educated urban speakers with origins in Iranian Azerbaijan with regard to their heritage language (Bulut 2018: 404-405).

In Iraq, Turkic speakers prefer to call their language *türkməncə* ‘Turkmen, Turcoman’ (Bulut 2018: 398). The use of the terms “Turkmen” or “Turcoman” in a West Asian context may cause confusion among non-specialists, as they refer to West Oghuz varieties, which are no more closely related to the Turkmen language of Turkmenistan than Turkish or Azeri are. The naming ambiguity stems from the medieval practice of using “Turkmen” or “Turcoman” as a general

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designation for non-Ottoman Oghuz Turks, rather than as an ethnolinguistic identifier tied exclusively to the Turkmen of Central Asia.

In this volume, the term “Azeri” is used to refer to the West Oghuz variety spoken across the continuous area covering the Caucasus and Iranian Azerbaijan (including the provinces of West and East Azerbaijan, Ardabil, and Zanjan, as well as dialectal outliers and migrant communities outside this area), with roughly 22 million speakers. All other varieties of the Azeri type will be referred to by their geographical or demonymic appellation (e.g., “Qashqai”, “Eastern Anatolian”, “Iraqi Turkic”, etc.). For the purposes of this volume, the Turkic varieties under study encompass all Oghuz varieties of the Azeri type—i.e., Azeri in the broadest possible sense of the term—which cover the following areas and total roughly 35 million speakers:

- Western Iran: provinces with a West Oghuz-speaking (absolute or relative) majority, i.e., West and East Azerbaijan, Ardabil, Zanjan, Hamadan; provinces with a significant West Oghuz-speaking minority, i.e., Gilan, Kurdistan, Kermanshah, Markazi, Qazvin, Qom, Alborz, Tehran, Semnan, Isfahan, Lorestan, Chahar Mahal and Bakhtiari, Fars, Kohgiluyeh and Boyer-Ahmad, Bushehr, Khuzestan, Yazd, and Kerman; and provinces with isolated West Oghuz-speaking communities, i.e., Mazandaran and Razavi Khorasan (see [Anonby et al. 2020](#) for further details);
- Caucasus
 - Azerbaijan;
 - Georgia: mainly the provinces of Kvemo Kartli, Kakheti and the capital Tbilisi, and to a lesser extent the provinces of Mtskheta-Mtianeti and Shida Kartli, prior to World War II also Samtskhe-Javakheti;
 - Dagestan (an administrative unit of Russia bordering Azerbaijan): mainly the districts of Derbent, Tabasaran, Rutul, and Kizlyar, and the capital Makhachkala);
 - until the fall of the USSR, Armenia: throughout;
- Iraq: the governorates of Arbil, Kirkuk, Nineveh, Saladin, Diyala, and Wasit;
- Eastern Anatolia: the provinces of Turkey east of the Euphrates and south of the Pontic Mountains (per the classification of [Karahan 1996](#); in some of them Turkic speakers form a minority), i.e., Gümüşhane, Bayburt,

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Erzincan, Erzurum, Tunceli, Elazığ, Urfa, Diyarbakır, Bingöl, Muş, Bitlis, Van, Ağrı, Kars, Iğdır, Ardahan, Artvin; some of these areas have Turkic-speaking populations descended from nineteenth- and twentieth-century migrants from the Caucasus and Iran who speak “Azeri proper”;

- Afghanistan: around Kabul.

1.2.5.4 Dialects

Classifying the varieties of Azeri-type West Oghuz is problematic for the same reasons as the classification of Oghuz varieties in general: complex migration and settlement patterns, regional convergence and divergence, and poor documentation. Varieties spoken in Azerbaijan and in Eastern Anatolia are relatively well studied, and their classifications, established by Şirəliyev (1967a) and Karahan (1996), are generally accepted. The classification of the varieties spoken in Iran found in the literature (Əzizov 1999: 289; Doerfer 2006; Johanson 2009) is almost never as detailed as that of the varieties spoken in the Caucasus and is often based on modern-day administrative divisions, which do not necessarily correspond to dialectal isoglosses. For instance, Anonby et al. (2020: 302-303) report that in Chahar Mahal and Bakhtiari, which has an estimated 160,000 Turkic speakers, a study carried out in just eight villages already made it clear that the Turkic varieties are not linguistically uniform. Still, it is possible to identify at least groups of dialects. While it is common to speak of the Azeri dialect clusters in terms of “North Azeri” and “South Azeri”, i.e., divided along the former Soviet–Iranian border, a quick look at the isoglosses reveals a pattern of dialectal differentiation that runs, at least in the Azeri proper zone, more strongly east–west than north–south. Thus, the Eastern dialect group, which includes varieties spoken along the Caspian shore in the Caucasus as far north as Derbent, continues into Iran and encompasses Ardabil and possibly Sarab, while the South-western group, which includes the varieties of Naxçıvan and Yerevan, also extends to Khoy and Urmia in Iran. The North-western dialect group, frequently referred to in this volume for comparative purposes, includes the varieties of the right bank of the Kura River, stretching from southern Georgia into Karabakh. The dialects of Şəki and Zaqatala–Qax are either grouped into a separate Northern group (following Şirəliyev 1967a) or described as transitional between the Eastern and North-western groups, aligning more with the latter (following Əzizov 1999).

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1.2.5.5 Status

Historically, Azeri was the language of the Safavid and, later, Qajar courts. As the language of the Qizilbash tribes, it long served as the language of the Persian army. It also functioned as a vehicular language of economic exchange throughout the Safavid realm, a role it to some extent retains to this day in Tehran (Bulut 2018: 401). For centuries, it acted as the lingua franca across northern Iran, much of the South Caucasus, southern Dagestan, Eastern Anatolia, and Kurdistan. During this period, it developed a substantial literary tradition (although not as extensive as that of Persian, Ottoman Turkish or Chaghatay), which declined in Iranian Azerbaijan in the eighteenth century but continued to flourish in the Caucasus (where Azeri surpassed Persian as the main language of literary expression), including in non-Turkic-speaking parts of southern Dagestan, where Azeri is attested in some of the earliest known poetic compositions. Following the demise of the Safavids, Azeri also became, alongside Persian, the main administrative language in the khanates north of the Araxes River, a status it maintained until the Russian conquest in the first decades of the nineteenth century.

Today, Azeri has the status of the official language of the Republic of Azerbaijan (as *Azərbaycan dili* ‘Azerbaijani language’, lit. ‘language of Azerbaijan’), where the literary norm is based on the Eastern dialects, mostly that of Şamaxı. The language was historically written in the Perso-Arabic script but switched to the Latin alphabet in the early 1920s, at the onset of the Soviet period, becoming one of the first languages of the Islamic world to do so. The Latinisation project was initially developed in 1919, during the first period of Azerbaijan’s independence (1918–1920), but was never officially adopted. Following the cooling of the relations between the USSR and Turkey, Azeri, like all Turkic languages of the USSR, switched to the Cyrillic alphabet, which remained in official use until 1991 (and in semi-official use until 2001), when the Latin alphabet (a modified version of the 1920s alphabet, much closer to the modern Turkish alphabet) was reintroduced.

The literary standard of Azerbaijan is used by the Azeri-speaking population of Georgia and Dagestan, in neither of which does Azeri have an explicitly defined official status (the Constitution of Dagestan designates “the languages of the peoples of Dagestan”, without naming them, as co-official alongside Russian). However, schools with Azeri as the medium of instruction (in Georgia) or with Azeri taught as a subject (in Dagestan), as well as state-funded Azeri-language media and cultural activities, do exist as a legacy of Soviet language policy, which promoted the official languages of the constituent republics in regions outside them where their speakers formed substantial communities.

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Azeri has no official status in Iran, where most of its speakers reside, although the written form of Azeri-language publications in Iran is based on the Tabriz variety or Istanbul Turkish. When in Azeri, the language continues to be written in Perso-Arabic script (Bulut 2018: 405), and many speakers are now familiar with the Latin alphabet through exposure to both Turkish and Azerbaijani media.

The current Constitution of Iraq recognises “the language of the Turkmen” as a minority language. In practice, however, the written standard of Iraqi Turkic is Istanbul Turkish, the literary norm of Turkey, which retains a strong cultural presence among the Turkic-speaking population of Iraq (Bulut 2018: 357).

The remaining “Azeri-type” Oghuz varieties exist solely in oral form.

1.2.5.6 **Language maintenance**

Despite their wide historical diffusion as both a first and second language and their considerable resilience over time, Azeri and related varieties entered a period of reduced prestige in the twentieth century across much of their historical range, as they increasingly came to compete with Persian in Iran and Russian in the Caucasus in the domain of administration and as a means of intercommunal communication.

In Azerbaijan, although Azeri was an official language throughout much of the Soviet period, its speakers were outnumbered in the capital city of Baku by Russian speakers, and Russian came to dominate administration while competing with Azeri as the language of public life, despite subtle but persistent resistance from the Azeri-speaking intelligentsia. It was not until the re-establishment of Azerbaijan’s independence in the 1990s that Azeri regained its role as the uncontested working language of the country.

Outside Azerbaijan, in areas where Azeri has no formal status, it has undergone varying degrees of replacement by other languages.

In Iran, with the rise of the Pahlavi dynasty, the consolidation of Persian as the sole national language and the language of universal education became a priority, leaving little room for other local languages to thrive. As a result, Azeri was relegated to the status of a low-prestige language, and its public use—or even the use of Persian with an Azeri accent—became associated with poor education (Bulut 2018: 405; Zandi 2023: 184). Despite legal provisions supporting minority languages in Iran, Turkic speakers have already begun to show signs of a shift toward Persian (Bani-Shoraka 2005; Anonby et al. 2020: 302), including in areas where they historically constitute a majority (Hawes & Mirvahedi 2013; Zandi 2023).

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In Iraq, Turkic varieties have relatively low prestige, and their speakers typically acquire Arabic and/or Kurdish. Nevertheless, Turkic continues to function as a regional language of communication among speakers of other low-prestige languages, including NENA (Bulut 2018: 357).

1.2.5.7 Turkic and NENA

Turkic–NENA contact across much of the area occurred via Kurdish as an intermediary, though not always. Direct contact between Turkic and NENA took place around Lake Van, where it nonetheless remained marginal, and in the area around Lake Urmia, where it was particularly strong. In Miandoab, contact had been so intense that the previously NENA-speaking Jewish community had shifted to Azeri as its spoken language by the beginning of the twentieth century (Hopkins 1999b: 317). Azeri literary and musical culture enjoyed popularity among Christian NENA speakers from Urmia, including after their departure from Persia. For a long time, the 1913 musical comedy *The Cloth Peddler* by Üzeyir Hacıbəyov remained the most frequently staged theatrical work in Assyrian communities in both the Soviet Union and the United States (Naby 2018: 507).

The South Caucasus attracted NENA-speaking Jews as early as the late eighteenth century, owing in part to the presence of established indigenous Jewish communities such as that of Quba, and, from the last quarter of the nineteenth century onwards, as a result of the rapid economic expansion of Baku, the future capital of Azerbaijan, where their settlement continued well into the early Soviet period. In Baku, NENA-speaking Jews established their own synagogue, where services were reportedly conducted in Aramaic as late as the 1930s. In the same decade, however, the community declined sharply following Soviet policies that resulted in the transfer of many of its members to Kazakhstan or, in the case of those without Soviet citizenship, their deportation to Persia. In Quba, the community, which traced its origins to Saqqez and was concentrated in the Ağacani neighbourhood, had largely assimilated into the local Iranian-speaking Mountain Jewish population, although speakers of Jewish NENA could still be found there as late as the 1950s (Krupnik & Kupoveckij 1988).

By contrast, Christian NENA speakers who moved into the Russian-controlled Caucasus settled in linguistically mixed areas, most of which are located in present-day Georgia and Armenia, with the exception of Siyaqut, the only Chaldean village in the Caucasus, which is located in the present-day Autonomous Republic of Naxçıvan in Azerbaijan. The Chaldean population of Siyaqut left the region soon after the fall of the Russian Empire. A major

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migratory wave of Christian NENA speakers into Azerbaijan occurred in the aftermath of World War I, when thousands of Christian refugees from Bohtan arrived in the Caucasus and settled in the German-populated colonies in the western part of the country, mainly in Helenendorf (present-day Göygöl, called Xanlar in the Soviet period) and Annenfeld (present-day Şəmki, called Şamxor in the Soviet period). In 1949, after a relatively stable early Soviet period during which the community even briefly benefited from state-funded schools with NENA as the language of instruction, the central authorities in Moscow ordered its deportation to Tomsk Oblast in Eastern Siberia, on the grounds that its members were still legally Turkish subjects and, with Turkey now a member of NATO, were considered “undependable”. Very few of them subsequently returned to Azerbaijan.

1.2.5.8 Urmia Azeri

In the description of grammatical phenomena relevant to the scope of this volume, priority is given to Urmia Azeri, which is in closer contact with NENA than any other Azeri variety. The overview relies heavily on [Doğan \(2010\)](#), a PhD dissertation on Urmia Azeri defended at Gazi University and the most detailed description of the variety to date.

It remains unclear whether Urmia Azeri constitutes a single homogeneous variety, as data collected even across the author’s sample of 36 locales in Urmia County reveal noticeable morphological variation.

Among the characteristic features of Urmia Azeri is a series of consonant shifts affecting dorsals, typical of South-western Azeri varieties (and extending as far as Kirkuk) and of non-Turkic languages of the region more generally (including NENA). Here, historical velar stops, already shifted to palatals in most Azeri varieties, are further fronted to postalveolar affricates /tʃ/ and /dʒ/, while historical postalveolar affricates are in turn fronted to alveolar affricates /ts/ and /dz/.

Bilingualism remains common in the Urmia region, and Azeri continues to function as a language of communication between speakers of different languages.

1.3 Methodology and terminology

Each chapter opens with a detailed overview of a particular feature of phonology, morpho-syntax or syntax across the entire NENA area. In most cases the feature in question manifests considerable diversity across the many dialects of the NENA area. Much of the data on NENA dialects has not been published

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elsewhere and is derived from fieldnotes and also data in the *North-Eastern Neo-Aramaic Database* (<https://nena.ames.cam.ac.uk/>), which has been compiled by research teams in Cambridge.⁵ The sections on NENA in the chapters are followed by sections on Iranian languages, Armenian, Arabic and, where relevant, Turkic, which present comparative data from these contact languages. This data includes phonological data and linguistic structures that match the NENA data and also cases where the exponence of a particular feature in a contact language is different in some way from the corresponding feature in NENA. As with the sections on NENA features, in many cases the data in these sections on contact languages has not been published previously and is drawn from fieldnotes of the authors. Each chapter closes with a section that discusses the correlations between the NENA feature in question and the corresponding feature in the contact languages and the extent to which NENA has undergone contact-induced change. The final chapter of the volume presents some concluding remarks about the contextualisation of NENA dialects within the language area of Western Asia and the processes of contact-induced change that they exhibit.

A few remarks are in order on the terminology we use in book in connection with contact-induced change. The term “borrowing” is generally reserved for “matter borrowing” (Matras & Sakel 2007; Matras 2009), which involves the copying of lexical or morphological material from a contact language to NENA. Since this volume does not include a study of loanwords, matter borrowing is only marginal to our main focus, which is concerned with the patterns of phonology, morpho-syntax and syntax. Where NENA develops innovations in structural patterns under the influence of contact languages, we use terms such as “replication” and “convergence”.

“Replication” (the term originates in Weinreich 1953: 30-31) denote processes of copying that do not involve phonetic substance. These could involve the copying of semantic patterns or the copying of the syntactic ordering of elements. They typically involve the extension of existing patterns to new contexts and, in some case, grammaticalisation (Heine & Kuteva 2010).

The term “convergence” is a scalar term and is used to refer to various degrees of approximation of patterns and systems of NENA with those of contact languages. It is typically used where the convergence of the internal NENA feature with the external feature of the contact language is not complete but only partial. This may result in the replica feature being less grammaticalised than

⁵Reseachers who have contributed so far unpublished data to the database from their fieldwork include Geoffrey Khan, Eleanor Coghill, Roberta Borghero, Lidia Tuwalska-Napiorkowska, Paul Noorlander and Hezy Mutzafi.

1.4 Transcriptions systems

the corresponding feature in the model language (Heine & Kuteva 2010: 94). Our use of the term “convergence” is in most cases unidirectional, i.e., we discuss the convergence of NENA with contact languages. We occasionally use the term to refer to the mutual approximation of the structures of two languages in contact.

In various places, features of NENA are said to “match” features in a contact language. This reflects a process that lays the ground for convergence and replication, whereby a particular feature in a contact language is perceived to correspond to a particular feature in NENA. This process is equivalent to what Matras & Sakel (2007) call “pivot matching” in the replication of syntax or morpho-syntax (cf. also Matras 2009: 240-43; 2010: 71-72). The process of pivot matching involves identifying a structure in the model language as the equivalent of a structure in the replica language and reorganising the inherited structure in the replica language in terms of grammatical and semantic meaning, and also distribution, to replicate those of the structure in the model language. In intense contact situations, the mechanism of pattern replication is characterised by adapting meanings and functions of inherited structures and enhancing them to carry out organisational procedures that are replicated from the model language (Matras 2009: 238), a process that leads to syncretising the “mental planning operations” while interacting in each language.

Hickey (2012: 12) refers to such a process as the search for categorial equivalence. In the contact of phonological systems it corresponds to the process of matching of particular phonetic tokens of one language with particular phonological prototypes in a contact language, as described by Blevins (2017a). Compare also the models of contact-induced change in bilinguals of Bolonyai (1998) and Myers-Scotton (2006: 271) as a combination, i.e., matching, of surface-level forms from one language with an underlying abstract structure from another language.

1.4 Transcriptions systems

We present here some details of the transcription systems used in the volume for NENA, Iranian and Armenian. Arabic and Turkic are transcribed with the normal conventions of the linguistic literature for these languages.

1.4.1 NENA

NENA dialects are represented with the transcription system used by Geoffrey Khan in his grammars of NENA.

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The transcription of consonants corresponds to IPA with a few deviations in accordance with the conventions of Semitic linguistics (IPA equivalents in square brackets): ʔ [ʔ], ʕ [ʕ], *h* [h], *š* [ʃ], *y* [j], *j* [dʒ]. Emphatic (pharyngealised) consonants are marked by a subscript dot, e.g., *ṣ* [sˤ], *ṭ* [tˤ], *ḷ* [lˤ], *ṛ* [rˤ], *ṣ̣* [ʃˤ]. Plain (unpharyngealised) voiceless stops and affricates are aspirated, e.g., *t* [tʰ], *p* [pʰ], *k* [kʰ], *č* [tʃʰ], although the VOT varies across dialects and is eliminated after fricatives. In some NENA dialects there is a series of unaspirated voiceless stop phonemes. These are represented by a diacritic, thus: *ṭ̣* [t], *ḵ̣* [k], *ḥ̣* [p], *č̣* [tʃ].

Vowel length is generally predictable from syllable structure. Vowels in open syllables are, in principle, long or semi-long, and vowels in closed syllables are short, e.g., C. Urmi *betan* [ˈbeːtʰan] ‘our house’. When vowel length follows these principles, which it does in the majority of cases, no diacritics are used to distinguish long and short vowels. In exceptional cases where a long vowel occurs in a closed syllable or a short vowel occurs in an open syllable, a macron diacritic is used to mark the long vowel and a breve diacritic to mark the short vowel, e.g., C. Barwar *brāt* [braːtʰ] ‘the daughter of’, *šāme* [ʃameː] ‘he hears.’

In most NENA dialects, word-initial vowels are obligatorily preceded by a glottal stop, e.g., C. Barwar *ʾawwa* [ˈʔawwa] ‘this’. The glottal stop does not occur in some dialects in the north-eastern sector of NENA.

Word stress is marked in some of the transcriptions by an acute accent, e.g., C. Urmi *bétan* [ˈbeːtʰan] ‘our house’. In Christian NENA dialects and Jewish *Lišana Deni* dialects word stress generally falls on the penultimate syllable. In Jewish Trans-Zab dialects, the default position of word stress is on the final syllable, e.g., J. Arbel *belán* [beːˈlan] ‘our house’. In some transcriptions of clauses the boundary of intonation groups is marked by the symbol ^ˆ and the nuclear stress of the intonation by a grave accent, e.g.,

- (1) C. Barwar (Khan 2008c: 602)
- | | | | |
|--------------------|-----------------|----------------|---------------------------|
| <i>qím-la</i> | <i>šáry-a</i> | <i>bānúd-a</i> | <i>diya.</i> ^ˆ |
| rise.PST-OBL.3SG.F | untie.PRS-3SG.F | bands-3SG.F | GEN.3SG.F |
- ‘She got up and untied her bands.’

1.4.2 Iranian

All data from Northern, Central, and Southern Kurdish, Zazaki, and Gorani are transcribed in the native Kurdish “Hawar” alphabet. In the phonology section IPA is used where relevant. Note that there are some variations in the way that different Kurdish varieties transcribe forms.

1.4 Transcriptions systems

In Northern Kurdish, there is a three-way stop contrast between voiced, voiceless unaspirated, and voiceless aspirated: *b, d, g, c; p, t, k, ç; p', t', k', ç'*. In contrast, the rest of the Kurdish varieties have a two-way stop contrast consisting of the voiced and voiceless aspirated series. However, as aspiration is not phonemic, it is not marked, *p, t, k, ç* (for NK *p', t', k', ç'*).

The vowel *e* in Kurdish surfaces as either [ɛ] or [ə], depending on stress or syllable structure. This is not a phonemic distinction. Additionally, the precise realisation of the stressed variant [ɛ] is lower as one moves southward and eastward, i.e., [æ].

All vowels in Kurdish occur with an prosthetic glottal stop when in word-initial position. This is not written in the Hawar Latin script. However, it is mandatory in the Soranî script. In Table 1.4, the vowels are given in their word initial form, e.g., <ʔ> not <|> for *a*.

1.4.3 Armenian

For dialect examples, and examples from the standard languages that are transcribed from recordings, IPA is used. IPA transcriptions are phonemic, except when fine phonetic detail is relevant to the analysis. For Classical Armenian examples, and examples from the modern standard languages which are transcribed from written texts rather than from recordings, the Hübschmann-Meillet system of transliteration is used, as shown in Table 1.4.5. The writing systems of the standard languages do not always correspond directly to the pronunciation.

1.4.4 Turkic

Turkic examples are transcribed using the Latin Azeri alphabet, with additional diacritics or characters where necessary. Examples taken from published sources may be adapted to the transcription system used here. Where phonemic or phonetic detail is relevant to the analysis of dialect data, IPA transcriptions are used.

1.4.5 Summary of transcriptions systems

Table 1.4: Summary of transcriptions systems

IPA	Soranî	Hawar	Armenian	Hübschmann-Meillet	NENA
ʔ	–	–	–	–	ʔ

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IPA	Soranî	Hawar	Armenian	Hübschmann- Meillet	NENA
ʕ	ع		–	–	ʕ
a	ا	a	ա	a	a
b	ب	b	բ	b	b
–	–	–	–	–	c
–	–	–	–	–	ʃ
dʒ	ج	c	ջ	ǰ	j
tʃ	چ	ç	ճ	č	ç
tʃ ^h	چ ^ه	ç'	չ	č ^c	č
tʃ ^s	–	–	–	–	č̣
dz	دز	dz	ծ	j	ǰ
ts	تس	ts	ծ	c	č̣
ts ^h	تس ^ه	ts'	ց	c ^c	č̣
ts ^s	–	–	–	–	č̣̣
d	د	d	դ	d	d
d ^s	د ^س	d	դ	d	ḍ
ð	–	–	–	–	ð
ð ^s	–	–	–	–	ð̣
e	ئی	ê	Է	ē	e
je	ئی	ê	Ե	e	–
ɛ~æ	ئه	e	–	–	ɛ
ə	ئه	e	ը	ə	ə
f	ف	f	Ֆ	f	f
g	گ	g	գ	g	g
ɣ	غ	ğ	ղ	ɫ	ğ
h	ه	h	հ	h	h
ħ	ه	ħ	–	–	ħ
i	ی	î	ի	i	i
ɪ	–	i	–	–	–
ʒ	ژ	j	ժ	ž	ž
k	ک	k	կ	k	k̂
k ^h	ک ^ه	k'	ք	k ^c	k
k ^s	–	–	–	–	–
l	ل	l	լ	l	l

1.5 Note on place names

IPA	Sorani	Hawar	Armenian	Hübschmann-Meillet	NENA
l	ل	l	–	–	l
m	م	m	մ	m	m
n	ن	n	ն	n	n
o	ئو	o	o	ō	o
(v)o	–	–	n	o	–
p	پ	p	պ	p	ḡ
p ^h	پ	p'	փ	p ^c	p
p ^ʕ	–	–	–	–	p̣
q	ق	q	–	–	q
r	ر	ř	ռ	ř	ṛ
r	ر	r	ր	r	r
s	س	s	ս	s	s
s ^ʕ	–	–	–	–	ṣ
ʃ	ش	ʂ	շ	š	š
t	ت	t	տ	t	ṭ
t ^h	ت	t'	թ	t ^c	t
t ^ʕ	–	–	–	–	ṭ
θ	–	–	–	–	θ
u	وو	û	ու	u	u
ʊ	و	u	–	–	u
v	ڤا	v	վ	v	v
w	و	w	ւ	w	w
x	خ	x	խ	x	x
j	ی	y	յ	y	y
z	ز	z	զ	z	z
z ^ʕ	–	–	–	–	ẓ

1.5 Note on place names

Several place names in the region have variant forms that are used in various languages, e.g., Urmi/Urmia, Salamas/Salmas/Salmast, Hewler/Arbil/Arbel. Rather

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than imposing a standard, we have retained in several places variant forms that are used by contributors to the volume.

Part II

Phonology

2 Interdentals

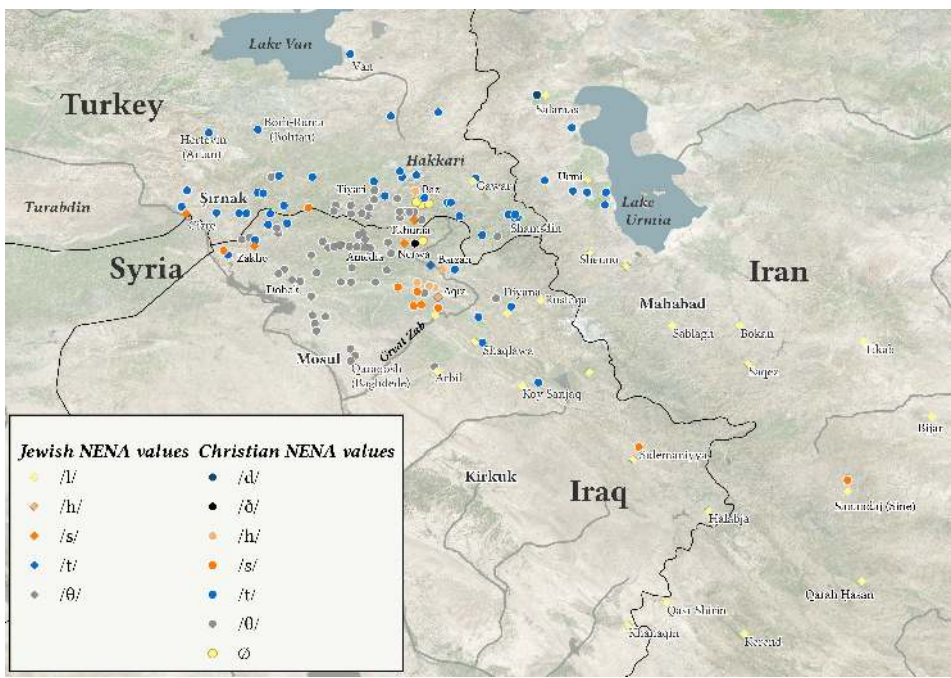


Figure 2.1: Move this to where you want it in the text and change caption.

2.1 Interdentals in NENA Dialects

2.1.1 Historical Background

The NENA interdental fricative consonants /θ/ and /ð/ are derived historically from postvocalic fricative allophones *[θ] and *[ð] of the stops */t/ and */d/ respectively in earlier Aramaic. In NENA these fricative allophones became phonemicized, with the result that minimal pairs are found with stops and fricatives,

2 Interdentals

e.g.,¹

- (1) C. Qaraqosh (Khan 2002a: 33, 35)
 šata ‘year’ *šaθa* ‘fever’
 guda ‘wall’ *guḏa* ‘churn’

This phonemicization has come about since the phonetic process of lenition of stops to fricatives after vowels ceased to operate at some point in the earlier history of NENA and subsequently by separate processes the stop consonants *bgdkpt* developed in word-initial and postvocalic position. The processes that had the outcome of a stop after a vowel include:²

- i. Cases where an originally geminated stop has lost its gemination, e.g.,
 C. Qaraqosh *šata* ‘year’ < **šattā*
- ii. The monophthongization of a diphthong before a stop, e.g.,
 C. Qaraqosh *ganota* ‘thief’ < **gannāwta*
- iii. Cases where a stop *bgdkpt* root consonant occurring after a consonant in a verbal inflection has been extended to all inflections of the verbal root, including inflections where this root consonant occurs after a vowel, e.g.,
 C. Qaraqosh *š-t-y štilə* ‘he drank’, but *šatə* ‘he drinks’
 Conversely an interdental occurring postvocally in a verbal inflection has sometimes been extended to all inflections of the verbal root, including to inflections where there is no preceding vowel, e.g.,
 C. Qaraqosh *ʔaθə* ‘may he come’, but *θilə* ‘he came’
 C. Qaraqosh *raθəx* ‘it boils’, but *marθəx* ‘he causes to boil’
- iv. The shift of a fricative *bgdkpt* to a stop by a process of dissimilation from an adjacent fricative, e.g.,
 C. Qaraqosh *ʔiḏa* ‘hand’, but *ʔidaθa* ‘hands’

The interdental phonemes /θ/ and /ð/ are not found in the consonant inventories of all NENA dialects. In many dialects they have been lost by a process of merger with other consonantal phonemes.

Figure 2.2: Map of primary reflexes of /θ/

¹Christian dialects are indicated by the abbreviation C. and Jewish dialects by the abbreviation J.

²Khan (2002a: 33-38)

2.1 Interdentals in NENA Dialects

Figure 2.3: Map of primary reflexes of /ð/

2.1.2 Reflexes of the Interdentals and their Distribution

(2) Reflexes of /θ/

/θ/

/ð/

/t/

/s/

/l/

/h/

/∅/

(3) Reflexes of /ð/

/ð/

/d/

/z/

/l/

The NENA dialects that have preserved the interdentals are mostly situated in the western half of the NENA region, including those of the Mosul plain, the Duhok region, Sapna valley and Barwar in Iraq, and those of the Țyare and Txuma regions of south-eastern Turkey. An outlier in the eastern sector of NENA is the dialect of Hawdiyan, which has preserved the interdentals.

In most areas in the eastern sector of NENA, the original interdentals have been lost. There are a variety of non-interdental reflexes of the historical interdental consonants across this eastern sector of NENA. The primary reflexes differ according to region and also according to religious community. In many cases the Christian dialects of one region have different reflexes of interdentals from those of the Jewish dialects.

The most common process involves merger of the interdentals with other consonant phonemes. These are typically stops ($\theta > t$, $\ð > d$), e.g., C. Urmi *beta* < **beθa* ‘house’, *ida* < **iða* ‘hand’, or sibilants ($\theta > s$, $\ð > z$), e.g., J. Zakho *besa* < **beθa* ‘house’, *iza* < **iða* ‘hand’.

Another process involves debuccalisation, i.e., the loss of articulation in the mouth, resulting in the interdental shifting to a laryngeal fricative *h* or zero (∅). This process affects mainly the voiceless interdental θ . It is found predominantly in Christian dialects in the north-eastern sector of NENA, e.g.

2 Interdentals

	θ (< * <i>xaθa</i> ‘sister’)	δ (< * <i>ʔiða</i> ‘hand’)
(4)		
C. Baz Maha Xtaya	<i>xaha</i>	<i>ʔida</i>
C. Baz Aruntus	<i>xa</i>	<i>ʔida</i>

The explanation as to why the voiced interdental was not debuccalised in parallel to that of the voiceless interdental is likely to be the difference in glottal setting of voiced and voiceless consonants. When voiced consonants are produced the glottal folds are brought together slightly, causing turbulence in the flow of air, which produces voice. By contrast, the glottal folds are further open when a voiceless consonant is produced, allowing for a free flow of air (Catford 2001: 35–38). The greater degree of closure of the glottal folds in a voiced interdental would have been a constraint against weakening its articulation to /h/ or zero.

Another form of lenition of * θ is its voicing. This is found in the C. Nerwa dialect. As a result, the reflexes of * θ and * δ are both /ð/, e.g., C. Nerwa *beða* < **bayθa* ‘house’, *ʔiða* ‘hand’ < **ʔiða*.

In the Trans-Zab Jewish Neo-Aramaic dialects both the interdentals * θ and * δ shift to the lateral /l/ (Mutzafi 2008c: 411–412; Khan 2018a: 314, 2018b: 486). There is evidence that in the Neo-Aramaic dialects concerned the two interdentals first shifted to the voiced stop **d* before finally becoming a lateral sonorant /l/ (Khan 2008b: 29–31). The sonorant /l/ can then be regarded as lenition of the **d*.

Within dialects, there are often secondary reflexes of the interdentals that are conditioned by specific phonetic environments and/or are restricted to specific lexical items. In Jewish Trans-Zab dialects, for example, where the primary reflex of * θ is /l/, the reflex of * θ is /h/ in some words, e.g., J. Arbel *ʔlaha* < **tlaθa* ‘three’, J. Urmi *ʔahra* < **ahra* < **aθra* ‘town’. This debuccalised reflex is associated with the environment of a preceding low vowel /a/ and the presence of pharyngealisation in the word. In some Jewish Trans-Zab dialects in the Kurdistan province of Iran and the adjacent region of Iraq the pharyngealisation in the word has developed into a pharyngeal feature that is added to the segment /h/, resulting in the shift of the /h/ to a voiceless pharyngeal approximant /ħ/, e.g., J. Sulemaniyya *tlaha*, J. Sanandaj *təlha* < **ʔlaha* < **tlaθa* ‘three’; J. Sulemaniyya, J. Sanandaj *ʔahra* < **ahra* < **aθra* ‘town’.

Conversely in the J. Barzani dialect (Mutzafi 2002a), /l/ is a secondary reflex of * θ and * δ , the primary reflexes being the debuccalisation of * θ to /h/ or zero and the shift of * δ to a stop, as in Christian dialects in the north-eastern sector of NENA, e.g.,

- (5) a. Primary reflexes:
gdada < **gðaða* ‘thread’

2.1 Interdentals in NENA Dialects

- | | | | |
|-------------|---|----------------|---------|
| <i>raha</i> | < | <i>*raθa</i> | ‘lungs’ |
| <i>momá</i> | < | <i>*momaθa</i> | ‘oath’ |
- b. Secondary reflexes:
- | | | | |
|--------------|---|-----------------|------------|
| <i>klawá</i> | < | <i>*kθawá</i> | ‘to write’ |
| <i>belta</i> | < | <i>*be ʾiða</i> | ‘sleeve’ |

In some dialects north of the Aqra mountain, where the primary reflex of **θ* is /h/, the reflex is emphatic (i.e., pharyngealised) /ʕ/ in some words that have adjacent pharyngealised segments, e.g.,

(6) C. Gerbish (Al-Zebari & Khan 2022)

- | | | | |
|--------------|---|---------------|-----------|
| <i>bəha</i> | < | <i>*bayθa</i> | ‘house’ |
| <i>ʾaʃra</i> | < | <i>*ʾaθra</i> | ‘country’ |
| <i>qarʃa</i> | < | <i>*qarθa</i> | ‘cold’ |

The pharyngealisation of the sibilant reflex /ʕ/ in these words is likely to have arisen by spreading of pharyngealisation from an adjacent pharyngealised consonant. The process of pharyngealisation involves the retraction of the tongue root resulting in the constriction of the upper pharynx with increased muscular tension. The constriction of the pharynx would have obstructed the flow of air of a laryngeal continuant /h/ and, moreover, the muscular tension would have strengthened the articulation of the grooved sibilant. Both of these articulatory conditions blocked the shift to /h/.

In the environment of emphatic /r/ in the C. Gerbish dialect the reflex of **θ* is the velar fricative /x/ in some words, e.g.,

(7) C. Gerbish

- | | | | |
|---------------|---|----------------|--------------|
| <i>qəxra</i> | < | <i>*qəθra</i> | ‘knot’ |
| <i>qəxʀre</i> | < | <i>*qəθrre</i> | ‘he knotted’ |
| <i>baxra</i> | < | <i>*baθra</i> | ‘behind’ |

A similar process is attested marginally in the J. Barzani dialect (Mutzafi 2002a) in the word

(8) *nxara* < **nθara* ‘to fall off (leaves)’

The explanation for this exceptional reflex also is based on the articulatory gestures of the adjacent pharyngealised consonant /r/. The constriction of the pharynx that takes place as a coarticulation of the /r/ would have narrowed the flow of air, which resulted in the velar fricative /x/ rather than /h/. In principle one may have expected the outcome to be /ʕ/ (i.e., **θ* > /ʕ/), as in the previous set of

2 Interdentals

examples, which was also induced by the pharyngealised environment. Here we can hypothesise the development $*\theta > *ʃ > /x/$ in a pharyngealized environment, i.e., an pharyngealized $*ʃ$ has lost its coronal articulation by a process of partial debuccalisation.

In some dialects of the Tiyare region a voiceless interdental $/\theta/$ shifts to the palato-alveolar $/ʃ/$ in certain environments (Talay 2008b: 66–69). These include after the vowel $/\epsilon/$, which is a contraction of the original diphthong $*ay$, e.g., Upper Tiyare $b\epsilonʃa < *b\epsilon\theta a < *bay\theta a$ ‘house’. Another environment is the feminine ending $*-i\theta a$ of verbal and nominal forms from final-weak roots, e.g., Upper Tiyare $mliʃa < *mli\theta a$ ‘full’, $x\tau iʃa < *x\tau i\theta a$ ‘sin’. These two environments both have in common contact with a high vowel, historically in the case of the diphthong $*ay$.

In many dialects, certain reflexes are restricted to specific lexical items. This applies to the complete lenition of voiceless θ to $\emptyset$. In dialects where the primary reflex of $*\theta$ is not $\emptyset$, the sound is reduced to $\emptyset$ in certain common words. In numerous Christian dialects in the northern sector of NENA, for example, where the primary reflex of the voiced interdental $*\delta$ is $/\delta/$ or $/d/$, this interdental is lenited in the independent genitive particle $*di\delta-$ (§11.1) and replaced by a glide $/y/$, resulting in forms such as $diy-$ (C. Barwar). In J. Sulemaniyya (Khan 2004: 30), where the primary reflex of θ is l , the sound is reduced to $\emptyset$ in the preposition $bar < *ba\theta ar$ ‘after’ and some common verbs, such as $k-e < *k-a\theta e$ ‘he comes’ and $k-me < *k-ma\theta e$ ‘he brings’. Even the voiced reflex $*\delta$ is reduced to $\emptyset$ in some common words, e.g., J. Sulemaniyya $k-\check{a}ye < *k-ya\delta e$ ‘he knows’. In all of these, the original interdental is preceded by a low vowel $/a/$.

When the interdentals merge with other consonants, in many cases this merger is symmetrical, e.g., all the interdentals merge with stops ($\theta, \delta > t, d$) or sibilants ($\theta, \delta > s, z$). In some dialects, however, there is asymmetry in the merger. In such cases, the outcome results in the reflex of the voiceless interdental $*\theta$ being weaker than that of the voiced interdental. In some dialects the voiceless interdental is preserved while the voiced one shifts to a stop, e.g., C. Ankawa $xa\theta a$ ‘sister’, $’ida (< *’i\delta a)$ ‘hand’, the voiceless becomes a sibilant, while the voiced becomes stop, e.g., J. Nerwa $xasa (< *xa\theta a)$ ‘sister’, $’ida (< *’i\delta a)$ ‘hand’, or the voiceless undergoes debuccalisation while the voiced becomes a stop or a sibilant, e.g., C. Baz Aruntus $xa (< *xa\theta a)$ ‘sister’, $’ida (< *’i\delta a)$ ‘hand’, C. Gerbish $beha$ ‘house’ ($< *bay\theta a$), $’iza (< *’i\delta a)$ ‘hand’.

2.2 Interdentals in Arabic Dialects

2.1.3 Summary

As remarked, the preservation of the interdentals /θ/ and /ð/ is found predominantly in dialects of the western sector of NENA, in the Mosul plain, the Duhok region, Sapna valley and Barwar in Iraq, and those of the Țyare and Txuma regions of south-eastern Turkey. The various innovative reflexes are more widely scattered, in the north-western sector and across the eastern sector.

The innovative debuccalisation of *θ to /h/ or Ø is located in a cluster of Christian dialects in the north-eastern sector of NENA, and in the geographically adjacent Jewish dialect of Barzan. The reflexes of stops (*t*, *d*) and sibilants (*s*, *z*) are widely distributed across the NENA area. There are a few geographical clusters of sibilant reflexes, e.g., in the Christian dialects of the Aqra area and the dialects of C. Sulemaniyya and C. Sanandaj.

There is evidence that some innovative reflexes of the interdentals went through various stages of historical development. The /l/ reflex of both the voiced and voiceless interdental in the Jewish Trans-Zab dialects appears to have developed from a transitional /d/ reflex. This is shown by isolated /d/ reflexes of both *θ and *ð in some dialects where the primary reflex of both is /l/, e.g., J. Urmi *ade* < **aθe* ‘he comes’, *ida* < **iða* ‘hand’. The reflex of *θ as /s/ in emphatic environments in C. Gerbish rather than as the debuccalised /h/, which is the normal reflex elsewhere, suggests that a sibilant reflex was a transitional stage of the development of /h/ (i.e., *θ > *s > /h/).

2.2 Interdentals in Arabic Dialects

Old Arabic had the following interdentals: *θ, *ð and *ḏ. Qəltu dialects as a whole have a variety of reflexes (Jastrow 1978: 34–39). Within the NENA region itself, we find three different reflexes: (i) a preservation of the interdentals, (ii) a shift to dental stops and (iii) a shift to sibilants. The shifts have occurred to the set as a whole: if one interdental has undergone the shift to a sibilant or stop, then all have (see Table 2.1). Just outside the NENA region, we find the same three realisations plus a shift to labio-dentals in the dialects of Siirt town and nearby villages, such as Tillo (see Table 2.2).

2.3 Lenition of /d/ in Iranian languages of the region

The Iranian languages of the region do not have interdental consonants. The process of lenition of the voiced stop /d/ that is attested in several Iranian varieties,

2 Interdentals

Table 2.1: Reflexes of Arabic interdentals within the NENA area

Mosul (M, C, J)	θ, δ, δ (Jastrow 1979: 38)
Bəh̄zāni (C, Y, M)	s, z, z (Jastrow 1978: 35–38)
Mērgī (C)	θ, δ, δ (Hassan 2023: 581)
Səndōr (J)	θ, δ, δ (Jastrow 1991b: 774)
Arbīl and ‘Aqra (J)	t, d, d (Jastrow 1990a: 27)
Khānaqīn (J)	θ, δ, δ (Bar-Moshe 2024: 108)

Table 2.2: Reflexes of Arabic interdentals on the periphery of the NENA area

Āzəx (C)	s, z, z (Wittrich 2001: 27)
Siirt town (M)	f, v, v (Biṭună 2016a: 32–33)
Tillo (M)	f, v, v (Lahdo 2009: 54)
Kirkūk (J)	t, d, d (Bar-Moshe 2024: 108)
Tūz Khurmātu (J)	t, d, d (Bar-Moshe 2024: 108)

however, is likely to have had an impact on the historical development of NENA interdentals.

The non-initial lenited voiced stop /d/ in Kurdish is derived from non-initial Old Iranian voiceless stops /t/ that develop into non-initial Middle Iranian voiced stops /d/. Following Honeybone (2012), a distinction is made between lenition as a diachronic process and lenition as a process that is synchronically active only in a subset of dialects. The main criterion for considering a lenition process as diachronic/old is that all varieties have undergone it. In contrast, synchronic lenition is an ongoing process, shown by the fact that it continues to be an active process in particular dialects while it has stopped being a phonological process in others. The lenition of /d/ is historically a process that is common to all major Kurdish varieties. The following description is based on Mohammadirad & Öpengin (2025).

2.3 Lenition of /d/ in Iranian languages of the region

		‘almond’ < Old Ir. * <i>vātāma</i>	‘God’ < Pth./MP <i>xwadāy</i>
	NK	<i>bahīv</i>	<i>xudē</i> ; WNK: <i>xwayē</i>
	NCK	<i>bādam</i> , <i>bāyam</i>	<i>xuwā</i> , <i>xudā</i>
(9)	SCK	<i>bāwim</i>	<i>xuwā</i>
	SK	<i>bāyim</i>	<i>xulā</i> , <i>xudā</i>
	G.	<i>vāmī</i> , <i>vāham</i> , <i>bāyam</i>	<i>xudā</i> , <i>xulā</i>
	Z.	<i>vām</i>	

The outcomes of lenition for cognates of ‘almond’ are debuccalisation (i.e., d > h) and approximantisation (i.e., *d > w), though with intermediate steps. In conservative Gorani dialects and Zazaki, the consonant is omitted altogether, hence *vām* < **vādam*. In some NCK dialects /d/ is not lenited, as seen in the variants *bādam* ‘almond’ and *xudā* ‘God’, though the picture is not entirely clear due to the cultural influence of Persian.

However, at the synchronic state of Kurdish, there is evidence that in more recent lexicon, post-vocalic /d/ still undergoes lenition. This is particularly true in SK, neighbouring CK dialects, and some Gorani dialects.

		‘much’ cf. P. <i>zīyād</i>	‘memory’ cf. P. <i>yād</i>	‘bad’
	NK	<i>zēda</i>		
	NCK	<i>zīyād</i>	<i>yād</i>	
(10)	SCK	<i>zīyāw</i>	<i>yād</i>	<i>baw</i>
	SK	<i>zīyāw</i> , <i>zīyāy</i>	<i>yāy</i>	<i>bay</i>
	G.	<i>zīyāy</i> , <i>zīyā</i>	<i>yāy</i>	<i>bay</i>

The same tendency for leniting /d/ occurs in other environments, including in the pre-consonantal slot, which is considered an unfavourable slot for lenition, or in Honeybone’s (2012) words a “strong position” disfavouring lenition.

		‘esteem’ < Ar. <i>qadr</i>
	Z.	<i>qadr</i>
	NK	<i>qadr</i>
(11)	NCK	<i>qadr</i>
	SCK	<i>qawr</i>
	SK	<i>qayr</i>
	G.	<i>qawr</i>

As can be seen, the synchronic lenition of /d/ within Kurdish is active in Kurdish varieties located at the historical Gorani zone of speech, see Figure 2.4. The synchronic lenition of /d/ in the south of the greater Kurdish zone remains a weakly conditioned change, since it involves the change in the phonetic realisation in

2 Interdentals

specific contexts and no contrastive phoneme has been added to the phonemic inventory of the language.

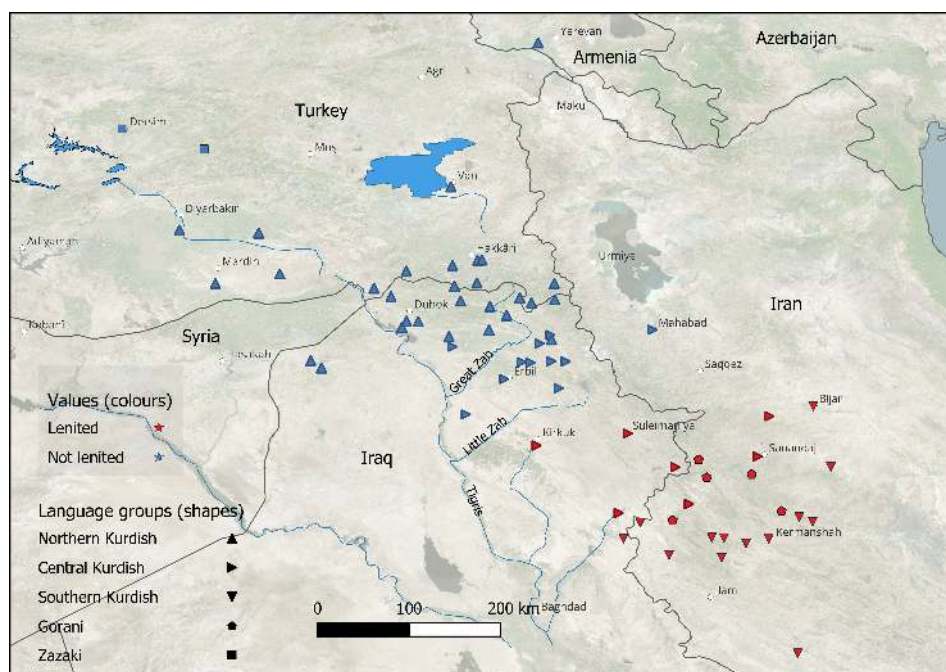


Figure 2.4: Intervocalic lenition of /d/ in the synchronic state of Kurdish

In some varieties of Gorani spoken in Hawraman lenited /d/ is realised as an interdental approximant. In the speech of migrants from Hawraman in Kermanshah it is realised as /y/ or as a lateral /l/ (Mahmoudveysi & Bailey 2019: §3.1). In Bakhtiari /d/ after a word-internal vowel or glide is realised as an interdental approximant or in some dialects as a sonorant /r/ (Anonby & Taheri-Ardali 2019: §2.1).

2.3.1 Lenition of /d/ in Turkic

In the Turkic dialect Sonqorî of the Zagros area postvocalic /d/ is weakened to the sonorant /r/ or sometimes l in combination with front vowels, e.g., [urin] < *odun ‘firewood’, [iri] < *idi ‘was’ (Bulut 2018: §2.1.2.4).

2.4 Armenian

Although some Armenian dialects have acquired various sounds that were not present in Classical Armenian, including pharyngeals and uvular stops (see Sections 6, 8), to the best of our knowledge, none have acquired interdentals.

2.5 Interdentals in Turkic

The existence of interdentals in Proto-Turkic, and more specifically the status of the voiced interdental /ð/ (rather than /d/) as the voiced counterpart of /t/, is a matter of debate. Several scholars argue that /ð/ was the original Proto-Turkic voiced alveolar (Erdal 2004: 67-69; Stachowski 2019; Johanson 2021: 24), taking into account its reflexes in Old and Middle Turkic—some of which feature /ð/—as well as its correspondences across modern Turkic languages, where it (or its presumed cognate) has evolved within roots into /y/, /d/, /t/, /z/ and /r/, depending on the genealogical branch (/y/ is the reflex in Oghuz, Kipchak and Karluk) and phonological position. Interdentals occur in very few modern Turkic languages, and wherever they do, their origin is typically unrelated to historical /ð/. The most notable cases are Turkmen and Bashkir, whose varieties generally possess both /ð/ and /θ/, which in these languages trace their origin to Turkic /z/ and /s/, respectively (Johanson 2021: 26).

In Azeri dialects, the voiced interdental fricative is attested but rare, and its phonetic status has not been thoroughly investigated. It has been reported in the north-western varieties, particularly Qazax and Gədəbəy, as well as in varieties previously spoken in north-eastern Armenia along the Armenian–Azerbaijani border (Karvansaray/Ijevan and Krasnoselsk/Chambarak) (Mämmädli 2019: 72; Bayramov 2022: 99). In these varieties, /ð/ appears to be an allophonic postvocalic realisation of /d/, e.g., /doðax/ ‘lip’ (cf. Standard Azeri *dodaq* [dodaχ]), /oð/ ‘fire’ (cf. Standard Azeri *od* [od]). The phenomenon is likely secondary, rather than inherited, since it is also observed in suffixes, e.g., /jeði/ <eat.PST.3> ‘(s/he) ate’ (cf. Standard Azeri *yedi* [jedi]), /jeðir/ <go.PRS.3> ‘(s/he) goes’ (cf. Standard Azeri *gedir* [jedir]), as well as loanwords, e.g., /dʒaðu/ ‘spell’ (cf. Standard Azeri *cadu* [dʒadu] < Persian *jādū*), /guða/ ‘parent of son- or daughter-in-law’ (cf. Standard Azeri *quda* [guda] < Middle Mongolian *quda*). In one cited loanword from Qazax Azeri (ibid.), /ð/ reflects an original /z/: /æðæli/ ‘initial’ (cf. Standard Azeri *əzəli* [æzæli] < Arabic *ʾazālī* ‘eternal’). In Gədəbəy, /ð/ is also reported word-initially, though only one example is attested, that being a kinship term: /ðæðæ/ ‘dad’ (cf. *dədə* [dædæ] in neighbouring varieties) (Bayramov 2022: 99). Finally, in the Sirik

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sub-variety of Cəbrayıl Azeri, spoken in Karabakh near the Iranian border and geographically removed from the north-western dialect area, /ð/ corresponds to what is typically realised as the voiced postalveolar affricate /dʒ/ in Standard Azeri or, post-vocally, the postalveolar fricative /ʒ/ in other Karabakh varieties, e.g., /ðan/ ‘soul’ (cf. Standard Azeri *can* [dʒan]), /mæðit/ (given name, cf. Standard Azeri *Məcid* [mædʒit]), /dʒið/ ‘foolish’ (cf. Standard Azeri *gic* [idʒ]) (Hacıyev 2018: 44).

2.6 Discussion

One factor that is likely to have contributed to the preservation of the interdentals in the western sector of NENA is contact with spoken Arabic dialects, many of which have interdentals in their sound inventories, e.g., Mosul, Mērgī, Səndōr (2.2). The Arabic dialect of Bəḥzānī on the Mosul plain has sibilant reflexes of historical interdentals, but this could be explained by the fact that the Christian inhabitants of this village were migrants from elsewhere (Fiey 1965: vol. 2, 461). Another factor is that some dialects in remote mountain villages where the interdentals were preserved, such as those in the Upper and Lower Ṭyare regions, appear to have had only limited contact with Kurdish or, indeed, any other language.

The innovative reflexes reflect a more intense contact with non-Semitic languages that do not have interdentals. Nowadays, the major contact language is Kurdish, though at an earlier period, the NENA dialects were in contact with Gorani. Gorani is now spoken only in a few isolated pockets in the region (Haig 2018: 297), but it was far more widespread some centuries ago (MacKenzie 1961b, Leezenberg 1992, Mohammadirad 2025b), and it has left its mark on NENA dialects that are now not in direct contact with it. As remarked, the loss of interdentals in various NENA dialects has brought their phoneme inventories closer to those of the non-Semitic contact languages, which lack interdentals or exhibit them only rarely, as is the case in some varieties of Azeri Turkic (§2.5). The most common process involves merger of the interdentals with other consonants that have a direct match in the inventories of the contact languages.

Several of the Jewish Trans-Zab dialects that exhibit the shift of the interdentals *θ* and *ð* to the lateral *l* were spoken in areas where in neighbouring Iranian and Turkic languages a /d/ following a vowel or sonorant undergoes lenition, known as ‘Zagros *d*’, resulting in it being realised as an approximant or as sonorant. Sonorant reflexes of lenited /d/ are marginally attested in Iranian varieties of the area (2.3) and also in the Turkic dialect of Sonqorî (2.3.1). As remarked,

2.6 Discussion

there is evidence that in the Neo-Aramaic dialects concerned the two interdentalals first shifted to the voiced stop **d* before finally becoming a lateral sonorant /*l*/ (Khan 2008b: 29-31). The sonorant /*l*/ can then be regarded as the lenition of the **d*. Such lenition, therefore, is likely to be due to the ‘perceptual magnet effect’ (Blevins 2017b) of the weakened Zagros *d*, whereby Neo-Aramaic speakers match this perceptually with the sonorant /*l*/ in their existing sound inventory. Since the more common reflex of lenition of /*d*/ in Iranian is debuccalisation rather than a sonorant, it is likely that the sonorant reflex /*l*/ of Jewish Trans-Zab NENA is not a direct replication of Iranian. Moreover, it should be noted that the lateral /*l*/ reflex of the historical interdentalals are found in Jewish Trans-Zab dialects outside the Zagros area, for example the Urmi region of north-western Iran and the Arbel plain in northern Iraq. This may be interpreted as evidence that the speakers of the Trans-Zab dialects in these areas migrated from the Zagros region at some earlier period.

It is noteworthy that Jewish Arabic dialects that were spoken in the region where Jewish Trans-Zab NENA was spoken in the twentieth century have different reflexes of historical interdentalals, e.g., J. Arbīl and ‘Aqra (*t, d, ɖ*), J. Khānaqīn (*θ, ð, ʔ*) (2.2). This could be interpreted as reflecting that the *l* reflex in Jewish Trans-Zab, although now widespread across the Trans-Zab area, diffused originally from a more restricted area in the Zagros region.

Interdentalals are not only prone to loss through contact but also do not spread by contact. This contrasts with some other sounds in languages that spread into the sound inventories of neighbouring languages. These include pharyngeals and unaspirated stops. Their diffusion has resulted in the enrichment of the consonantal inventories of the languages of the region. A factor that may have facilitated their spread is their salience (Blevins 2017a). The failure of interdentalals to spread, in contrast, can be correlated with their lack of salience (Maddieson 2013). There is, moreover, a priming to lose interdental sounds as they are acquired late in language acquisition, and this is reflected by their typological rarity.

3 Bilabials *w and lenited *b

3.1 *b and *w in NENA

In the history of Aramaic post-vocalic *b underwent lenition, which resulted in a weak labial that merged with the reflexes of the historical semi-vowel *w. In the NENA dialects, this is realised as /w/ (labio-velar approximant), /v/ (labio-dental fricative), /β/ (bilabial fricative) or phonetic variants of these. The geographical distribution of the various reflexes are discussed below.

3.1.1 Lenition of *b

Examples of lenition of *b after vowels:

- (1) C. Barwar (Khan 2008a: 40)

genawa ‘thief’ < **gannābā*
ṣliwa ‘cross’ < **ṣlībā*

In some words the *b has been lenited where there is now no preceding vowel. Generally the explanation is that the vowel had been elided at some earlier period, e.g.,

- (2) J. Arbel (Khan 1999a: 27)

xalwa ‘milk’ < < **ḥalḇā* < **ḥalabā*

An exceptional outcome of the lenition of *b as the bilabial sonorant /m/ is found in the phrase *biš-ṭam* ‘better’ (< **ṭāḇ*), which is attested in Trans-Zab Jewish dialects (Khan 1999a: 28).

As is the case with other members of the set of the *bgdkpt* consonants, the lenited form of *b was originally an allophone conditioned by the preceding vowel (represented here in historical reconstructions by the symbol *ḇ*), but in the NENA dialects this allophone has now become phonemicised and merged with the phoneme that is the reflex of *w. This phonemicisation is reflected by the existence of minimal pairs such as the following:

- (3) J. Sulemaniyya (Khan 2004: 28)

3 Bilabials *w and lenited *b

dwiql̥e ‘he held’ *dbiq̥le* ‘he stuck’

(4) C. Urmi (Khan 2016a: vol. 1, 161)

zavən (< *zabən*) ‘he buys’ *zabən* (< **mazabbən*) ‘he sells’

(5) C. Barwar (Khan 2008a: 40)

ʔarwan ‘our sheep (pl)’ *ʔarban* ‘our sheep (sg)’

This phonemicisation has come about since the phonetic process of lenition of stops after vowels ceased to operate at some point in the earlier history of NENA and subsequently, by separate processes, *bgdkpt* stops developed in word-initial and post-vocalic position. The processes that had the outcome of a stop after a vowel include:

(i) Where an originally geminated stop has lost its gemination, e.g.,

(6) C. Qaraqosh (Khan 2002a: 31)

xabuša ‘apple’ < **ḥabbūšā*
gib ‘next to’ < **gebb* < **genb*
šabθa ‘week’ < **šabbəṭā*

(7) C. Urmi (Khan 2016a: vol. 1, 162)

zabən ‘he sells’ < **mazabbən*

An exception to this is the lenition in many Jewish dialects of the geminated **bb* in the SG.M form of the adjective **rabbā* ‘big’, e.g.,

Table 3.1: Reflexes of **rabbā* ‘big’ with lenition

<i>ruwwa</i>	J. Arbel, J. Koy Sanjak, J. Sulemaniyya, J. Sanandaj, J. Saqəz
<i>ruwwa</i>	J. Arbel, J. Koy Sanjak, J. Sulemaniyya, J. Sanandaj, J. Saqəz
* <i>rəwwa</i>	J. Urmi
<i>ruwa</i>	J. Zakho, J. Dohok
<i>ʔurwa</i>	J. Amedia, J. Aradhin, J. Challa, J. Nerwa
<i>ʔərwa</i>	J. Betanure

3.1 *b and *w in NENA

This may have been a strategy to differentiate the adjective from the adverbial *raba* ‘much’ (< **rabbat*) in these dialects. It is significant that the f.sg form of the adjective has the expected /b/ reflex of the historically geminated /b/, e.g., J. Arbel *rabta*, *rubta* (< **rabbta*).

Several Christian dialects use the form *raba* (< **rabbā*), or phonetic variants of this, as a sg.M adjective meaning ‘big’, e.g.,

Table 3.2: Reflexes of **rabbā* ‘big’ without lenition

<i>raba</i>	C. Ankawa, C. Shaqlawa, C. Bēspən, C. Harbole, C. Derabun, C. Hassana, C. Hertevin, C. Karəmləsh, C. Tisqopa, C. Umra d-Shish
<i>ṛaba</i>	C. Qaraqosh, C. Barəṭle, C. Shiyuz
<i>ṛabv</i>	C. Telkepe
<i>roba</i>	C. Bohtan
<i>ṛoba</i>	C. Jənnət

In most of these Christian dialects the adverbial ‘many’ is expressed by another lexical item, which is generally a loanword, e.g.,

Table 3.3: Lexical items denoting ‘many’

<i>kābira</i>	C. Ankawa, C. Karamleş
<i>kābirv</i>	C. Telkepe
<i>gābira</i>	C. Hassana
<i>gālak</i>	C. Shaqlawa, C. Jənnət
<i>gallak</i>	C. Umra d-Shish
<i>ḥelə</i>	C. Qaraqosh, C. Barəṭle
<i>xābərri</i>	C. Bohtan

(ii) Where the stop occurs after a historical diphthong that has been monophthongised, e.g.,

- (8) C. Qaraqosh (Khan 2002a: 31)

mobəl ‘he brings’ < **mawbel*

3 Bilabials *w and lenited *b

(iii) Where the vowel preceding /b/ is an epenthetic, e.g.,

(9) C. Qaraqosh (Khan 2002a: 32)

<i>ʔabra</i>	‘son’	<	* <i>brā</i>
<i>ʔabʔurā</i>	‘dung’	<	* <i>bʔūrē</i>
<i>kaləbθa</i>	‘bitch’	<	* <i>kalbθa</i> (cf. <i>kalba</i> ‘dog’ m.)
<i>ʔaqubra</i>	‘mouse’	<	* <i>ʔuqbrā</i> (attested in Syriac)

(iv) Where a stop *b* root consonant occurring after a consonant in a verbal inflection has been extended to all inflections of the verbal root, including inflections where this root consonant occurs after a vowel, e.g.,

(10) C. Barwar

<i>ṭ-b-ʔ</i>	<i>ṭbile</i>	‘he sank’	<i>ṭābe</i>	‘he sinks’
<i>g-b-y</i>	<i>gbele</i>	‘it frothed’	<i>gabe</i>	‘it froths’

In these cases, the process may have been strengthened by the need to distinguish the roots from homophonous or near-homophonous roots with a medial /w/:

(11) C. Barwar

<i>ṭbaya</i>	‘to sink’	(< * <i>ṭ-b-ʔ</i>)	<i>ṭwaya</i>	‘to roast’	(< <i>ṭ-w-y</i>)
<i>gbaya</i>	‘to froth’	(< * <i>g-b-b</i>)	<i>gwaya</i>	‘to beg’	(< * <i>g-b-y</i>)

Similar root integrity (i.e., the preservation of an invariant form of the root) is exhibited by verbal roots that have a stop *b* consonant as their first radical, in that the stop remains through all inflections and derivations, including when prefixes that end in a vowel are attached to it, e.g.,

(12) C. Qaraqosh

(13)	<i>b-h-r</i>	‘to be bright’	<i>bahər</i>	‘it is bright’	<i>mabhər</i>	‘it shines’
	<i>b-x-y</i>	‘to weep’	<i>baxə</i>	‘he weeps’	<i>mabxə</i>	‘he causes to weep’

(14) C. Barwar

<i>b-r-q</i>	‘to shine’	<i>barəq</i>	‘it shines’	<i>mabrəq</i>	‘it causes to shine’
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Root integrity also conditions the preservation of the stop /b/ in cases such as the following:

(15) C. Barwar

3.1 *b and *w in NENA

<i>dabəš</i> I	‘it sticks (intr.)’	< * <i>dābeš</i>
<i>mdabəš</i> II	‘he sticks (tr.)’	< * <i>mdabbeš</i>

Here the transitive pattern II form originally had a geminated /b/ and the stop was, therefore, preserved. The stop was extended by analogy to the intransitive pattern I form, although historically it was ungeminated.

Root integrity across different verbal patterns is sometimes not found when there is a considerable semantic distinction between the forms, e.g.,

(16) C. Barwar

<i>zawən</i> I	‘he buys’	< * <i>zāben</i>
<i>mzabən</i> II	‘he sells’	< * <i>mzabben</i>
<i>gawe</i> I	‘he begs’	< * <i>gābē</i>
<i>mgabe</i> II	‘he elects’	< * <i>mgabbē</i>

This levelling applies to stems of nouns where a historical vowel before *b that originally conditioned its lenition has been elided, e.g.,

(17) J. Arbel

<i>kalba</i>	‘dog’	<i>kalbe</i> (< * <i>kalbē</i> < * <i>kalabē</i>)	‘dogs’
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The plural form of such words, however, has retained its lenited *b in some dialects, e.g.,

(18) C. Qaraqosh

<i>kalba</i>	‘dog’	<i>kalwə</i> (< * <i>kalbē</i> < * <i>kalabē</i>)	‘dogs’
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(v) Lenition of *b after vowels does not take place in loanwords, e.g.,

(19) J. Arbel

<i>baba</i>	‘father’ (Kurd.)
<i>zəbla</i>	‘dung’ (Arab.)
<i>’eba</i>	‘fault, shame’ (Kurd. < Arab.)

This applies to some loaned verbal roots, e.g.,

(20) J. Arbel

<i>d-b-ḥ</i>	<i>dabəḥ</i>	‘he slaughters’ (< Arab.)
<i>ṭ-b-l</i>	<i>ṭabəl</i>	‘he soaks, dips’ (< Heb.)

(vi) Lenition of *b after vowels does not take place where a word beginning with /b/ has been bonded prosodically with a preceding constituent that ends in vowel, e.g.,

3 Bilabials *w and lenited *b

(21) C. Qaraqosh

<i>ʾibə</i>	‘there is in it’	<	*ʾi + bə
<i>lebə</i>	‘there is not in it’	<	*la + i + bə
<i>ʾu-baθər</i>	‘and after’		

(22) C. Barwar

xo-bəθa ‘under the house’

(vii) Lenition does not take place where the *b was originally preceded by a vowelless consonant, which has now been elided, e.g.,

(23) C. Barwar

<i>sebuθa</i>	‘old age’	<	*saʾbuθa
<i>mašaboðe</i>	‘to enslave’	<	*mašaʾboðe

(viii) In these various contexts where the stop /b/ occurs after vowels, the stop is sporadically lenited to a bilabial fricative [β] in some dialects, e.g.,

(24) J. Arbel (Khan 1999a: 17)

raba [ʾræ:βæ] ‘many’

(25) C. Qaraqosh (Khan 2002a: 26)

da-baθra [dæʾβaθræ] ‘for behind’

The devoicing of the stop *b to /p/ has become lexicalised in some words. This devoicing has been conditioned by direct contact with an unvoiced consonant in one its inflections and then this has been extended to other inflections, e.g.,

(26) C. Barwar

<i>gupta</i>	‘cheese’	<	*gubbtā	<	*gubntā
<i>gupyaθa</i>	‘cheeses’				

In a few words word-initial *b has been devoiced to /p/ when an unvoiced sibilant occurs in the coda of the syllable, e.g.,

(27) J. Sulemaniyya (Khan 2004: 24), J. Arbel (Khan 1999a: 29), J. Urmi (Khan 2008b: 563), C. Qaraqosh (Khan 2002a: 32)

pəsra ‘meat’ < *bəsra

Contrast C. Barwar *bəsra*, C. Urmi **busra*.

3.1 **b* and **w* in NENA

- (28) J. Sulemaniyya (Khan 2004: 24)

pəʃla ‘onion’ < **bəʃla*Contrast C. Barwar *bəʃla*, C. Qaraqosh *bəʃəlθa*, C. Urmi **busla*.

In some dialects **b* shifts to unaspirated unvoiced *ʃ* (§4.1.3.3) when in contact with a laryngeal /ʔ/. The sequence **bʔ* coalesces to the unaspirated segment /*ʃ*/, e.g.,

- (29) C. Urmi (Khan 2016a: vol. 1, 142)

ʃ-ida < **b-ʔida* ‘by hand’
arʃa* < *arʔa* < **arʔā* ‘four’

A pharyngealised /*ʃ*/ has developed as a phoneme in some dialects. This has been motivated, it seems, to fulfil the functional need of creating semantic contrasts, e.g.,

- (30) J. Amedia (Greenblatt 2011: 15)

ʃaza ‘hawk’ *b-aza* ‘she will go’3.1.2 Phonetic Realisation of Lenited **b* and **w*

As remarked above, lenited **b* in NENA dialects merged with the reflexes of the phoneme **w*. There are differences in phonetic realisation of this phoneme across the NENA area.

3.1.3 Christian and Jewish Dialects with Default [w]

In Christian NENA dialects in Iraq, south-eastern Turkey and in Sanandaj in Iran the phoneme is generally realised as a labio-velar approximant [w]. This, likewise, is the realisation of the segment in *Lišana Deni* Jewish NENA dialects west of the Zab, e.g.,

- (31) C. Qaraqosh (Khan 2002a: 30)

sawa [sa:wa] ‘grandfather’

- (32) C. Barwar (Khan 2008a: 40)

sawa [sa:wa] ‘grandfather’

- (33) C. Armota (Khan, fieldnotes)

sawa [sa:wa] ‘grandfather’

3 Bilabials *w and lenited *b

- (34) C. Sulemaniyya (Khan, fieldnotes)

sawa [sa:wa] ‘grandfather’

- (35) C. Sanandaj (Khan, fieldnotes)

dewa [ˈde:wa] ‘gold’

- (36) C. Txuma (Jacobi 1973: 5)

dawa [ˈda:wa] ‘gold’

- (37) C. Van (Khan, fieldnotes)

sawuna [sa:ˈwu:na] ‘grandfather’

In the environment of /y/ or high front vowels it is sometimes realised as a labio-palatal [ɥ], e.g.,

- (38) C. Qaraqosh (Khan 2002a: 26)

maywša [ˈmeɟʏʃæ] ‘she dries’

In dialects in which the realisation of the phoneme is normally a labio-velar [w] and which are in contact with Northern Kurdish, loanwords from Northern Kurdish contain the labio-dental [v], whereas the semi-vowel in native NENA words is realised as the labio-velar [w], e.g.,

- (39) C. Barwar (Khan 2008a: 30-)

šāvana (NK) ‘shepherd’

ʾažvana (NK) ‘miller’

zvara (NK) ‘to turn’

In NENA dialects that are in contact with Northern Kurdish, however, a labio-dental realisation of the semi-vowel is occasionally found in native NENA words. In C. Barwar (Khan 2008c: 36), for example, the word *jvoṭa* ‘straight’, which is apparently of NENA etymology (cf. Syriac *šabbet* ‘to straighten’), is regularly pronounced with labio-dental /v/. The /v/ would be a reflex of lenited *b. In this dialect, moreover, the words *niwa* (< *nībā) ‘molar tooth’ and *gawana* (< *gaḇānā) ‘boundary’, which are also of Aramaic stock, the /w/ is realised as either a labio-velar or a labio-dental: [ˈni:wa] ~ [ˈni:va], [ga:ˈwa:na] ~ [ga:ˈva:na]. In these words also the /w/ is a reflex of a lenited *b.

3.1 **b and *w in NENA*

3.1.4 Jewish Trans-Zab Dialects

In Trans-Zab Jewish dialects the phoneme is generally realised either as a labio-velar [w], as labio-dental with very little friction, which may be represented by the labio-dental approximant symbol [ʋ], or as a labio-dental fricative [v].

3.1.4.1 J. Arbel

In J. Arbel the labio-dental realisation occurs in the environment of the close front vowels /e/ and /i/:

(40) J. Arbel (Khan 1999a: 18)

<i>qrawá</i> [qra:'wa]	'conflict'
<i>yaltá=weli</i> [jæl't ^h æve:li]	'I was a girl'
<i>sawilóx</i> [sævi:'lɒx]	'your pillow'

In J. Arbel after a close front vowel in the same syllable in word-internal position, it is realised either as a labio-palatal glide [ɥ] or as a labio-dental fricative [v]:

(41) J. Arbel (Khan 1999a: 18)

<i>híwlan</i> ['hy:ɥlæn]	'we gave'
<i>qliwtá</i> [qli:v'tæ]	'fine'

The labio-dental may be devoiced before a voiceless consonant:

(42) J. Arbel (Khan 1999a: 18)

<i>qiw'tá</i> [qəftæ]	'strength'
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3.1.4.2 J. Sulemaniyya

In J. Sulemaniyya the realisations of the semi-vowel are generally [w], [v] or [ʋ], and these are often in free variation:

(43) J. Sulemaniyya (Khan 2004: 20)

<i>šəmmáw</i> [ʃim'maw]	'her name'
<i>babáw</i> [ba:'bæv]	'her father'
<i>dábqawale</i> ['dæbqa:wale]	'she stuck it'
<i>xazyáwale</i> [xæz'yævæ:le]	'she saw him'
<i>willu</i> ['willu]	'they did'
<i>willú</i> [vɪ'lú]	'they did'

3 Bilabials *w and lenited *b

The labio-dental realisation is often devoiced when in contact with a following unvoiced consonant:

- (44) J. Sulemaniyya (Khan 2004: 20)
šwawtán [ʃwɑftæn] ‘our neighbour’

The labio-dental realisation occurs in loanwords from Kurdish, whereas in the contemporary local Central Kurdish dialects the consonant is realised as a labio-velar:

- (45) J. Sulemaniyya (Khan 2004: 20)
ʔašawán [ʔɑ:ʃæ'va:n] ‘miller’ (cf. CK [ɑ:ʃawa:n])

Examples of approximant labio-dental [ʋ]:

- (46) J. Sulemaniyya (Khan 2004: 20)
hiyén-awa [hi:'je:næʋæ] ‘they came back’
ziliwa ['zi:liʋæ] ‘they went back’

After /i/ the consonant /w/ is occasionally realised as a labio-palatal glide [ɥ]:

- (47) J. Sulemaniyya (Khan 2004: 20)
hiwle ['hyɥle] ‘he gave’

3.1.4.3 J. Sanandaj

In J. Sanandaj the phoneme is realised as a labio-dental [ʋ] in most cases, e.g.,

- (48) J. Sanandaj (Khan 2009: 19)
šiwá [si:'va] ‘wood’
hāwe [ha've:] ‘may he be’
hewalé [he'va:l'e:] ‘(that) he could’

The friction is sometimes reduced and it is pronounced as a labio-dental approximant [ʋ]. This is heard mainly after back consonants, e.g.,

- (49) J. Sanandaj (Khan 2009: 19)
dóqwa ['dɔ:qʋa] ‘he used to hold’
gwárté=ya [gʋər'te:ja] ‘he has married her’

It tends to be realised as a labio-velar continuant [w] when in contact with a sibilant, when it is between two instances of the low vowel /a/, or when it is adjacent to back rounded vowels:

3.1 *b and *w in NENA

(50) J. Sanandaj (Khan 2009: 19)

<i>šwawá</i> [ʃwɔːwʌ]	‘neighbour’
<i>šatāwáe</i> [ʃɑːthaˈwaːe]	‘years’
<i>rūwá</i> [ruːwa]	‘big’

3.1.4.4 J. Urmi

In J. Urmi the realisation of this phoneme freely varies between a labio-dental [v] and a bilabial [β] with very little friction, e.g.,

(51) J. Urmi (Khan 2008b: 20)

<i>widén</i> [viˈden ~ βiˈden]	‘I have done’
<i>wéli</i> [ˈveːli ~ ˈβeːli]	‘I was’
<i>hawé</i> [haˈveː ~ haˈβeː]	‘that he be’
<i>kwál</i> [ˈkvɪl ~ ˈkβɪl]	‘he gives’

It is in some cases realised as a bilabial approximant [w]. This is found predominantly when it is the second member of a cluster following /š/ or after /u/, e.g.,

(52) J. Urmi (Khan 2008b: 20)

<i>ušwá</i> [ʏʃwa]	‘seven’
<i>šwéle</i> [ʃweːle]	‘it was worth’
<i>juwé</i> [dʒuˈwe]	‘different’
<i>ˈlúwa</i> [ˈluːwa]	‘inside’

The realisation [w] is found occasionally elsewhere, especially between two low /a/ vowels, e.g.,

(53) J. Urmi (Khan 2008b: 20)

<i>šwawá</i> [ʃwaˈwa]	‘neighbour’
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3.1.5 Christian Dialects in the Region of Urmi

The most common realisation of the phoneme in C. Urmi, C. Salamas and C. Gavilan in north-western Iran is a labio-dental fricative [v], e.g.,

(54) C. Urmi (Khan 2016a: vol. 1, 103)

3 Bilabials *w and lenited *b

<i>vədlə</i> [ˈvɪdlɪ]	‘he did’
* <i>ʿav</i> [ˈʔav]	‘he’
<i>səvlux</i> [sɪvɫʊx]	‘you have grown old’
<i>ləlavatə</i> [ləːlaˈvaːtʰɪ]	‘nights’

Its articulation is sometimes weakened to a labio-dental approximant [ʋ], e.g.,

(55) C. Urmi (Khan 2016a: vol. 1, 103)

* <i>ʿav</i> [ˈʔav]	‘he’
* <i>šammánəvən</i> [ˈʃamˈmaːnəvən]	‘I am a listener’
<i>švávʋ</i> [ʃvaːvʋː]	‘their neighbour’

In some cases there is no perceptible dental articulation and it is realised as a labio-velar approximant [w], e.g.,

(56) C. Urmi (Khan 2016a: vol. 1, 103)

* <i>xoravatə</i> [ˈxoraˈwaːtʰə]	‘friends’
* <i>varída</i> [ˈwaːriːda]	‘flower’
<i>ju-bétə=vax</i> [juːbeːtʰɪwax]	‘you (fs.) are in the house’
<i>savunvátə</i> [savʊnˈwaːtʰɪ]	‘grandfathers’

3.1.6 Christian Dialects in the Targawar and Margawar Region

In the Christian dialects of the mountainous region of Targawar and Margawar, situated to the west of the Urmi plain, such as those of Mawana, Darband, Qalla and Tazacand, the phoneme is frequently pronounced as a bilabial continuant [β̞] or, when pronounced strongly, as a bilabial fricative [β], e.g.,

(57) C. Mawana (Khan, fieldnotes)

<i>wāhane</i> [ˈβahaːneː]	‘these’
<i>wéle</i> [ˈβeːle]	‘there he is’
<i>lwázta</i> [ˈlβɪsta]	‘emaciated’
<i>pardúwwa</i> [parˈdɪββaː]	‘lath’
<i>lawáza</i> [laːβaːza]	‘thin’

Its F2 is high and this correlates with the front rounded vowel IPA [y], which exists within the vowel system of these dialects (represented by the symbol /ü/ in transcriptions). Such a realisation of the phoneme, therefore, can be considered to be a weak form of front /ü/. The front realisation is accompanied by a lower degree of lip-protusion than in the articulation of IPA [w], which is a weak form of the back high vowel IPA [u].

Occasionally it is realised as a labio-dental continuant [ʋ̞], or, when pronounced strongly, as a labio-dental fricative [v̞].

3.2 Bilabials in Iranian languages

3.1.7 Summary

The predominant reflex of **w* and lenited **b* in the NENA dialects is a labio-velar /w/. This is its reflex in the Christian dialects of Iraq, south-eastern Turkey and Sanandaj in Iran and the *Lišana Deni* Jewish NENA dialects west of the Zab. In the Christian dialects in the region of Urmi in north-western Iran the reflex is a labio-dental /w/. In the Christian dialects of the Targawar and Margawar region west of the Urmi plain, the predominant realisation is a bilabial continuant [β] or, when pronounced strongly, as a bilabial fricative [β]. The Jewish Trans-Zab dialects are distinct from other NENA dialects in that they exhibit labio-dental realisations of the phoneme as far west as Arbel in Iraq. There is, however, some areal diversity within the Jewish Trans-Zab subgroup, in that the labio-dental fricative and the bilabial fricative realisations are more prevalent on the eastern periphery, e.g., in J. Sulemaniyya, J. Sanandaj and J. Urmi.

3.2 Bilabials in Iranian languages

3.2.1 Historical development

In the contemporary Iranian languages of the region, the bilabial stop /b/ derives historically from several different sources. These include Proto Indo-Iranian **bh* (word-initially), **p* intervocally, and **w* word-initially before low vowels. The bilabial semi-vowel /w/ derives historically both from **w* and from post-vocalic **b* by a process of lenition, with an intermediate stage **v*. The outcome of post-vocalic **b* in some varieties has not progressed beyond the intermediate stage /v/.

There is a general tendency for the lenition of voiced stops, and /b/ in particular in the Iranian languages of the region. In the Southern Kurdish zone, this phenomenon is part of the ‘active phonology’ of the languages. As one goes further north into the Central Kurdish region, the lenition of /b/ passes out of ‘active phonology’ into the realm of ‘historical sound change.’ As can be observed in the Central Kurdish variety spoken in Kerkuk in (58), in which the /b/ in the root *bûn* ‘to be’ is not lenited when prefixes are added. However, the shift from a historical post-vocalic **b* to /w/ is present in monomorphemic words, e.g., *aw* ‘water,’ *şwan* ‘shepherd,’ etc. cf. Persian *āb* ‘water,’ *šabān* ‘shepherd.’

(58) Central Kurdish (Kerkûk)

e-b-in

bi-b-in

na-b-in

IND-be.PRS-3PL:S SBJV-be.PRS-3PL:S NEG.IND-be.PRS-3PL:S

3 Bilabials *w and lenited *b

ne-b-in

NEG.SBJV-be.PRS-3PL:S

‘they are’ ‘(that) they were’ ‘they are not’ ‘(that) they were not’

As remarked, this lenition occurred with an intermediate stage of *v. This is demonstrated by northern and western varieties of Northern Kurdish, which maintain a distinction between /w/ and /v/, the latter being an intermediate stage in the development from *w < *b. Note that these varieties do not have lenition as an active phonological rule, as is the case also in Central Kurdish, see (59), in which /b/ is not lenited after the vowels of verbal prefixes. However, the shift from a historical post-vocalic *b to /v/ is present in monomorphemic words, e.g., *av* ‘water,’ *şivan* ‘shepherd,’ etc., cf. Persian *āb* ‘water,’ *šabān* ‘shepherd.’

(59) Northern Kurdish (Diyarbekir)

di-b-in

bi-b-in

na-b-in

IND-be.PRS-3PL:S SUB-be.PRS-3PL:S NEG.IND-be.PRS-3PL:S

ne-b-in

NEG.SBJV-be.PRS-3PL:S

‘they are’ ‘(that) they were’ ‘they are not’ ‘(that) they were not’

Much is known about the earlier phonology of Iranian languages. As such, we can recognise that this lenition of *b is part of a much deeper cycle of historical sound changes. The proto-phoneme referred to here as *b is the result of a previous lenition, *p > *b post-vocalically, and a (less relevant to the current discussion) modality shift, *b^h > *b. Additionally, the proto-phoneme *m behaves just like *b, being retained word-initially but lost post-vocalically. This change affects *sm clusters as well. Additionally, the changes from *b and *m > /w/ are likely to be more recent as they affect older loanwords but not the most recent ones, e.g., CK *tewaw* ‘finished/complete,’ *kitêw* ‘book,’ cf., Arabic *tamām* and *kitāb*. See Table 3.4 for a comparison of inherited forms. Note that inherited *w is /w/ in all Kurdish varieties.

The following section focuses on the distribution of the synchronic Iranian phonemes /b/, /w/, and v.

3.2.2 Synchronic

The lenition of non-initial /b/ in Kurdish exhibits similar tendencies for lenition as seen for /d/ in §2.3. In the early stages of Kurdish, a period roughly following Middle Kurdish, the reflexes of /b/ are usually fricativisation (i.e., the labio-dental

3.2 Bilabials in Iranian languages

Table 3.4: Some inherited reflexes of *p, *b^h, *m, and w

Language	*p	*šm	*b ^h
Avestan	<i>āp(am)</i>	<i>čašman</i>	^(H2) <i>ā-b^(h)arəta</i>
NK	<i>av</i>	<i>çav</i>	<i>*bir</i>
CK	<i>aw</i>	<i>çaw</i>	<i>hawird</i>
SK	<i>aw</i>	<i>çaw</i>	<i>hawird</i>
Hewramî	<i>aw(i)</i>	<i>çam</i>	<i>awird</i>
Zazaki	<i>aw</i>	<i>çim</i>	<i>*bera</i>
Persian	<i>ab</i>	<i>češam</i>	<i>avord</i>
Gloss	‘water’	‘eye’	‘bring’

/v/) in NK and Zazaki and approximantisation (i.e., the bilabial /w/) in the rest of Kurdish, see (60).

	MP <i>ābād</i> ‘prosper’	MP <i>šubān</i> ‘shepherd’	MP <i>nibēs-</i> ‘to write’	MP <i>nibastan</i> ‘to sleep’
(60) NK	<i>āvā</i>	<i>šivān</i>	<i>nivis-</i>	SENK <i>nivistin</i>
CK	<i>āwā</i>	<i>šuwān</i>	<i>nūs-</i>	NCK <i>nūstin</i>
SK	<i>āwā</i>	<i>šuwān</i>	<i>nūs-</i>	
G.	<i>āwā</i>	<i>šuwāna</i>	<i>nvīs-, nwīs</i>	
Z.	<i>āvā</i>	<i>šiwāna</i>	<i>nus-</i>	

In the synchronic state of Kurdish, the lenition of /b/ presents more sub-local variation than the lenition of /d/. With regard to Kurmanji Kurdish, western dialects and, to some extent, northern dialects lenite /b/ in intervocalic and post-vocalic slots, thus forming part of the active phonology, while the rest of NK retains /b/ in these contexts. Within Gorani, lenition of /b/ is a feature of peripheral dialects. The core Hewramî varieties generally resist leniting /b/ in the aforementioned contexts. It is also notable that the northern dialects of CK resist lenition along with the neighbouring NK dialects to the north, see Figure 3.1.

	‘news’ < Arab. <i>xabar</i>	‘bad’ < Arab. <i>kharaba</i>
(61) NK	<i>xabar</i> ; NNK <i>xavar</i>	<i>xarāb</i> ; NNK <i>xirāv</i>
CK	NCK <i>xabar</i> ; SCK <i>xawar</i>	NCK <i>xirāp</i> ; SCK <i>xirāw</i>
SK	<i>xawar</i>	<i>xirāw</i>
G.	<i>xawar</i> ; Hawr. <i>xabar</i>	<i>xirāw</i> ; Hawr. <i>xirāb</i>
Z.	<i>xabar</i>	

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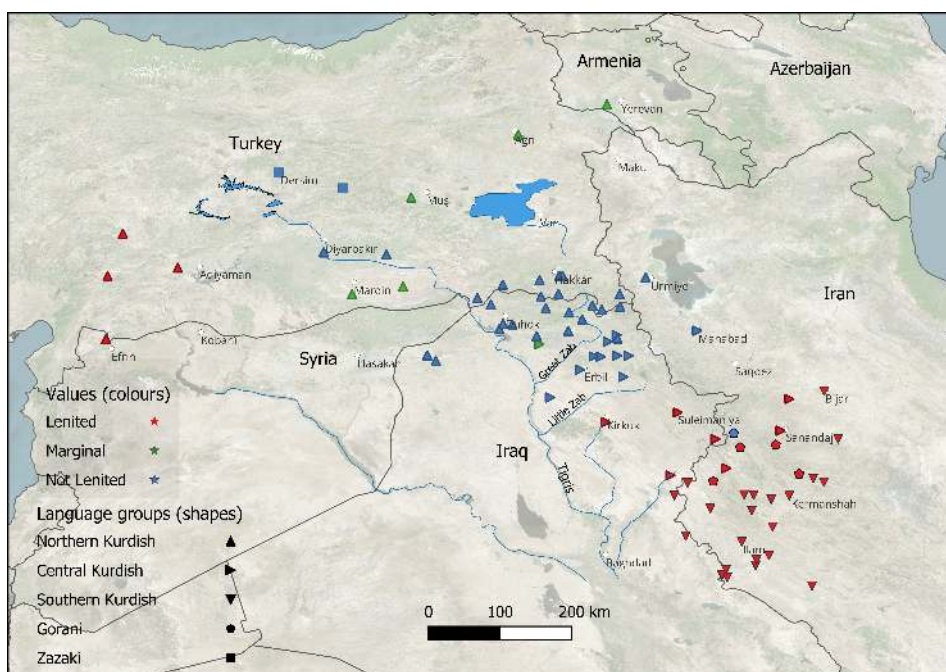


Figure 3.1: Post-vocalic lenition of /b/ in the synchronic state of Kurdish

As can be seen from Figures 2.4, and 3.1, in the synchronic status of the major Kurdish varieties, lenition of the voiced stops /b/ and /d/ is a feature of the south of the greater Kurdish zone, comprising the historical Gorani speech zone, which is now the region of the southern dialects of Central Kurdish, Southern Kurdish and the remaining Gorani dialects. It is thus possible that Kurdish dialects of the region developed lenition by matching non-initial /b/ with the lenited realisation of Gorani through the mechanism Blevins (2017b) refers to as the “perceptual magnet effect”.

However, the scenario described above only accounts for part of the data. It cannot explain why, in the synchronic state of dialects, conservative Gorani dialects do not lenite /b/ in some contexts, e.g., in intervocalic and pre-consonantal position. By contrast, peripheral Gorani dialects exhibit the same effect of lenition as neighbouring CK and SK dialects. For example, intervocalic /b/ undergoes lenition in peripheral Gorani dialects but not in the highland Hewramî dialects. This suggests that lenition was innovated in Southern Kurdish and spread northward to peripheral Gorani dialects and neighbouring CK dialects, but failed to spread to the highland Hewramî dialects. This reflects the effect of geographical

3.3 Bilabials in Armenian

barriers on the diffusion of innovation (see [Mohammadirad & Öpengin 2025](#) for explanations).

3.2.3 Bilabial /w/ vs. labio-dental /v/ in the Synchronic State of Iranian languages of the region

As remarked, most Northern Kurdish varieties maintain a distinction between *w* and *v*, the latter being an intermediate stage in the development from **w* < **b* (fortition of /w/ to /b/). Central Kurdish, on the other hand, has fully lenited **b* as /w/, e.g., NK *av* vs. CK *aw* ‘water’. The realisation of lenited **b* as /w/ is a feature of Central Kurdish, including in the immediate trans-Zab dialects such as Hewlêr, Rewanduz, Shaqlawa, etc.

Within Gorani, the lenited **b*, which historically is the result of a previous lenition **p* > **b* has merged with reflexes of **w*, e.g., *wa* < **wāta* ‘wind’. This observation is true of peripheral Gorani dialects in Iran (e.g., Gorani Zarda) and Iraq (e.g. Gorani Bajalani). However, in conservative Hewramî /w/ is sometimes realised as [v]. Both [w] and [v] realisations of /w/, which is a merger of **w* and lenited **b*, can be heard in the speech of the same speaker.

- (62) *avî* [avi] ‘water’ < **ap*
 wamî [vami] ‘almond’ < Middle Persian *wādām*

Intervocally, the realisation is close to a labio-dental approximate [v], e.g., *aweyanî* [avæjani] ‘prosperity’.

Figure 3.2 shows the distribution of /w/ and /v/ in the current state of Iranian languages of the region.

3.3 Bilabials in Armenian

3.3.1 /b/ in Armenian

In general, Armenian lacks the types of changes associated with lenition of /b/. However, a possible parallel is found in Van ([Acharyan 1952](#): 82), where there are isolated changes of /p/ > /v/, e.g., *spitak/siptak* ‘white’ > *sivtak* (although this may have taken place through analogy with *sev* ‘black’).

3.3.2 /v/ and /w/ in Armenian

Classical Armenian has separate graphemes that are transliterated as *v* (վ) and *w* (լ). These are “in complementary distribution: վ occurs word-initially and after

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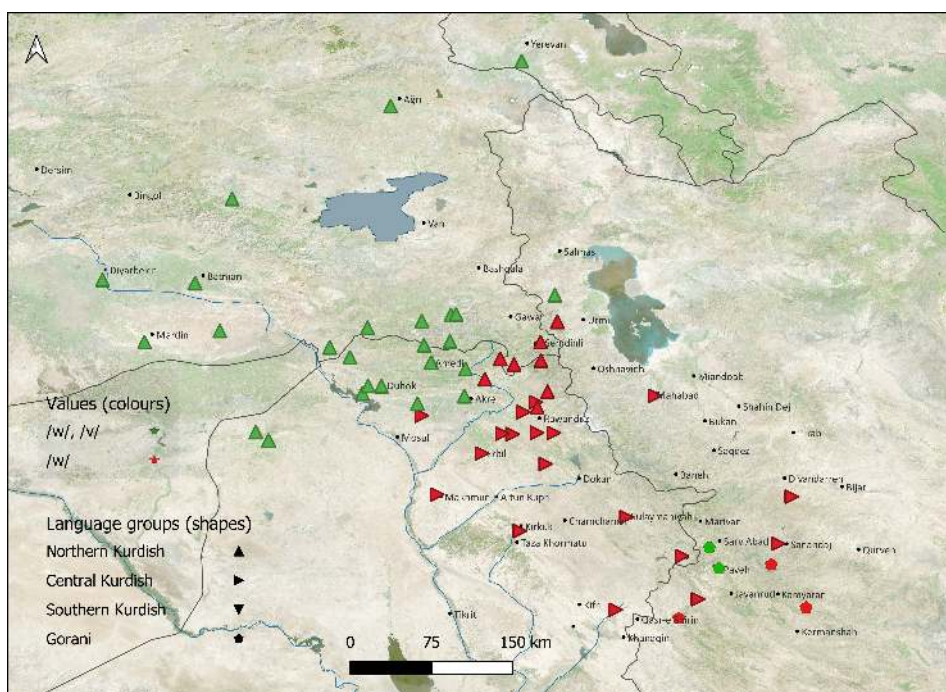


Figure 3.2: The distribution of /w/ and /v/ in the current status of Iranian languages and their merger coloured in red

n (o), while L occurs elsewhere. ... The digraph nL (ow), which represents [u], also renders [v] < *[w] when post-consonantal after the reduction of the weak vowels /u/ or /i/. ... These facts suggest that L and ʟ represent /u/ when syllabified consonantly as a bilabial OA [Old, i.e., pre-Classical Armenian] *[w] (> CA [v])” (Macak 2017: 1040-1). Thus, Macak (2017) implies that this consonantal allophone of [u] was pronounced [v] in Classical Armenian, but [w] in an earlier phase of the language.

Most modern Armenian dialects, including Van (Acharyan 1952), Khoy/Urmia (Asatryan 1962), and Salmast (Vaux 2022) are reported to have only [v], not [w]. However, there are some dialects in which inherited /v/ may be pronounced as [w]. According to Jahukyan (1972), this feature is found in Shatakh and Moks in the Van dialect area, Maragha in northern Iran (in the same dialect group as Urmia), and also Tigranakert (Diyarbakir), Hazzo and Gelieguzan (Sasun), considered by Jahukyan (1972) to be part of the wider Mush dialect area.

3.4 Bilabials in Turkic

In word-initial position, Azeri /b/ is reconstructed by some authors to Proto-Turkic **p*, which is assumed by some authors to have undergone voicing at a relatively early stage (Gadžieva 1984: 182-183). Accordingly, the word-initial /p/ in inherited Azeri lexemes, e.g., *pas* [pas] ‘rust’ < **bas* or *poz-* [poz-] ‘to disrupt’ < **boz-*, reflects a (secondarily) devoiced earlier **b*. Onomatopoeic roots like *partla-* [part:a-] ‘explode’ or *parilda-* ‘to shine’ [parulda-] do not necessarily follow this reconstruction.

In addition, /b/ displays a strong tendency to nasalise when followed by a nasal consonant, either as the coda of the same syllable or as the onset of the following syllable, a development common to Turkic languages in general (Rentzsch 2014: 138). Given the Proto-Turkic constraint against word-initial nasals, most instances of word-initial /m/ in inherited Turkic lexemes in Azeri are attributable to this process, e.g., *mən* [mæn] ‘I’, *min* [min] ‘thousand’, *min-* [min-] ‘to mount, to board’, all of which are reconstructed with initial **b*.

Standard Turkish and many closely related varieties, as well as Gagauz, are among the more conservative Turkic varieties (along with Chuvash) in preserving the archaic consonant, e.g., *ben* ‘I’, *bin* ‘thousand’, *bin-* ‘to mount, to board’. Within Oghuz Turkic, this distribution constitutes a well-known isogloss separating Turkish from Azeri. Eastern Anatolian varieties, however, are divided between nasalised and non-nasalised forms (Rentzsch 2014: 139), and nasalised variants are also documented in Thracian Turkish (Kalay 1998: 44) and Western Anatolian varieties (Rentzsch 2014: 139). Conversely, non-nasalised forms are virtually unattested in Oghuz varieties spoken in the Caucasus and Iran, with only a few exceptions, e.g., Standard Azeri *bəniz* [bæniz] ‘face’ (cf. Turkmen *meñiz*) and its derivatives, e.g., *bənək* [bænək] ‘red birthmark’ (cf. Turkmen *menek*).

Further developments affecting /b/ can be observed at the dialectal level. In many, if not most, Azeri dialects, nasalisation extends to the oblique forms of the demonstrative *bu* ‘this (one)’, which uniformly contain reflexes of the Turkic pronominal *n*, e.g., Urmia Azeri *munun* ‘this.GEN’, *mīnī* ‘this.ACC’ (Doğan 2010: 77). In some cases, the change is not directly conditioned by the presence of a nasal either within the same syllable or in the onset of the following syllable, e.g., Gədəbəy Azeri *moyun* [mojun] ‘neck’; cf. Standard Azeri *boyun* (Şirəliyev 1967b: 81), and may affect even lexical copies, e.g., Culfa Azeri and Meghri Azeri *mahna* [mahna] ‘excuse’ < Persian *bahāna* (Səfərova 2019: 70).

In most northern and north-western dialects, underlying /b/ undergoes lenition in the onset of non-initial syllables (Şirəliyev 1967b: 81,93), e.g., Qazax Azeri [gurvən] ‘sacrifice’ (cf. Standard Azeri *qurban*), Şəki Azəri [χælβir] ‘colander’

3 Bilabials *w and lenited *b

(cf. Standard Azeri *xəlbir*), as well as in postvocalic positions, e.g., Qazax Azeri [gavvurka] ‘rib’ (cf. Standard Azeri *qabırğa*), [gav] ‘container’ (cf. Standard Azeri *qab*), [jælif] ‘come.CVB’ (cf. Standard Azeri *gəlib* [jælip]), including in recent loanwords, e.g., [juſka] (< Russian *jubka* ‘skirt’). Outside this dialect area, the phenomenon is not systematic and is only sporadically attested in other dialects, such as Muğan, Naxçıvan (*Şirəliyev* 1967b: 81,93) and Urmia (*Doğan* 2010: 91), e.g., Urmia Azeri [galmuɟuɟ] ‘s/he has not stayed’, [juɟuɟuɟla] ‘they have gathered’ (cf. Standard Azeri *qalmayıb*, *yığışblar* respectively).

The bilabial fricative /v/ is generally stable except when preceded by mid and open vowels, in which case it may contribute to the formation of a diphthong or a long vowel, e.g., Urmia Azeri, e.g. [o:ɬu] ‘hunter’ (cf. Standard Azeri *ovçu*, Standard Turkish *avcı*), [dø:læt] ‘state’ (cf. Standard Azeri *dövlət* ultimately from Arabic *dawla*[t]), or simply cause the rounding of the adjacent vowel, e.g., Urmia Azeri [ovuf] ‘palm’ (cf. Standard Azeri *ovuc*, Standard Turkish *avuç*), Şahsevən Azeri [søvün-] ‘rejoice’ (*Çam* 2021: 146; cf. Standard Azeri *sevin-*). Sometimes both realisations are found, e.g., Urmia Azeri [gøvvardula] or [gourardula] ‘they would fry’ (*Doğan* 2010: 41, 65; cf. Standard Azeri *qovurardılar*, Standard Turkish *kavururdular*).

In addition, [v] ~[w] is a common reflex of historical /ɣ/ in many eastern and some central Azeri dialects, e.g., Baku Azeri [bajram(u)vuz] ‘holiday.POSS:2PL’ (cf. Qazax Azeri [bajramuɟuz]), Khalkhal Azeri [ælyvæ] ‘hand.POSS:2SG.DAT’ (*Karini* 2009: 98; cf. Qazax Azeri [æliɣæ]). The south-westernmost extent of this isogloss is Sonqor Turkic in Kermanshah (*Bulut* 2014: 37). In Urmia Azeri, as well as in the Bayat varieties of Hamadan and Saveh and and further south in Isfahan (*Bulut* 2014: 37), the corresponding reflex is [j], e.g., [bajramuɟuz] (*Doğan* 2010: 81). Isolated instances of this development are found as far east as Ardabil (*Karini* 2009: 99) and Lerik in south-eastern Azerbaijan (*Mämmədli* 2019: 70). Iraqi Turkic varieties display both patterns, with those spoken west of Kirkuk exhibiting [j] (*Bulut* 2014: 37).

The labial-velar approximant [w] is found in situations of intense contact with Arabic and Kurdish, most commonly in etymological contexts, as described by *Bulut* (2014: 32-33), e.g., Tuzxurmatı Turkic (Iraq) [wɔxɪd] ‘time’ (< Arabic *waqt*), Sonqor Turkic (Iran) [wæɬh] ‘face’ (ultimately from Arabic *wajh*); but also often non-etymologically, as a reflex of Turkic /v/, e.g. Tuzxurmatı Turkic [wer-] ‘give’ (cf. Standard Azeri *ver-*), Sonqor Turkic [war] ‘there is’ (cf. Standard Azeri *var*). On the northern Azeri periphery, it is found in the same way in varieties showing strong East Caucasian substrate, e.g., Zaqatala–Qax Azeri [war] ‘there is’, [watan] ‘homeland’ (*Mämmədli* 2019: 71; cf. Tsakhur [watan] vs. most Azeri [vætæn], ultimately from Arabic *waṭan*).

3.5 Discussion

The labio-velar reflex [w] of historical *w and lenited *b is attested in a wide range of NENA dialects in the areas of both Northern and Central Kurdish. Whereas the reflex of lenited *b in Northern Kurdish is [v], in NENA dialects in contact with Northern Kurdish across northern Iraq and south-eastern Turkey the reflex of lenited *b is [w]. This could be explained by the hypothesis that the lenition of *b to [w] is an internal development of Aramaic and not induced by contact. One possibility, however, is that NENA dialects in the northern sector originally matched Northern Kurdish with *b leniting to the labio-dental [v], and subsequently, internally, this was levelled phonetically and phonemically with [w] < *w. One piece of evidence supporting this hypothesis is that in NENA dialects in the northern sector there are a small number of native Aramaic words in which the reflex of *b is /v/ or is a /w/ (depending on the dialect) that is realised phonetically either as [w] or [v]. Further support for such a hypothesis is that in the Mlaḥso dialect of Aramaic, which is spoken west of the Tigris in the Ṭur ʿAbdin region, the reflex of lenited *b is regularly /v/ whereas the reflex of *w is /w/, exactly as in Northern Kurdish (Jastrow 1994a: 19). Mlaḥso may preserve a more archaic stage in the development of bilabials that reflects more transparently the matching of the Aramaic sound system in dialects in the region with that of Northern Kurdish.

The Christian NENA dialects spoken in the area of Central Kurdish that have /w/ as the reflex of both lenited *b and *w match the synchronic state of Central Kurdish in this respect. It is not necessary, therefore, to posit such an internal levelling of *v with /w/.

The labio-dental ([v], [ʋ]) and bilabial ([β], [ɸ]) reflexes of *w and *b in several NENA dialects are likely to have been induced by contact.

The labio-dental fricative reflex [v] in the Christian dialects of the Urmi region is likely to have developed due to contact with Armenian and Azeri Turkic dialects in this region, which have /v/ in their sound inventories. This would have developed by a process of perceptual matching of [w] with [v]. It is relevant to note that some cases of [v] in Azeri Turkic resulted by lenition of *b, which would parallel the process of lenition in NENA (§3.4).

The bilabial ([β], [ɸ]) reflexes that are characteristic of the Christian NENA dialects of the Targawar and Margawar region are linked to the fronting of /u/ to /ü/ [y] in these dialects. Fronted /ü/ is an areal feature of the region shared by the Kurdish dialects in south-eastern Turkey and north-western Iran, and in the Behdîni Kurdish dialect of northern Iraq (Haig 2019b: §4.1.2).

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In the Jewish Trans-Zab dialects, labio-dental realisations of the phoneme are found as far west as Arbel in Iraq, although they are slightly more predominant in the dialects of this sub-group on the eastern periphery in Iran and near the Iranian border. In recent times, the Jewish Trans-Zab dialects have been mainly in contact with Central Kurdish, in which the labio-dental [v] has shifted to labio-velar [w]. It is possible that the phoneme in Jewish Trans-Zab dialects converged with a labio-dental reflex of lenited **b* that would have existed at some earlier period in areas where Central Kurdish is now spoken. This historical situation would have been, presumably, like the synchronic situation in Northern Kurdish, in which /w/ and /v/ co-exist, the latter being the intermediate stage of lenition of **b*. A difficulty with this argument, however, is that the labio-dental realisation in Jewish Trans-Zab dialects is exhibited by the reflexes of both lenited **b* and historical **w*. It is more likely that the labio-dental realisation arose by contact with Gorani. In Hawraman, Gorani dialects have labio-dental realisations ([v] or [ʋ]) of the merged reflexes of **w* and lenited **b*. Indeed, it has been shown in [Khan & Mohammadirad \(2024b\)](#) that Gorani is likely to have been a substrate of Jewish Trans-Zab NENA dialects.

4 Unaspirated stops and affricates

4.1 Unaspirated stops and affricates in NENA

4.1.1 Unaspirated stops with phonemic status

Unaspirated voiceless non-emphatic (i.e., non-pharyngealised) stop and affricate phonemes are not inherited from earlier Aramaic but have entered NENA either through loanwords or through innovative internal processes.

The non-inherited unaspirated phonemes include the following (marked in the transcription by the diacritic _ˁ):

/p̰/ [p]
 /t̰/ [t]
 /k̰/ [k]
 /t͡ʃ̰/ [t͡ʃ]

In some dialects of the north-eastern sector, such as C. Urmi, there is a general fronting dorsals by a process of palatalisation. The voiceless velar stop /k̰/ is advanced to the unaspirated palatal /ç̰/ [ç] or further to the alveolar affricate /t͡ʃ̰/ [t͡ʃ], and /t͡ʃ̰/ [t͡ʃ] is advanced to t̰̂ [t̰̂]. The place of /k̰/ is taken by the original uvular /q̰/, which was unaspirated due to its being originally a member of the emphatic (pharyngealised) series of consonants. This can be represented as follows:

(1) C. Urmi (Khan 2016b)

*q̰ > /k̰/
 *k̰ > /ç̰/, /t͡ʃ̰/
 *t͡ʃ̰ > /t̰̂/

The unaspirated phonemes /p̰/ [p], /t̰/ [t], /k̰/ [k] have a much shorter Voice Onset Time (VOT) before vowels (typically averaging 10ms) than the corresponding voiceless aspirated consonants /p̰̰̰/ [pʰ], /t̰̰̰/ [tʰ], /k̰̰̰/ [kʰ], /t͡ʃ̰̰̰/ [t͡ʃʰ] (typically averaging 50ms). In the articulation of unaspirated stop segments, voicing of the vowel begins almost immediately after release of the occlusion. During the occlusion of unaspirated segments, the vocal folds are far more constricted than they are during the occlusion stage of aspirated segments and there is greater tension of

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the vocal tract walls. They may be described as glottalised. In the variety of C. Urmi NENA spoken in Georgia, they are sometimes realised as glottal ejectives, which results from the raising of the larynx.

The emphatic, i.e., pharyngealised, stop /t/ and the emphatic affricate /č/, are also generally pronounced without aspiration. They combine the tense glottal setting of unaspirates with pharyngealisation through retraction of the tongue into the upper pharynx (§7.1).

This section is concerned with the non-inherited unaspirated stop and affricate phoneme segments that do not include pharyngealisation within the complex of their segmental features. The phonemic distinctness of these from the corresponding voiceless aspirated stops is demonstrated by minimal pairs such as:

- (2) C. Urmi (Khan 2016a: vol. 1, 99-100)

<i>ḥatḥat</i>	‘he whispers’	:	<i>patpat</i>	‘he cards (wool)’
<i>t-ilə</i>	‘who is’	:	<i>tilə</i>	‘he came’

- (3) C. Barwar (Khan 2008a: 31)

<i>ḥura</i>	‘one-year-old goat’	:	<i>kura</i>	‘furnace’
<i>ḥoma</i>	‘pile’	:	<i>koma</i>	‘black’

- (4) J. Arbel (Khan 1999a: 19)

<i>ṭāra</i>	‘fruit’	:	<i>tāra</i>	‘door’
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The section is not concerned with phonetically conditioned unaspirated voiceless allophones. These occur in all NENA dialects as phonetic allophonic alternants of aspirated stops when they are preceded by a fricative in a cluster. This is seen in inflections of verbal roots such as *š-t-y* in the following:

- (5) J. Sanandaj

<i>štele</i> [ʃteːleː]	‘he drank’	vs	<i>šāte</i> [ʃatʰeː]	‘he drinks’
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Aspiration is lost also across the dialects when the stop is followed by a fricative or continuant in a cluster, e.g.,

- (6) J. Arbel (Khan 1999a: 19)

4.1 Unaspirated stops and affricates in NENA

tre [tre:] ‘two’

Furthermore, the section is concerned only with unaspirated phonemes that contrast with aspirated phonemes. The VOT of voiceless stops and affricates varies across dialects that do not have this contrast. Indeed, in some dialects without the contrast, the VOT of inherited voiceless stops is as short as the VOT of unaspirated phonemes in dialects that have the contrast. This is the case, for example, in J. Sulemaniyya, where the VOT of /p/, /t/, /k/ and /č/ can be 10ms or below, e.g.,

(7) J. Sulemaniyya (Khan 2004: 19, 462)

čáran [tʃ(VOT 10)a:ran] ‘our fortunes’
píli [p(VOT 7)i:li] ‘they fell’

A similar short VOT has been documented by Coghill (2003: 26) for the C. Alqosh dialect.

4.1.2 Geographical distribution of unaspirated stop and affricate phonemes

The unaspirated voiceless stop and affricate phonemes are found in dialects in the northern sector of the NENA area. The furthest south they are attested is in J. Arbel (Khan 1999a: 19) (see Figure 4.1).

4.1.3 Processes of development

4.1.3.1 Loanwords

In many NENA dialects of the northern sector voiceless unaspirated stops are attested in loanwords from Northern Kurdish, e.g.,

(8) C. Barwar (Khan 2008a: 31-32)

ḫərruš ‘treacherous’
ḫura ‘one-year-old goat’
ḫoma ‘pile’
ṭarra ‘fresh’

(9) J. Arbel (Khan 1999a: 19)

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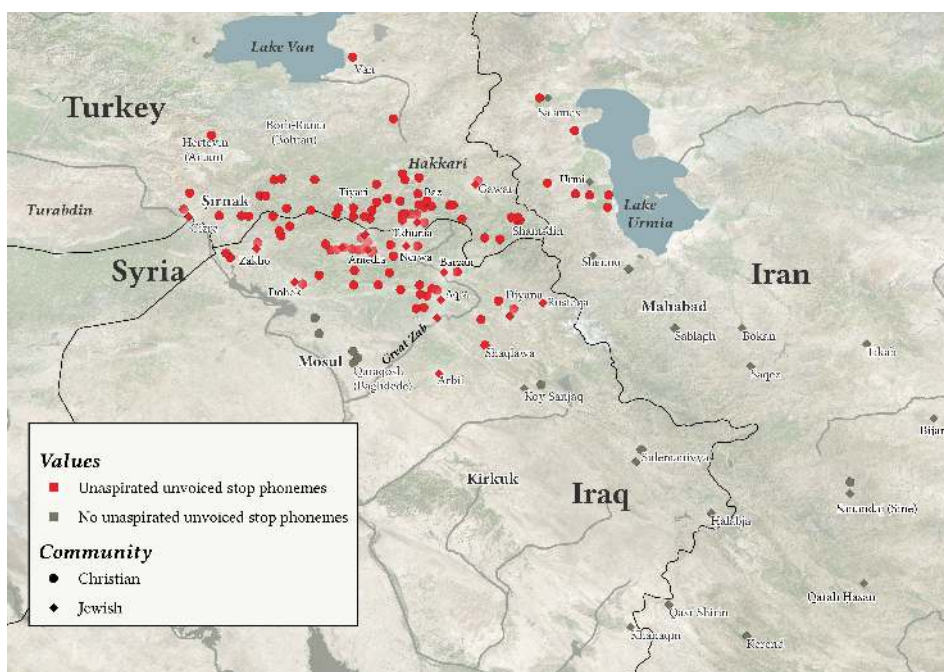


Figure 4.1: Distribution of unaspirated stops and affricates in NENA dialects

<i>ṭāra</i>	‘fruit’
<i>taḵāna</i>	‘only son’
<i>māṭ</i>	‘astounded’
<i>kurtaḵ</i>	‘Kurdish jacket’

(10) C. Urmi (Khan 2016a: vol. 3)

<i>ḥurṭila</i>	‘puppy’	< NK <i>ḥurṭ</i> ‘short’
<i>ḥarṭa</i>	‘curl’	< NK <i>ḥurṭ</i> ‘short’
<i>ḥaḥḥuna</i>	‘father’ (Arzni)	< NK <i>ḥeḥ</i> ‘elder brother’
<i>ḥura</i>	‘blind’	< NK <i>ḥur</i>
<i>ḥaḥḥə</i>	‘weapons, arms’	< NK <i>ḥek</i>
<i>ḥəḥḥə</i>	‘girl’	< NK <i>ḥeḥ</i>

4.1.3.2 Phonemicisation of phonetic de-aspiration after fricatives

As mentioned above, when the first consonant of a cluster is a fricative and the second is a voiceless stop, the stop is realised phonetically as unaspirated in

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all NENA dialects by a well-known process of de-aspiration after fricatives. A common place for such clusters is the past perfective inflection of verbs, e.g.,

(11) C. Qaraqosh

Past perfective	Present
/xpər-re/ [xpərre]	/xapər/ [xap ^h ər]
dig.PST-OBL.3SG.M	dig.PRS.3SG.M
‘he dug’	‘he digs’

Verbs in NENA, as is common in Semitic, have non-concatenative roots, which are merged with morphological vocalic patterns expressing TAM and voice distinctions. The verb ‘to dig’ in C. Qaraqosh has the root *x-p-r* ‘to dig’. When not preceded by fricatives, the voiceless stop is always realised as aspirated, e.g., in the present inflection /xapər/ [xap^hər].

In the C. Urmi dialect, by contrast, the unaspirated allophone in the cluster of the past perfective forms has been re-analysed as an underlying unaspirated phoneme, i.e., [xp̂] > /xp̂/. As a result, the unaspirated stop has come to be analysed as a radical of the morphological root, i.e. *x-p̂-r*. In other inflections of the verb where the stop is separated from the fricative by a vowel, therefore, the stop is still unaspirated:

(12) C.Urmi (Khan 2016a: vol. 1, 93-94)

Past perfective	Present
/xp̂ər-re/ [xp̂ərre]	/xap̂ər/ [xap̂ər]
dig.PST-OBL.3SG.M	dig.PRS.3SG.M
‘he dug’	‘he digs’

4.1.3.3 Phonemic merger of a stop and a laryngeal Stop

In this process, there is reanalysis of a cluster of a voiced stop and laryngeal stop as a unitary voiceless unaspirated phoneme. This is attested in the sequence of a voiced /d/ or /b/ followed by /ʔ/.

4.1.3.3.10 /d/ + /ʔ/ > /t̪/

(13) C.Urmi (Khan 2016a: vol. 1, 142)

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$d + 'ilə > t-ilə$

SBRD + COP.3SG.M

‘that he is’

$d + 'atə > t-atə$

FUT + come.PRS.3SG.M

‘he will come’

4.1.3.3.11 /b/ + /ʔ/ > /p̂/

(14) C. Urmi (Khan 2016a: vol. 1, 142)

$b + 'ida > p̂-ida$

‘by hand’

$*b'urta > p̂urta$

‘droppings of sheep’ (cf. Syriac *bā'ūra* ‘droppings’)

(15) C. Mawana (Khan, fieldnotes)

$*'arb'a > 'arp̂a$

‘four’

When this process occurs in inflections of verbal roots in forms in which a voiced stop and a laryngeal are adjacent, the resulting voiceless unaspirate may be reanalysed as a root letter and extended to other inflections, e.g.,

(16) C. Mawana (Khan, fieldnotes)

$*d'ere > tere$ ‘he returned’

$*da'ər > tayər$ ‘he returns’

$*b'ele > p̂ele$ ‘he loved’

$*ba'e > p̂äye$ ‘he loves, he wants’

4.1.3.3.12 /č/ + /ʔ/ > /č̣/

An unaspirated affricate /č̣/ can develop from the coalescence of voiceless aspirated /č/ with a laryngeal, as in:

(17) C. Urmi (Khan 2016b)

$*'əč'i > 'əč̣'i$

‘ninety’

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4.1.3.4 Assimilation

A voiced stop shifts to a voiceless unaspirated stop when preceding an unaspirated stop in a cluster, e.g.,

- (18) C. Mawana (Khan, fieldnotes)

**b_ɬ-awe* > *ḫ_ɬ-awe*
 FUT-be.PRS.3SG.M
 ‘he will be’

4.1.3.5 Reanalysis of pharyngealised segments

The consonantal phoneme inventories of earlier Aramaic included consonantal segments with a secondary coarticulatory feature of pharyngealisation, known in the Semitic philological literature as ‘emphatic consonants’. These were inherited as pharyngealised segments by many NENA dialects (§7.1), e.g.,

- (19) C. Qaraqosh

/t/ [t̤]
 /s/ [s̤]

The fundamental articulatory gesture of pharyngealisation of such segments in dialects such as Qaraqosh involves the constriction of the upper pharynx with increased muscular tension. This is a non-primary articulation that accompanies the primary articulation of the consonants. It is accompanied by a variety of other features including the retraction of the tongue from the primary point of articulation. The muscular tension and constriction of the upper pharynx also brings about some degree of glottal constriction, with the consequence that the stop /t/ is realised as a pharyngealised unaspirated stop.

In Qaraqosh the pharyngealisation can be identified as the feature [+pharyngeal] within the feature matrix of the phoneme segment. The feature matrix can also be assumed to contain the feature [+tense] reflecting glottal constriction, i.e.,

- (20) C. Qaraqosh

/t/ [+coronal, +pharyngeal, +tense]

The coarticulatory gesture of pharyngealisation spreads to adjacent vowels, which are retracted, e.g.,

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(21) C. Qaraqosh

/t̪a/ [t̪ˢɑˢ] vs /ta/ [tʰæ]

In several dialects in the north-eastern sector of the NENA area, a historical change has taken place, whereby the pharyngealisation feature of the inherited emphatic consonant segments and the associated phonetic coarticulatory pharyngealisation of adjacent segments have been reanalysed as a phonemic suprasegmental feature whose domain is the entire word. This can be represented as follows:

(22) C. Urmi (Khan 2016a: vol. 1, 108)

Word level:		>	/ʕ/ +Pharyngeal
Segment level:	/t̪ˢ/ +Pharyngeal +Tense		/t̪/ +Tense

As a result, the coarticulatory pharyngealisation of adjacent segments has become phonemicised (the initial ⁺ in the transcription represents a suprasegmental phonemic feature.):

/t̪a/ [t̪ˢɑˢ] > ⁺t̪a

On a phonetic level, C. Urmi /⁺t̪/ has the same realisation as Qaraqosh /t̪/, but there is a difference in underlying phonemic form whereby the pharyngeal feature is now a ‘floating’ feature at the word level. As a result of this reanalysis, the stop is now the unaspirated phoneme /t̪/, since the pharyngeal feature has been disengaged from it on a segmental level.

In some sporadic cases the /t̪/ develops further into the unaspirated affricate /t͡ʃ/, e.g.,

(23) C. Mawana (Khan, fieldnotes)

⁺⁺t̪āme > ⁺t͡ʃāme ‘he extinguishes’
⁺⁺k̪āte > ⁺k̪āt͡ʃe ‘he pulls off’

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4.1.3.6 Induced by pharyngealisation

An original aspirated stop may become an unaspirated segment in pharyngealised words in dialects with suprasegmental pharyngealisation. This is attested with regard to the labial stop /p/, e.g.,

(24) C. Urmi (Khan 2016a: vol. 1, 163)

^{**}*parmə* > ⁺*parmə* ‘he understands’
^{**}*pərʔa* > ⁺*pərʔa* ‘a tear, a rip’

4.1.3.7 Loss of segmental pharyngeal feature

An unaspirated stop in some cases has developed in plain (i.e., non-pharyngealised) words in dialects with suprasegmental pharyngealisation by the elimination of the pharyngealisation in the word, e.g.,

(25) C. Mawana (Khan, fieldnotes)

^{**}*ʔaləḵ* > *ʔaləḵ* ‘he disappears’
^{**}*ʔayəp* > *ʔayəp* ‘he folds’

(26) C. Urmi (Khan 2016a: vol. 1, 178–179)

^{**}*ʂəʔəx* > *ʂəʔəx* ‘he spreads out’

4.1.4 Sound symbolism

In dialects that contain unaspirated phonemes, these phonemes have been introduced into some words by a process of sound symbolism. Sound symbolism involves the correspondence between sound and meaning. Particular sounds, clusters of sounds or phonological features may be associated with a particular domain of meaning. Such phonological features have been termed ‘ideophones’ (Voeltz & Kilian-Hatz 2001) or ‘phonesthemes’ (Reay 2006). The unaspirated consonants are used symbolically to express a diminutive degree. The key ideophonic feature of the unaspirated consonants is their higher pitch compared to the corresponding aspirated consonants. The high pitch matches with the semantic feature of diminutive.¹

¹The high pitch of glottalised consonants has been found to express a diminutive function in various other languages, e.g., Nichols (1971), who drew attention to this phenomenon in North American Indian languages.

4 Unaspirated stops and affricates

In nouns expressing small items, particles expressing smallness, or verbs expressing actions of inherently short duration or diminutive sound, voiceless stops and affricates that were originally aspirates are made unaspirated.

This is found in C. Urmi in words that are NENA in origin and also loanwords, in which the voiceless stops are aspirated in the source language, e.g.,

(27) C. Urmi (Khan 2016a: vol. 1, 178–181)

<i>təlpa</i>	‘eyelash’	<	* <i>təlpa</i> (cf. Syriac <i>tallīpē</i> , Turoyo <i>tlafine</i>)
<i>taḫtəp</i>	‘to blink; to flicker’	<	* <i>taptəp</i> (cf. C. Barwar <i>tarəp</i> , <i>mtarpəθ</i> , <i>rapəθ</i> ‘to blink’)
<i>ḫərtuxə</i>	‘crumbs’	<	* <i>pərtuxə</i> (cf. <i>p^hart^həx</i> ‘to crumble’, C. Barwar <i>pərtoxe</i>)
<i>ḫərtə</i>	‘crumb’	<	* <i>pərtə</i> (cf. C. Barwar <i>pərtika</i> ‘splinter’)
<i>ḫəçḫəç</i>	‘to whisper’		cf. Azeri <i>pıçıldamaq</i> [p ^h ɪɾɪɫɪɫdamaq]
<i>xəççə</i>	‘a few, a small amount’	<	* <i>ḫad</i> + Turkic suffix <i>-çə</i>
<i>ḫurçəfə</i>	‘locks of hair’	<	NK <i>pirç</i> [p ^h ɪɾɪɾɪɫɪɫ]

In C. Urmi the adjective *šəpīra* ‘beautiful’ has a non-etymological unaspirated /p/ (< **šappīrā*), which is not found in most other NENA dialects, e.g., C. Barwar *šəpīra*. This may have developed by a similar process of diminutive sound symbolism, in this case the diminutive being associated with endearment.

We should include here the anomalous occurrence of a tense /t/ in the feminine ending of the C. Urmi adjective *sura*:

(28) C. Urmi (Khan 2016a: vol. 1, 178–179)

surta ‘small (sg.f)’

The normal realisation of the feminine-marking suffix of nouns and adjectives is *-ta* with an aspirated /t/.

It should be noted that the use of unaspirated consonants for sound symbolism is used more extensively in the north-eastern sector of NENA dialects. It has not developed in any significant way in other dialects that have unaspirated consonants in their sound inventories. This can be seen in various comparisons of C. Urmi to C. Barwar above. One possible case of introduction of unaspirated

4.1 Unaspirated stops and affricates in NENA

stops for sound symbolism in a word in C. Barwar that is not a Kurdish loan is the following:

(29) C. Barwar

to̞to ‘old woman, grandmother’

4.1.5 Further phonetic aspects of voiceless unaspirated stops

The unaspirated consonants /*k̥*/, /*p̥*/, /*t̥*/ and /*ç̥*/ are pronounced with a greater degree of muscular tension than the corresponding aspirated consonants. In Kurdish loanwords in C. Barwar this is sometimes accompanied by a certain amount of pharyngealisation and consequent backing of the point of articulation, resulting in their pronunciation as emphatic consonants, e.g.,

(30) C. Barwar

<i>laṭṭa</i>	‘shallow’	<	NK <i>lat</i> [la:t]
<i>saqqat̪</i>	‘crippled’	<	NK <i>seqet</i> [saqat]
<i>ṭanga</i>	‘saddle-girth’	<	NK <i>teng</i> [tang]
<i>p̪ena</i>	‘kick’	<	NK <i>p̪en</i> [pe:n]
<i>ç̪anga</i>	‘leather gloves used in harvesting’	<	NK <i>ç̪eng</i> [tʃang]

In dialects with suprasegmental pharyngealisation, this process results in the whole word being pronounced pharyngealised, e.g.,

(31) C. Urmi

⁺ <i>laṭṭa</i>	‘shallow’	<	NK <i>lat</i> [la:t]
⁺ <i>p̪uza</i>	‘snout’	<	NK <i>poz</i> [po:z]

(32) C. Mawana

⁺ <i>p̪ena</i>	‘kick’	<	NK <i>p̪en</i> [pe:n]
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By the same process, the point of articulation of unaspirated /*k̥*/ in C. Barwar is sometimes moved back to that of the uvular stop /*q̥*/, which belongs to the class of emphatic consonants in Semitic, e.g.,

(33) C. Barwar (Khan 2008a: 32)

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<i>qaška</i>	‘cake of dried yoghurt curds’	<	NK <i>keşk</i> [kaʃk]
<i>gisqa</i>	‘young male goat’	<	NK <i>gîsk</i> [gi:sk]

Conversely an original uvular /q/ in Kurdish is sometimes brought forward to a velar /k/, e.g.,

(34) C. Barwar

<i>kaška</i>	‘wheat husk’	<	NK <i>qaşik</i> [qaʃik]
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In this case, the stimulus for the shift may have been the need to distinguish the word from *qaška* ‘cake of dried yoghurt curds’, which in C. Barwar acquired the /q/ by a backing of the place of articulation. An example of fronting of Kurdish /q/ to an unaspirated /ç/ in C. Urmi is as follows:

(35) C. Urmi

<i>çutta</i>	‘short; dwarf’	<	NK <i>qut</i>
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This may have been motivated by sound symbolism to express a diminutive. It would have involved the elimination of the pharyngeal feature in /q/. Elimination of pharyngealisation is a strategy for the expression of the diminutive in this dialect (§4.1.4).

In J. Arbel, in intervocalic position an unaspirated /k/ is sometimes pronounced voiced, e.g.,

(36) J. Arbel (Khan 1999b: 25)

<i>kurtaqe</i> [k ^h ürt ^h a:ge:]	‘Kurdish jackets’
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4.2 Unaspirated stops and affricates in Kurdish

4.2.1 Overview

Northern Kurdish, but also some other varieties (see below), have a three-way distinction between stops:

- voiceless aspirated: /p/ [p^h], /t/ [t^h], /k/ [k^h], /ç/ [tʃ^h]
- voiceless unaspirated: /p/ [p], /t/ [t], /k/ [k], /ç/ [tʃ]
- voiced stops: /b/, /d/, /g/, /j/

4.2 Unaspirated stops and affricates in Kurdish

There are thus two sets of voiceless stops as aspirated and unaspirated (Kahn 1976: 21-29). The phonemic status of the distinction appears in the following near-minimal pairs (The underlined characters are unaspirated):

Table 4.1: Three-way phonemic distinction in stops in Northern Kurdish

Voiceless Aspirated		Voiceless Unaspirated		Voiced	
<i>tîx</i>	‘blade’	<i>ṭîj</i>	‘bitter’	<i>dîn</i>	‘crazy’
<i>pare</i>	‘money’	<i>ṭar</i>	‘last year’	<i>bar</i>	‘burden’
<i>çê</i>	‘good’	<i>çêl</i>	‘cow’	<i>jêr</i>	‘bucket’
<i>ker</i>	‘donkey’	<i>ķer</i>	‘deaf’	<i>ger</i>	‘stroll (n.)’

Additionally, there are emphatic/pharyngealised versions of the *p* and *t*, as in *pez* ‘sheep’, *ṭeyr* ‘bird’. All emphatic realisations of these sounds are unaspirated, but these form another class with a number of other emphatic consonants and hence will be treated also separately (see §7.3) .

4.2.2 Zazaki

Unaspirated stops are a characteristic feature of the Northern Zazakî Dialects, which exhibit a three-way phonation distinction for stops (including affricates) in initial position: voiceless aspirated, voiceless unaspirated, and voiced. Asatrian (1995) provides a lexical pair *çîm* ‘eye’ and *çîçeg* ‘flower’ demonstrating contrastive aspiration for the northern group and Paul (1998: 184) confirms this pattern. Some rare occurrences have been claimed for Central Zazakî as well (cf. Keskin 2008: 33). This distinction is entirely absent in the Southern peripheries of the Zazakî speech area. Where they occur, unaspirated stops are limited to the voiceless obstruents *p*, *t*, *k*, and *ç* [tʃ] / [ts], represented as *k̰*, *p̰*, *t̰*, *ç̰*. Finding actual minimal pairs, however, is difficult (Table 4.2). There is no phonological rule governing their distribution, and there is some dialectal variation, so the articulation is not universal. Still, unaspirated stops appear predominantly in syllable onsets, although word-final instances, especially *p̰* and *ç̰*, have been documented (Selcan 1998: 148-158).

The exact nature of the articulation is also debated. While most authors argue for an aspirated vs. unaspirated contrast, Selcan (1998: 148) instead proposes ejective plosives. Unfortunately, no comprehensive acoustic-phonetic analysis exists to settle the debate. It has, however, been observed that some speakers articulate these sounds with a more glottalic airstream.

4 Unaspirated stops and affricates

Table 4.2: (Near-)minimal pairs in aspirated and unaspirated stops

Aspirated		Unaspirated	
<i>kal</i>	‘raw’	<i>ḵal</i>	‘old’ (human)
<i>kêf</i>	‘mood’	<i>ḵêz</i>	‘blonde’
<i>pîze</i>	‘stomach’	<i>pîz</i>	‘pole’
<i>por</i>	‘hair’	<i>ḵonc</i>	‘five’
<i>tam</i>	‘complete’	<i>ṭam</i>	‘taste’
<i>tüye</i>	‘mulberry’	<i>ṭüye</i>	‘(a type of) owl’
<i>çakuç</i>	‘hammer’	<i>çapole</i>	‘slap’
<i>çerme</i>	‘leather’	<i>çem</i>	‘river’

It has been suggested that the distinction is the result of contact influence from Armenian (see below; cf. also [Haig 2017](#): 292, [Vahman & Asatrian 1990](#): 268), since Armenian was dominant in the north. Surprisingly, in the south Zazaki area, where Northern Kurdish is more dominant, this distinction is lacking, which is further support for the Armenian-influence as the source of the contrastive aspiration in Zazaki since it is in this area that until late 20th century significant Armenian communities were living. Interestingly, Semitic loanwords such as *ṭeyr* ‘bird’ appear unaspirated in Northern Zazakî, whereas they remain emphatic (*ṭeyr*) in Central and Southern Zazakî.

4.2.3 Evolution

Most descriptions treat this situation as the “addition” of an aspirated stop series, and transcribe this series with the relevant diacritic (^h). However, from diachronic as well as synchronic perspectives, the aspirated voiceless stops in Kurdish correspond to the modal fortis stops /p/, /t/, /k/, /č/ in other varieties of Iranian. It is thus the voiceless unaspirated stops which are innovated in Kurdish.

[Lecoq \(1997: 34\)](#) views the emergence of this three-way distinction (i.e., the addition of unaspirated stops to the Kurdish phoneme inventory) as the result of the situation of (Northern) Kurdish within a “Caucasian” linguistic area prior to the arrival of Islam, since it shares this feature with Armenian and Ossetic, in addition to some varieties of Zazaki.

Related to its innovatory status in Northern Kurdish, the distribution of the unaspirated stop series is restricted in a number of ways.

First, phonemic unaspirated voiceless stops are limited to morpheme-initial position; they are not found intervocally within stems, and while they can be

4.2 Unaspirated stops and affricates in Kurdish

found in consonant clusters and word-finally, contrast with their aspirated and voiced counterparts is neutralised there.

Secondly, in initial position, they are most often followed by high and mid vowels /i/, /ê/, /u/, /û/ (e.g., *îr* ‘arrow’, *tê* ‘she comes’, *tu* ‘you (sg)’, *tû* ‘mulberry’). This tendency suggests that they emerged in the language through an allophonic alternation. That is, the unaspirated counterparts were favoured before high vowels while in other environments the aspirated counterparts were used. Today, however, there are many counterexamples to this pattern as seen in Table 4.1, which confirms their contrastive status.

Thirdly, on balance, unaspirated voiceless stops remain less commonly distributed in the lexicon, and there is significant interdialectal variation in the assignment of this feature to specific words (e.g., *tirs* ~ *ṭirs* ‘fear’, and see below); see map and discussion in Anonby (2022: 91-93).

4.2.4 Distribution

Voiceless aspirated and unaspirated stops are found primarily in Northern Kurdish, but also in the central dialects of Zazaki, and some north-eastern Central Kurdish dialects. All of the Northern Kurdish dialects have unaspirated voiceless stops. However, there is extensive variation among dialects as to whether a lexical item is pronounced with or without aspiration. Some lexical items do not show any such variation. For example, the words *pare* ‘money’ vs. *ḫar* ‘last year’ or *çê* ‘good’ vs. *çem* ‘river’ (with slightly differing dialectal forms) are pronounced in this way across all Northern Kurdish varieties. But the items in 4.3 show how dialects differ from each other, with only limited internal patterning in the realisation of voiceless stops in one of the three options available (i.e., aspirated, unaspirated, emphatic unaspirated):

Table 4.3: Variation in aspiration across NK dialects

Gloss	Zakho	Shemdinan	Amedi	Batman	Bayazid
‘eye’	çav	çaw	çav	çav	çav
‘back’	pişt	pişt	pişt	pişt	pişt
‘dress’	ḫiras	ḫiras	ḫiras	ḫiras	ḫiras
‘fear’	ṭirs	ṭirs / tirs	ṭirs	tirs	ṭirs

Aspirated vs unaspirated pronunciations vary sporadically between dialects within the same locality between two speech communities (e.g., neighborhoods,

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tribes, sub-districts). In Shemdinan region, for instance, the words *tirş* ‘sour’ and *kanî* ‘fountain’ are pronounced with emphatic, unaspirated, and aspirated initial stops, respectively, but in the speech of the Harki tribe (as also found in their nomadic communities in Soran/Erbil provinces) these are pronounced as aspirated non-emphatic and unaspirated stops, respectively.

Proportionally, however, the dialects closer to Central Kurdish speech zone (e.g., Shamdinan, Soran, Mergasor/Barzan, Akre) have more aspirated stops than other Northern Kurdish dialects.

Akre and Barzan/Mergasor dialect have a lot more aspirated pronunciations such as *kew* ‘partridge’ which is unaspirated as *ķew* ‘partridge’ in all other Northern Kurdish dialects, nevertheless it does have a lot of unaspirated initial stops too.

(37) Akre

<i>ķezan</i>	‘terebinth tree’
<i>îçek</i>	‘a little’
<i>îenîr</i>	‘cheese’
<i>îenî</i>	‘thirsty’

There are thus no known NK dialects that lack unaspirated stops, but in bilingual contexts with CK as the dominant language of socialisation (e.g., in Erbil city), the aspirated pronunciation replaces the unaspirated pronunciation of young speakers. Thus, the data on the Mergasor/Barzan variety obtained from a young adult postgraduate student who grew up in a suburb of Erbil very often have aspirated pronunciations.

(38) Barzan/Mergasor

<i>penîr</i>	‘cheese’
<i>spî</i>	‘white’
<i>ker</i>	‘deaf’
<i>kevil</i>	‘hide’
<i>tîr</i>	‘arrow’
<i>tû</i>	‘mulberry’
<i>çêl</i>	‘cow’
<i>çerm</i>	‘skin’

It seems that the unaspirated stops /p/ and /ç/ exist in these dialects in close contact with CK, as illustrated by *îenc* ‘five’ and *çem* ‘river’, but they may be entirely lost with velar and alveolar counterparts /k/ and /t/ as there are no unaspirated realisations of them in the data.

4.3 Unaspirated stops and affricates in Armenian

Finally, some CK dialects in the northern area of Erbil province such as Soran (Diyana), Surchi and Masif seem to have some unaspirated initial consonant pronunciations, e.g., *tirs* ‘ear’ in Surchi and Soran, and *teng* ‘narrow’ and *tezî* ‘cold’ in Soran. Strikingly, the contrast is relevant only with the alveolar stop /t/, and not with the rest of the sets /p/, /k/, /ç/, which are only found aspirated. The extent and nature of unaspirated stops and affricates in Central Kurdish needs further exploration.

4.3 Unaspirated stops and affricates in Armenian

Many Armenian dialects have a series of voiceless unaspirated stop and affricate phonemes: /p̥/ [p̥], /t̥/ [t̥], /k̥/ [k̥], /t͡s̥/ [t͡s̥], /t͡ʃ̥/ [t͡ʃ̥]. These exist in a three-way system of glottal settings of consonants consisting of voiceless aspirated, voiceless unaspirated, and voiced. The voiceless unaspirated series is characterised by a longer voiceless closure compared to its voiced and voiceless aspirated counterparts (Xaçatryan 1988; Seyfarth & Garellek 2018). Some Eastern Armenian speakers, from Iran as well as Armenia, have been reported to pronounce this series with glottal ejection or a glottalic airstream (e.g., Allen 1950; Fleming 2000; Ladefoged & Maddieson 1996; Schirru 2012). This ejective articulation is not universal or obligatory for plosives (Seyfarth et al. 2024: 8, but may be more common and more pronounced for affricates; (Seyfarth et al. 2024: 11) note that the word-final voiceless unaspirated affricates in their recordings of speakers from Yerevan are “audibly ejective”.

The voiceless unaspirated series in modern Armenian dialects has two main sources. Many dialects, including all Eastern dialects, but also much of the eastern part of the Western Armenian dialect area, including Van, Mush, and Karin (Erzurum), preserve the Classical Armenian voiceless unaspirated series, at least in word-initial position. This is the outcome of the PIE voiced unaspirated series. The preservation of this series is therefore a conservative feature in Armenian. In many Western Armenian dialects, including Standard Western Armenian, this series has changed to voiced unaspirated, although some speakers of Standard Western Armenian, particularly those living in Anglophone areas, have been recorded producing voiceless unaspirated realisations, especially in utterance-initial position. This is perhaps due to influence from English, where “voiced” plosives are regularly pronounced as voiceless unaspirated when not between voiced segments (Seyfarth et al. 2024: 9).

The other main source of voiceless unaspirated stops and affricates in Armenian dialects is the devoicing of the Classical Armenian voiced series (which

4 Unaspirated stops and affricates

continues the PIE voiced aspirated series). This is an innovation that is found in word-initial position throughout the Urmia dialect area, in nearly all Van and Karabagh subdialects, the Kharberd-Erzinka group of Asia Minor dialects, and most of the Extreme South-West group (Kesab-Svedia, Beylan). The Mush, Ararat, and Agulis-Meghri groups are split, with some members having this feature, others not. The feature is partially present in some members of the Polis subdialect group (Nikomedia, Partizak, Adapazar), and is also found in some isolated members of the Asia Minor dialect group: Syolyoz, Sivrihisar, Kyurin. It is reported in one Hamshen subdialect (Martil), but not the others, and partially in one dialect of Cilicia (Hachən) (see [Jahukyan 1972](#)).

Thus, to our knowledge, voiceless unaspirated stops and affricates are found in all the Armenian dialects spoken in the NENA-speaking area.

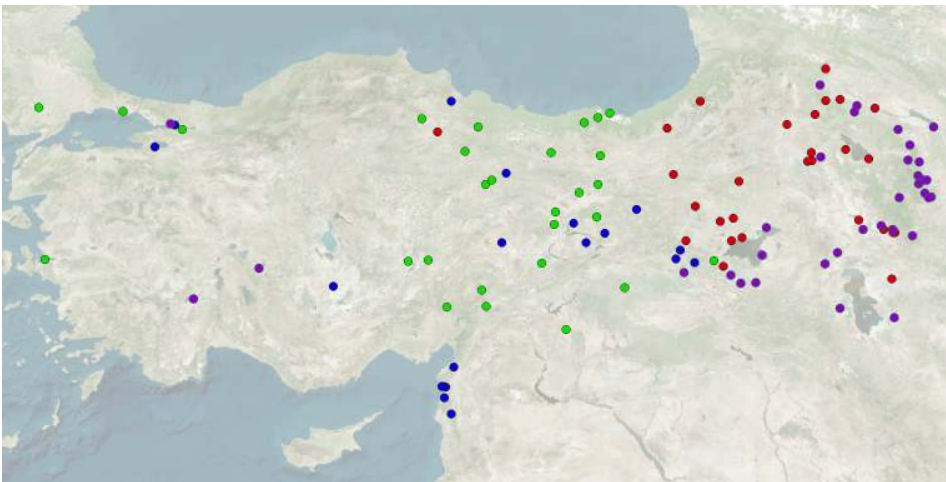


Figure 4.2: Voiceless unaspirated stops in Armenian dialects: green (absent), red (inherited), blue (from devoicing of inherited voiced stops), purple (both inherited and devoiced) (according to [Jahukyan 1972](#))

4.4 Unaspirated stops and affricates in Turkic

In most Azeri varieties, and more broadly in most Turkic varieties of the Caucasus, Iran and Anatolia, voiceless stops are underlyingly aspirated, and the distinction between aspirated and unaspirated stops and affricates is phonetic. Unaspirated stops have phonemic values contrasting with aspirated stops only in peripheral Azeri varieties spoken under conditions of intensive contact with East

4.5 Discussion

Caucasian languages, where a similar distinction exists. A notable case is the Azeri variety of Nizhny Katruk, spoken in Dagestan (Džidalaev 2008: 7).

Unaspirated consonants ([p, t, k, c ~tʃ, tʃ ~ts]) in Azeri occur in word-non-final environments. An obstruent is realised as unaspirated when it follows a voiceless consonant, e.g., *asta* [asta] ‘slow’, *hafta* [hæftæ] ‘week’, including when the stop itself is etymologically voiced, e.g., *tərafdar* [tʰæræftar] ‘supporter’, *heçdən* [heftæn] ‘out of nothing’. If an obstruent, including an etymologically voiced one, precedes a voiceless consonant, it is realised as unaspirated only if the following consonant is a voiceless obstruent, e.g., *bitki* [bitʰi] ‘plant’, *qeydkeş* [gejtʰeʃ] ‘caring’ vs. *qətfə* [gæftʰæ] ‘bath towel’. When a voiceless consonant is followed by an etymologically voiced obstruent, the former causes the latter to devoice, and both are realised as unaspirated, e.g., *rütbə* /rytpæ/ ‘rank’, *tutqun* [tʰutkun] ‘overcast’.

All geminate obstruents are unaspirated, e.g., *həttə* [hæt:a] ‘even’, including those resulting from assimilation, e.g., *gözətçi* [ʒəzæftʃ:i] ‘watchman’.

In fast speech in most dialects, when an etymologically voiced stop is followed by an unstressed close front vowel and a non-final voiceless consonant, it is devoiced during the release but remains nevertheless unaspirated, e.g., *dişim* [tʃim] ‘my tooth’ vs. *diş* [diʃ] ‘tooth’.

4.5 Discussion

Voiceless unaspirated stop and affricate consonants are a feature of the northern sector of NENA. This coincides, broadly, with the geographical distribution of voiceless unaspirated stops and affricates in Kurdish, which are characteristic of Northern Kurdish but not Central Kurdish. The ultimate source of the feature in both NENA and Kurdish seems to be Armenian, in which they are an inherited component of the sound system. The same applies to unaspirated stops in Zazaki. Some Zazaki dialects were not in contact with Kurdish and so they must have acquired them by direct contact with Armenian.

The distribution of unaspirated stops and affricates in NENA dialects differs across the northern sector of the NENA area. In Iraq and the Hakkari region, they are mainly restricted to loanwords from Northern Kurdish. Their presence, therefore, would have arisen principally by contact with Northern Kurdish rather than directly with Armenian. In NENA dialects of the north-eastern sector, such as the Christian dialects of the Urmi region, by contrast, they are found both in loanwords from Northern Kurdish and in the native NENA lexicon. The voiceless unaspirated stops and affricates in the Christian dialects of the Urmi region

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match closely the set of unaspirated stops and affricates in Northern Kurdish and Eastern Armenian, e.g.,

(39) C. Urmi

Eastern Armenian	N. Kurdish	C. Urmi NENA
/p̥/	/p̥/	/p̥/
/t̥/	/t̥/	/t̥/
/k̥/	/k̥/	/k̥/
/t͡ʃ̥/	/t͡ʃ̥/	/t͡ʃ̥/
/t͡ʃ̥̟/		/t͡ʃ̥̟/

It should be noted that there are no clear cases of Armenian loanwords in the C. Urmi NENA dialect spoken in the Urmi region. Speakers of C. Urmi, however, had contact with Armenian on the Urmi plain and there was intermarriage between NENA speakers and Armenian speakers. The lack of Armenian loanwords appears to have been the result of a conscious lexical exclusion with a sociolinguistic motivation related to community identity (Khan 2020a: 389). The unaspirated stops and affricates in native NENA words in C. Urmi developed by internal processes within the dialect, which have been described above (§4.1.3). These internal processes resulting in the addition of unaspirated stop and affricate phonemes to the NENA sound inventory must have been facilitated by a matching of the sound system of NENA with those of Northern Kurdish and Armenian. Although unaspirated stops and affricates originated historically in Armenian, it is likely that the NENA dialects of the north-eastern sector acquired the feature by convergence with the areal feature that was shared by both Armenian and Northern Kurdish. It is generally not possible to establish whether they were acquired directly from specifically Armenian or Kurdish. The tendency of NENA speakers who migrated to Georgia to pronounce unaspirated stops as glottalised is an areal shared by Georgian and Armenian in this region.

In Armenian, in addition to the inherited unaspirated stops, innovative unaspirated stops and affriates have developed by the devoicing of the Classical Armenian voiced series (§4.3). This may have facilitated one of the processes of internal development of unaspirated stops in NENA involving devoicing of voiced consonants in contact with laryngeals (§4.1.3.3).

The acquisition of unaspirated stop and affricate phonemes in NENA dialects of the north-eastern sector led to an enrichment of their sound systems, in that they were exploited to perform functional purposes such as sound symbolism (Khan 2022a). This, indeed, may have been a factor motivating their acquisition from languages in contact. Another factor that is likely to have facilitated the internal development of unaspirated stops in NENA dialects of the north-eastern

4.5 Discussion

sector such as C. Urmi is the development of suprasegmental pharyngealisation. The pharyngeal feature of the original emphatic (i.e., pharyngealised) stop **t* was disengaged from the feature matrix of the consonant segment and became a suprasegmental feature, resulting in the segment being analysed as an unaspirated stop /*t*/ (§7.1.3).

It should be noted that a voiceless unaspirated stop and affricate series did not develop in the Jewish Trans-Zab NENA dialect of Urmi or in Jewish Trans-Zab dialects of the surrounding region. This suggests that the Jewish speakers of these dialects migrated from the south where there was originally no contact with Armenian or Northern Kurdish (Khan 2024).

NENA shares with Northern Kurdish the association of unaspirated stops and affricates in some words with an emphatic (pharyngealised) pronunciation (§4.1.3.6, §4.2.1, table 4.3).

In Northern Kurdish voiceless unaspirated stops and affricates are found in the native lexicon. The distribution of these unaspirated consonant phonemes in Northern Kurdish, however, differs from their distribution in the native lexicon of NENA dialects in the north-eastern sector of NENA. In Northern Kurdish, they are restricted to morpheme-initial position and are not found intervocally within stems. A second feature of their distribution in Northern Kurdish is the tendency for unaspirated stops to be followed by the high and mid vowels /*i*/, /*ê*/, /*u*/, /*û*/. These features of distribution do not apply to unaspirated phonemes in the NENA dialects. This reflects the fact that the internal processes of development of the unaspirated phonemes in NENA and Kurdish were different.

Furthermore, NENA dialects do not exhibit the same degree of variation of the occurrence of unaspirated consonants in specific lexical items across the dialects as is found in Northern Kurdish. What does vary across dialects in NENA, however, is the distribution of suprasegmental pharyngealisation in the lexicon (see, e.g., 16).

5 Palatalisation of consonants

5.1 Palatalisation in NENA

5.1.1 Palatalisation of the Velar Stops **k* and **g*

5.1.1.1 C. Urmi

In the C. Urmi dialect, the voiceless velar stop **k* has been fronted to a voiceless palatal stop /c/ or a palatal affricate /č/. The voiced dorsal stop has been fronted to a voiced palatal stop /j/ or a voiced palatal affricate /j/. These different degrees of palatalisation can be correlated with different regional varieties of the dialect.

In the northern variety of C. Urmi, spoken in villages on the Urmi plain north of Urmi, the reflex of **k* is normally an aspirated palatal stop [c^h] or an aspirated postpalatal stop [c^h], transcribed as /c/. This is the realisation also in the C. Salamas and C. Van dialects spoken further north. In the southern variety of C. Urmi, spoken in villages on the Urmi plain south of Urmi, on the other hand, the reflex of **k* is normally a voiceless an aspirated palato-alveolar affricate [tʃ^h] or an aspirated alveolo-palatal affricate [tʃ^h], transcribed as /č/. In the variety of C. Urmi spoken in the Caucasus (Armenian and Georgia), the reflex of **k* is normally /č/, as in the southern Urmi villages:

- (1) C. Urmi (Khan 2016a: vol. 1, 101–102)

	Northern C. Urmi, C. Salamas, C. Van	Southern Urmi/Caucasus
<i>*karma</i> ‘orchard’	<i>carma</i>	<i>čarma</i>

There is a corresponding geographical distribution of stop and affricate reflexes of voiced **g*. In the northern variety, it is normally pronounced as a voiced palatal stop [j] transcribed /j/. This is the realisation also in the C. Salamas and C. Van dialects spoken further north. In the southern variety, it is normally realised as a voiced palato-alveolar affricate [dʒ] or alveolo-palatal affricate [dʒ], transcribed /j/. In the Caucasus it is realised as /j/, as in the southern Urmi vil-

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(2) C. Urmi (Khan 2016a: vol. 1, 102)

	Northern C. Urmi, C. Salamas, C. Van	Southern Urmi/Caucasus
* <i>garma</i> ‘bone’	<i>ɟarma</i>	<i>jaɾma</i>

Unaspirated voiceless **k*, which is largely restricted to loanwords from Kurdish and Azeri, is likewise fronted to an unaspirated palatal stop /*ç*/ (northern C. Urmi, C. Salamas, C. Van) or an unaspirated affricate /*ç̣*/ (southern Urmi and the Caucasus):

(3) C. Urmi (Khan 2016a: vol. 1, 102)

	Northern C. Urmi, C. Salamas, C. Van	Southern Urmi/Caucasus
* <i>karra</i> ‘deaf’	<i>çarra</i>	<i>ç̣arra</i>

In a number of words the sounds /*ç*/ and /*j*/ are an inherited feature due to sound shifts that applied to NENA dialects in general. They also appear in loanwords. In the northern varieties of C. Urmi, C. Salamas and C. Van, the sounds in these words remain palato-alveolar/alveolo-palatal affricates /*ç*/ and /*j*/. Speakers of the southern varieties of C. Urmi realise the offset of the affricate as an alveolar, viz. [tʰ] and [dʒ], transcribed as /*ç*/ and /*j*/. In most villages in the Caucasus they are realised as /*ç*/ and /*j*/, though in the village of Arzni in Armenia, they are realised as as /*ç*/ and /*ç̣*/.

(4) C. Urmi (Khan 2016a: vol. 1, 101–102)

C. Barwar	Northern C. Urmi, C. Salamas, C. Van	Caucasus	Southern Urmi/ Arzni
<i>çeri</i> ‘autumn’	<i>çiriyyə</i>	<i>çiriyyə</i>	<i>çiriyyə</i>
<i>javəj</i> ‘he moves’	<i>javəj</i>	<i>javəj</i>	<i>javəj̣</i>

The same variation is found in the realisation of the voiceless unaspirated affricate /*ç̣*/, which is found in loanwords:

(5) C. Urmi (Khan 2016a: vol. 1, 101–102)

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C. Barwar	Northern C. Urmi, C. Salamas, C. Van	Caucasus	Southern Urmi/ Arzni
<i>çaḵke</i> ‘weapons’	<i>çiriyyə</i>	<i>çaççə</i>	<i>çaççə</i>

5.1.1.2 Christian Dialects of the Targawar Region

In Christian NENA dialects of the Targawar region, both the aspirated palatal stop and aspirated affricate reflexes of **k* and **g* are found. Unlike in C. Urmi, however, these reflexes correlate with specific lexical items rather than with regional varieties. The stop reflex is more frequent than the affricate reflex. The affricate realisation occurs predominantly before an /a/ vowel:

(6) C. Mawana (Khan, fieldnotes)

Words with /c/ < **k*

<i>caca</i>	‘tooth’
<i>cale</i>	‘he stands’
<i>cănüşta</i>	‘broom’
<i>care</i>	‘becomes short’
<i>cărüba</i>	‘angry’
<i>malca</i>	‘king’

Words with /č/ < **k*

<i>čalba</i>	‘dog’
<i>čäləbta</i>	‘bitch’
<i>čära</i>	‘butter’
<i>čarma</i>	‘vineyard’
<i>časa</i>	‘stomach’

Words with /ɟ/ < **g*

<i>ɟawər</i>	‘he marries’
<i>ɟüda</i>	‘wall/churn’
<i>ɟare</i>	‘roof’
<i>daɟala</i>	‘liar’
<i>ɟinawa</i>	‘thief’
<i>talɟa</i>	‘snow’
<i>+ɟora</i>	‘big’

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Words with /j/ < *g

<i>jarma</i>	‘bone’	
<i>⁺jarüsa</i>	‘big’	
<i>⁺jəra</i>	‘husband’	(< <i>⁺⁺jawra</i>)
<i>jeba</i>	‘side’	
<i>⁺jira</i>	‘arrow’	

The reflex of unaspirated voiceless **k̰* is always a palatal stop /ç/ and is never affricated:

(7) C. Mawana (Khan, fieldnotes)

<i>çaççe</i>	‘weapons’	<	NK <i>čaḵ</i>
<i>çära</i>	‘deaf’	<	NK <i>ḵar</i>
<i>çal</i>	‘unripe’	<	NK <i>ḵal</i>

The sounds /ç/ and /j/ that are an inherited feature due to sound shifts that applied to NENA dialects or that appear in loanwords remain palato-alveolar affricates ([tʃ^h], [dʒ]) and the offset of the affricate is never fronted to an alveolar, viz. [ts] and [dʒ]:

(8) C. Mawana (Khan, fieldnotes)

<i>çeriye</i>	‘autumn’	
<i>çu</i>	‘none’	< NK <i>ču</i>
<i>jüwab</i>	‘answer’	< Arab. <i>jawāb</i>

5.1.1.3 Baz (Argab, Bi-Sillem, Maha Xtaya, Şawwa), Rekan

Most of the Christian dialects of the Baz region in south-eastern Turkey and the closely related Rekan dialect exhibit lexically specific palatal affrication of **k* and **g*, predominantly before /a/, in a range of lexical items that corresponds closely to that of C. Mawana, with some variations. When these dorsal consonants remain stops, they are articulated in the region of the velum and so are transcribed /k/ and /g/.¹

¹The following data are mainly taken from fieldnotes. Some of the data for the Maha Xtaya dialect are taken from [Mutzafi \(2000\)](#).

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		<i>*karma</i> ‘orchard’	<i>*kalba</i> ‘dog’	<i>*kawda</i> ‘liver’
(9)	Argab	<i>čarma</i>	<i>čalwa</i>	<i>čawda</i>
	Bi-Sillem	<i>čarma</i>	<i>čalwa</i>	<i>čawda</i>
	Maha Xtaya	<i>čarma</i>	<i>čalwa</i>	<i>čawda</i>
	Šawwa	<i>čarma</i>	<i>čalwa</i>	<i>čawda</i>
	Rekan	<i>čarma</i>	<i>čalwa</i>	<i>čawda</i>

		<i>*kalθa</i> ‘daughter-in-law’	<i>*kaka</i> ‘tooth’
(10)	Argab	<i>čāla</i>	<i>čeka</i>
	Bi-Sillem	<i>čāla</i>	<i>čeka</i>
	Maha Xtaya	<i>čāla</i>	<i>čeka</i>
	Šawwa	<i>čāla</i>	<i>čeka</i>
	Rekan	<i>čāla</i>	<i>keka</i>

		<i>*garma</i> ‘bone’	<i>*+gawra</i> ‘man’
(11)	Argab	<i>jarma</i>	<i>+jawra</i>
	Bi-Sillem	<i>jarma</i>	<i>+jawra</i>
	Maha Xtaya	<i>jarma</i>	<i>+jawra</i>
	Šawwa	<i>jarma</i>	<i>+jawra</i>
	Rekan	<i>jarma</i>	<i>+jawra</i>

In most other lexical items /k/ and /g/ do not undergo affrication, e.g.,

		‘king’	‘place’	‘snow’	‘roof’
(12)	Argab	<i>malka</i>	<i>duka</i>	<i>talga</i>	<i>gare</i>
	Bi-Sillem	<i>malka</i>	<i>duka</i>	<i>talga</i>	<i>gare</i>
	Maha Xtaya	<i>malka</i>	<i>duka</i>	<i>talga</i>	<i>gare</i>
	Šawwa	<i>malka</i>	<i>duka</i>	<i>talga</i>	<i>gare</i>
	Rekan	<i>malka</i>	<i>duka</i>	<i>talga</i>	<i>gare</i>

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5.1.1.4 **Baz Aruntus and Jilu**

In the Christian dialects of the village of Aruntus, which lies to the south of the Baz region, and in the adjacent Christian dialect of Jilu, palatal affrication of **k* and **g* does not take place. In this respect, therefore, they resemble the dialects of northern Urmi, Salamas and Van, which also lack affrication:

(13)

	‘orchard’	‘dog’	‘liver’	‘daughter- in-law’	‘tooth’
Aruntus	<i>karma</i>	<i>kalba</i>	<i>kawda</i>	<i>kāla</i>	<i>kika</i>
Jilu	<i>kerma</i>	<i>kelba</i>	<i>kewda</i>	<i>kella</i>	<i>kika</i>

(14)

	‘bone’	‘man’
Aruntus	<i>garma</i>	⁺ <i>gawra</i>
Jilu	<i>germa</i>	⁺ <i>gawra</i>

5.1.1.5 **Šamizdīn Region**

Napiorkowska (2015: 18–20) has found that in Christian dialects originally spoken in the Šamizdīn (Turkish: Şemdinli; Kurdish: Şemzînan) region in south-eastern Turkey to the west of Targawar and surrounding areas now in northern Iraq, such as Zariwaw, palatalisation of dorsals was more systematic across the lexicon than in C. Mawana and the Baz dialects. Moreover, there was a wide range of realisations of the palatalisation, ranging from a palatal offset ([*kʲ*], [*gʲ*]), to palato-alveolar affricates ([*tʃʰ*], [*dʒ*]) and alveolar affricates ([*ts*], [*dz*]), e.g.,

- (15) *čalba* [*kʲalba*] ~ [*tʃʰalba*] ~ [*tsælba*] ‘dog’
 jəlla [*gʲəlʰa*] ~ [*dʒəlʰa*] ~ [*dzəlʰa*] ‘grass’

Talay (2008b: 101) reports the existence of affrication of velar dorsals also in the dialects of Marbišo, Nočiya and Iyyəl in the Šamizdīn region.

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5.1.1.6 Hakkari Region

Affrication of dorsals is found in Christian dialects of the Hakkari region. It is not, however, systematic across the lexicon in these dialects and the sets of words in which it occurs correspond broadly to the sets in which it occurs in the Baz and Targawar dialects, e.g.,

- (16) Upper Tyare, C. Walto (Talay 2008b: 100, 2009: 32–50)

With Affrication	Without Affrication
<i>čalba</i> ‘dog’	<i>kale</i> ‘he stands’
<i>čalθa</i> ‘daughter-in-law’	<i>’akkara</i> ‘farmer’
<i>jawra</i> ‘man’	<i>gawa</i> ‘inside’
<i>jarma</i> ‘bone’	<i>ga</i> ‘time’

- (17) Lower Tyare, C. Ashitha (Borghero 2005: 19–20)

With Affrication	Without Affrication
<i>čalba</i> ‘dog’	<i>kale</i> ‘he stands’
<i>čalθa</i> ‘daughter-in-law’	<i>’akkara</i> ‘farmer’
<i>čarma</i> ‘orchard’	<i>goma</i> ‘ground-floor stable’
<i>jərma</i> ‘bone’	<i>garoma</i> ‘rolling pin’

According to Jacobi (1973: 80), there is variation between affricates and stops in a small set of words in the Txuma dialect.

- (18) Txuma Gawaya (Talay 2008b: 100–102)

<i>čarma</i> ~ <i>karma</i>	‘orchard’
<i>čepa</i> ~ <i>kepa</i>	‘stone’
<i>jora</i> ~ <i>gora</i>	‘husband’

Elsewhere /k/ and /g/ in Txuma dialects are pronounced as stops with palatal offglides ([kʲ], [gʲ]) (Jacobi 1973: 10, 12).

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5.1.1.7 C. Bədyal

In the Christian dialect that was spoken in the village of Bədyal, in the Barzan region of northern Iraq south of the Hakkari, one finds sporadic palatalisation of the dorsal velar stops /k/ and /g/ to the affricates /č/ and /j/ in some specific lexical items, e.g.,

(19) C. Bədyal (Khan, fieldnotes)

<i>čipa</i>	‘stone’	< * <i>kipa</i>
<i>čalo</i>	‘‘bride’	< * <i>kālo</i>
<i>jera</i>	‘arrow’	< * <i>gera</i>
<i>julpa</i>	‘wing’	< * <i>gulpa</i>
<i>jəsra</i>	‘bridge’	< * <i>gəsra</i>
<i>jəlla</i>	‘grass’	< * <i>gəlla</i>

In the indicative of the verb ‘to know’ an affricate /č/ has developed by the coalescence of the indicative prefix *k-* with the /y/ at the beginning of the verbal stem:

(20) C. Bədyal (Khan, fieldnotes)

<i>č-ăde</i>	‘he knows’	< * <i>k-yăde</i>
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Most dorsal velars, however, are not palatalised in the dialect, e.g.,

(21) C. Bədyal (Khan, fieldnotes)

<i>karma</i>	‘orchard’	<i>gare</i>	‘roof’
<i>kawutra</i>	‘breakfast’	<i>garəš</i>	‘he pulls’
<i>kawa</i>	‘window’	<i>talga</i>	‘snow’
<i>kanušta</i>	‘broom’	<i>gura</i>	‘man’

5.1.1.8 C. Barwar

In the C. Barwar dialect, spoken in villages south of the Țyare region, the stops /k/ and /g/ have not developed into affricates but are sometimes pronounced as stops with palatal offglides before non-rounded vowels. This is not regular and the set of words it occurs in includes the set of words that exhibit affrication of the velar dorsals in the Hakkari, Baz and Targawar dialects, e.g.,

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(22) C. Barwar (Khan 2008a: 30)

<i>kerma</i>	[c ^h ɛrmæ]	‘orchard’
<i>kalo</i>	[c ^h æ:lo]	‘bride’
<i>germa</i>	[ʃ ^h ɛrmæ]	‘bone’
<i>gawra</i>	[ʃ ^h awɾˤɑ]	‘man’

There are a few cases of palatalisation of *g > j as a fixed feature of words in C. Barwar. Some may be loanwords from Tyare dialects, e.g., *jawatta* ‘chopping tool’ (< *g-ˤd < *g-d-ˤ ‘to hew off’). Some appear to have developed to distinguish their meaning from homophones with /g/, e.g.,

(23) C. Barwar (Khan 2008a: 31)

<i>jwina</i>	‘side’	<i>gwina</i>	‘eyebrow’
<i>jlw</i>	‘to kidnap’	<i>g-n-w</i>	‘to steal’

5.1.1.9 Christian Dialects in the Western and South-western Sectors of NENA

In the Christian dialects in the western and southern sectors of NENA, ranging into western Iran, palatalisation and affrication of the velar dorsals is only found very marginally in a few lexical items.

The affricate /č/ deriving from /k/ occurs in the prefix of the indicative present form of the verbs ‘to know’ and ‘to give’ in C. Shaqlawa and C. Koy Sanjaq/Armota. This has developed by coalescence of the prefix *k-* with the following /y/ of the verbal present stem. A similar development is attested in the C. Bədyal dialect, which was spoken in Barzan in northern Iraq (see §5.1.1.7). In the present indicative form of the verb ‘to give’ in C. Koy Sanjaq/Armota, a secondary *k-* prefix has been added before the /č/:

(24)		C. Shaqlawa	C. Koy Sanjaq/Armota
	‘he knows’	č-ade (< <i>k-yade</i>)	č-ǎde (< <i>k-yede</i>)
	‘he gives’	č-awəl (< <i>k-yawəl</i>)	<i>k-č-awəl</i> (< <i>k-yawəl</i>)

In C. Koy Sanjaq/Armota, there are a few sporadic cases of palatal affrication of velar dorsal stops before the vowel /e/, e.g.,

(25) C. Koy Sanjaq/Armota (Khan, field notes)

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čepa (< *kepa*) ‘stone’
jera (< *gera*) ‘arrow’

In C. Shaqlawa and C. Koy Sanjaq/Armota original affricates /č/ [tʃ^h] and /j/ [dʒ] in loanwords are pronounced as alveolar affricates /ć/ [ts] and /ǰ/ [dz], e.g.,

(26) C. Shaqlawa (Khan, fieldnotes)

baxća (< *baxča*) ‘garden’
ći (< *či*) ‘none’
ǰwān (< *jwān*) ‘beautiful’
ǰüle (< *jüle*) ‘clothes’

(27) C. Koy Sanjaq/Armota (Khan, field notes)

ćamća (< *čamčā*) ‘spoon’
ćale (< *čale*) ‘ditch’
ǰwān (< *jwān*) ‘beautiful’
ǰāmi (< *jāmi*) ‘they gather’

In sporadic cases, an original /g/ in a word with Aramaic etymology shifts to /ǰ/ to distinguish meaning:

(28) C. Koy Sanjaq/Armota (Khan, field notes)

ǰānawa ‘thief’ *ǰānawa* ‘pickpocket’

5.1.1.10 Jewish Dialects

In both the *Lišana Deni* and the Trans-Zab subgroups of the Jewish dialects the velar stops are, in principle, not palatalised. This applies to dialects that were spoken in areas where Christian dialects exhibit palatalisation, such as Urmi:

(29)

	J. Urmi	C. Urmi (northern)	C. Urmi (southern)
‘vineyard/orchard’	<i>karma</i>	<i>carma</i>	<i>čarma</i>
‘bone’	<i>garma</i>	<i>ǰarma</i>	<i>jarma</i>

In some Jewish Trans-Zab dialects in Iraq, the indicative verbal prefix *k-* coalesces with the initial /y/ of the present stem of the verb ‘to know’, which results in the affricate /č/:

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- (30) J. Arbel (Khan 1999a: 101), J. Koy Sanjaq (Mutzafi 2004a: 90)

čəl (< *k-ya'əl*) 'he knows'
 čex (< *k-ya'ex*) 'we know'

There are some sporadic cases of palatalisation as a fixed feature in isolated lexical items. One such case is the word for 'chopper/billhook' (< **g-^hd* < **g-d-^h* 'to hew off'), which also exhibits this lexically-specific shift in Christian dialects, such as C. Barwar *jawətta*:

- (31) Mutzafi (2002b: 45)

J. Sandu	J. Aradhin	J. Nerwa
<i>jawoda</i>	<i>ja'oda</i>	<i>jahoda</i>

Another case is the palatalisation of **g* > /*j*/ in one member of a doublet to distinguish meaning:

- (32) J. Arbel (Khan 1999a: 34)

g-n-w 'to steal' *jnw* (< **gnw*) 'to kidnap'

5.1.2 Palatalisation of NENA Alveolar Stops **t* and **d*

The palatalisation of the alveolar stops /*t*/ and /*d*/ to the palato-alveolar affricates /č/ [tʃ^h] and /j/ [dʒ] is attested in the C. Bədyal dialect, in the Barzan region of northern Iraq. The palatalisation of these stops takes place before the high vowels /*i*/ and /*ɨ*/. The central high rounded vowel /*ʊ*/ has developed by fronting of an original back high rounded vowel **u*.

- (33) C. Bədyal (Khan, fieldnotes)

beči (< **beti*) 'my house'
čiqa (< **'atiqa*) 'old'
'iji (< **'idi*) 'my hand'
mənji (< *məndi*) 'thing'
čɨna (< **tuna*) 'straw'
bečɨta (< **betuta*) 'household'
jɨša (< **duša*) 'honey'
zarjɨta (< *zarduta*) 'yellowness'

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In such contexts /č/ and /j/ are occasionally realised as [tj] and [dj], in which the affricates have a voiced palatal approximant [j] has their second component rather than a post-alveolar [ʃ] or [ʒ].

The palatalisation may take place where an originally /i/ has been elided, e.g.,

- (34) C. Bədyal (Khan, fieldnotes)
jexun (< **diyexun*) ‘yours (pl.)’
jeni (< **diyeni*) ‘ours’

The palatalisation of /t/ and /d/ does not take place where a /i/ has developed historically from **e*, e.g.,

- (35) C. Bədyal (Khan, fieldnotes)
tina (< **tena*) ‘fig’

It does not take place where /i/ and /u/ have developed by assimilation to an adjacent vowel, e.g.,

- (36) C. Bədyal (Khan, fieldnotes)
tili ‘I came’ < **тели*, cf. *tele* ‘he came’
dili ‘I knew’ < *дели*, cf. *dele* ‘he knew’
šatutun ‘you drink’ < **šatetun*
dəqun! ‘hold (pl.)!’ < **duqun* < **doqun*

It does not take place before the high vowel /u/, which is derived historically from **o*, e.g.,

- (37) C. Bədyal (Khan, fieldnotes)
túnele ‘he spoke’ < **tónele*
betux ‘your house’ < *betox*
šadure ‘to send’ < **šadore*

Elsewhere the affricates /č/ and /j/ have developed by the coalescence of /t/ and /d/ with the palatal consonants /y/ or /š/, e.g.,

- (38) C. Bədyal (Khan, fieldnotes)
k-əčča ‘she comes’ < **k atya*
k-mačča ‘she brings’ < **k-matya*
’ačča ‘this’ < **’atya*; cf. *’anya* ‘these’
’ačcata ‘this year’ < **’at-šāta*
’əjjə, ’əjjə ‘today’ < **’ədyu*
kəjjum ‘every day’ < *kud-yum*
Bəjjal ‘Bədyal’ < *Bədyal*

5.2 Palatalisation of Velars in Iranian languages of the Region

5.1.3 Map of geographical distribution of palatalisation

The geographical distribution of the various processes of palatalisation described above are represented in Figure 5.1

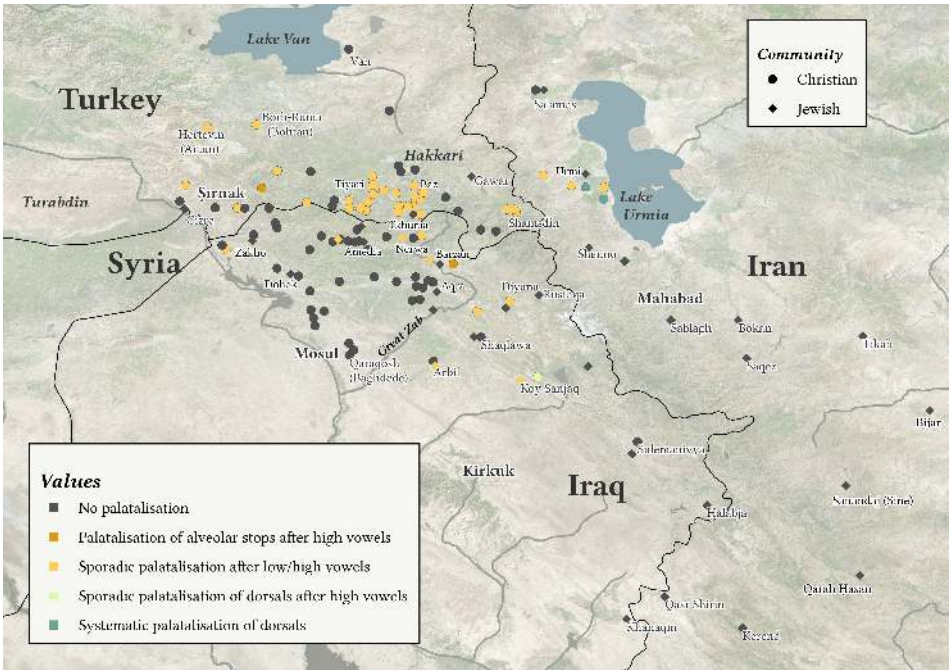


Figure 5.1: Distribution of palatalisation processes in NENA dialects

5.2 Palatalisation of Velars in Iranian languages of the Region

In most Central Kurdish dialects located close to the Northern Kurdish speech zone, the velar consonants /k/ and /g/ exhibit palatalisation. The exact phonological contexts for the palatalisation of velars have not been systematically investigated so far. Our data suggest that palatalisation is most common in the environment of front vowels [i], [e] and [ø]. There is variation in palatalisation of /k/ and /g/. The resulting realisations can range between [c] and [tʃ] for /k/ and [ɕ] and [dʒ] for /g/. The following examples are from the Mukri variety of Mahabad (south of lake Urmia, north-western Iran) (Öpengin 2016: 30), where the velars are affricated before high front vowels:

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(39)

<i>kefæ</i>	[tʃe:fæ]	‘problem’
<i>kør</i>	[tʃør]	‘blind’
<i>kif</i>	[tʃi:f]	‘bag’
<i>gezær</i>	[dʒe:zær]	‘carrot’
<i>gøz</i>	[dʒøz]	‘walnut’
<i>bargin</i>	[ba:rdʒi:n]	‘carhouse’

The affrication of velar stops is uncommon adjacent to the open vowels [a] and [a:] and adjacent to the back vowels [u] and [u:], e.g.,

(40)

Palatalisation favoured			Palatalisation not favoured		
<i>kö</i>	[tʃø]	‘where’	<i>ganəm</i>		‘wheat’
<i>kīž</i>	[tʃi:ʒ]	‘girl’	<i>kon</i>		‘old’
<i>gö</i>	[dʒø]	‘ear’	<i>kānī</i>		‘spring’
			<i>kuř</i>		‘son’

The fronting of velars in Northern dialects of CK, including Mahabad, Shaqlawa, Erbil, and Rewanduz, creates a push-chain effect leading to the alveolar articulation of post-alveolar affricates /č/ and /j/, with the resultant allophones being [tʃ] and [dʒ], respectively.

(41)

<i>čə</i>	[tʃə]	‘what’
<i>čīrok</i>	[tʃi:rok]	‘tale’
<i>jē</i>	[dʒe:]	‘place’

In the neighbouring Kurdish varieties, velar sounds may get fronted before front and mid-open vowels, but the result is a palatalised and not affricated sound.

(42)

CK Sulemaniyya	<i>g^jyān</i>	‘dear’		
CK Halabja	<i>g^jö</i>	‘ear’,	<i>k^jisat</i>	‘tortoise’
NK Hakkari	<i>dik^jiřim</i>	‘I buy’		

Figure 5.2 illustrates the areal distribution of affrication/palatalisation of /g/ word-initially in the word *gisk*, *gêsk* ‘kid-goat’ across Kurdish. As can be seen, the realisation as a voiced alveolar affricate [dʒ] is specific to CK dialects surrounding Erbil, extending into the neighbouring CK Mukri dialects in Iran; see also [Asadpour \(2021\)](#) for a recent discussion.

Many other Kurdish dialects have palatalised velars before all vowels except the back vowels /a/, /o/ and /u/. In fact, it is only sporadically in the dialect of

5.2 Palatalisation of Velars in Iranian languages of the Region

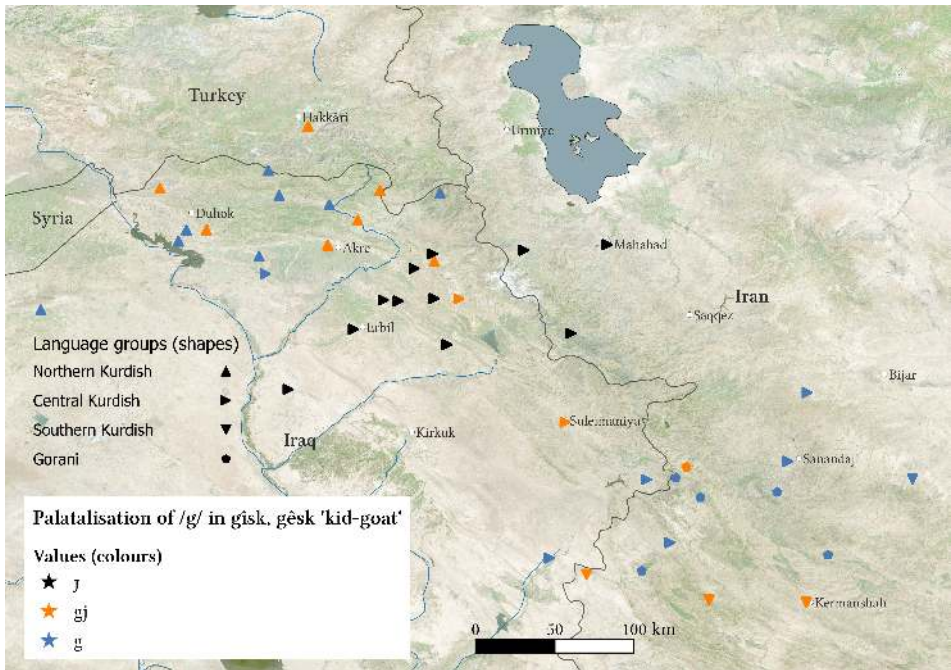


Figure 5.2: Palatalisation of /g/ across Kurdic

some tribes (such as nomadic Harki tribes in Akre, the north of Rawanduz and Shemdinli regions) that such palatalisation does not occur.

- (43) Targawar, Shemdinli Harki
- | | | |
|-------------|-------------|---------|
| <i>kar</i> | <i>kar</i> | ‘piece’ |
| <i>gawr</i> | <i>gawr</i> | ‘gray’ |
| <i>kir</i> | <i>kir</i> | ‘did’ |
| <i>gir</i> | <i>gir</i> | ‘gross’ |

The velar affrication that is found in the Central Kurdish dialects close to the Northern speech zone, such as Mukri and Erbil, is not found in Northern Kurdish dialects. In the Urmi region, affrication is found in the Mukri dialects south of the city of Urmi. In dialects spoken to the north or west (e.g., Targawar) of Urmi, however, velars are palatalised but not affricated.

5.2.1 Zazaki

Palatalisation of the velar consonants /k/ and /g/ is a common phenomenon across many dialects of Zazakî. A distinction must be made between diachronic

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sound change that has affected certain varieties - most prominently Northern Zazakî as well as the variety spoken in Şankuş in the Southern Dialect area - and active phonological processes that are still at work in some dialects, such as those spoken in Dêrsim.

A few exceptions notwithstanding (e.g., some villages of Xozat and Kêxî; cf. Keskin 2008: 98; 178), most varieties of Northern Zazakî underwent a historical sound change whereby velar consonants palatalised before the front vowels ê [e], î [i], and occasionally ü [y] (see Table 5.1). As is often the case, there is some interdialectal variation. Depending on the variety, the specific lexemes that are affected may differ.

Table 5.1: Palatalisation in Northern Zazakî

Gloss	Northern Zazakî	CZ and SZ
‘home’	çê	keye / kê
‘daughter’	çêna	kêna / keyna
‘grudge’	çîn	kîn
‘feces’	cî	gî
take.PRS	cên-	gên-

In parts of Şankuş (turk. Çünküş), this shift in velar consonants is systematic and not conditioned by the following vowel (cf. Qetekonij 2018: 15–18; see Table 5.2). In this variety, nearly all velar consonants appear as affricates.

Table 5.2: Palatalisation in Şankuş Qetekonij (2018)

Gloss	Şankuş	Other varieties
do.PRS	çan- / çen-	ken-
‘Kurd’	çurd	kurd
‘sugar’	şeçer	şeker
‘ox’	ca	ga
‘wolf’	varc	verg

By contrast, palatalisation of velar consonants (and also sibilants) is still operative in Northern Zazakî, particularly in the dialects of the Dêrsim region. This is especially evident with the high front vowel -î, which functions both as the oblique case marker and the plural suffix. When attached to a word ending in /k/

5.3 Palatalisation of Velars in Armenian Dialects of the Region

or /g/, these consonants shift to /ç/ and /c/ respectively. However, this is not uniform and depends both on dialect and on speaker: older speakers tend to apply the process more consistently than younger ones (cf. [Paul 2024](#): 185).

5.3 Palatalisation of Velars in Armenian Dialects of the Region

Palatalisation of velar stops is found in many Armenian dialects, including those spoken in the Van and Urmia areas. In most cases, this results in palatal stops rendered in the dialectological literature as g^j , k^j , k^{hj} , but usually pronounced as [j], [c], and [c^h] respectively in the speech of speakers of the Urmia dialect from Armenia, whose ancestors appear to come from villages in northern area of the Urmi plain (Hodgson field notes). In some areas, palatalisation results in affrication: $g > dʒ$, $k > tʃ$, $k^h > tʃ^h$ ([Jahukyan 1972](#) refers to Ashtarak, near Yerevan, and Maragha in northern Iran, but this is also recorded in folk tales in the dialect of Maku in northern Iran published in [Hayrapetyan 2016](#)).

[Jahukyan 1972](#) presents palatalisation of velar stops as a single feature (Feature 8), distinguishing only between those dialects where it results in palatal stops, and those where it results in affricates. However, a closer examination reveals that palatalisation of velars is conditioned by different environments in different dialect areas. In Tigranakert (Diyarbakir), according to [Haneyan 1978](#): 47, 48, 49, palatalisation only occurs preceding the low front vowel [æ], e.g.,

- (44)
- | | | | |
|--------------------------|-----------|---|--------------|
| <i>gari</i> | ‘barley’ | > | $k^{hj}æri$ |
| <i>kat^h</i> | ‘milk’ | > | $g^jæt^h$ |
| <i>k^hayel</i> | ‘to pick’ | > | $k^{hj}æyil$ |

It does not occur preceding other front vowels, e.g.,

- (45)
- | | | | |
|-------------------------|-----------------|---|-------------|
| <i>getin</i> | ‘ground’ | > | k^hedin |
| <i>gini</i> | ‘wine’ | > | k^hini |
| <i>kes</i> | ‘half’ | > | <i>ges</i> |
| <i>oski</i> | ‘gold’ | > | <i>osgi</i> |
| <i>k^heni</i> | ‘wife’s sister’ | > | k^heni |
| <i>mak^hi</i> | ‘ewe’ | > | $mæk^hi$ |

However, in the texts in the dialect of Maku in [Hayrapetyan 2016](#) (recorded from speakers resident in the village of Ddmashen, Kotayk province, whose ancestors migrated from Maku in 1828-9), we find palatalisation (affrication) of velars before the front vowels [i], [e], and [æ], as well as before [ə] when this results from reduction of the front semivowel j+V:

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(46)	$\text{ʃ}^h\text{it}^h$	<	$k^h\text{it}^h$	‘nose’
	ʃes	<	kes	‘half’
	$\text{ɟerut}^h\text{jan}$	<	gerut^hjan	‘captivity (gen.)’
	$\text{het}^h\text{æ}^h\text{t}^h$	<	hek^hiat^h	‘folk tale’
	$\text{ʃ}^h\text{əsib}$	<	$k\text{jasib}$	‘poor’

Urmia (Asatryan 1962), Salmast (Vaux 2022 and Atayan 1906), Van (Acharyan 1952), and Shatakh (Muradyan 1962) have a completely different pattern of palatalisation, which is shared, with minor differences, by all these dialects (and probably others in the wider Van-Urmi area, for which analytic data is lacking). Agulis (data from Ačafyan 1935) also shows a similar pattern.

In these dialects, *g* (including [k] resulting from devoicing of *g* in initial position) is usually palatalised in all environments.

k resulting from CA *k* is usually not palatalised. In Urmia and Salmast, *k* is usually palatalised when it appears in word-final position: $\text{ax}\text{ʃ}^h\text{ik}^j < \text{ay}\text{ɟ}\text{ik}$ ‘girl’, $\text{pæ}\text{k}^j < \text{bak}$ ‘yard’, $\text{ayek}^j < \text{ayek}$ ‘good’, $\text{poxk}^j < \text{boyk}$ ‘radish’, but this is not the case in Van and Shatakh.

In Van and Shatakh, the reflex of CA *k* is palatalised when it changes to *g* or k^h : $\text{ing}^j\text{er} < \text{ə}n\text{ker}$ ‘friend’, $\text{xung}^j < \text{xunk}$ ‘incense’ (after nasal), $\text{ʃapik}^{hj} < \text{ʃapik}$ ‘shirt’.

In Agulis, *-k* is palatalised in word-final position following *i*, including [i] < *e*, and also [ay], [a] or [e] < *i*: $\text{ax}\text{ʃ}^h\text{ik}^j < \text{ay}\text{ɟ}\text{ik}$ ‘girl’, $\text{ayik}^j < \text{ayek}$ ‘good’, $\text{kə}n\text{ak}^j$, $\text{kə}n\text{ajg}^j$, $\text{kə}n\text{ekj} < \text{knik}$ ‘woman’, but not [e] < *e*: $\text{ʃek} < \text{ʃek}$ ‘blond’. In the Tsghna subdialect, unlike in Agulis itself, *k* is also palatalised in word-final position after [æ]: $\text{pæ}r\text{æ}\text{k}^j < \text{barak}$ ‘thin’, $\text{æ}r\text{ts}^h\text{æ}\text{k}^j < \text{ardzak}$ ‘loose, untied’.

k^h is usually palatalised in Van and Shatakh, and also in Urmia with some exceptions (especially in initial position preceding *a*: Urmia $k^h\text{ayak}^h$ ‘city’ vs. Van $k^{hj}\text{ayak}^{hj}$).

In Van, when k^h is immediately preceded by *d*, t^h , *t*, *ts*, ts^h , *s*, *f*, *ʒ*, ʃ^h , it is often deaspirated to *k* and not palatalised: $\text{a}\text{ʃ}\text{k} < \text{a}\text{ʃ}^h\text{k}^h$ ‘eye’ etc. The same occurs in Shatakh, though *t* and *d* are not mentioned by Muradyan (1962).

In Agulis, k^h is usually not palatalised, except in word-final position *-ik^h* and *-ink^h*: $\text{hə}r\text{sani}\text{k}^{hj} < \text{harsani}\text{k}^h$ ‘wedding’, $\text{irik}^{hj} < \text{erek}^h$ ‘three’, $\text{ergin}\text{k}^{hj} < \text{erkin}\text{k}^h$ ‘sky, heaven’, etc.

5.4 Palatalisation in Azeri

Azeri, a Turkic language of the West Oghuz branch and closely related to Turkish, is spoken in a linguistic area characterised by a series of fronting phenomena,

5.4 Palatalisation in Azeri

resulting in the progressively palatal realisation of dorsal consonants when they occur syllable-initially or, in monosyllabic words, word-finally.

In inherited words, Azeri dorsal stops and fricatives trace back to the Proto-Turkic velar consonants **k* and **g* (Johanson 2021: 272). Before the arrival of Turkic languages in West Asia, these Proto-Turkic consonants developed a velar–uvular allophonic distribution, whereby **k* was realised as [k] next to front vowels and [q] next to back vowels, while **g* was realised as [g] next to front vowels and [ɣ] next to back vowels.

Likely due to the influence of numerous loanwords—whose adaptation disrupted the historical allophonic distribution by triggering velar realisations in back-vowel contexts and uvular realisations in front-vowel contexts—as well as substrate effects, Azeri at an early stage reanalysed its reflexes of the four aforementioned allophones as phonemes. This phonemic distribution is preserved in Standard Azeri today, e.g., *kim* ‘who’ (inherited) vs. *kal* ‘unripe’ (Persian loanword); *göz* ‘eye’ vs. *gor* ‘tomb’ (Persian loanword); *qar* ‘snow’ (inherited) vs. *qərar* ‘decision’ (Arabic loanword), etc.

Stilo (1994: 79ff.) identifies the first stage of palatalisation (which he terms “Stage One Fronting”), a process that must have affected all Azeri dialects at some point, whereby /k/ and /g/ developed secondary palatalisation before front vowels. It is important to note that this phonetic development is not restricted to dorsal consonants. In Azeri, all consonants exhibit positional variants characterised by varying degrees of palatalisation when they occur syllable-initially before a front vowel or syllable-finally after a front vowel, e.g., *dal* ‘behind’ [dalʲ] vs. *dil* ‘tongue’ [dʲilʲ]. In many Azeri dialects, this palatalisation extends beyond secondary articulation, resulting in the emergence of /ç/ and /ʃ/, respectively.

The second stage of fronting identified by Stilo (1994) involves the palatalisation of historical syllable-initial /k/ and /g/ in all positions, including back-vowel contexts (thus also affecting loanwords), yielding [kʲ ~ c] and [gʲ ~ ɟ], e.g., *kim* [kʲim ~ cim] ‘who’, *gor* [gʲor ~ ɟor] ‘tomb’. This fronting is characteristic of the varieties that formed the basis for Standard Azeri, as well as those of Ardabil and Zanjan (all classified by Stilo as “Eastern” Azeri). It does not occur in what he terms “Peripheral” Azeri varieties, which ostensibly include the dialects of Hamadan and Saveh, spoken in regions where Azeri-like (West Oghuz) varieties transition into Qashqai-like (South Oghuz) varieties, and where palatalisation has stopped at the first stage. In addition to these, one may also include among the dialects that do not exhibit this palatalisation process the northern periphery of the Azeri-speaking area, namely the dialects of Quba in Azerbaijan (Rüstəmov 1961: 62) and Derbent in Russia (Axundov et al. 2009: 26). This fronting process

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likely also encompasses the parallel palatalisation of word-final /k/ in polysyllabic words, realised as the voiceless palatal fricative [ç], e.g., *ürək* [yræç], typical of Standard Azeri as well as of all but the “Eastern” and “Peripheral” Azeri dialects.

The third stage of fronting, as described by Stilo (1994), involves the historical syllable-initial uvular /q/ shifting to velar [g] (likely passing through an intermediate [ɟ] stage). This change was facilitated by the phonological restructuring that followed the second stage of fronting, which had palatalised the original velars, thereby freeing the velar slot. The process affects both inherited words and loanwords, yielding e.g., *qar* [gar] ‘snow’, *qərar* [gærar] ‘decision’. This fronting is absent in Stilo’s “Peripheral” Azeri dialects, as well as in the Derbent and Quba Azeri varieties and adjacent dialects extending into the western outskirts of Baku, where this consonant is typically realised as /c/. It is also not observed in some Iraqi Turkic varieties (Bulut 2018: 360), where it surfaces as either [ɟ] or [q]. The process may have been completed as late as the early twentieth century, as suggested by contact phenomena: languages that have borrowed vocabulary from Azeri generally preserve a uvular realisation rather than a velar one, e.g., dialectal Armenian *ḡonaḡ* [ɸoˈnax] ‘guest’ < Azeri *qonaq*, Georgian *q’ačayi* [q’atʃaxi] ‘robber’ < Azeri *qaçaq*, Lezgian *qužaχ* [quʒaχ] ‘bosom’ < Azeri *qucaq*, Tat *quş* /ɟuʃ/ ‘bird’ < Azeri *quş*.

The fourth stage of fronting represents a further development of the second stage and is characterised by the evolution of the palatal stops [kʲ ~ c] and [gʲ ~ ɟ] into [tʃ] and [dʒ], respectively, e.g., *kim* [tʃim] ‘who’, *göz* [dʒœz]. This phenomenon is particularly prominent in the dialects of Tabriz and Urmi in Iran, the north-eastern dialects of Iraqi Turkic, and the dialects of Naxçıvan and Ordubad in Azerbaijan (Ritter 1921: 181; Caferoğlu & Doerfer 1959: 290; Stilo 1994; Bulut 2018: 360; Ragagnin 2022a: 243–244). It is also observed in some varieties spoken in Şəki (Karakoç & Abbasova 2023: 223–224), Zəngilan (Behbudov 1966: 10), Upper Şirvan, and Qazax (Şirəliyev 1967b: 83–84), extending as far as the variety spoken in Dmanisi, Georgia (Džangidze 1965: 59). Aşmarin (1926: 20–21), who produced the earliest detailed description of the urban variety of Şəki Azeri, characterises the historical velars in this dialect as “strongly palatalised” and nearly indistinguishable from affricates. A recent study of Şəki Azeri describes inherited velar stops in word-initial, word-medial, and word-final positions in monosyllabic words as having developed into aspirated affricates, while the original affricates remain unaffected, suggesting a merger (Karakoç & Abbasova 2023: 223–224), e.g., *kim* /tʃim/ ‘who’ vs. *çim* /tʃim/ ‘bathe’. This convergence or near-convergence, often recognised by ordinary Azerbaijanis as a characteristic feature of Şəki Azeri, has curiously escaped the attention of prominent Soviet-era

5.5 Discussion

dialectologists. Şirəliyev (1967b: 83–84) does not include Şəki among the Azeri dialects exhibiting a palatal shift of historical velars. Similarly, İslamov (1968), who collected data from over 20 settlements in the Şəki region in the mid-1950s and dedicated a substantial section of his study to consonantal divergences between Standard Azeri and Şəki Azeri (op. cit. 19–28), makes no mention of palatalisation, not even in passing.

Finally, a great number of dialects participating in the fourth stage also exhibit the fifth stage of fronting, which, according to Stilo, occurs simultaneously with the former. In this stage, the inherited affricates /tʃ/ and /dʒ/, whose phonemic scope has been “invaded” by the progressively palatalising historical velars, are fronted to [ts] and [dz] respectively, thus preventing the merger observed in Şəki Azeri and similar varieties, e.g., *kim* [tʃim] ‘who’ vs. *çim*- [tsim] ‘bathe’; *cəngəl* [dzændʒæl] ‘jungle’ vs. *cəncəl* [dzændzæl] ‘muddle, dilemma’. This development is typical of the south-western part of the Azeri-speaking area, most notably the varieties of Tabriz and Urmi (Ritter 1921: 181), much of Naxçıvan (Səfərova 2019: 65–66), Ordubad and Meghri (Əliyev 2003: 119), with small pockets found in Cəbrayıl and Upper Şirvan (Şirəliyev 1967b: 78). The fifth stage is particularly characteristic of the Ayrım dialect, spoken in the District of Gədəbəy (ibid.), which lies outside the area of greatest concentration of the phenomenon.

Səfərova (2019: 65), in her description of the consonantism of the Culfa variety of the Naxçıvan dialect, alleges that historical velars in Tabriz Azeri, which have developed into alveolar affricates /tʃ/ and /dʒ/ (stage four), are further palatalised to /ts/ and /dz/. Unfortunately, she does not provide references to support this claim, which, if accurate, would imply a sixth stage of fronting and a new type of consonant merger, with historical velars and affricates merging as /ts/ and /dz/.

Outside of the Caucasus and Iran, the palatalisation of both velars and affricates occurs in Black Sea Turkish, specifically in the varieties of Rize and, to some extent, Trabzon (Brendemoen 2022: 227). Caferoğlu (1963: 14, 24) notes that among the Anatolian varieties, the former shift also occurs in Erzurum, while the latter is characteristic of the Rize–Trabzon area only.

A non-Turkic variety, in close contact with Azeri and exhibiting the fourth and fifth stages of fronting, is Şirvan Tat, which is rarely mentioned as part of this West Asian isogloss (Suleymanov 2020: 78–81).

5.5 Discussion

The geographical distribution of palatalisation in NENA dialects exhibits close parallels to the geographical distribution in Kurdish dialects. In the Urmi region,

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the advancement of the dorsals **k* and **g* to the postalveolar affricates [tʃ] and [dʒ] and the historical postalveolar affricates **tʃ* and **dʒ* to the alveolar affricates [ts] and [dz] is a feature of NENA dialects and Central Kurdish dialects spoken south of the town of Urmi. In NENA dialects and Kurdish dialects spoken north of Urmi there was no such affrication. Azeri Turkic dialects in north-western Iran exhibit similar advanced palatalisation resulting in these two types of affricates, but this phenomenon is not restricted to Azeri spoken south of Urmi, but is a feature also of Azeri varieties spoken to the north of Urmi, as far north as Azerbaijan and Georgia. It is possible, however, that the affrication in NENA dialects in the southern Urmi region was reinforced by contact with Azeri. This is because in southern C. Urmi NENA and Azeri the affrication occurs in all phonetic contexts, whereas in Kurdish dialects of the region it seems to be largely restricted to occurrence after front vowels. In the northern portion of the Urmi plain the palatalisation of velar dorsals to palatal stops [c] and [j] is matched by the palatalisation process that is found in Armenian dialects of this area (§5.3).

In Christian NENA dialects in the Central Kurdish region of Koy Sanjak and Shaqlawa, outside the area of Azeri but within the area of Kurdish affrication, there is sporadic affrication of **k* to [tʃ] after front vowels and also fronting of original /tʃ/ to /ts/. This, therefore, resembles the Kurdish patterns of affrication more closely.

In some Christian NENA, outside of the area of Kurdish affrication, in Targawar (§5.1.1.2), Baz (§5.1.1.3) and Hakkari (§5.1.1.6 in south-eastern Turkey there is affrication of dorsals in a specific set of lexical items, which is remarkably stable across these dialects. In the Northern Kurdish dialects of these areas there is no affrication, though dorsals may develop a palatal offglide by a process of palatalisation. It appears that affrication in these NENA dialects is an internal development, catalysed by palatalisation in Northern Kurdish. Lexical specificity of palatalisation, which sometimes plays a role in distinguishing otherwise homophonous forms, is a feature of some other NENA dialects (cf. example 28).

The further fronting of unaspirated post-velar /*k̤*/ in NENA dialects in the Baz region to the unaspirated affricate [tʃ] appears to be an internal development of these NENA dialects, since it does not have a specific parallel in other languages of the region. It may, however, have been indirectly facilitated by the general process of palatalisation of dorsals in Northern Kurdish, in that it was an internal extension of this process, as was the case with the affrication of dorsals in the NENA dialects of this region.

The fronting of **q* to post-velar unaspirated /*k̤*/ in Christian dialects of the Urmi region and further north has no clear analogue in Kurdish. This may have been simply due to an internal pull-chain effect after the fronting of the velar **k*.

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A possible external model for the fronting, however, is Azeri, in which uvular /q/ was fronted to velar /g/ in some dialects.

There are no clear parallels in Kurdish to the affrication of alveolar **t* and **d* to /tʃ/ and /dʒ/ before high vowels in the NENA dialect of C. Bədyal (Barzan region). Palatalisation of alveolar stops is attested in Azeri, but it is not clear how this could have induced the process in C. Bədyal, since the NENA dialect was not in contact with Turkic. It is more likely that this too was an internal development of extension of the process of palatalisation of dorsals in Northern Kurdish.

6 Pharyngeals

6.1 Pharyngeals in NENA

The historical Aramaic pharyngeal consonants $*^c$ (voiced pharyngeal fricative) and $*h$ (voiceless pharyngeal fricative) generally undergo the following shifts in NENA dialects:

(1)

$*^c$ > /ʔ/ or Ø
 $*h$ > /x/

A shared innovation of all NENA dialects that distinguishes the subgroup from the neighbouring Turoyo subgroup of Neo-Aramaic is the loss of the voiced velar fricative $*g$ through a process in which the velar first shifted to a pharyngeal fricative before weakening to a laryngeal and zero (Tsereteli 1990).

(2)

$*\bar{g}$ > /ʕ/ > /ʔ/ or Ø

The reflex of historical $*g$ is, therefore, the same as the reflex of historical $*^c$. The reflex /ʔ/ is attested predominantly in dialects of Iraq:

(3) C. Qaraqosh (Khan 2002a)

<i>zra'a</i>	<	<i>*zra'ā</i>	'cultivation'
<i>da'ar</i>	<	<i>*dā'er</i>	'he returns'
<i>bə'ta</i>	<	<i>*bē'tā</i>	'egg'
<i>šamə'</i>	<	<i>*šāme'</i>	'he hears'
<i>tar'a</i>	<	<i>*tar'ā</i>	'door'
<i>šra'a</i>	<	<i>*šrāgā</i>	'lamp'
<i>ra'əš</i>	<	<i>*rāgēš</i>	'he wakes'
<i>lə'ma</i>	<	<i>*laḡmā</i>	'jaw'
<i>palə'</i>	<	<i>*pāleḡ</i>	'he divides'
<i>pal'a</i>	<	<i>*pālḡā</i>	'she divides'

6 Pharyngeals

(4) C. Barwar (Khan 2008a)

<i>zraya</i>	<	* <i>zrā</i> ‘ā	‘cultivation’
<i>dayər</i>	<	* <i>dā</i> ‘er	‘he returns’
<i>bita</i>	<	* <i>bē</i> ‘tā	‘egg’
<i>šāme</i>	<	* <i>sāme</i> ‘	‘he hears’
<i>tāra</i>	<	* <i>tar</i> ‘ā	‘door’
<i>šraua</i>	<	* <i>šrāgā</i>	‘lamp’
<i>rayəš</i>	<	* <i>rāgēs</i>	‘he wakes’
<i>pela</i>	<	* <i>pāglā</i>	‘radish’
<i>pāle</i>	<	* <i>pālēg</i>	‘he divides’
<i>pala</i>	<	* <i>pālḡā</i>	‘she divides’

(5) C. Diyana-Zariwaw (Napiorkowska 2015)

<i>zraya</i>	<	* <i>zrā</i> ‘ā	‘cultivation’
<i>ṭayər</i>	<	* <i>dā</i> ‘er	‘he returns’
<i>bita</i>	<	* <i>bē</i> ‘tā	‘egg’
<i>šāme</i>	<	* <i>sāme</i> ‘	‘he hears’
<i>taṛa</i>	<	* <i>tar</i> ‘ā	‘door’
<i>šreya</i>	<	* <i>šrāgā</i>	‘lamp’
<i>rayəš</i>	<	* <i>rāgēs</i>	‘he wakes’
<i>lama</i>	<	* <i>laḡmā</i>	‘jaw’
<i>pāle</i>	<	* <i>pālēg</i>	‘he divides’
<i>pāla</i>	<	* <i>pālḡā</i>	‘she divides’

(6) J. Amedia (Greenblatt 2011)

<i>pqa’a</i>	,	* <i>pqā</i> ‘ā	‘explosion’
<i>da’ər</i>	<	* <i>dā</i> ‘er	‘he returns’
<i>be’ta</i>	<	* <i>bē</i> ‘tā	‘egg’
<i>šame’</i>	<	* <i>sāme</i> ‘	‘he hears’
<i>tar’a</i>	<	* <i>tar</i> ‘ā	‘door’
<i>šra’a</i>	<	* <i>šrāgā</i>	‘lamp’
<i>ra’əš</i>	<	* <i>rāgēs</i>	‘he wakes’
<i>pe’la</i>	<	* <i>pāglā</i>	‘radish’
<i>pale’</i>	<	* <i>pālēg</i>	‘he divides’
<i>pal’a</i>	<	* <i>pālḡā</i>	‘she divides’

(7) J. Koy Sanjak (Mutzafi 2004a)

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<i>da'ər</i>	<	* <i>dā'ər</i>	‘he returns’
<i>be'e</i>	<	* <i>bē'ē</i>	‘eggs’
<i>beta</i>	<	* <i>bē'tā</i>	‘egg’
<i>šame</i>	<	* <i>šāme'</i>	‘he hears’
<i>tāra</i>	<	* <i>tar'ā</i>	‘door’
<i>zo'a</i>	<	* <i>zōgā</i>	‘pair’
<i>ṛā'əš</i>	<	* <i>rāgēš</i>	‘he wakes’
<i>pella</i>	<	* <i>paḡlā</i>	‘radish’
<i>pāle</i>	<	* <i>pāleḡ</i>	‘he divides’
<i>palya</i>	<	* <i>pālḡā</i>	‘she divides’

A laryngeal continuant /h/ reflex of intervocalic *^c is sporadically attested, e.g.,

(8) C. Qočanəş (Talay 2008a: 79)

bihe < **bē'ē* ‘eggs’

(9) J. Rustaqa (Khan 2002b: 397)

yahəl < **ya'əl* < **yāḏe'* ‘eggs’

The development of the voiceless pharyngeal **ħ* to the velar fricative /x/ is an innovation of the NENA subgroup and is exhibited by all dialects except those in the north-western periphery, e.g.,

(10)

		C. Barwar	C. Urmi	J. Sanandaj
* <i>ħmārā</i>	‘ass’	<i>xmara</i>	<i>xmara</i>	<i>xmara</i>
* <i>ḥāṭā</i>	‘sister’	<i>xaṭa</i>	<i>xata</i>	<i>xālāsta</i>

In a small group of NENA dialects in the Bohtan region on the north-western periphery of NENA, including Derabün in north-western Iraq, the speakers of which migrated from the Bohtan region, the reflex of **ħ* is /ħ/ (Jastrow 1994b; Talay 2008a: 44-45). In such dialects a velar fricative **x* has shifted to the pharyngeal /ħ/, so the pharyngeal reflex of **ħ* may have resulted in the development **ħ* > **x* > **ħ*:

(11) C. Hertevin (Jastrow 1988b)

* <i>ħmārā</i>	‘ass’	<i>ħmara</i>
* <i>baxe</i>	‘he weeps’	<i>baḥe</i>

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In some of the NENA dialects, the original voiced and voiceless pharyngeal are preserved in words containing emphatic, i.e., pharyngealised, consonants, including /q/. This is found in most dialects except those in the Hakkari and north-western Iran, e.g.,

(12) C. Qaraqosh (Khan 2002a)

<i>ṭaʿən</i>	‘he lifts’	<	*ṭāʿen
<i>raḥoqa</i>	‘distant’	<	*rāḥōqā

(13) J. Zakho

<i>ʿapša</i>	‘gallnut’	<	*ʿapšā
<i>ʿamoqa</i>	‘deep’	<	*āmōqā
<i>raḥuqa</i>	‘distant’	<	*rāḥōqā

(14) C. Sanandaj (Panoussi 1990)

<i>ʿarəq</i>	‘he flees’	<	*ʿāreq
<i>raḥuqa</i>	‘distant’	<	*rāḥōqā

(15) J. Koy Sanjak (Mutzafi 2004a)

<i>təʿna</i>	‘load’	<	*ṭaʿnā
<i>ḥānəq</i>	‘he chokes’	<	*ḥāneq

This preservation has come about by a process of ‘long-distance consonant agreement’ (Khan 2013). As can be seen above, in Jewish Trans-Zab dialects such as J. Koy Sanjak (15), the preservation of the pharyngeal /ʿ/ entailed the neutralization of the pharyngealization of the emphatic *ṭ (see §7.1.2). Note, however, that the /q/ is retained. The process here involved the segmentalisation of the pharyngealisation and associated flat resonance (low F2) of pharyngealised consonants such as /ṭ/ into the pharyngeal segment /ʿ/. This resulted in the conversion of emphatic consonants that produce flat resonance into plain consonants. The consonant /q/ is considered to be an emphatic consonant in Semitic (§8.1), but it typically does not have the acoustic property of inducing flat resonance, i.e., reduced F2 (Khan 2016a: vol. 1, 116). Since /q/ does not produce flat resonance, it is not affected by the process.¹

A voiced pharyngeal that has evolved from an original voiced velar fricative *ḡ has, likewise, been preserved in emphatic environments in some dialects, e.g.,

¹Note that acoustic differences between the pharyngeals and /q/ were also found for Arabic, Kurdish and Armenian (Ariyae et al. 2024).

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(16)

J. Amedia J. Koy Sanjaq

**l̥ə ǧawwāyā* ‘inside’ *l-‘oya* *lo‘a*

Conversely, a pharyngeal in a word may cause an etymological velar **k* to shift to a /q/, as in Jewish Arbel dialect:

(17) J. Arbel (Khan 1999a)

ḥaqen ‘I speak’ < **ḥky* (Arabic)

In some dialects pharyngeals are preserved in a wider range of contexts, including words containing sonorants and labials, e.g.,

(18) C. Harbole (Khan, fieldnotes)

zar‘a ‘seed’ < **zar‘ā*
‘əllaya ‘upper’ < **‘əllāyā*
paraḥta ‘bird’ < **pāraḥtā*
raḥme ‘mercy’ < **raḥmē*
dahle ‘field’ < **daxla* ‘grain’

A voiced pharyngeal that has been preserved in an emphatic environment may become devoiced, e.g.,

(19) J. Rustaqa (Khan 2002b: 397)

ḥiqta ‘distress’ < **‘iqtā*

Non-etymological pharyngeals have developed in the environment of emphatic consonants (see 7.1.2) and/or long /a/ vowels. A voiced /ʕ/ has developed by this process in some Jewish Trans-Zab dialects, e.g.,

(20) J. Sanandaj (Khan 2009)

tma‘nisar ‘eighteen’ < *tṃanisar*
tma‘ni ‘eighty’ < *tṃani*

(21) J. Sulemaniyya (Khan 2004: 35)

ma‘e ‘water’ < *ṃa‘e* < *maye*
ma‘danusi ‘parsley’ < *ṃadanusi*

Non-etymological *ḥ* has developed by this process in some NENA dialects in the north-western sector, e.g.,

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(22) C. Derabun

kaḥka ‘tooth’ < *kaka* < *kakka*

A non-etymological *ḥ* may also develop by strengthening of *h* in words with emphatic consonants and/or the low vowel *a*, e.g.,

(23) J. Sanandaj (Khan 2009: 23)

ʾaḥra ‘town’ < **ʾahra* < **ʾaθra*
təlḥa ‘three’ < **tḷaha* < **tḷaθa*

(24) C. Derabun

bəḥra ‘light’ < **bahra*

In Jewish Trans-Zab dialects, the voiced pharyngeal /ʕ/ is in some cases realised with less muscular tension as a laryngeal [ʔ] (25). Conversely a laryngeal /ʔ/ is sometimes realised as a pharyngeal [ʕ] (26):

(25) J. Sanandaj (Khan 2009: 29)

tamʕa [t^hamʕa t^hamʕa] ‘she tastes’
ʾəṣʕa [ʔiʃʕa ʔiʃʕa] ‘nine’

(26) J. Arbel (Khan 1999a: 26)

hulaʕe [hu:laʕe: hu:laʕe:] ‘Jews’

In Christian dialects pharyngeals have been preserved in a number of words associated with the church and religion. In such cases the pharyngeal pronunciation may be an archaism influenced by Classical Syriac, e.g.,

(27) C. Qaraqosh (Khan 2002a: 41)

mšiḥa ‘the Messiah’
roḥəd qudša ‘the Holy Spirit’
šliḥə ‘apostles’
šabtəd ḥaša ‘Week of Suffering (Holy Week)’

In some dialects on the Mosul plain and adjacent regions, such as Peshabur (Coghill 2013) and Shiyuz, and original *h* has been strengthened to a pharyngeal /ḥ/ in 3rd person singular pronominal suffixes. This is a strategy to prevent the elision of the *h*, which would have made the suffixes indistinguishable from the nominal inflectional vowels, e.g.,

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- (28) C. Qaraqosh (Khan 2002a: 41)

3sg.m -əḥ < -əh
 3sg.f -aḥ < -ah

In the system of demonstrative pronouns in some dialects, an original /h/ is strengthened to the pharyngeal fricative /ḥ/ in far deixis and/or very far deixis forms (see 10.1), e.g.,

- (29) C. Qarawilla

ʾāwaha sg.m far deixis
 ʾāwaḥa sg.m very far deixis

- (30) C. Haşsan

ʾāwaha sg.m far deixis
 ʾāwaḥa sg.m very far deixis

- (31) C. Peshabur (Coghill 2013)

ʾāwaha sg.m far deixis
 ʾāwaḥa sg.m very far deixis

In addition to the occurrence of pharyngeals in native NENA words that has been described above, pharyngeals are also attested in loanwords from Arabic and Kurdish, e.g.,

- (32) J. Sanandaj (Khan 2009: 27)

ḥamām ‘bath’
 ḥana ‘henna’
 ḥqy ‘to speak’ < Arabic ḥky
 maʿlām ‘teacher’
 ʾayza ‘good’ < Arabic ʾazīz

The pharyngeals in these loanwords, however, are only preserved in NENA dialects that have pharyngeals in their sound systems, so they do not appear in loanwords in the NENA dialects of the north-eastern sector. In latter subgroup of dialects, the reflex of the voiceless pharyngeal ḥ in loanwords is typically /h/ and reflex of the voiced pharyngeal is /ʔ/, zero or, in some cases, results in suprasegmental pharyngealisation, e.g.,

- (33) C. Urmi (Khan 2016a: vol. 1, 170-171)

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<i>hamla</i>	‘attack’	< Arabic <i>ḥamla</i>
<i>halqa</i>	‘ring’	< Arabic <i>ḥalqa</i>
<i>ʿazizi</i>	‘my dear’	< Arabic <i>ʿazīzī</i>
<i>*jamma</i>	‘she gathers’	< <i>*mjamʿa</i> (< Arabic <i>j-m-ʿ</i>)

6.2 Pharyngeals in Arabic

The Old Arabic pharyngeals **ʿ* and **ḥ* are usually preserved in the Qəltu Arabic dialects, including in the NENA region, with the exception of the Sason dialects (Jastrow 1978: 43). In some Anatolian dialects /ʿ/ undergoes devoicing to /ḥ/ under certain conditions. While in much of Anatolian Arabic the pharyngeal articulation of **ʿ* and **ḥ* is ‘weaker’, to the extent that they come close to the laryngeals /ʔ/ and /h/ respectively, this is not the case in or near the NENA-speaking region, i.e., among the Tigris and Mardin dialects (Jastrow 1978: 44).

6.3 Pharyngeals in Kurdish

6.3.1 Distribution

The pharyngeals /ḥ/ [h] and /ʿ/ [ʕ] exist in Kurdish. Given that the sounds originate in Semitic, they are found mostly in Arabic-origin words. The following table illustrates the presence of these phonemes in Arabic loanwords in the Central and Northern Kurdish varieties:

Table 6.1: Pharyngeals in Arabic loanwords in Kurdish

Arabic	NK	CK	Gloss
<i>ʿarab</i>	<i>ʿereb</i>	<i>ʿereb</i>	‘Arab’
<i>maʿlūm</i>	<i>meʿlūm</i>	<i>meʿlūm</i>	‘evident’
<i>ʿadāla(t)</i>	<i>ʿedalet</i>	<i>ʿedalet</i>	‘justice’
<i>maḥall</i>	<i>miḥele</i>	<i>maḥal</i>	‘neighborhood’
<i>maḥṣar</i>	<i>meḥşer</i>	<i>meḥşer</i>	‘resurrection (day)’
<i>ḥākim</i>	<i>ḥakim</i>	<i>ḥakim</i>	‘judge, governor’
<i>ḥammām</i>	<i>ḥemam</i>	<i>ḥemam</i>	‘bath’
<i>baḥr</i>	<i>beḥr</i>	<i>beḥr</i>	‘sea’

The pharyngeals are also found in a restricted number of native Iranian words in Kurdish:

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- (34) Northern Kurdish
- | | |
|---|------------|
| <i>heft</i> ~ <i>heft</i> | ‘seven’ |
| <i>heşt</i> ~ <i>heşt</i> | ‘eight’ |
| <i>hişk</i> ~ <i>hişk</i> | ‘dry’ |
| <i>hemî</i> ~ <i>hemî</i> | ‘all’ |
| <i>biheşt</i> ~ <i>bihişt</i> | ‘paradise’ |
| <i>esman</i> ~ <i>azman</i> | ‘sky’ |
| <i>ezman</i> ~ <i>ziman</i> , <i>hizmān</i> | ‘tongue’ |
| <i>ewr</i> ~ <i>hewr</i> | ‘cloud’ |

There is significant variation in the realization of pharyngeals in Kurdish dialects. As the data in (34) shows, all the native words have non-pharyngeal realizations in other dialects.

Secondly, regardless of the native vs. loanword status of the item, the Central Kurdish dialects show a great deal of fluctuation between the voiced and voiceless pharyngeal realization of a given sound. Thus, the Mukri dialect often replaces a voiced pharyngeal (in the Arabic source or other Central Kurdish dialects) in a syllable onset with the voiceless counterpart:

- (35)
- | Mukri | | Other CK dialects | Gloss |
|----------------|---|-------------------|----------|
| <i>hejayêb</i> | < | <i>‘ecayêb</i> | ‘weird’ |
| <i>hemr</i> | < | <i>‘emr</i> | ‘order’ |
| <i>herz</i> | < | <i>‘erz</i> | ‘ground’ |
| <i>cemaheţ</i> | < | <i>cema‘et</i> | ‘people’ |
| <i>hafret</i> | < | <i>afret</i> | ‘woman’ |
| <i>hasman</i> | < | <i>asman</i> | ‘sky’ |

Thirdly, some pharyngeal sounds of Arabic loanwords in Kurdish have been replaced with non-pharyngeal counterparts /h/, in the case of the voiceless pharyngeal, and compensatory vowel lengthening, in the case of the voiced pharyngeal. Thus the Arabic word *haqq* ‘right’ is found in all Kurdish dialects as *haq*. Likewise, the Arabic word *ta‘m* ‘taste’ is found without a pharyngeal in CK and Behdînî dialects of NK as *tām* and *ṭām*, respectively.

Conversely, some Arabic loanwords that do not have a pharyngeal in the source language have developed a pharyngeal in Kurdish.

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(36)	Arabic	NK	Gloss
	<i>jur'at</i>	<i>cur'et</i>	'daring'
	<i>'arḍ</i>	<i>'erd</i>	'earth'
	<i>jāhil</i>	<i>caḥêl</i>	'young' (in Behdîni dialects only)
	<i>qur'ān</i>	<i>qur'an</i>	'Koran'
	<i>'awil</i>	<i>'ewil</i>	'first'
	<i>qahwa</i>	<i>qeḥwe</i>	'coffee'

A unique development in the northern dialects (Diyarbakir, Erzurum, Yerevan) of Northern Kurdish concerns the emergence of a voiced pharyngeal in a subset of native words with similar syllabic structures. As seen in the examples below, where these northern dialects have a mid-open vowel /e/, in the rest of Kurdish, where no such pharyngeal has emerged, the words have a long back vowel /a/:

(37)	Central Kurdish / Southern NK	Northern NK	Gloss
	<i>masî</i>	<i>me'sî</i>	'fish'
	<i>çaw / çav</i>	<i>çe'v</i>	'eye'
	<i>mar</i>	<i>me'r</i>	'snake'
	<i>tarî</i>	<i>ta'ri</i>	'dark'

This is probably a purely Kurdish-internal development. The forms in question are likely to have developed from a syllabic structure where a complex coda with an initial glottal fricative /h/, e.g., *çehv* 'eye', which is still attested in some central N. Kurdish dialects, is changed to a pharyngeal. This fricative is lost in the south-eastern NK dialects and Central Kurdish with the compensatory lengthening of the syllable nucleus to [a:].

Northern NK dialects are geographically farthest from direct Arabic/Semitic contact but close to Caucasian languages which also possess pharyngeals. This introduction of pharyngeals is thus more likely to be the result of language-internal processes, though facilitated by contact with either Arabic or Caucasian languages, or both.

As for the distribution of pharyngeals, they are found in all major Kur-dic varieties (Zazaki, NK, CK, Gorani, SK) except for Laki. Gorani dialects of Hawraman (core and conservative) have pharyngeals, while peripheral dialects in Iran spoken within the SK speech zone do not have pharyngeals. SK dialects in Iraq have both pharyngeals. Geographically, the pharyngeal phonemes are most robustly present in the central areas of the Northern and Central Kurdish speech zones. Their presence in Arabic loan-words is weakened towards the extreme northern and southern peripheries in heavy contact with Turkish and Persian, respectively (see Map 1.27 in

6.4 Pharyngeals in Armenian

the Manchester Kurdish Database, which illustrates such weakening of pharyngeals at the peripheries through the distribution of the Arabic loanword *heywan* ‘animal’ <http://kurdish.humanities.manchester.ac.uk/pharyngeal-retentionloss-animal/> (accessed 31/10/2024).

6.3.2 Evolution

Given that pharyngeals are not found in any Iranian languages other than Kurdish varieties and the Kumzari language in Oman, which is in close contact with Arabic, it is safe to conclude that their introduction into Kurdish phonology has been via contact with Semitic (Barry 2019; Anonby 2020). Although in theory the Semitic language does not have to be Arabic, given the fact that the pharyngeals in Kurdish are predominantly from Arabic borrowings strengthens this possibility. A more empirical piece of support, however, comes from the layers of borrowing in Kurdish. The Arabic word *jamā‘at* ‘community’ (root *j-m-‘*) is found in Kurdish in two forms: *civat* and *cima‘et*, with the former possessing the more specialized sense of ‘company, assembly’ while the latter having a more direct semantic correspondence with the source meaning in Arabic. In the former, the pharyngeal is lost, resulting in compensatory lengthening of the vowel (see above for more on this phenomenon), whereas in the latter the pharyngeal is retained. The borrowing of the former must have taken place earlier, at a time when pharyngeals were not tolerated in the phonology of Kurdish and when the shift of intervocalic *m* to *v* was active (dated to the immediately post-Islamic period, see Paul 2008). The semantic specialization of the former item further supports its antiquity. The latter form dates from more recent times when the pharyngeals were established in the phonological system of the language. The earliest written records of Kurdish from the late 16th century make abundant use of the pharyngeals, both in Arabic loanwords and some native words.

Still, the pharyngeals should be considered as peripheral to the phonological system (cf. Haig 2007; Anonby 2020), since, as noted by Haig 2007: 167, they are restricted to individual lexical items, their functional load is very limited, and there is considerable cross-speaker and cross-dialectal variability in the extent of their presence.

6.4 Pharyngeals in Armenian

Classical Armenian did not possess pharyngeals, nor are they present in most modern Armenian dialects. However, Vaux (1992: 13), referring to Haneyan

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(1985), states that the so-called “voiced h”, which is mainly derived from Classical Armenian initial /j/, or from a later process of anaptyxis of [h] in initial position before vowels, is a voiced laryngeal sonant in all of the dialects in which it appears (which include Mush, Karin (Erzurum), and Tigranakert (Diyarbakir)), except in the Shatakh dialect, where it is a voiced pharyngeal sonant. However, Muradyan (1962), in her grammar of the Shatakh dialect, although she gives some examples of words with anaptyxis of initial [h], such as *hænæpæt* < *anapat* ‘wilderness, desert, monastery’, *hundytk^{hj}* < *andund* ‘abyss’, *hənt^hatak* < *ant^hatak* ‘armpit’ (Muradyan 1962: 94), does not mention anything unusual about the pronunciation of this /h/, nor does she refer to the presence of a voiced and/or pharyngeal *h*-type sound in this dialect in the phonology section of the grammar. Note that inherited /h/ generally changes to [x] in the Shatakh dialect (*xam* < *ham* ‘taste’, *paxel* < *pahel* ‘to keep’ etc., see Muradyan (1962: 50), a feature that is characteristic of and particular to the area around lakes Van and Urmia. “Voiced *h*” also appears in place of a pharyngeal in some dialects in loanwords from Arabic, e.g., *hæləb* ‘Aleppo’, *hæsræt^h* ‘nostalgia, longing’, *hajf* ‘regret’, etc. These examples are from Tigranakert (Diyarbakir), given by Haneyan 1978: 228, who describes this sound as “a lower-throat voiced fricative” (Haneyan 1978: 17), presumably laryngeal rather than pharyngeal, given that in Haneyan (1985) she states that “voiced h” is pronounced as a pharyngeal only in Shatakh.

6.5 Pharyngeals in Turkic

Pharyngeal consonants are not typical of Turkic phonological systems but are encountered in varieties spoken in areas of intense language contact or strong substratum influence.

The most notable cases are found in Iraqi Turkic varieties, where the phenomenon is described in detail by Bulut (2014: 32-33). In these varieties, Arabic lexical copies are commonly integrated with their original pharyngeals (/ħ/, /ʕ/) preserved, e.g., [hæwɒləd] ‘transfer, referral’, [‘qjɪb/] ‘shame’ (< Arabic *hawāla*[t], ‘ayb, respectively). Exceptions do occur, however. In some cases, a pharyngeal is replaced by /h/ in copies, as is typical of most West Oghuz varieties, e.g., [halædde:] ‘state.LOC’ (< Arabic *hāla*[t] ‘state’). More strikingly, there are also instances in which an etymological /h/ is realised as a pharyngeal in an inherited lexeme, e.g., [ħæpʃ] ‘all’ (cf. Standard Turkish *hepsi*).

Sonqor Turkic (Bulut 2014: 32-33), spoken in Kermanshah, which is not in direct contact with Arabic but is in intense contact with Central and Southern Kurdish, likewise displays pharyngeals in words of Arabic origin, both etymolog-

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ically, e.g., [narahætr] ‘inconvenience’ (ultimately from Arabic *rāḥa*[t] ‘comfort’), and non-etymologically, e.g., [hæwas] ‘folly, frenzy’ (< Arabic *hawas*).

On the basis of her data, **Bulut** (2014: 33) concludes that copied Semitic phonemes in Turkic function as allophonic variants of their non-pharyngeal counterparts rather than as independent phonemes and today are nothing more than a stylistic feature or perhaps a way to emulate the language of prestige. Since the same pattern is observed in Kurdish varieties (including the presence of non-etymological pharyngeals in Semitic copies, bearing in mind that pharyngeals are not inherited in Iranian), she suggests that while in Iraq such features may have been transmitted through direct contact with Arabic, the situation in Sonqor likely reflects a multi-stage copying process, whereby Semitic features were first incorporated into Kurdish—possibly through Aramaic substrate influence—and only subsequently transmitted to Turkic varieties via Kurdish.

In other areas of contact with Semitic, pharyngeals are not preserved in lexical copies, as illustrated by the following examples from Urmia Azeri (all from **Doğan** 2010). The typical reflex of /h/ is /h/, as in most Azeri dialects, e.g. [hesap] ‘count’ (ultimately from Arabic *ḥisāb*). As for /ʕ/, it is lost word-initially, e.g., [ɑ:dæt] ‘tradition’ (ultimately from Arabic *ʿāda*[t]), often accompanied by vowel lowering when the etymological vowel is close, e.g., [elm] ‘science’, [øhdæ] ‘responsibility’ (ultimately from Arabic *ʿilm*, *ʿuhda*[t] respectively). Word-medially, it is dropped, with compensatory lengthening of the preceding vowel, e.g., [dæ:væt] ‘invitation’ (ultimately from Arabic *daʿwa*[t]), accompanied again by vowel lowering if the latter is etymologically close, e.g., [mø:tæbær] ‘reliable, influential’ (ultimately from Arabic *muṭabar*).

Outside the areas of contact with Kurdish and Semitic, /h/ in Azeri is attested in the Quba dialect, where it occurs primarily in words of Arabic origin, e.g., [hisab] ‘count’ (**Rüstəmov** 1961: 61). This is noteworthy given that the variety has had only limited direct contact with Arabic, and at a time when Arabic was no longer a dominant spoken language in the Caucasus. The situation is, therefore, comparable to that of Sonqor Turkic, in that lexemes containing /h/ appear to have been transmitted via an Iranian substratum, namely Tat. Formerly widespread in what is now north-western and central Azerbaijan, Tat was very likely in contact with Arabic at a period when the latter still functioned as a spoken prestige language in the region, and it has not only preserved /h/ and /ʕ/ in Arabic lexical copies (cf. Northern Tat [hisab] ‘count’) but has also incorporated these sounds into its phonemic inventory, where they occur in inherited words.

Pharyngealised vowels are rare in Turkic languages and typically occur in contact situations, most frequently with East Caucasian (Nakh-Dagestani) lan-

6 Pharyngeals

guages. Zaqatala Azeri, for instance, possesses the phonemic vowels /a^ʕ/, /o^ʕ/, /u^ʕ/ and /u^ʁ/ not only in loanwords, e.g., /a^ʁrxa^ʁna^ʁ/ ‘whey’ (< Tsakhur a^ʁχa^ʁne), but also, as a result of substrate influence, in inherited vocabulary, e.g., /o^ʕč/ ‘three’, /o^ʕlu^ʕ/ ‘dead’ (cf. üç [ytʃ], ölü [øly] in non-contact varieties respectively; [Aslanov 1966](#): 204). Tabasaran, an East Caucasian language spoken in Dagestan, Russia, with a history of intense contact with Azeri, has the pharyngealised vowels /a^ʕ/ and /u^ʕ/ ([Alekseev & Šixaliev 2003](#): 29). Under Azeri influence, it has developed front-back vowel harmony, which also accounts for pharyngealisation, e.g., *gažin* ‘jug’ > *gažn-ar* ‘jugs’, *düd* ‘throat’ > *düd-er* ‘throats’, *ma^ʁrx* ‘sledge’ > *ma^ʁrx-a^ʁ* ‘sledges’ ([Alekseev & Šixaliev 2003](#): 32). However, younger Tabasaran speakers increasingly neutralise the opposition between /u^ʕ/ and /y/, pronouncing both as [y], which apparently reflects both Azeri influence and the standard Tabasaran orthography, which does not distinguish the two graphically (Gilles Authier, p.c.).

6.6 Discussion

The pharyngeal consonants are found in the sound systems of all NENA dialects except those in the north-eastern sector. This geographical distribution of pharyngeals corresponds, broadly speaking, to the geographical areas where Arabic dialects are spoken, all of which have pharyngeals in their sound inventories. In the north-eastern sector, contact with Arabic was more remote. The geographical distribution also corresponds broadly to the areas where pharyngeals are robustly attested in Kurdish.

NENA dialects in the north-eastern sector that lack pharyngeal consonants also lack emphatic, pharyngealised, consonants. Pharyngealisation in these dialects is a suprasegmental feature at the level of words. Elsewhere in NENA, emphatic consonants play a role in preserving pharyngeal consonants within a word (see examples [12](#), [13](#), [14](#), [15](#)). In words without emphatic consonants pharyngeals have generally been weakened (**h* > /x/ and **ʕ* > ’ or zero).

Some features of pharyngeal consonants in NENA have parallels with pharyngeals in Kurdish and most of these features apply also to pharyngeals in Turkic varieties in the region (§6.5). These features include the following.

Variation in the voicing of pharyngeals, which is attested across Kurdish dialects (see §6.3.1), is also found in NENA (see example [19](#)).

Instability of pharyngeals involving phonetic interchange between etymological pharyngeals and laryngeals, and vice versa, which is a feature of Kurdish and Turkic, is attested within individual Jewish Trans-Zab dialects of NENA, e.g., examples ([25](#)) and ([26](#)).

6.6 Discussion

As seen in (37), in some varieties of Kurdish non-etymological pharyngeals have developed in some words, e.g., NK *çe‘v* ‘eye’ < *çav*. The process seems to involve the segmentalisation of flat (low F2) resonance of a low back vowel /a/ into a pharyngeal segment (Barry 2019). A parallel to this process is attested in some dialects of NENA, including the Trans-Zab Jewish dialects and Christian dialects in the north-western sector of NENA (see examples 20 and 22).

In some cases this process of segmentalisation in NENA dialects involves the strengthening of a laryngeal /ʔ/ or /h/ to a pharyngeal /ħ/ or ʕ respectively (see examples 23, 21 and 24), which could be compared to the strengthening of etymological /h/ in some Kurdish dialects to ʕ in some words (see 34). This was due, it seems, to a perceived acoustic similarity between laryngeals and pharyngeals. A similar process is found in some Turkic dialects of the region (§6.5). A typologically related process may be the strengthening of voiced /h/ to a pharyngeal in some Armenian dialects. This feature of strengthening segments to a pharyngeal is attested in the Iranian languages Kurdish and Gorani and in Turkic within the Trans-Zab area, e.g., Gorani Takht *hawt* [ħæwt] ‘seven’ (cf. Persian *haft*). Note also Gorani Takht *əna* ‘buttocks’ (cf. NK and CK *qûn*), where the flat resonance is induced by the back rounded vowel and a /q/ has shifted to a pharyngeal. Also relevant to note is the strengthening of etymological /h/ to the pharyngeal /ħ/ in demonstrative pronouns and pronominal suffixes in some NENA dialects, which could have been facilitated by the aforementioned strengthening of /h/ to ʕ in Kurdish.

The development of the voiceless pharyngeal /ħ/ from the velar fricative /x/, including where the velar fricative is etymologically a velar, in NENA dialects in the north-western sector (see example 11), may have been facilitated by the development of non-etymological pharyngeals in Kurdish from /h/ (see 34).

The strengthening of /h/ to a pharyngeal in demonstrative pronouns in some NENA dialects (29, 30, 31) as an iconic strategy to express distant deixis appears to be unique to NENA.

7 Emphatic (Pharyngealised) Consonants

7.1 Emphatic Consonants in NENA

The NENA dialects contain in their sound systems pharyngealised consonants, known traditionally as ‘emphatic’ consonants, or at least pharyngealisation that has arisen from the historical presence of such emphatic consonants in a dialect’s sound system. Pharyngealisation has developed historically in various ways across the dialect group. Three typologies of pharyngealisation can be identified, which are associated with three sub-groups of dialects, viz., (i) the dialects of Iraq and the adjacent areas of south-eastern Turkey, (ii) the dialects spoken in western Iran south of the Urmi region, and (iii) the dialects of north-western Iran in the Urmi region. There are no absolute boundaries between these groups with regard to pharyngealisation typology, but this classification will be useful to give a clear presentation of the different types of pharyngealisation that have developed.

7.1.1 Dialects of Iraq and the Adjacent Areas of South-Eastern Turkey

The dialects of this group include the Christian dialects and Jewish *Lišana Deni* dialects of Iraq and of south-eastern Turkey from the north-western periphery of NENA up to the Tsuma region. In this group pharyngealisation corresponds broadly to the type that is found in Arabic, though it is weakened in dialects on the northern periphery in south-eastern Turkey, such as Bohtan (Fox 2009b: 20) and Hertevin (Jastrow 1988b: 5). It is a secondary feature of various oral consonants, which involves the constriction of the upper pharynx with increased muscular tension. This is a non-primary articulation that accompanies the primary articulation of consonants and vowels. It is accompanied by a variety of other features including the retraction of the tongue from the primary point of articulation of coronal and dorsal consonants. The articulation of vowels tends to be retracted and the acoustic resonance made ‘flat’ by the lowering of F2. There

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tends to be a lip rounding gesture. Labial consonants when pharyngealised are pronounced with a degree of protrusion and rounding of the lips.

In the Iraq dialect group, pharyngealisation is a feature of the historical emphatic consonants of Aramaic, viz., the coronals /t/, /s/ and the uvular /q/. These are the pharyngealised counterparts of /t/, /s/ and /k/. Several dialects have a pharyngealised affricate /č/, which is the emphatic counterpart of the plain /č/. This pharyngealised affricate has developed from sibilants or stops, e.g.,

(1) J. Amedia (Greenblatt 2011)

čə'a	'smooth'	<	*šə'a
qoč	'pick!'	<	*qot

The pharyngealised stop /t/ and affricate /č/ are not kept clearly apart from their plain counterparts with regard to their glottal setting. Outside of the context of a preceding fricative, where all stops and affricates are unaspirated, the /t/ and /č/ are realised either aspirated or unaspirated, whereas plain /t/ and /č/ are aspirated. Pharyngealisation, therefore, carries the main burden of contrast.

Pharyngealisation is also associated with a group of 'non-historical' emphatic consonants that includes the sonorants /r/, /l/ and the labials /b/, /m/, /p/, also /w/. The forms in this latter group developed in certain words containing the non-emphatic consonants, in some cases, it seems, to fulfil a functional need to create semantic contrasts, e.g.,

(2) J. Amedia (Greenblatt 2011)

mlele	'he filled'	:	mlele	'it sufficed'
baza	'hawk'	:	b-aza	'she will go'
bərke	'pool'	:	bərke	'his knee'

In dialects on the northern periphery such as Bohtan (Fox 2009b: 20) and Hertevin (Jastrow 1988b: 5), in which the pharyngealisation of the inherited coronal emphatics /t/ and /s/ has been considerably weakened, that of the secondary labial and sonorant emphatics is more robust.

The Iraqi dialects generally also have the voiced pharyngealised segments /z/, /d/ and /ð/ in loanwords from Arabic.

In the Iraqi NENA dialects of this group, the pharyngealisation of pharyngealised consonants tends to spread to adjacent segments in ways that are similar to what has been documented in Arabic dialects. As in some Arabic dialects (Hoberman 1989, Watson 1999), the spreading tends to be restricted to segments in adjacent syllables and is blocked by high front vowels, e.g.,

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(3) J. Zakho (Khan, fieldnotes)

<i>tālāb</i>	‘he requests’	[^h t ^h ɑː (F1 562, F2 1109)lɪb]
<i>tāxər</i>	‘he remembers’	[^h t ^h ɑː (F1 965, F2 1747)xɪr]
<i>mšále</i>	‘he prays’	[m ^h ɑː (F1 849, F2 1281)le]
<i>mšálya</i>	‘she prays’	[m ^h s ^h ɑː (F1 800, F2 1292)lja (F1 647, F2 1945)]
<i>p-sápəs</i>	‘it will rot’	[psɑː (F1 743, F2 1758)pɪs]

It can be seen that the F2 after the emphatics /t/ and /s/ is considerably lower (by 500-600 Hz). The F2 reading of the final vowel of *mšálya* shows that the pharyngealisation does not spread throughout the word.

By contrast, this pharyngealisation is not detectable in vowels adjacent to /q/ and /h/:

(4) J. Zakho (Khan, fieldnotes)

<i>qála</i>	‘voice’	[qaː (F1 983, F2 1719)la]
<i>nḥáqa</i>	‘to touch’	[n ^h ḥɑː (F1 1029, F2 1610)qa]
<i>ḥábəs</i>	‘he incarcerates’	[ḥɑː (F1 899, F2 1707)bɪs]

The F2 values of the /a/ vowels here are roughly equivalent to those of the vowel after plain consonants. The data from J. Zakho relating to the voiced pharyngeal /ʕ/ indicate that adjacent /a/ vowels have a range of values that are roughly intermediate between those of /a/ after /t/ and /s/ and those found adjacent to plain consonants:

(5) J. Zakho (Khan, fieldnotes)

<i>‘ájəq</i>	‘he becomes elated’	[‘ʕɑː (F1 895, F2 1477)ɖɪq]
<i>‘ájəz</i>	‘he becomes tired’	[‘ʕɑː (F1 950, F2 1398)jɪz]
<i>‘áqəl</i>	‘intelligent’	[‘ʕɑː (F1 904, F2 1407)ɪl]
<i>‘áyəč</i>	‘he smears’	[‘ʕɑː (F1 846, F2 1360)jɪtʃ ^h]
<i>‘ájab</i>	‘wonder’	[‘ʕɑː (F1 943, F2 1439)ɖʒab]
<i>‘ájam</i>	‘Persia’	[‘ʕɑː (F1 856, F2 1412)ɖʒam]

In sum, the adjacent /a/ vowels are clearly and consistently pharyngealised adjacent to /t/ and /s/, have intermediate pharyngealisation adjacent to /ʕ/ and are not pharyngealised adjacent to /h/ and /q/.

The reason for the preservation of the pharyngeal in a word such as J. Zakho *nḥaqa*, therefore, is not any suprasegmental spread of pharyngealisation, which

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would have affected all segments, but rather some form of consonant assimilation. The process, it seems, involved the harmonising of the pharyngeal consonant with the pharyngealised emphatic consonants, without necessarily affecting other intervening segments. This is what [Rose & Walker \(2004\)](#) term ‘Long Distance Consonant Agreement’, which operates in many languages on consonants that share a considerable degree of similarity in articulation and gestures, but skips intervening, less similar segments.

7.1.2 Jewish Trans-Zab Dialects of Iraq and Western Iran

This harmonisation between pharyngeals and pharyngealised emphatics (/t/, /ʕ/, /q/) took place also in the trans-Zab Jewish NENA dialects of Iraq and western Iran, as shown by [Hoberman \(1985\)](#). In the trans-Zab dialects in Iraq, the pharyngealisation of the oral emphatics is generally preserved, though often eliminated or considerably weakened. This has been documented in the Jewish dialects of Arbel ([Khan 1999a](#)), Sulemaniyya and Ḥalabja ([Khan 2004](#)), and Koy Sanjaq ([Mutzafi 2004a](#)).

In the Jewish Trans-Zab dialects of western Iran, such as J. Sanandaj ([Khan 2009](#)), the pharyngeal has been preserved in some words containing historical pharyngealised consonants, but the emphatic coronals /t/ and /ʕ/ have regularly lost their emphasis and have become plain consonants, e.g.,

(6) J. Sanandaj ([Khan 2009](#))

<i>tamʕa</i>	‘she tastes’	<	* <i>tāmʕā</i>
<i>taʕna</i>	‘load’	<	* <i>tāʕnā</i>
<i>ʕaqəwra</i>	‘scorpion’	<	* <i>ʕaqəbrā</i>
<i>ḥānəq</i>	‘he drowns’	<	* <i>ḥānəq</i>

In J. Sanandaj, pharyngeals are also preserved in environments where other consonants adjacent to the pharyngeal became emphaticised at some stage of development. This is found especially in the environment of the sonorant consonants /m/, /b/, /l/ and /r/, e.g.,

(7) J. Sanandaj ([Khan 2009](#))

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<i>dəm'e</i>	'tears'	<	* <i>dəṃ'e</i>	<	* <i>dəm'e</i>
<i>gul'a</i>	'kernel of fruit'	<	* <i>gul'a</i>	<	* <i>gul'ā</i>
<i>ma'lela</i>	'eve of festival'	<	* <i>ma'lela</i>	<	* <i>ma'lē 'edā</i>
<i>pərta'na</i>	'flea'	<	* <i>pərta'na</i>	<	* <i>purta'nā</i>
<i>za'ra</i>	'barley'	<	* <i>za'ra</i>	<	* <i>s'ārā</i>
<i>z'r</i>	'to plant'	<	* <i>zr'</i>	<	* <i>z-r'</i>
<i>bl'</i>	'to swallow'	<	* <i>bl'</i>	<	<i>b-l'</i>
<i>'əč'a</i>	'nine'	<	* <i>'əč'a</i>	<	* <i>tš'ā</i>

In some cases a labial or sonorant became emphaticised independently. Non-etymological pharyngeals have even developed in the environment of such emphaticised labials and sonorants. In the examples in (8) the pharyngeal has developed from a long /a/ vowel:

(8) J. Sanandaj (Khan 2009)

<i>tma'nisar</i>	'eighteen'	<	<i>tṃanisar</i>
<i>tma'ni</i>	'eighty'	<	<i>tṃani</i>

Similar examples are found in the neighbouring Jewish dialect of Suleimaniyya, e.g.,

(9) J. Sulemaniyya (Khan 2004: 35)

<i>ma'e</i>	'water'	<	<i>ṃa'e</i>	<	<i>maye</i>
<i>ma'danusi</i>	'parsley'	<	<i>ṃadanusi</i>		

In the examples in (10) the pharyngeal has developed from an original interdental *θ by strengthening of a laryngeal /h/, which developed by debuccalisation of /θ/:

(10) J. Sanandaj (Khan 2009: 23)

<i>'ahra</i>	'town'	<	* <i>'ahra</i>	<	* <i>'aθra</i>
<i>təlḥa</i>	'three'	<	* <i>təlḥa</i>	<	* <i>tlaθa</i>

7.1.3 North-Western Iran and Adjacent Areas in South-Eastern Turkey

In dialects in the Urmi region north-western Iran and in adjacent areas in south-eastern Turkey from the Txuma region eastwards pharyngealisation is not a phonological feature of individual consonant segments, as it is in the NENA dialects of Iraq and in Arabic. The historical emphatic consonant phonemes *ṭ and *š no longer exist as segments, but rather pharyngealisation has become

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detached from the feature matrix of these segments and has become a ‘floating’ suprasegmental feature. The phonological domain of emphasis, therefore, is a word rather than a segment. In the transcription this is marked by a ⁺ sign at the beginning of the word.

When the historical emphatic *ṣ lost the feature of pharyngealisation from its segmental profile, it merged on the segmental level with historical *s resulting in one phoneme /s/.

In Christian dialects of the Urmi region and adjacent areas in south-eastern Turkey, when the historical emphatic *ṭ lost the feature of pharyngealisation from its segmental profile, it retained a glottal tension that distinguished it from the historical plain *t, creating the opposition of unaspirated tense /t̚/ to lax aspirated /t/. Unlike in the opposition of emphatic /t̚/ to plain /t/ in the Iraqi dialects, the two stops /t̚/ and /t/ in C. Urmi and adjacent Christian dialects were distinguished primarily by their different glottal settings. On the phonetic level the first segment in a word such as C. Urmi *ṭamər ‘he buries’ would sound similar to the corresponding emphatic segment in its historical antecedent *ṭamər. On a phonological level, however, there has been a reanalysis, which can be represented thus:

(11) C. Urmi (Khan 2016a: vol. 1, 108)

Word level:		>	/ʕ/ +Pharyngeal
Segment level:	/t̚/ +Pharyngeal +Tense		/t̚/ +Tense

As a result of the reanalysis of segmental emphasis and suprasegmental emphasis, a ternary system of stops emerged:

Voiceless aspirated	Voiceless unaspirated	Voiced
/t/	/t̚/	/d/

Not all occurrences of the segment /t̚/ in emphatic words in the synchronic state of dialects of this group derive historically from *ṭ. In emphatic words /t̚/ sometimes derives from a coalescing of *d and a laryngeal *, e.g.,

(12) C. Urmi (Khan 2016a: vol. 1, 112)

*yatt̚a	‘she knows’	<	*yadʔa	<	*yadʕa
*ṭata	‘sweat’	<	*dʔata	<	*dʕata

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As can be seen, in both cases in (12) the laryngeal is ultimately derived from a pharyngeal.

Furthermore, /t/ occurs in a few plain words in C. Urmi that, at some stage of development, appear to have contained an emphatic *t. All available examples are nouns denoting short thin objects and verbs denoting shortness or sounds that are short and/or shrill with high pitch, e.g.,

(13) C. Urmi (Khan 2016a: vol. 1, 112–113)

- a. *ṭanṭunə* ‘to sing in a whisper to oneself’ (cf. **ṭanṭunə* ‘to sing in a low tone, to buzz.’)
- b. *ṭupra* ‘tail’; cf. C. Qaraqosh *bar-ṭupra* ‘strap for securing saddle tied around the back of an animal’ (literally: ‘behind the tail’), in which the first segment of *ṭupra* is emphatic.
- c. *ṭuvvulta* ‘worm’; cf. C. Qaraqosh *ṭolə’θa*, C. Alqosh *ṭoleθa*, in which the first segment is emphatic.

The emphasis has been eliminated from these words, through a process of sound symbolism whereby there is an iconic association between the higher pitch of plain sounds and the thinness and lightness of objects and sounds.

Another manifestation of sound symbolism is the existence of pairs of terms in which the term denoting largeness is emphatic and the term denoting smallness is plain, e.g.,

(14) C. Urmi

Emphatic		Plain	
* <i>jura</i>	‘big’	<i>sura</i>	‘small’
* <i>yarixa</i>	‘long’	<i>ḵərya</i>	‘short’
* <i>raba</i>	‘much’	<i>xačča</i>	‘small amount’

Although the emphasis has been removed, the glottal tension of the originally emphatic *t has been retained. This is evidence that in C. Urmi the feature of emphasis is on a level that is independent of that of the segment and that at the segment level the segment is a tense /t/ unspecified for emphasis. Emphasis is word-level pharyngealisation, independent of the laryngeal settings of segments. It occurs not only in words with unaspirated segments, e.g., **ṭala* ‘she plays’, **ṭamra* ‘she buries’, but also in those with aspirated segments, e.g., **tala* ‘fox’, **tama* ‘there’. As can be seen, there are oppositions between aspirated and unaspirated segments within emphatic words, e.g.,

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(15) C. Urmi (Khan 2016a: vol., 100)

⁺*tala* ‘fox’ : ⁺*ṭala* ‘she plays’

The distribution of suprasegmental emphasis in the lexicon is different across the various Christian dialects of the region. In the C. Mawana dialect of Tergawar in the mountains to the west of the Urmi plain, for example, there are several emphatic words that are plain in C. Urmi and vice versa, e.g.,

(16) C. Mawana (Khan, fieldnotes)

	C. Mawana	C. Urmi
‘village’	⁺ <i>mata</i>	<i>mata</i>
‘tree’	⁺ <i>’ilana</i>	<i>’ilana</i>
‘gum’	<i>data</i>	⁺ <i>data</i>
‘lamp’	<i>šraya</i>	⁺ <i>šraya</i>

Also, in the J. Urmi dialect and other Jewish dialects in the Urmi region, emphasis has become a suprasegmental word-level feature, but, unlike Christian dialects such as C. Urmi, there are no tense stops in their phoneme inventory, so when emphasis was removed from the historical emphatic stop segment ⁺*t*, it became an aspirated stop:

(17)

	C. Urmi	J. Urmi
⁺ <i>ṭina</i> ‘mud’	⁺ <i>ṭina</i> [t ^ʰ i:n ^ʰ ɑ ^ʰ]	⁺ <i>tina</i> [t ^h ʰi:n ^ʰ ɑ ^ʰ]

In the J. Salamas dialect the suprasegmental pharyngealisation feature has been lost in all words that originally used to contain it. The phonetic affect of this feature on segments has been preserved, however, in one particular case, namely that of pharyngealised long /a/ vowels, which were backed and rounded to the quality /o/, e.g., the penultimate vowels in the following, which were pronounced long:

(18)

	J. Salamas	J. Urmi
‘very’	<i>róba</i>	⁺ <i>rába</i>
‘back’	<i>xosá</i>	⁺ <i>xasá</i>

The /o/ phoneme in these examples has the segmental features of [+back] and [+round]. These qualities can be considered to have been originally phonetic consequences of the suprasegmental phonological [+pharyngeal] feature but have now been reanalysed as phonological features of the segment. This has occurred

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only in vowels that are a stable feature of words and that do not vary in different morphological inflections. It occurs in nominal lexical bases, but not in word-final inflectional endings of nouns:

(19)			Lexical base	Inflectional ending
	'back'	<i>xosa</i>	<i>xos-</i>	<i>a</i> (nominal singular)
	'player'	<i>tyolana</i>	<i>tyol-</i>	<i>ana</i> (agentive singular)

It does not occur in some /a/ vowels that form part of word-internal inflectional patterns of verbal roots, e.g., forms of the present base:

(20)	J. Salamas	J. Urmi	
	<i>palət</i>	⁺ <i>palət</i>	'he goes out'
	<i>paləx</i>	<i>paləx</i>	'he opens'

This applies to inflections of a paradigm in which the length of the /a/ vowel varies. In the 3sg.m present form *palət* 'he goes out', for example, the /a/ vowel in the open syllable is long, [pa:'lɪt], but elsewhere in the paradigm it occurs in a closed syllable and is short, e.g.,

(21)	3sg.m	<i>palət</i>	[pa:'lɪt]	'he goes out'
	3sg.f	<i>palta</i>	[pal'ta]	'she goes out'
	3pl	<i>palti</i>	[pal'ti:]	'they go out', etc.
	2sg.m	<i>paltet</i>	[pal'te:t]	
	2sg.f	<i>paltat</i>	[pal'tat]	
	2pl	<i>paltetun</i>	[pal'te:tun]	
	1sg.m	<i>palten</i>	[pal'te:n]	
	1sg.f	<i>paltan</i>	[pal'tan]	
	1pl	<i>paltex</i>	[pal'te:x]	

When a verbal paradigm has an invariably long *a vowel, in an open syllable, in a verb that was originally emphatic, the vowel is realised as /o/ as in nominal lexical roots, e.g., in the present progressive forms:

(22)	J. Salamas	J. Urmi	
	3sg.m	<i>tyolé</i>	⁺ <i>tyalé</i> ‘he is playing’
	3sg.f	<i>tyolá</i>	⁺ <i>tyalá</i> ‘she is playing’
	3pl	<i>tyolú</i>	⁺ <i>tyalu</i> ‘they are playing’

7 Emphatic (Pharyngealised) Consonants

7.2 Emphatic Consonants in Arabic

The Old Arabic emphatic (velarised or pharyngealised) consonants are preserved as emphatic consonants in the Qəltu dialects generally (Jastrow 1978: 48–49) and thus also in the NENA-speaking region, e.g., Mosul (Procházka 2019: 247) and Jewish Arbīl and ‘Aqra (Jastrow 1990a: 26). The near-universal merger of Old Arabic *ǧ and *ḍ that occurred in Arabic dialects (Versteegh 2011: 544) resulted apparently in *ǧ in the ancestors of Qəltu Arabic, although this has since undergone various shifts in different dialects (Jastrow 1978: 34). The other two historical emphatics that are preserved are *š and *ṭ (Jastrow 1978: 48).

Emphasis is suprasegmental in Qəltu as in other modern Arabic dialects, covering at least the following vowel but usually a whole word (Jastrow 1978: 47–52; Davis 2011: 637). Whole-word emphasis was demonstrated for the Mḥallami Qəltu dialect of south-eastern Turkey by Sasse (transcribed by a prefixed + symbol),¹ e.g.,

(23) Mḥallami Arabic (Sasse 1971: 32, 56)

+ära	‘he saw’
+bäwašəq	‘falcons’

Sasse (1971: 35) notes that the distribution of suprasegmental emphasis across the lexicon differs in the various villages that speak the dialect.

In some cases new emphatic consonants have acquired a marginal phonemic status, most commonly these are /r/ and /l/, but sometimes also /b/, /m/, /w/, /n/ and /z/ (Jastrow 1978: 49–51; 1990a: 26). Emphatic consonants, some of them new, have also been introduced into Qəltu dialects via Turkish and Kurdish loanwords (Jastrow 1978: 51).

7.3 Emphatic Consonants in Kurdish

Several pharyngealised emphatic consonants such as /ṣ/, /ẓ/ and /ṭ/ [ṣ, ẓ, ṭ] exist in Kurdish along with the marginally pharyngealised pronunciations of /ḍ/ [ḍ] and /p̣/ [p̣].

Northern Kurdish makes widespread usage of /ṣ/, /ẓ/ and /ṭ/, both in Arabic borrowings and native words:

¹Whole-word emphasis is the usual case, but emphasis is sometimes restricted to only part of a word, whether one or more syllables (Sasse 1971: 35).

7.3 *Emphatic Consonants in Kurdish*

- (24) *şaf* ‘naïve’
 zəlîm ‘oppressor’
 şed ‘hundred’
 pez ‘sheep’
 teyr ‘bird’
 tarî ‘dark’

The Shamdinan dialect has pharyngeal pronunciations of /d/ as well, as in:

- (25) *damanê* ‘checkers’
 dewat ‘wedding’

Most of Bahdinan and Hakkari dialects have pharyngealised /p/, albeit in a very restricted set of words:

- (26) *pez* ‘sheep’
 pe!û ‘paste’
 pi!epi! ‘nagging’

Note that the phoneme is accompanied by other emphatic (i.e., pharyngealised) consonants in the word, indicating that the pharyngealisation may indeed be applying to the whole word as a suprasegmental phenomenon.

Similarly, the dialects of the Hakkari and the region north of Urmi have a pharyngealised pronunciation of the word for ‘cheese’ (noted to be suprasegmental in Kahn 1976):

- (27) *penîr* ‘cheese’

Within NK dialects, there is substantial variation in the assignment of emphatic pronunciation to specific words. Thus, Shamdinan, and most Bahdinan dialects, have the words *beş* ‘enough’, *mezin* ‘big’ *ţam* ‘taste’, while in the more western dialects (e.g., Diyarbakir) these have plain consonants, viz., *bes*, *mezin* and *te‘m*.

In general, within NK, emphatics are most widespread in the southern and south-eastern dialects of NK (southern Kurmanji dialects around Mardin-Sinjar, Behdînî dialects around Duhok-Zakho, and dialects of the Hakkari-Shamdinan-Urmi area) (Blau 1989: 329). This tendency is weakened northwards in dialects of Erzurum-Kars, while MacKenzie (1961a: 43) notes that they are altogether absent in the dialect of Yerevan.

This distribution is of course consistent with a language-contact scenario, in the sense that in the northern dialects away from Semitic influence the language either did not develop emphatics or lost them as a result of contact with and

7 Emphatic (Pharyngealised) Consonants

bilingualism in Armenian, Turkic and Caucasian languages, which do not possess such emphatics. This is also consistent with the fact that the nomadic Harki tribe of the areas of Akre-Rawanduz-Shamdinan make very limited use of the emphatics, if at all, indicating that due to their life style and disconnection from settled populations their dialect did not develop emphatics.

Central Kurdish and Gorani make very limited usage of the two consonants: /ʒ/, /z/, /t/, though the latter is only found in the Sanandaj dialect and not in the Erbil and Suleimaniya dialects:

- (28) *şeg* ‘dog’ (CK)
 şet ‘hundred’ (CK, Gorani)
 ţal ‘bitter’ (Sanandaj dialect)
 mizl ‘house’ (CK dialect of Mahabad)

In addition to language contact, it is also possible that language-internal developments have played a role in the emergence of emphatics in Kurdish. Of the emphatic obstruents, the fricative pair (/ʒ/, /z/) are found both in Northern and Central Kurdish (though less often in the latter), while the stops (/t/, /d/) are found only in Northern Kurdish, with the voiced counterpart being extremely rare. The fact that the voiceless emphatic stop is widespread only in Northern Kurdish most probably has to do with the presence of unaspirated (in addition to aspirated) voiceless stops in the language. The unaspirated stops are probably intermediary in the development of emphatics. This is further reinforced by the fact that in Northern Kurdish the bilabial voiceless stop /p/ also has an emphatic version, as in the native words *peʒ* ‘sheep’ and *penîr* ‘cheese’ (in some dialects; cf. Kahn (1976: 27)).

7.4 Armenian

The phenomenon in Armenian that seems to correspond most closely to the pharyngealisation found in NENA, and creates the conditions for the development of vowel harmony in some Armenian dialects, is Acharyan’s law (Acharyan 1952), by which back vowels are fronted following a voiced obstruent (/a/ > /æ/, /o/ > /ø/, /u/ > /y/):

Vaux (1998: 176) proposes that this results from a rule spreading [+ATR] (advanced tongue root) from voiced obstruents to following vowels.

As for the chronology and geographic spread of this change, we know that Acharyan’s Law preceded the process of devoicing of voiced stops and affricates

Table 7.1: Acharyan’s law (see Martirosyan 2019: 81-82)

ClArm/SEA	Van	Khoy/Urmia	Karabagh	Krzen (no devoicing)
<i>das</i> ‘order’	<i>tæs</i>	<i>tæs</i>	<i>tæs/tes</i>	<i>dæs</i>
<i>tas(n)</i> ‘ten’	<i>tas</i>	<i>tas</i>	<i>tasə</i>	<i>tasə</i>

in initial position found in Van, Khoy/Urmia, Karabagh etc. (for details the distribution of this phenomenon, see §4), which is considered to have taken place around the 7th century AD. Acharyan (1952) dated his law between the 7th and 11th centuries, based on the behaviour of loanwords from Arabic (affected) and loanwords from Turkish (not affected).

Together with other isoglosses identified by Weitenberg (1996) and Weitenberg (1999), Acharyan’s law seems to separate a SE area (Van, Khoy/Urmia, Agulis-Meghri, Karabagh) from the central and western areas, a division that cuts across the main East-West boundary, and quite possibly precedes it. It corresponds quite closely to the border established in 591 AD after the Byzantine Emperor Maurice’s defeat of the Sasanian Persians, shown in Figure 7.1.



Figure 7.1: Armenian lands around 387 and 591 AD. Beleg, Wikimedia Commons https://commons.wikimedia.org/wiki/File:Armenian_lands_387_591.png.

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As discussed by [Vaux \(1998\)](#), [Hopkins & Vaux \(2024\)](#), and others, in many of the dialects affected by Acharyan’s Law, which creates new front vowels [æ ø y] and thus an opposition between [a o u] and [æ ø y], systems of vowel harmony have developed, whereby the vowels in a given root share the same value for [back]. It can be seen how the [-back] quality acquired due to Acharyan’s Law by the vowel that originally followed a voiced obstruent spreads to the other vowels of the root in the following examples from Salmast given by [Hopkins & Vaux \(2024: 6\)](#):

	CA	Acharyan’s law	(Devoicing)	Vowel harmony
(29)	<i>barak</i> ‘thin’	> <i>bærak</i>	(> <i>pærak</i>)	> <i>pæræk</i>
	<i>bazuk</i> ‘beetroot’	> <i>bæzuk</i>	(> <i>pæzuk</i>)	> <i>pæzyk</i>

According to [Hopkins & Vaux \(2024: 7\)](#), root harmony appears in seven of the major groups of Armenian dialects identified by [Jahukyan \(1972\)](#): the extreme south-western group (Kesab, Aramo), the Cilician group (Marash, Zeytun), the southern-central group (Tigranakert (Diyarbakir)), the southern intergroup (Van, Moks, Ozmi, Shatakh), the south-eastern group (Khoy, Maragha, Urmia, Salmast), the extreme north-eastern group (Burdur, Karabagh, Shamakhi, Goris) and the Agulis/Meghri group.

In some dialects, there is harmony for the feature [round] as well as [back], especially for epenthetic vowels, and in some, affixes as well as roots are affected, so that the domain of harmony can be described as the prosodic word. The following Table 7.2 is presented in [Hopkins & Vaux \(2024: 9\)](#). We have added the information that backness and roundness harmony for epenthetic vowels is also found in Hadrut ([Poghosyan 1965](#)), and in the Agulis subdialects of Tanakert, Tsghna, Ramis, and Paraka ([Hodgson 2022a](#)).

7.5 **Emphatic consonants in Turkic**

[Bulut \(2014: 33\)](#) provides examples of words containing pharyngealised consonants beyond Arabic loanwords attested in contact varieties, such as Iraqi Turkic and Sonqor Turkic. Specifically, a limited set of non-Semitic vocabulary in the Bayat dialect of Hamadan—a variety not in contact with Arabic—features such consonants, e.g., [dʰɑʃ] ‘stone’, [dʰuz] ‘salt’ (cf. *daş* [daʃ], *duz* [duz] in most other varieties respectively), and [sʰɑːvɛ] ‘(the town of) Sâveh’. In the case of [dʰ], the author supports Doerfer’s (1975–1976: 120–121) explanation that [dʰ] may represent an intermediate stage in the transition of an older word-initial */t/ to /d/, thus reflecting an internal development.

7.5 Emphatic consonants in Turkic

Vowel harmony is a well-known trait of most Turkic languages. Standard Azeri prescribes rules of vowel harmony that do not show any exceptions. These rules create a predictable pattern whereby vowels within a word assimilate to one another. Two primary types exist: frontness-backness harmony and roundedness harmony.

The frontness-backness harmony rule categorizes vowels in Standard Azeri into two distinct groups based on tongue position: front vowels (*e* [e], *ə* [æ], *i* [i], *ö* [ø], *ü* [y]) and back vowels (*a* [a], *ı* [w], *o* [o], *u* [u]).

The roundedness (or labial) harmony rule refines the vowel category by distinguishing between unrounded front (*e* [e], *ə* [æ], *i* [i]), unrounded back (*a* [a], *ı* [w]), rounded front (*ö* [ø], *ü* [y]) and rounded back (*o* [o], *u* [u]) vowels.

This section will only address vowel harmony in the context of agglutination, excluding root-internal harmony, which ceased to be rigorous in Turkic at an early stage.

Suffixes containing vowels in Standard Azeri belong to two types based on their allomorphic variation: those showing a two-way open-vowel variation *a* vs. *ə* and those showing the four-way close-vowel variation *i* vs. *ı* vs. *ü* vs. *u*. Mid vowels (*e*, *o*, *ö*) never occur in suffixes in Standard Azeri. In addition, suffixes containing a dorsal obstruent display further variation: a back-vowel nucleus triggers a uvular allomorphic realisation whereas a front-vowel nucleus combines with a palatal one, e.g., *qul* [gul] ‘slave’ > *qul-luq* [gul:uχ] ‘service’ vs. *küt* [kyt] ‘blunt’ > *küt-lük* [kyt:yç] ‘dullness’ (with progressive assimilation of *l*).

Following conventional Turkic linguistic notation, capital *A* represents an open vowel that alternates between *a* and *ə*, while capital *X* represents an archiphonemic close vowel that surfaces as *i*, *ı*, *ü*, or *u*. If the suffix is attested only with non-labial realisations, the vowel is noted as *I*; if only with labial ones, it is noted as *U*. A suffixal dorsal obstruent is noted as *Q*.

Few, if any, Azeri dialects exhibit flawless vowel harmony, both due to inherited phenomena (some suffixes belonging to the *X*-type in the standard norm historically exhibited only rounded or only unrounded harmony; in addition, Old Anatolian Turkic, the direct ancestor of Azeri, passed through an “indifference stage” of suffix vocalism (see [Johanson 1981](#)) and as a result of internal development and innovations (including substrate influence).

Urmia Azeri shows considerable vowel disharmony in suffixes.

One such instance, typical of a much larger area covering Eastern Anatolia, Iraq, much of Iranian Azerbaijan, and some adjacent (and even more remote) areas in the Caucasus, involves weak or no frontness-backness harmony in suffixes containing a dorsal consonant. With *A*-type suffixes, speakers use only the back variant, which can only yield uvular consonant allomorphy, e.g., Urmia

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Azeri [oχu-maχ] ‘to read’ vs. [dzij-maχ] ‘to put on’ (cf. Standard Azeri [oχu-maχ] vs. [ʃej-mæç]). With *X*-type suffixes, vowel harmony is more consistent (though when it does not occur, the back variant is preferred). However, the dorsal consonant often manifests itself as a uvular regardless of the vowel, Urmia Azeri [baχur-uχ] ‘we are watching’, [verir-iχ] ‘we are giving’, [dʒedir-uχ] ‘we are going’, [dyfyr-uχ] ‘we are falling’ (cf. Standard Azeri [baχur-uχ], [verir-iç], [ʃedir-iç], [dyfyr-yç], respectively; Doğan 2010: 33–34, 160).

Urmia Azeri, like many Eastern Anatolian, Iraqi and most Eastern Azeri varieties, shows some restriction on word-final rounded vowels. This restriction sometimes leads to the accusative suffix *-l*, the third-person possessive suffix *-(s)l* and other high-vowel-final suffixes to display only unrounded variants (Doğan 2010: 39–40), e.g., [syɬ] ‘milk’ > [syɬ-i] ‘milk-ACC’ or ‘milk-POSS.3’ (cf. Standard Azeri [syd-y]), which can also be fronted, e.g., [jɔɬ] ‘road’ > [jɔɬ-i] ‘road-ACC’ or ‘road-POSS.3’ (cf. Standard Azeri [jɔɬ-u]).

Urmia Azeri, or some of its sub-varieties, has inherited from an earlier stage of Oghuz development the unrounded-only vocalism of the perfect suffix *-mɪʃ* and the rounded vocalism of the jussive suffix *-sʊn*. In addition, as in Eastern Azeri varieties, the develarisation of historical [ŋ] often results in the rounding of the adjacent vowel (Doğan 2010: 40), e.g., [sæn-yn] ‘yours’, [dʒed-yn] ‘go! (PL)’ (cf. North-western Azeri [sæn-in], [ged-in]).

7.6 Discussion

The inherited emphatic (pharyngealised) consonant segments of Aramaic have been retained in NENA dialects in Iraq and the north-western sector in Turkey, together with innovative emphatic sonorants, labials and rhotics. A similar range of emphatic consonants are found in adjacent Arabic dialects. Emphatic consonants have been acquired by many Kurdish dialects through contact with Semitic. In all of these languages pharyngealisation of emphatic consonants spreads beyond the segment to the adjacent syllable and, in some case, to the entire word. In some Arabic and Kurdish varieties it is possible to analyse emphasis as a word-level suprasegmental phenomenon rather than spreading from an emphatic consonant segment, though this does not seem to apply systematically to all words with pharyngealisation. This suprasegmental type of emphasis has become a systematic phonological feature of NENA dialects in the north-eastern sector of the dialect area. The roots of this appear to be in the phonetic spread of pharyngealisation from emphatic consonant that is attested elsewhere in NENA. Its development into a suprasegmental feature is likely to have been facilitated by several factors.

7.6 Discussion

One factor is the emergence of pharyngealisation that could be analysed as suprasegmental in Kurdish dialects in the region (the attested Arabic dialects with suprasegmental emphasis are not in this region).

Another factor could be contact with Armenian in the region, which, as shown above (§7.4), exhibit word-level harmony of front vowels (+ATR) resulting from Acharyan's law. In NENA dialects in the north-eastern sector in words that do not have suprasegmental pharyngealisation, it is possible that the lack of pharyngealisation has come to be analysed as +ATR, a word-level harmonic feature matching this phenomenon in Armenian. This may have facilitated the analysis of pharyngealisation as a word-level harmonic feature.

Another factor may have been the lack of pharyngeal consonant segments in the NENA dialects of the north-eastern sector. In Kurdish dialects of this region, non-etymological pharyngeal segments develop from the suprasegmental 'flatness' of a word manifested in a back vowel, e.g., *çav* > *çeʿv* 'eye' (§37), and in some cases these alternate in the same dialect (Kahn 1976: 49-52). The loss of a pharyngeal in NENA dialects of this region often leaves a legacy of pharyngealisation, e.g., C. Urmi *jam'a* > *ʔamma* 'she gathers'. The development of this into a word-level suprasegmental feature could have been facilitated by matching of the relation between pharyngeal segments and word-level flat resonance.

A further factor may have been the development of unaspirated stop phonemes in NENA dialects in the north-eastern sector by convergence with the sound system of Armenian (§4.5). In the case of the emphatic stop /t̤/, this facilitated the disengagement of the pharyngeal feature from the the emphatic segment and its reanalysis as a floating word-level feature. This is because the erstwhile emphatic segment was reanalysed as an unaspirated stop and its surface phonetic realisation in a word with suprasegmental pharyngealisation did not change.

Prima facie, one may think that Turkic vowel harmony played a role in the development of word-level harmony in NENA dialects of the north-eastern sector. It should be taken into account, however, that in the varieties of Azeri Turkic in the region vowel harmony has broken down (§7.5). Also, the phenomenon of suprasegmental word-level emphasis extends geographically to NENA dialects of the Hakkari region in south-eastern Turkey outside the area where NENA was in contact with Turkic. Finally, the nature of the harmonic process in NENA is different from that of Turkic. In the Turkic system only the vowels are affected, with the exception of the velar consonants, which are retracted adjacent to back vowels (Vaux 2009). In C. Urmi there is only harmony of backness of vowels, not of roundness, since, unlike in Turkic, words can contain high rounded and unrounded vowels. Moreover, in Turkic, all vowels participate in the front-back

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harmony system, whereas in C. Urmi this is, in effect, restricted to long and short /a/ and to the short central vowel /ə/. This is more characteristic of the Iraqi NENA dialects and Arabic, in which emphasis spread does not affect or is blocked by high vowels, especially long /i/. In some village dialects of C. Urmi, e.g., Sardarid (Younansardaroud 2001), a clear front-back distinction is extended to more vowels, notably to the /u/ and /o/ vowels, which are noticeably fronted in plain words. This would represent a greater convergence with Turkic. The Jewish Urmi dialect also exhibits the distinct fronting of /u/ and /o/ in plain words, but high front vowels are not backed in emphatic words as they are in Turkic (Khan 2008b: 23). All this suggests that contact with Turkic may not have played such a central role in inducing the phenomenon of suprasegmental pharyngealisation in NENA dialects of the north-eastern NENA sector.

One characteristic of suprasegmental pharyngealisation in NENA dialects of the north-eastern sector is the fact that its distribution across the lexicon differs across different dialects. This can be regarded as a reflection of the fact that it is not bound to inherited emphatic consonants. A similar phenomenon of cross-dialectal variation across the lexicon has been documented by Sasse for the Mḥal-lami Arabic dialects (Sasse 1971: 35).

The development of pharyngealisation into a pharyngeal segment, in some cases a non-etymological pharyngeal, in Jewish Trans-Zab NENA dialects (examples 8, 10, 9) has a parallel in the development of a pharyngeal segment in Kurdish from the suprasegmental 'flatness' of a word, e.g., *çav* > *çeʿv* 'eye'. A feature of this development in these NENA dialects is the transfer of all pharyngealisation and associated acoustic flatness in the word to the pharyngeal segment, so that emphatic consonants become plain, e.g., **tmanisar* > *tmaʿnisar* 'eighteen' (8). This matches the development of pharyngeals by this process in Kurdish, in which no more than one pharyngeal or pharyngealised segment is tolerated in a word. So, when Arabic loanwords are integrated into the Kurdish dialects, pharyngealisation is dropped for all but one segment, e.g., Iraqi Arabic *ṣuḥba(t)* 'conversation' > Kurdish *sibhet* **šibḥat* (*ṣ* has become plain), Iraqi Arabic *qūṭiyya* 'box' > Kurdish *qotî* **qoṭî* (*t* has become plain) (Kahn 1976: 44). According to Kahn's explanatory model, this is due to the fact a word can contain only one underlying pharyngeal segment. Developments such as *çav* > *çeʿv* 'eye' in Kurdish are attested in dialects outside the area of Jewish Trans-Zab NENA. One may say, however, that the association between pharyngeal segments and flat resonance is a general feature across Kurdish dialects (Barry 2019) and so the Trans-Zab NENA dialects reflect a convergence with Kurdish in this respect.

7.6 Discussion

Table 7.2: A typology of vowel harmony in Armenian (adapted from Hopkins & Vaux 2024: 9)

Harmonic domain	Harmonic feature	
	back	round
Root	Agulis, Aresh, Goris, Karabagh, Karchevan, Khoy, Maragha, (Marash?), Meghri, Moks, Ozmi, Salmast, Shamakhi, Tigranakert, Urmia, Van, Zeytun	(Marash?)
Affix	Agulis, Aresh, Goris, Karchevan, Kesab, Khoy, Maragha, Meghri, Shamakhi, Tigranakert, Zeytun	Maragha
Epenthetic vowel	Agulis (some speakers), Hadrut, Goris, Karchevan, Marash, Maragha, Paraka, Ramis, Shamakhi, Tanakert, Tsgghna, Zeytun	Hadrut, Goris (partial), Karchevan, Marash, Maragha, Paraka, Ramis, Shamakhi, Tanakert, Tsgghna, Zeytun

8 Uvular stops

8.1 Uvular stop in NENA

8.1.1 Western and Southern Sectors of NENA

In the majority of NENA dialects, the reflex of **q* is an voiceless unaspirated uvular stop /q/.

In Christian dialects in the western and southern sectors of NENA, in Jewish *Lišana Deni* dialects and in Jewish Trans-Zab dialects in Iraq and western Iran, /q/ belongs phonologically to the emphatic (pharyngealised) consonants, i.e., the emphatics of earlier Aramaic /š/, /t/, and pharyngealised labials and sonorants such as /m/, /b/, /r/. In dialects of these sectors, the presence of /q/ or one of the emphatic consonants in a word containing a historical pharyngeal **ʕ* or **ħ* preserves the pharyngeal (Hoberman 1985). In other contexts these historical pharyngeals typically lose their pharyngeal feature, resulting in the shifts **ʕ* > /ʔ/, Ø and **ħ* > x.

- (1) C. Qaraqosh (Khan 2002a: 40–41)

<i>ʕpša</i>	‘gallnut’
<i>ʕaʕn</i>	‘he lifts’
<i>ʕmoqa</i>	‘deep’
<i>ʕtiqa</i>	‘ancient’
<i>raħoqa</i>	‘distant’
<i>ħaziqa</i>	‘strong’

- (2) C. Harbole (Khan, fieldnotes)

<i>raħuqa</i>	‘distant’
<i>ʕmiquta</i>	‘depth’
<i>ʕiqa</i>	‘narrow’

- (3) C. Shaqlawa, C. Koy Sanjaq/Armota, C. Sulemaniyya (Khan, fieldnotes)

<i>ħaruqa</i>	‘distant’ < <i>raħuqa</i>
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8 Uvular stops

- (4) C. Koy Sanjaq/Armota (Khan, fieldnotes)

ḥamuqa ‘deep’ < ‘*amuqa*’

- (5) Jewish
- Lišana Deni*
- (Khan 2016b: vol. 1, 115; Molin 2024: 25)

raḥuqa ‘distant’

‘*amuqa*’ ‘deep’

‘*apša*’ ‘gallnut’

nḥaqa ‘to touch’

This reflects the fact that /q/ in these dialects belongs to the pharyngealised emphatic series of consonants. The /q/ typically does not induce a suprasegmental spread of pharyngealisation like other emphatic consonants. The pharyngeal segments /ʕ/ and /ħ/ have apparently been preserved, therefore, by a process of harmonising the pharyngeal consonant with the pharyngealised emphatic consonants, without necessarily affecting other intervening segments. This is what Rose & Walker (2004) term ‘long distance consonant agreement’, which operates in many languages on consonants that share a considerable degree of similarity but skips intervening, less similar segments.

Pharyngeal consonants have been preserved by this process even where the pharyngealisation of the emphatic consonants /ʕ/, /ħ/ and of the emphatic labials and sonorants was subsequently weakened. This is the case in Jewish Trans-Zab dialects of Iraq up to the border with Iran (including the border locations of Rustaqa and Sulemaniyya) and in those of western Iran. In such dialects the /q/ retains its uvular articulation, e.g.,

- (6) J. Koy Sanjaq (Mutzafi 2004a: 28–33)

ḥānəq ‘he chokes’

‘*arəq*’ ‘he flees’

‘*aqla*’ ‘leg’

‘*ātuqa*’ ‘old’

təʕna (< **ṭaʕna*) ‘load’

səwəʕ (< **ṣāḇeʕ*) ‘he dyes’

- (7) J. Sanandaj (Khan 2009: 28)

‘*aqərwa*’ ‘scorpion’

taʕna (< **ṭaʕna*) ‘load’

tamʕa (< **ṭamʕa*) ‘she tastes’

8.1 Uvular stop in NENA

In J. Kerend, in western Iran lying south of Sanandaj, pharyngeals have been preserved in words that historically contained pharyngealised consonants but do not preserve pharyngeals in words with /q/:

- (8) J. Kerend (Khan, fieldnotes)

<i>'aqərwa</i> (< * <i>'aqərwa</i>)	'scorpion'
<i>'aq̣la</i> (< * <i>'aq̣la</i>)	'leg'
<i>bālə</i> (< * <i>ḅalə</i>)	'he swallows'
<i>dəm'e</i> (< <i>dəm'e</i>)	'tears'
<i>lā'el</i> (< <i>lā'el</i>)	'above'

In NENA dialects in which the presence of /q/ preserves original pharyngeals, it may analogically induce non-etymological pharyngeals, e.g.,

- (9) C. Shaqlawa (Khan, fieldnotes)

qaḥrana 'sad' < Arabic *q-h-r*

- (10) J. Koy Sanjaq (Mutzafi 2004a: 28–33)

<i>ḥāləq</i>	'he locks'	< Arabic <i>ḡ-l-q</i>
<i>'āsəq</i>	'he ascends'	< * <i>'asəq</i>

Likewise, a /q/ may be innovated from an original **k* when in contact with a pharyngeal, e.g.,

- (11) J. Sulemaniyya (Khan 2004: 34)

ḥāqe 'he speaks' < Arabic *ḥ-k-y*

In some Jewish Trans-Zab dialects, /q/ is occasionally lenited to an uvular fricative after vowels or /w/, e.g.,

- (12) J. Arbel (Khan 1999a: 26)

<i>baqu</i> [ba:χu:]	'to them'
<i>la qabliwa</i> [læ χæb'li:wa]	'the did not accept'

- (13) J. Sanandaj (Khan 2009: 21)

šəwqa=y [ʃɪfχaj] 'he has left'

In J. Urmi, when occurring between vowels, between a vowel and /y/ or between a sonant and a vowel, /q/ is voiced and is pronounced either as a stop or a fricative, e.g.,

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- (14) J. Urmi (Khan 2008b: 20)

maqé [mΛ'geː ~ mΛ'ʁeː] 'he will speak'
mqí ['mgiː ~ 'mʁiː] 'speak!'

8.1.2 Hakkari and north-western Iran

In NENA of the Hakkari region, including dialects of adjacent areas in the north of Iraq, and in dialects of north-western Iran the pharyngeal consonants *ʕ and *ħ have been lost in native Aramaic words. The presence /q/ does not preserve the pharyngeals:

- (15) C. Čāl (Talay 2008b: 71)

'amuquθa 'depth'
'aqərwa 'scorpion'

- (16) C. Səra (Talay 2008b: 71)

'umqa 'depth'
'iqarwa 'scorpion'

- (17) C. Qočanəş (Talay 2008b: 71)

'umqa 'depth'
'eqarwa 'scorpion'

- (18) C. Sat (Khan, fieldnotes)

'emuqa 'deep'
'etiqa 'old'

- (19) C. Zariwaw (Napiorkowska 2015: 18)

raxqa 'distant'
'amuqa 'deep'

- (20) C. Bədyal (Khan, fieldnotes)

rəqqa 'distant'
yāmuqa 'deep'

According to Jacobi (1973) many of the words containing /q/ that also had a historical pharyngeal are pronounced with suprasegmental pharyngealisation in C. Txuma:

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- (21) C. Txuma (Jacobi 1973)

⁺ <i>ʼumqa</i>	‘deep’
⁺ <i>rəxqa</i>	‘distant’
⁺ <i>ʼaqla</i>	‘leg’
<i>ʼaqubra</i>	‘mouse’

In the dialect of C. Barwar, in northern Iraq on the southern periphery of the Hakkari region, pharyngeals are, in principle, lost in native Aramaic words, but a relic of pharyngeal /ħ/ is retained as a variant realisation in the word ‘distant’:

- (22) C. Barwar (Khan 2008a: 53-54)

<i>ʼaqərwa</i>	‘scorpion’
<i>ʼaqla</i>	‘leg’
<i>ʼamuqa</i>	‘deep’
<i>rəħqa ~ rəxqa</i>	‘distant’

In the Jewish Trans-Zab dialects J. Urmi and J. Salamas, in north-western Iran, pharyngeal consonants have not been preserved in any context. In J. Urmi, the word ‘distant’ (**rəħqa*) containing /q/ and a historical pharyngeal is pronounced with suprasegmental pharyngealisation:

- (23) J. Urmi (Khan 2008b: 33)

<i>⁺rəqqa</i> (< <i>*rəħqa</i>)	‘distant’
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Most words containing /q/ with a historical pharyngeal are pronounced plain, without suprasegmental pharyngealisation:

- (24) J. Urmi (Khan 2008b: 34)

<i>aqla</i> (< <i>*ʼaqla</i>)	‘leg’
<i>aqubra</i> (< <i>*ʼaqubra</i>)	‘mouse’
<i>aqərwa</i> (< <i>*ʼaqərwa</i>)	‘scorpion’

In the J. Salamas dialect, words containing /q/ and a historical pharyngeal are all pronounced plain:

- (25) J. Salamas (Khan, fieldnotes)

<i>rəqqa</i> (< <i>*rəħqa</i>)	‘distant’
<i>aqla</i> (< <i>*ʼaqla</i>)	‘leg’
<i>əqra</i> (< <i>*ʼəqra</i>)	‘bottom’

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In the Christian dialects of the north-eastern sector of NENA, the uvular **q* is fronted to a post-velar or velar articulation, which is represented here /*k̠*/. It tends to be post-velar before back vowels and velar before front vowels. This fronting of **q* is linked to the general palatalisation of the velar dorsals in these dialects (**k* > *c/č*, **g* > *ʃ/j*). The /*k̠*/ is unaspirated, like **q*, and so remains distinct in glottal setting and VOT from the aspirated reflexes of **k* ([*cʰ*], [*tʃʰ*]). The Christian dialects where the fronting of **q* is attested include C. Urmi, C. Salamas, C. Van, Tergawar, Jilu, Baz, Rekan. In all of these dialects, pharyngeal consonants have been lost. Most words containing a historical **q* and a historical pharyngeal are plain without suprasegmental pharyngealisation.

(26) C. Urmi (Khan 2016a: vol. 1, 171, 197)

ʔa _{k̠} la (< <i>*ʕaqla</i>)	‘leg’
ʔam _{k̠} u (< <i>*ʕamoqa</i>)	‘deep’
ʔa _{k̠} ubra (< <i>*ʕaqubra</i>)	‘mouse’
ʔarə _{k̠} (< <i>*ʕarəq</i>)	‘he runs’
rə _{k̠} k̠a (< <i>*rəḥqa</i>)	‘distant’

A few such words have suprasegmental pharyngealisation in C. Urmi. In the following examples the pharyngealisation may have been induced by the historical presence of a pharyngealised rhotic sonorant **r̥* that had developed at some earlier period (see §7.1) rather than due to the **q*:

(27) C. Urmi (Khan 2016a: vol. 1, 170–171)

+ʔa _{k̠} ərva (< <i>*ʕaqərḃa</i>)	‘scorpion’
+ʔə _{k̠} ra (< <i>*ʕqra</i>)	‘root’

In other dialects where **q* has been fronted to /*k̠*/, the distribution of suprasegmental pharyngealisation is sometimes different from what is found in C. Urmi in this set of words:

(28) Tergawar, C. Mawana (Khan, fieldnotes)

ʔam _{k̠} ü (< <i>*ʕamoqa</i>)	‘deep’
ʔarə _{k̠} (< <i>*ʕarəq</i>)	‘he runs’
+ʔə _{k̠} ra (< <i>*ʕqra</i>)	‘root’
+ʔa _{k̠} ərwa (< <i>*ʕaqərḃa</i>)	‘scorpion’
+ʔa _{k̠} übra, +ʔabü _{k̠} ra (< <i>*ʕaqubra</i>)	‘mouse’
+rə _{k̠} k̠a (< <i>*rəḥqa</i>)	‘distant’

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- (29) C. Salamas (Khan, fieldnotes)

'imuḵa (< *'amoqa) 'deep'
 rəxḵa (< *'rəḥqa) 'distant'
 'iḵərva (< *'aqərḥa) 'scorpion'

- (30) C. Van (Khan, fieldnotes)

rəḵḵa (< *'rəḥqa) 'distant'
 'imuḵa (< *'amoqa) 'deep'
 'aḵla (< *'aqḷa) 'leg'
 'əḵra (< *'əqra) 'root'
 *'iḵarwa (< *'aqərḥa) 'scorpion'

- (31) C. Jilu (Fox 1997)

rəxḵa (< *'rəḥqa) 'distant'
 'emuḵa (< *'amoqa) 'deep'
 'aḵla (< *'aqḷa) 'leg'

- (32) Baz C. Aruntus (Khan, fieldnotes)

yəḡmuḵa (< *'amoqa) 'deep'
 rəxḵa (< *'rəḥqa) 'distant'
 'aḵla (< *'aqḷa) 'leg'
 yəḡḵərwa (< *'aqərḥa) 'scorpion'
 yəḡḵubra (< *'aqubra) 'mouse'

In the Baz dialects of Maha Xtaya, Argab, Bi-Silləm and Šawwa and the dialect of Rekan, a *q is fronted in some words further to an unaspirated palato-alveolar affricate /ç/ [tʃ], mainly before /a/. This is a lexically-specific shift the conditioning of which is not yet clear. In the majority of words, the reflex of *q is /ḵ/. In some dialects of this group, words containing the /ḵ/ or /ç/ reflexes of *q are pronounced with suprasegmental pharyngealization, while in other dialects they are pronounced plain:

- (33) Baz Maha Xtaya (Mutzafi 2000; Khan, fieldnotes)

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⁺ <i>rəxḵa</i> (< * <i>rəḥqa</i>)	‘far’
⁺ <i>amuḵa</i> (< * <i>‘amoqa</i>)	‘deep’
⁺ <i>aḵla</i> (< * <i>‘aqla</i>)	‘leg’
⁺ <i>čam</i> (< * <i>qam</i>)	‘before’
⁺ <i>čāra</i> (< * <i>qarθa</i>)	‘cold’
⁺ <i>čadamta</i> (< <i>qadamta</i>)	‘morning’
⁺ <i>čamxa</i> (< <i>qamxa</i>)	‘flour’
<i>čāra</i> (< <i>qar‘a</i>)	‘gourd’

(34) Baz C. Argab (Khan, fieldnotes)

⁺ <i>ḵa</i>	‘narrow’
⁺ <i>amuḵa</i>	‘deep’
<i>rəxḵa</i>	‘distant’
<i>čedamta</i>	‘morning’
<i>čamxa</i>	‘flour’
<i>čāma</i>	‘first’
<i>čam</i>	‘before’
<i>ḵawra</i>	‘grave’

(35) Rekan (Khan, fieldnotes)

⁺ <i>amuḵa</i>	‘deep’
<i>rəḵxa</i>	‘distant’
⁺ <i>čāma</i>	‘first’
⁺ <i>čarha</i> (< * <i>qarθa</i>)	‘cold’
⁺ <i>čāra</i> (< * <i>qar‘a</i>)	‘gourd’
⁺ <i>əl-‘čama</i>	‘forwards’

An isolated case of affrication of **q* in a lexical item is found in the C. Bədyal dialect, spoken in Barzan, northern Iraq:

(36) Bədyal (Khan, fieldnotes)

<i>pəqqačə</i> (< * <i>pəqqaqe</i>)	‘frogs’
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8.2 Uvular stop /q/ in Arabic

The voiceless uvular stop is usually preserved in Qəltu Arabic dialects (Talay 2011: 913), as implied by the name Qəltu (literally: ‘I said’ from the root *q-w-l*),¹ and

¹It is so commonly assumed that **q* is preserved that it often goes unstated, but its preservation can in any case be deduced from examples, such as forms of the verb *qāl* ‘to say’.

8.3 Uvular stop /q/ in Kurdish

no other reflex is attested in the NENA-speaking region. Just outside its borders, however, /q/ is found in apparently free variation with another reflex, /ʔ/, in Siirt town and Siirt villages (Jastrow 1978: 40–41). In the Siirt village of Tillo, the original realization, as preserved by women, is /ʔ/, while /ʔ/ alternates with /q/ in the speech of men, who have undergone more external dialectal influence (Lahdo 2009: 50–52). The reflexes *x* and *ġ* are also found in certain positions.

Table 8.1: Uvular stops inside the NENA area

Mosul (M, C, J)	<i>q</i> (Jastrow 1979: 39)
Bəḫzāni (C, Y, M)	<i>q</i> (Jastrow 1978: 40)
Mērgī (C)	<i>q</i> (Hassan 2023: 580)
Səndōr (J)	<i>q</i> (Jastrow 1991b: 775)
Arbīl and ‘Aqra (J)	<i>q</i> (Jastrow 1990a: 26)
Khānaqīn (J)	<i>q</i> (Bar-Moshe 2024: 113)

Table 8.2: Uvular stops on the periphery of the NENA area

Āzəx (C)	<i>q</i> , word-finally <i>x</i> (Wittrich 2001: 12)
Siirt town (M)	<i>q</i> (Biṭună 2016a: 23)
Tillo (M)	’ ~ <i>q</i> , <i>x</i> and <i>ġ</i> (Lahdo 2009: 5–52)
Kirkūk (J)	<i>q</i> (Bar-Moshe 2024: 113)
Tūz Khurmātu (J)	<i>q</i> (Bar-Moshe 2024: 113)

8.3 Uvular stop /q/ in Kurdish

The uvular stop /q/ did not exist in Proto-Iranian. Words containing /q/ were borrowed from Semitic languages (and possibly Turkic) into Kurdish varieties over a long period. Remodeling strategies changed as the /q/ phoneme was nativized. An early borrowing was the Arabic word for time *waqt*, which surfaces in Kurdish a *wext*. The consonant [q] shifts to [x], losing its uvular place of articulation and spirantizing; there are not other clear examples of this process. It is possible that after modifying the place of articulation, the resulting [kt] cluster was also impermissible. One of the earliest common Iranian sound changes was the pre-consonantal spirantization of voiceless stops: *p*, *t*, *k* > *f*, *θ*, *x* / _C. It is unlikely

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that this word was borrowed into Kurdish before this shift, i.e., before 1600 BC.²

It is notable that Persian has remodeled the voiceless uvular stops of Arabic into the voiced counterpart [g] in most varieties. The original *q is preserved before voiceless consonants, e.g., Persian *vaqt* ‘time’. Additionally, the voiced uvular is spirantized in some varieties post-vocalically, e.g., Persian (Tehran) *čay* ‘fat’.

More recent borrowings leave the Arabic sounds intact, e.g., *qşe-kirdin* ‘to talk.’ This has been borrowed from Arabic *qişsa(t)* ‘story’, preserving the original consonant phonemes but not the vowel. Finally, there are very few native words that have acquired an uvular pronunciation from an original /k/. There is at least one minimal pair between *kun* ‘hole’ and *qûn* ‘buttocks’, which may have a unified etymon. In some varieties, they are homophones, e.g., Kerkuk: *kun* ‘hole’ and *kun* ‘buttocks’.

Another word worth mentioning here is the curse *qurzelqurt* ‘fie, curse.’ This is a possible example of Turkish borrowing of the uvular. The folk wisdom surrounding this word is that it is from the Turkish *qızıl qurt* or *kızıl kurt* ‘red worm’ referring to a poisonous insect that could be used to kill. The version with the uvulars was conditioned by surrounding back vowels in Old Turkic. We can only speculate about whether this variant was borrowed very late into Kurdish or it was borrowed as the more common *kızıl kurt* and it became uvular due to sound symbolism.

In the native Kurdic lexicon, there is variation regarding the realisation of /q/ that is derived historically from a /k/. By way of example, word-initial /k/ in the environment of the back vowel *û* in the word *kûn* ‘buttocks’ results in different segments across different Kurdish varieties. At any rate, /q/ does not undergo fronting.

- It is retained as an voiceless velar stop in NK dialects of Dirî and Harkî tribes in Gavar (Yuksekovâ) and Shamzîdîn, where the forms *kûn*, *kûn* occur.
- It is realised as a uvular stop in CK and NK dialects with the form is realised as *qûn*. In Southern dialects of Central Kurdish, peripheral Gorani dialects and Southern Kurdish the form *qîñ* occurs.
- It is realised as a voiced pharyngeal /ʕ/, perhaps through the intermediary of /q/ in Hewramî varieties, where the word is realised as *ʕəna*.

²These [xt] sequences later shifted to [ht] then [t], supporting the idea that a *waxt* form was not part of the language at the time of the shift, cf. Old Iranian *yuxta* ‘yoked’ > (**yuxt* > **juxt* > **juht* >) Kurdish *jût* ‘yoke.’ However, it is the case that this shift left the language without [kt] sequences, which needed to be dealt with in some way.

8.4 Uvular stops in Armenian

- Further examples of *k* – *q* alternations in Kurdish: Shamdinan dialect *kur* ‘mud’ is found as *qur* in neighbouring Akre N. Kurdish but also in Central Kurdish dialects. Likewise, the word for ‘sesame’ is *kuncî* in most Kurmanji dialects but *quncî* in Shamdinan and in the neighbouring CK dialects of Erbil.

8.4 Uvular stops in Armenian

Classical Armenian did not possess uvular stops. However, they are found in several modern Armenian dialects. Their distribution indicates that they are likely to have been acquired by language contact, as they seem to be found only in dialects which are in contact with languages possessing these sounds. They appear all along the southern fringe of the Armenian-speaking area in dialects in contact with Arabic and Persian, including the whole area surrounding Lake Urmia, in eastern Azerbaijan (Shamakhi), and in areas where there was contact with Kartvelian languages, namely in dialects spoken in the Black Sea region, where there is contact with Laz, and in Tbilisi, where there is contact with Georgian.

Most of the words where uvular stops appear are loans from Turkic, Semitic, or Iranian languages, as seen in the following examples from Urmia dialect given by [Asatryan \(1962\)](#):

- (37) a. Turkic: *qaz* ‘goose’, *qazan* ‘cauldron’ etc.
 b. Arabic (although it is possible that some or all of these were borrowed into Armenian through Turkic): *qadər* ‘worth, respect’, *qabyl* ‘(preverb) to accept’, *qalam* ‘pencil’
 c. Persian: *qaləb* ‘mould’ (for shaping), *qumarbaz* ‘gambler’ etc.

In Armenian dialects that do not possess uvular stops, these words are usually rendered with a uvular or velar fricative transcribed as <ɣ>, i.e., *yaz*, *yazan*, etc., rather than a uvular stop.

However, there are some examples where [q] appears in place of unaspirated *k* in inherited Armenian words, as in the following examples from Urmia dialect given by [Asatryan \(1962\)](#):

- (38) Urmia dialect ([Asatryan 1962](#))

<i>qadəyel</i>	< <i>katayel</i>	‘to be furious’
<i>qum</i>	< <i>kum</i>	‘gulp, sip, mouthful’
<i>qul tal</i>	< <i>kul tal</i>	‘swallow’
<i>qərunq</i>	< <i>kərunk</i>	‘crane’

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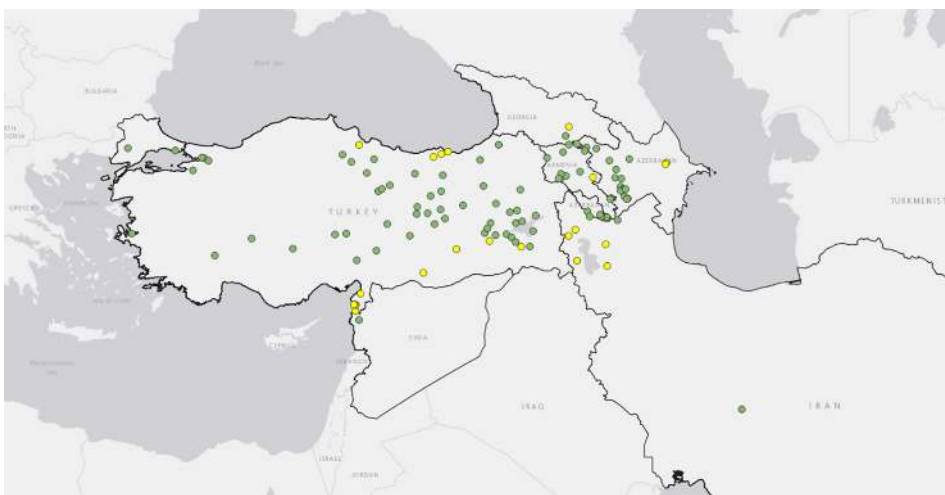


Figure 8.1: Uvular stops in Armenian dialects: present (yellow), absent (green) (according to [Jahukyan 1972](#))

8.5 Uvular stops in Turkic

In inherited vocabulary, Azeri dorsal stops and fricatives ultimately continue the Proto-Turkic velars **k* and **g* ([Johanson 2021](#): 272). Prior to the westward expansion of Turkic languages, these consonants had developed a systematic velar–uvular alternation conditioned by vowel harmony: **k* was realised as a velar [k] next to front vowels and as a uvular [q] next to back vowels, while **g* similarly alternated between [g] in front-vowel contexts and [ɣ] in back-vowel contexts. This distribution, originally allophonic, formed the basis of later developments in the Oghuz branch.

In Azeri, this inherited pattern was disrupted at an early stage. Extensive lexical borrowing—particularly from Persian and Arabic—introduced forms whose phonotactics did not conform to the inherited vowel-conditioned alternation, resulting in velar realisations in back-vowel environments and uvular realisations in front-vowel environments. Together with possible substrate influences, this led to a reanalysis of the former allophones as distinct phonemes. The resulting four-way opposition is maintained in Standard Azeri, e.g., *kim* ‘who’ (inherited) vs. *kal* ‘unripe’ (Persian loanword), *göz* ‘eye’ (inherited) vs. *gor* ‘tomb’ (Persian loanword), and *qar* ‘snow’ (inherited) vs. *qərar* ‘decision’ (Arabic loanword).

Subsequent developments affecting uvular consonants show considerable dialectal variation. In the majority of Azeri varieties, the historical syllable-initial uvular stop /q/ no longer surfaces as such, having shifted to voiced velar /g/,

8.6 Discussion

likely via an intermediate uvular stop /g/, both in inherited lexemes and loanwords, e.g., *qar* [gar] ‘snow’ and *qərar* [gərar] ‘decision’. Nevertheless, uvular realisations remain robust in a number of peripheral and transitional varieties. In the Sonqor variety of Kermanshah and the Bayat varieties of Hamadan and Saveh (Stilo 1994; Bulut 2014: 84–99, *passim*), located in an area of transition between West Oghuz (Azeri-like) and South Oghuz (Qashqai-like) varieties, the historical voiceless uvular is retained, as it is in Iraqi Turkic, where /q/ and /g/ co-exist (Bulut 2019: 360). The same holds for the Derbent and Quba varieties and for neighbouring dialects on the northern Azeri periphery, extending toward the western outskirts of Baku. In this area, the consonant is typically realised as /g/ and displays [q] as an allophone, for instance, in geminate environments, e.g., Quba Azeri [sq:al] ‘beard’, although Rüstəmov (1961: 61) cites for Quba Azeri phonemic realisations of /q/ in loanwords containing an etymological voiceless velar stop next to a back vowel, e.g., /qoma/ ‘hut’, /qasub/ ‘poor’ (ultimately from Arabic *kūma*, *kāsib*, respectively).

Additional evidence for the relatively recent fronting of uvular consonants in much of Azeri comes from contact-induced lexical transmission. Languages that have borrowed vocabulary from Azeri tend to reflect uvular rather than ‘velar’ consonants, suggesting that uvular realisations were still salient at the time of borrowing, e.g., Armenian *ḡonaḡ* [xoˈnax/] ‘guest’ < Azeri *qonaq*, Georgian *q’ačayī* [q’atʃaxi] ‘robber’ < Azeri *qaçaq*, Lezgian *qužax* [quʒax] ‘bosom’ < Azeri *qucaq*, Tat *quš* [guʃ] ‘bird’ < Azeri *quş*, and NENA *qūlūḡ* [quːluːʁ] ‘service’ < Azeri *qulluq*.

8.6 Discussion

The inherited voiceless uvular stop /q/ is widely preserved in NENA dialects. The fronting of the segment to unaspirated /k̟/ in the NENA dialects of the north-eastern sector may have been facilitated by contact with Armenian. In some Armenian words [q] appears in place of the inherited Armenian unaspirated /k/, reflecting an association between the two sounds. The fronting of /q/ > /k̟/ in NENA dialects in the north-eastern sector was also facilitated by the palatalisation of dorsal stops (/k/ > /c/ > /č/), which was induced by contact with Kurdish and Turkic varieties in the region (§5.1). Another facilitating factor may have been the fronting of syllable-initial uvular voiceless stops to voiced velar /g/ in Azeri varieties (§8.5), probably under the influence of Persian.

The further fronting of /k̟/ to an unaspirated palato-alveolar affricate /ç/ [tʃ], which is attested in some NENA dialects of the Baz region of south-eastern

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Turkey, appears to be an extension of this process that is internal to NENA. The fact that the NENA dialects of the north-eastern sector are not in contact with Arabic also no doubt facilitated the fronting of the segment. Arabic dialects in the NENA region have preserved the inherited /q/. It should also be noted that the NENA dialects in the north-eastern sector lack pharyngeal consonants. Moreover, they lack emphatic, i.e., pharyngealised, segments and pharyngealisation has developed into a suprasegmental feature (§7.1.3). These developments may also have facilitated the fronting of /q/, since it was originally a member of a set of pharyngealised segments (§7.1), and could be regarded as an emphatic *k*, manifested by its backing to the uvular position. Indeed, as seen in (11), the presence of a pharyngeal consonant in some dialects can induce the shift of an inherited /k/ to /q/. In the NENA dialects of the north-eastern sector, where historical **q* has lost its emphatic feature, its fronting would have been facilitated.

Other phonetic developments of NENA /q/, such as its fricativisation in some contexts in some dialects and/or its voicing, especially in the eastern sector of NENA, have parallels in Iranian, Armenian and Turkic varieties of western Iran.

The association of /q/ with pharyngeals, which is a feature of a number of NENA dialects, has a parallel in the realisation of /q/ in the word *qûn* 'buttocks' as a voiced pharyngeal /ʕ/ in Hewrami varieties, where the word is realised as ʕəna. This may be more than a typological parallel, since it occurs in an area in which NENA dialects exhibit this association of /q/ with pharyngeals, e.g., the J. Sanandaj dialect.

Part III

Nominal Morphology and Syntax

9 Case morphology and adpositional marking

9.1 Introduction

This chapter introduces the case morphology and adpositional marking strategies found in the languages under investigation. Following functional typological literature on case (Givon 1984, Croft 2001, Andrews 2007, Haspelmath 2005), both nominal case inflection and adpositional marking are subsumed under ‘case marking’ or ‘flagging’ in this volume, since both are morphological means for indicating the grammatical functions of the nominal argument, either by directly manipulating the form of the noun itself or by using an adposition to assign the noun its syntactic role. What follows here is a brief description of the case systems and adpositional marking typologies in the relevant languages. The syntactic functions of the core case-markers are discussed separately in greater detail in Chapter ???. In the Iranian languages considered here, nominal case inflection closely interacts with gender and number marking morphology. Grammatical gender is notably absent in Armenian (and Turkic). In Neo-Aramaic and Qeltu-Arabic, case only plays a role in the form of adpositional marking.

Table 9.1: Nominal inflection in C. Marga illustrated with *xmara* ‘donkey’

	Nominative	Accusative	Old Dative	New Dative
M.SG	<i>xmar-a</i>	<i>=xmar-a</i>	<i>l-xmar-a</i>	<i>ta-xmar-a</i>
M.PL	<i>xmar-e</i>	<i>=xmar-e</i>	<i>l-xmar-e</i>	<i>ta-xmar-e</i>
F.SG	<i>xmar-ta</i>	<i>=xmar-ta</i>	<i>l-xmara</i>	<i>ta-xmar-ta</i>
F.PL	<i>xmar-yasa</i>	<i>=xmar-yasa</i>	<i>l-xmar-yasa</i>	<i>ta-xmar-yasa</i>

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Table 9.2: Nominal inflection in J. Sanandaj illustrated with *axona* ‘brother’

	Bare	Determined	Genitive pronominal suffix
SG	<i>axon-a</i> ‘(a/the) brother’	<i>axon-ake</i> ‘the brother’	<i>axon-i</i> ‘my brother’
PL	<i>axon-awale</i> ‘(some/the) brothers’	<i>axonawal-ake</i> ‘the brothers’	<i>axonawal-i</i> ‘my brothers’

Table 9.3: Core NENA case markers for regular lexical nouns

	Nominative	Accusative	Dative
J. Dohok	<i>xmara</i>	= <i>xmara</i>	<i>ta-xmara</i>
J. Koy Sanjaq	<i>xmara</i>	’ <i>əl-xmara</i>	<i>ta-xmara</i>
C. Telkepe	<i>xmara</i>	<i>ta-xmara</i> =	<i>ta-xmara</i>
C. Nerwa	<i>xmara</i>	= <i>xmara</i>	<i>qa-xmara</i>
C. Lewen	<i>xmara</i>	<i>qa-xmara</i> =	<i>qa-xmara</i>

Table 9.4: Neo-Aramaic case forms for the first person singular pronoun

	Nominative	Accusative	Dative	Genitive
J. Dohok	’ <i>ana</i>	=’ <i>ana</i>	<i>tali</i>	<i>didi</i>
J. Koy Sanjaq	’ <i>ana</i>	’ <i>əlli</i>	<i>baqi</i>	<i>didi</i>
C. Urmi	’ <i>ana</i>	<i>ḵa-diyyi</i> =	<i>ḵa-diyyi</i>	<i>diyyi</i>
C. Barwar	’ <i>ana</i>	=’ <i>ana</i>	’ <i>əlli, ṭlali</i>	<i>diyi</i>
Ṭuroyo (Midyat)	<i>ūno</i>	= <i>ūno</i>	<i>l-ūno</i>	<i>diḏi</i>
Ṭuroyo (Miden)	<i>ono</i>	= <i>ono</i>	<i>eli</i>	<i>diḏi</i>
Mlahso	<i>ono</i>	<i>eli</i> =	<i>eli</i>	<i>uzi</i>

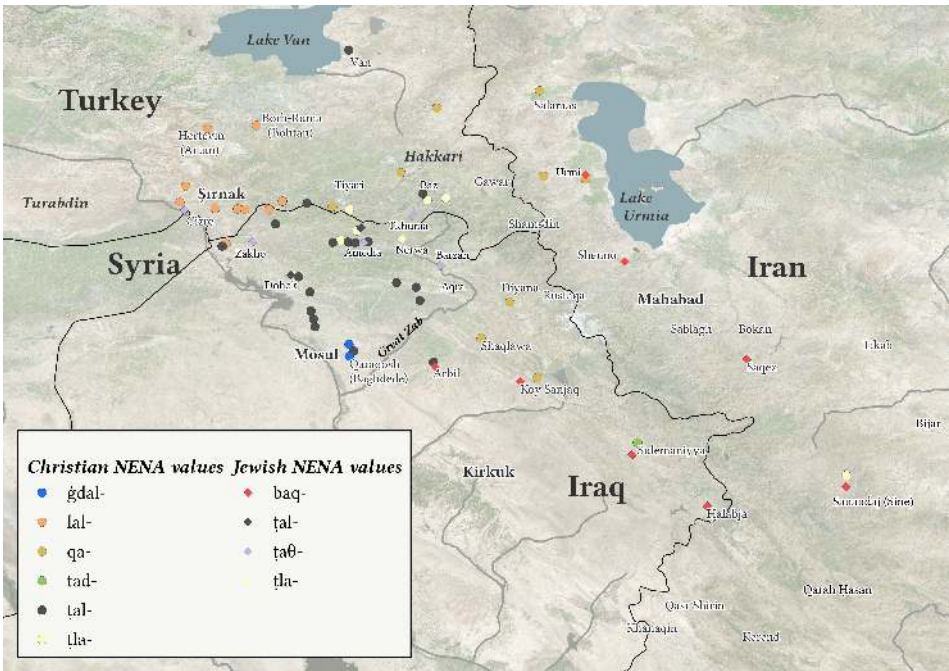


Figure 9.1: Distribution of dative prepositions across NENA dialects

9.2 Case marking in NENA

Generally, in Neo-Aramaic, the noun is declined by suffixes that mark only number and gender, as shown in Table 9.1. The nominal endings in the NENA dialects derive historically from the postposed definite article which was attached to determined nouns. In the ancestors of the eastern dialects of Neo-Aramaic, the original opposition between non-determined and determined forms of the noun had collapsed already in antiquity, and the non-determined forms became largely obsolete. Western Neo-Aramaic dialects, by contrast, did not entirely lose this opposition (see Arnold 1990). Trans-Zab Jewish NENA dialects located in eastern Iraqi Kurdistan and Iranian Kurdistan imported the Kurdish definite article as an invariant suffix *-ake* directly attached to the stem of the noun (Khan & Mohammadirad 2024c: 104-107). Even though NENA dialects did not develop a definite article, they did develop a so-called neo-construct state, i.e., head marking morphology in genitive constructions (see Chapter 11). Contrary to the majority of NENA dialects, ʦuroyo and Mlaḥso Neo-Aramaic innovated preposed definite articles, converging with the preposed definite article *al-* / *əC-* in local Arabic dialects.

9 Case morphology and adpositional marking

As in most Semitic languages, the original three-way case system collapsed, and case declensions were lost entirely by the time Aramaic took shape. Case-marking is virtually always prepositional in Neo-Aramaic, as it has been since its beginnings. This is a robust tendency, apart from a handful of postpositions that entered Neo-Aramaic through contact with neighbouring languages with postpositions (see also Fox 2009a; Noorlander 2024; Garbell 1965a). Postpositional marking in the strict sense seems to only occur among Jewish varieties of Iranian Azerbaijan under Turkic influence, as shown in 1, and possibly also among Christian varieties in the Bohtan region, as shown in 2.

- (1) J. Urmi (Garbell 1965c: 82)
sot-ew=inan *son-ew*
 grandmother.F-POSS.3SG.M=with grandfather.M.SG
 ‘his grandmother and (lit. with) his grandfather’
- (2) C. Bohtan (Fox 2009a: 126.139)
üzü-Ø-wa *matwota=ləl*
 go.PRF-3SG.M:S-PST village.PL=to
 ‘He had gone to the villages.’

Still, even in those dialects that have arguably undergone the most intense influence from Turkic, postpositions can be imported as prepositions, as in the case of the Azeri postposition *qabağ*, which serves as a preposition in the Jewish dialects of Iranian Azerbaijan, as shown in 3.

- (3) J. Urmi (Garbell 1965c: 126)
qabağ tara
 front door.M.SG
 ‘in front of the door’

Moreover, at no time in its history did Aramaic ever develop a system of nominal case declension, nor did it ever develop an unconditioned accusative case. A simplified overview of various case systems in NENA dialects is given in Table 9.3. Accusative case-marking is always dependent on inherent and/or contextual DOM and often entirely optional (see ??). The unmarked, i.e., nominative, form of nouns is widely used to express both subjects and direct objects. As such, one could say that many NENA dialects lack a dedicated accusative case marker (or a dedicated ergative case marker). A few NENA dialects have preserved an ergative case derived from a dative case marker *l-* that is only used to express the agent in the past perfective, for example, the Christian dialect of Marga (see §??).

9.2 Case marking in NENA

The same ergative-dative case syncretism occurs more widely among speakers of ʿTuroyo, a subgroup closely related to NENA. The usage of this preposition closely parallels the so-called ‘oblique’ case in Iranian (see Sections ?? and ??). In the majority of North-Eastern and Central Neo-Aramaic dialects, however, this preposition has been absorbed into the verbal person marking system and become part of the verbal inflection (see §??).

Generally, unmarked, i.e., nominative, free pronouns cannot be inflected for case, and can be used to express various syntactic functions (see §??), reflecting a process of neutralisation. Historically, Aramaic dialects distinguished free subject pronouns from a series of free object pronouns. In the Aramaic varieties of Anatolia and Mesopotamia, the free object pronouns were based on the dative case-marker *l-* (see also below), which also served as a differential object marker, thus Syriac nominative *ʾeno* ‘I’ existed alongside dative-accusative *li* ‘(to/for) me’. Several NENA dialects, mainly Christian dialects in the northern sector and Jewish dialects in the eastern sector, have dative-accusative pronouns, which are ultimately derived from the same dative case-marker, e.g., J. Koy Sanjaq nominative *ʾana* ‘I’ vs. dative-accusative *ʾalli* ‘(to/for) me’. Similarly, Mlaḥso Neo-Aramaic (Jastrow 1994a) also has dative-accusative pronouns distinct from their nominative equivalents, as shown in 9.2. In other Neo-Aramaic dialects, there is no dedicated series of object pronouns.

In Semitic in general, prepositions and attributive modifiers such as *-əd* and *d-* (see 11.1.8) are inflected by the addition of pronominal suffixes, as illustrated in (4a) and (4b), which are also attached to nouns to express the possessor, as shown in (4c).

(4) J. Dohok

- a. *tal-i*
to-1SG
‘to me’
- b. *did-i*
GEN-1M.SG
‘(that one) of mine’
- c. *xmar-i*
donkey.SG.M-POSS.1SG
‘my donkey’

Nouns and prepositions can often also be combined with free genitive pronouns in a similar way to nominal attributes, as shown in (5).

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(5) J. Koy Sanjaq

- a. *xmar-əd* *didi*
 donkey.SG.M.-CSTR GEN-1M.SG
 ‘donkey of mine’
- b. *baq-əd* *did-i*
 to-CSTR GEN-1M.SG
 ‘(that one) of mine’

This parallels the use of Iranian ezafe, see Chapter 11 on nominal annexation. Across NENA dialects, genitive pronouns of the third person feminine and masculine singular are typically used to distinguish between the homophonous nominal singular morpheme *-a* and the 3FSG. possessive suffix *-a* as well as between the nominal plural morpheme *-e* and the 3MSG. possessive suffix *-e*. Particularly in the northern and eastern sectors, we find prepositions where the genitive pronouns have become fully integrated into the inflection, such as C. Urmi *ka-*, the full, heavy form of which combines with genitive pronouns, e.g., *ka-diyy-ax* ‘for/to you (F.SG)’ with the phonetically reduced variant *ka-t-ax* or *k-āx*, which consists of the preposition *ka* and the genitive pronoun *diyyax* ‘your’ (F.SG.).

Several NENA dialects, primarily Christian dialects in the northern sector and Jewish dialects in the eastern sector, developed accusative case markers as part of their DOM system. With the exception of a handful of dialects, all of these accusative case markers are ultimately derived from dative case markers, and this history can still be reflected in their usage. The extension from the marking of recipient arguments to patient arguments resulted in dative-accusative case syncretism, which is partly resolved by the emergence of new dative prepositions.

The NENA dialects use a variety of forms for the dative preposition ‘to, for’. The forms and their distributions are shown in Figure 9.1. The dative prepositions in many NENA varieties, particularly those with pronominal suffixes, involve what were historically multiple case-markers.

Some dialects preserve a preposition derived from the erstwhile Aramaic dative prepositions **l-* and **’el-* expressing ‘to, for’. In several NENA dialects, the locative preposition **’al-* ‘on, upon’ merged with the aforementioned dative preposition. In most NENA dialects, the locative preposition was replaced by the grammaticalisation of *reša* ‘head’ from an NP to a preposition. Generally, prepositions show a complementary distribution of one allomorph combined with pronominal suffixes and another allomorph combined with full nominals. Thus, the preposition *l-* before full NPs alternates with *’all-* before pronominal suffixes in all NENA dialects that preserved this preposition. In Mlaḥso and Țuroyo Neo-Aramaic, closely related to NENA, the preposition *l-* has the allomorph *el-*.

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Many dialects have alternative or more complex forms of dative prepositions. The prepositions *ṭla-* with allomorphs *ta-/ṭal-/ṭaθ-/ṭlal-/ṭlāθ-*, and *qa-* (or *ḵa-*) with allomorphs *qat-/qal-*, have a widespread distribution, with *ğḏa-* also occurring alongside *ḏa-* and *ğḏal-* in NENA. The element *l-* included in some forms of these prepositions comes from the original dative preposition *l-*. This would mean that such forms also arose historically through the combination of two prepositions. This is in line with the complementary distribution found with this preposition in some NENA varieties. Nominals tend to lack the additional *l-*, e.g., J. Dohok *ta malka* ‘for the king’, while pronominal suffixes are preceded by *l-*, e.g., *tal-e* ‘for him’. It is possible that the same preposition *ṭlal-* is reflected in forms such as *lal-* and *lol-* (< **lāl-* as an allomorph of *l-*, since **lāl-* could be a simplification of earlier *ṭlāl-*). It is also possible that this is simply a case of adding two instances of the preposition *l-*. The *lal-* forms are typical of the Christian dialects in the western sector, such as: Umra, Umra d-Shish, Jənnət and some Hakkari dialects (e.g., Mēr).

Historically, *qa-*, *ḵa-* and *ğḏa-* clearly go back to **qḏām* ‘in front of, for, before’. It is conceivable that *ṭa-* and *ta-* ultimately go back to the same preposition *qḏām*, like the preverb *tam-* found in dialects such as C. Sanandaj, which is cognate with *qam-* found in other dialects. The etymology of the preposition *ṭla-* remains uncertain, but Mutzafi (Mutzafi 2008a: 388) proposes that *ṭla-* is cognate with Syriac *meṭṭol* and Jewish Babylonian Aramaic **amṭūl* meaning ‘because of, on account of, for the sake of’.

Finally, many of the Jewish Trans-Zab dialects feature the preposition *ba-* or *baqa-*. The latter shares its reduplicated character with many of the forms discussed above, but both *ba-* and *baqa-* stand out for their apparently (partly) Iranian origin, cf. also *muj* ‘because of’ borrowed from Kurdish (Garbell 1965a). The segment *ba-* seems to derive from the Kurdish-Gorani preposition *bo* ‘to, for, because of’ (Garbell 1965a: 166). The preposition *bo-* is attested in J. Urmi where it always combines with the genitive extension *-d* reflecting the Iranian ezafe construction (see also Chapter 11), e.g., *bo-d ma* ‘wherefore? why?’ consisting of *bo-* ‘for’ + *-d* ‘of’ + MA ‘what’ and *bo-d did-i* ‘because of me’ consisting of *bo-* ‘for’ + *-d* ‘of’ + *didi* ‘mine’, (Garbell 1965a: 166). The sound correspondence *ba* : *bo* is not a regular one; the vowel *o* in Kurdish does not typically correspond to *a* in NENA (including Trans-Zab Jewish), nor do we find an *o* > *a* shift in these NENA varieties. In this subgroup, too, *ba* is often combined with the Aramaic segment *qa-* < **qḏām* ‘before’, used elsewhere for the dative, for instance C. Urmi NENA *ḵa* (see above).

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9.3 Case marking in Arabic

Case marking in Mesopotamian Arabic shows similar typological traits to NENA and is chiefly prepositional. Mesopotamian Arabic has varieties that developed accusative case-markers out of former dative case-markers, which are conditioned by the differential treatment of the direct object (see ?? in Chapter ??). In the majority of Mesopotamian Arabic dialects, however, core arguments are not case-marked.

9.4 Case marking in Iranian

9.4.1 Nominal case inflection

Case marking in Kurdish exists on a continuum, with the Northern varieties distinguishing case, number, and gender, and the Southern varieties only distinguishing number. In Table 9.5, we present some general patterns, while acknowledging the fact that there is much variation in these precise realisations of these forms. In terms of case declension, modern Iranian languages generally distinguish between an unmarked case, the so-called ‘direct’ case, which is a reflex of the Old Iranian nominative, and a marked case, the so-called ‘oblique’ case, which most likely continues the Old Iranian genitive. Already in Old Persian, the former distinction between dative and genitive case is lost, and the polyfunctional genitive has been extended at the expense of the dative (Haig 2008: 45-46). The modern Iranian oblique case is thus the outcome of genitive-dative case syncretism, ultimately also absorbing the accusative. Moreover, the oblique case serves as a complement of any adposition. Neo-Iranian languages have preserved the oblique case alongside its direct equivalent in different contexts and to different degrees, with total loss of case declension in Southern Kurdish and modern Persian. Modern Persian developed an accusative case-marker out of a former dative posposition, which serves as differential object marker of definite and specific indefinite animate referents.

In Northern Kurdish, case, number, and gender are all represented in the nominal inflection. The two genders, masculine and feminine, are distinguished in the singular but collapsed in the plural. Case distinctions are also observed in the pronominal system, illustrated in 9.6 for the first and second person pronouns in Behdîni.

Within Northern Kurdish, subtypes can be discerned. First, there is the “Muş” Type, where gender and number are collapsed in the direct case declension, but

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singular and plural are still distinguished in the oblique case inflection. This contrasts with the “Behdîni” Type, also known as south-eastern Northern Kurdish, where there are unique markers for masculine singular, feminine singular, and common plural in the oblique case inflection. The third type, the “Bingol” Type, also known as western Northern Kurdish, represents the dominant Northern Kurdish (Standard Kurmancî) system. Like the other two systems, gender and number collapse in the direct case declension, but there is no distinct oblique case marker in the inflection of masculine singular nouns. This contrasts with the presence of unique oblique case markers for both feminine singular and common plural. In these varieties, however, some masculine singular nouns are inflected for oblique case in the form of umlaut, e.g., *şivan* [shepherd.M.SG.DIR] ~ *şivên* [shepherd.M.SG.OBL].

Within Central Kurdish, in turn, there are two main subtypes. First, in Iranian Kurdistan, there is the “Mukrî” Type, which has preserved the direct and oblique case opposition in the singular, but has neutralised completely or partially all gender distinctions. There are only a few relics of an original feminine singular oblique marker in a small class of feminine nouns, e.g., *aw-ê* [water-OBL], *jin-ê* [woman-OBL]. These endings, however, have been decoupled from biological gender in animate referents, and speakers often use the oblique suffixes *-ê* and *-î* interchangeably with these nouns (see Öpengin 2016: 52). Here, case is neutralised in the plural where the oblique ending *-an* has been generalised. In Iraqi Kurdistan, stretching into Iranian Kurdistan, there is the “Suleymani” Type, where case and gender distinctions are completely lost. Number is distinguished throughout: the singular is unmarked, while the old oblique plural suffix *-an* has been generalised.

Southern Kurdish is similar to the varieties of Central Kurdish spoken in Iraqi Kurdistan. In the “Qorwe” Type, the inherited plural ending *-an* is only used, when a noun is definite, usually with an additional definite suffix, e.g., *kor-ef-an* [boy-DEF-PL]. Elsewhere, the de-collective plural marker is used, e.g., *kor-eyl* [boy-PL]. Further south, in the “Beyrey” Type, the de-collective plural marker is used throughout.

In Goranî, the conservative dialects of Hewraman distinguish case, number and gender in a way that is similar to Northern Kurdish. There is a single set of endings presented in Table 9.7 as the “basic” forms. There are unique forms in all case, number and gender combinations, except for [F.SG.OBL] and [PL.DIR]. This syncretism is common in Iranian in general (following Stilo 2008). The same set of endings added to nouns with the historically diminutive extension **-ag* result in slightly different outcomes, neutralising case in the feminine singular, and, similarly, when **-ag* is combined with *î*-final feminine nouns.

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Table 9.5: Kurdish case endings for regular lexical nouns

		M.SG		F.SG		PL	
		DIR	OBL	DIR	OBL	DIR	OBL
NK	Muş Type	-Ø	-ê	-Ø	-ê	-Ø	-an
	Behdînî Type	-Ø	-î	-Ø	-ê	-Ø	-an
	Bingol Type	-Ø	-Ø	-Ø	-ê	-Ø	-an
CK	Mukrî Type	-Ø	-î	-Ø	-î/-ê	-an	-an
	Suleymanî Type	-Ø	-Ø	-Ø	-Ø	-an	-an
SK	Qorwe Type	-Ø	-Ø	-Ø	-Ø	-an/-eyl	-an/-eyl
	Beyrey Type	-Ø	-Ø	-Ø	-Ø	-eyl	-eyl

Table 9.6: Behdînî NK case forms of first and second person pronouns (MacKenzie 1961a: 171)

		DIR	OBL
NK Barwar	1SG	<i>ez</i>	<i>min</i>
	2SG	<i>to</i>	<i>te</i>
	1PL	<i>em</i>	<i>me</i>
	2PL	<i>hing</i>	<i>hewe</i>
NK Zakho	1SG	<i>ez</i>	<i>min</i>
	2SG	<i>to</i>	<i>te</i>
	1PL	<i>em</i>	<i>me</i>
	2PL	<i>hûn</i>	<i>we</i>

Table 9.7: Hewramî case endings for regular lexical nouns

		M.SG		F.SG		PL	
		DIR	OBL	DIR	OBL	DIR	OBL
Luhon	basic	-Ø	-î	-e	-ê	-ê	-a
	*-ag	-e	-ey	-ê		-ê	-a
	-î			-î	-ê	-î	-a

9.4 Case marking in Iranian

In Zazaki, there is an interplay between case, number and gender as well as between attribution, possession, animacy, and definiteness. First of all, indefinite forms are not marked for case or for gender. Similarly to Northern Kurdish and Hewramî, gender is only distinguished in the plural. In southern Zazaki, feminine nouns only optionally mark case with the suffix *-er* borrowed from familial nouns, e.g., *wa* [sister.F.SG.DIR] vs *wa-r* [sister.F.SG-OBL]. No systematic study has ever been undertaken to ascertain the conditions under which the optional suffix is used. Masculine nouns distinguish case, even when part of an *ezafe* construction involving an attributive adjective. However, case is neutralised when the noun is part of an *ezafe* construction involving a genitive possessor. A masculine noun that is inanimate is always unmarked, regardless of its syntactic role, while the plural is always marked for case, regardless of the presence or the type of *ezafe* construction. The syncretism between singular oblique and plural direct also occurs here, but with masculine nouns (as opposed to feminine in Hewramî). All of these facts are equally true of northern Zazaki, except that the feminine does not have the optional oblique suffix *-er*.

9.4.2 Adpositional marking

The Iranian languages present us with a considerably mixed typology in terms of adpositional marking strategies. This mixed typology is also reflected in the distribution of adpositional types of marking in corpora. Table 9.8 displays the results of a corpus-based study of the frequencies of prepositionally-flagged nominal obliques, with the highest frequencies on the right of the spectrum, exemplified by modern Persian, and the lower frequencies towards the left, with Zazaki having the lowest frequency of prepositionally-flagged nominal obliques.

Table 9.8: Frequencies of prepositionally-flagged nominal obliques taken from Haig et al. (2025)

% of preposition-flagged	0%	20%	53%	56%	87%	87%	98%
Language	Zazaki	Zazaki	NK	CK	SK	Gorani	Persian
Dialect	(SiWereg)	(Cewlig)	(Muş)	(Sanandaj)	(Bijar)	(Gawraju)	(New)

In Northern Kurdish varieties, where case marking is robust, the oblique case is used with prepositions, postpositions, and circumpositions, as (6) illustrates.

(6) Northern Kurdish

- a. *yek ji wan*
one from 3PL.OBL
'one of them'

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- b. *di vî* *war-î* *da*
 in DEM.PROX.M.SG.OBL regard-M.SG.OBL in
 ‘in his regard’

The same holds true for Zazaki and Gorani as well as the Central Kurdish varieties that have preserved case marking, as illustrated by the Mukrî Central Kurdish forms in (7). In Zazaki and Gorani, however, oblique case assignment is further conditioned by definiteness, animacy, gender, number, and other discourse-related factors; see [Dogan \(2022\)](#) and [Mohammadirad \(2025a\)](#) for more information.

- (7) CK Mukrî ([Öpengin 2016](#): 101)
 'alem-e *de mizgewt-ê* *d-êt=e* *der-ê*
 people-DEF in mosque-SG.OBL IND-come.PRS.3SG=to outside-SG.OBL
 ‘The people in the mosque come outside.’

One additional idiosyncrasy of Central Kurdish is that oblique pronominal forms have become part of the set of derivational verbal affixes. These are the result of the contraction of a prepositional third person singular form. These forms attach as the first element in the verbal complex and allow an additional indexed argument. This distribution led [Karim & Salehi \(2022\)](#) to classify these markers as applicatives. The malefative construction in (8) shows how the particle *lê-* highlighted in bold is to be interpreted as the verbal index that serves as the complement of the originally adpositional element. This phenomenon is limited to Central Kurdish.

- (8) CK Mukrî ([Öpengin 2013](#): 191)
 *pişîle-ke=yan **lê**-kuşt-im*
 cat-DEF.SG=3PL:A MAL-kill.PST-1SG:R
 ‘They killed my cat, and I perceived this as indirectly affecting me in a negative manner.’

9.5 Case marking in Armenian

9.5.1 Nominal case inflection

The Classical Armenian case system had essentially the same cases that Proto-Indo-European is proposed to have had: nominative, accusative, genitive, dative, ablative, instrumental and locative. According to [Meillet \(1962: 53\)](#), “la langue

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Table 9.9: Zazaki case endings for regular lexical nouns

		M.SG		F.SG		PL	
		DIR	OBL	DIR	OBL	DIR	OBL
SZ	INDF	<i>-e(n)</i>					
	DEF	<i>-Ø</i>	<i>-î</i>	<i>-Ø</i>	<i>(-er)</i>	<i>-î</i>	<i>-an</i>
	EZ.POS	<i>-ê</i>		<i>-a</i>		<i>-ê</i>	<i>-anê</i>
	EZ.ATT	<i>-o</i>	<i>-ê</i>				
NZ	INDF	<i>-e(n)</i>					
	DEF	<i>-Ø</i>	<i>-î</i>	<i>-e</i>		<i>-î</i>	<i>-u(n)</i>
	EZ.POS	<i>-ê</i>		<i>-a</i>		<i>-ê</i>	<i>-unê</i>
	EZ.ATT	<i>-o</i>	<i>-ê</i>				

a conservé la déclinaison avec un fidélité presque unique”. However, Classical Armenian totally lacked gender marking, as do all modern forms of the language. Classical and modern Armenian have two number categories: singular (unmarked) and plural. Some of the Classical Armenian case forms combine case and number in one ending, as seen in Table 9.10. In modern Armenian, these have mainly been replaced by agglutinative-type morphology, with separate endings for case and number, of which number is closer to the stem than case, as seen in Table 9.12.

Modern Armenian dialects possess nominative (unmarked), genitive, dative, ablative, and instrumental cases. Most Eastern dialects, including SEA, also possess locative case, although in some dialects, including SEA, this is not used with animate referents. A small number of dialects have preserved the Classical Armenian accusative case marker *z-*, used with definite/specific objects (see Chapter ??), and more, including SWA, have preserved separate accusative forms of some pronouns, as seen in Table 9.13.

In general, dedicated accusative forms are absent from Eastern dialects, in which direct objects take nominative case if inanimate or non-specific, and dative case if animate and specific (see §??). There is a tendency for syncretism between genitive and dative, with the endings being generally identical for lexical nouns even in Classical Armenian as well as modern Armenian, as seen in Tables 9.10 and 9.12, although most dialects preserve separate forms for pronouns, as seen in Tables 9.13 and 9.14.

Many dialects occupying what could be described as a “central” area in the Armenian-speaking world (Ararat including SEA, Mush, Van, Urmia) also distinguish genitive and dative by the fact that the former is incompatible with the

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Table 9.10: Classical Armenian case forms of regular lexical nouns

Singular	<i>a</i> -stem	<i>i</i> -stem	<i>o</i> -stem	<i>u</i> -stem
NOM	<i>azg</i> ‘people’	<i>ban</i> ‘word’	<i>cov</i> ‘sea’	<i>beran</i> ‘mouth’
ACC	(<i>z</i> -) <i>azg</i>	(<i>z</i> -) <i>ban</i>	(<i>z</i> -) <i>cov</i>	(<i>z</i> -) <i>beran</i>
GEN-DAT	<i>azg-i</i>	<i>ban-i</i>	<i>cov-u</i>	<i>beran-oy</i>
ABL	<i>y-azg-ē</i>	<i>i ban-ē</i>	<i>i cov-ē</i>	<i>i beran-oy</i>
INS	<i>azg-aw</i>	<i>ban-iw</i>	<i>cov-u</i>	<i>beran-ov</i>
LOC	<i>y-azg-i</i>	<i>i ban-i</i>	<i>i cov-u</i>	<i>i beran-oy</i>
Plural				
NOM	<i>azg-k^c</i>	<i>ban-k^c</i>	<i>cov-k^c</i>	<i>beran-k^c</i>
ACC	(<i>z</i> -) <i>azg-s</i>	(<i>z</i> -) <i>ban-s</i>	(<i>z</i> -) <i>cov-s</i>	(<i>z</i> -) <i>beran-s</i>
GEN-DAT	<i>azg-ac^c</i>	<i>ban-ic^c</i>	<i>cov-uc^c</i>	<i>beran-oc^c</i>
ABL	<i>y-azg-ac^c</i>	<i>i ban-ic^c</i>	<i>i cov-s</i>	<i>i beran-s</i>
INS	<i>azg-awk^c</i>	<i>ban-iwk^c</i>	<i>cov-uk^c</i>	<i>beran-ovk^c</i>
LOC	<i>y-azg-s</i>	<i>i ban-s</i>	<i>i cov-s</i>	<i>i beran-s</i>

Table 9.11: Classical Armenian case forms of personal pronouns

	1SG	2SG	3SG	1PL	2PL	3PL
NOM	<i>es</i>	<i>du</i>	<i>na</i>	<i>mek^c</i>	<i>duk^c</i>	<i>nok^ca</i>
ACC	<i>z-is</i>	<i>z-k^cez</i>	<i>z-na</i>	<i>z-mez</i>	<i>z-jez</i>	<i>z-nosa</i>
GEN	<i>im</i>	<i>k^co</i>	<i>nora</i>	<i>mer</i>	<i>jer</i>	<i>noc^ca</i>
DAT	<i>inj</i>	<i>k^cez</i>	<i>nma</i>	<i>mez</i>	<i>jez</i>	<i>noc^ca</i>
ABL	<i>y-inēn</i>	<i>i k^cēn</i>	<i>i nmanē</i>	<i>i mēnĵ</i>	<i>i jēnĵ</i>	<i>i noc^canē</i>
INS	<i>inew</i>	<i>k^cew</i>	<i>aynuk^c</i>	<i>mewk^c</i>	<i>jewk^c</i>	<i>novawk^c</i>
LOC	<i>y-is</i>	<i>i k^cez</i>	<i>i nma</i>	<i>i mez</i>	<i>i jez</i>	<i>i nosa</i>

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definite article. This fact is mentioned in Mkhitar Sebastatsi's grammar of 1727, who uses the presence or absence of the article to distinguish between genitive and dative cases (see [Abajyan 2006](#): 257).

Table 9.12: Case forms of SEA (and SWA) regular noun

NOM	<i>k'alak^c</i> 'city'	<i>k'alak^c-ner</i>
GEN-DAT	<i>k'alak^c-i</i>	<i>k'alak^c-ner-i</i> (SWA <i>-u</i>)
ABL	<i>k'alak^c-ic^c</i> (SWA <i>-ē(n)</i>)	<i>k'alak^c-ner-ic^c</i> (SWA <i>-ē(n)</i>)
INS	<i>k'alak^c-ov</i>	<i>k'alak^c-ner-ov</i>
LOC (not SWA)	<i>k'alak^c-um</i>	<i>k'alak^c-ner-um</i>

Table 9.13: SWA case forms of personal pronouns

	1SG	2SG	3SG	1PL	2PL	3PL
NOM	<i>es</i>	<i>dun</i>	<i>an</i>	<i>menk^c</i>	<i>duk^c</i>	<i>anonk^c</i>
ACC	<i>z-is</i>	<i>k'ez</i>	<i>z-ayn</i>	<i>mez</i>	<i>jez</i>	<i>z-anonk^c</i>
GEN	<i>im</i>	<i>k'u</i>	<i>anor</i>	<i>mer</i>	<i>jer</i>	<i>anonc^c</i>
DAT	<i>inc-i</i>	<i>k'ez-i</i>	<i>anor</i>	<i>mez-i</i>	<i>jez-i</i>	<i>anonc^c</i>
ABL	<i>inj-m-ē</i>	<i>k'ez-m-ē</i>	<i>an-k-ē</i>	<i>mez-m-ē</i>	<i>jez-m-ē</i>	<i>anonc^c-m-ē</i>
INS	<i>inj-m-ov</i>	<i>k'ez-m-ov</i>	<i>a-m-ov</i>	<i>mez-m-ov</i>	<i>jez-m-ov</i>	<i>anonc^c-m-ov</i>

Table 9.14: SEA case forms of personal pronouns

	1SG	2SG	3SG	1PL	2PL	3PL
NOM	<i>es</i>	<i>du</i>	<i>na</i>	<i>mek^c</i>	<i>duk^c</i>	<i>nrank^c</i>
GEN	<i>im</i>	<i>k'o</i>	<i>nra</i>	<i>mer</i>	<i>jer</i>	<i>nranc^c</i>
DAT	<i>inj</i>	<i>k'ez</i>	<i>nran</i>	<i>mez</i>	<i>jez</i>	<i>nranc^c</i>
ABL	<i>inj-(a)n-ic^c</i>	<i>k'ez-(a)n-ic^c</i>	<i>nran-ic^c</i>	<i>mez-(a)n-ic^c</i>	<i>jez-(a)n-ic^c</i>	<i>nranc^c-ic^c</i>
INS	<i>inj-(a)n-ov</i>	<i>k'ez-(a)n-ov</i>	<i>nran-ov</i>	<i>mez-(a)n-ov</i>	<i>jez-(a)n-ov</i>	<i>nranc^c-ov</i>

9.5.2 Adpositional marking

In Armenian, a shift took place from predominantly prepositional Classical Armenian to predominantly postpositional modern Armenian. Traditional Armenian

9 Case morphology and adpositional marking

grammar distinguishes two types of adpositions: a small number of “true” prepositions, referred to by Acharyan (1965) as *naxdir*, and a second, larger category, referred to by Acharyan (1965) as *naxadrut’yun*, which are mainly derived from nouns. The former were exclusively prepositional in Classical Armenian, and are no longer used productively in most modern Armenian dialects. The main exception is the preposition *z*, which was used to mark definite/specific accusative objects, and is still used in this way in Armenian dialects spoken in Mush, Syria, and Romania/Hungary (Acharyan 1965: 103), to which Jahukyan (1972) adds Ordu on the Black Sea coast (see §??, e.g., example (??)). It is worth noting that in each of these areas, there was contact with languages that use prepositions (Kurdish in Mush, Arabic in Syria, German and Romanian in Romania/Hungary, Pontic Greek in Ordu). The preposition *i/y* has apparently survived only in Mush (9) and Syria (10) (Acharyan 1965: 105).

- (9) Mush Armenian (Acharyan 1965: 105)

k-ert^h-am j-art
PTCL-go-PRS.1SG to-field.
‘I am going to the field.’

- (10) Syrian Armenian (Acharyan 1965: 105)

i ton ari
to house come.2SG.IMP
‘Come to the house./Come home.’

The other category of adpositions, referred to by Acharyan (1965) as *naxadrut’yun*, which are mainly derived from nouns, were also usually prepositional in Classical Armenian (11), (12). However, there are some examples, from as early as the 5th century AD, where they appear as postpositions (13).

- (11) Classical Armenian (Acharyan 1965: 115)

i veray meroy tan
PRP on our.GEN house.GEN
‘on our house’

- (12) Classical Armenian (Acharyan 1965: 115)

i miji meroy tan
PRP in our.GEN house.GEN
‘in our house’

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- (13) Classical Armenian (Acharyan 1965: 115)

i gayl-oc' mēj en
 PRP wolf-GEN.PL in be.PRS.3PL
 'They are among wolves.'

Acharyan (1965: 115) notes that as we approach the modern age, the postpositional uses increase, and eventually become dominant in most forms of Armenian, probably promoted by the influence of Turkish. However, in Mush dialect, where historically the main contact language was Kurdish, rather than Turkish, and there was also contact with Semitic languages, including NENA, which was spoken in parts of the dialect area, many adpositions that appear only as postpositions in other dialects (15), (17), (19) may also appear as prepositions (14), (16), (18).

- (14) Mush Armenian (Acharyan 1965: 115)

vər mər tan
 on 1PL.GEN house.DAT
 'on our house'

- (15) SEA

mer tan vra
 1PL.GEN house.GEN on
 'on our house'

- (16) Mush Armenian (Acharyan 1965: 115)

məʃ^h mər tan
 in 1PL.GEN house.DAT
 'in our house'

- (17) SEA

mer tan mej
 1PL.GEN house.GEN in
 'in our house'

- (18) Mush Armenian (Acharyan 1965: 115)

mət əndzi
 near 1SG.DAT
 'near me'

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- (19) SEA
inj mot
 1SG.DAT NEAR
 ‘near me’

Grigoryan (1957: 372) observes that, in Mush dialect, when these elements appear as postpositions, the complement takes genitive case (20), while when they appear as prepositions, it takes dative case (21).¹

- (20) Mush Armenian (Grigoryan 1957: 372)
gey-i mietʰ
 village-GEN in
 ‘in the village’

- (21) Mush Armenian (Grigoryan 1957: 372)
matʰ gey-i-n
 in village-DAT-DEF
 ‘in the village’

Similar examples where these elements appear as prepositions with dative complements are also found in the Armenian dialects of Van (22), (23) and Dersim (24), again in areas where contact with Kurdish was more intense than that with Turkish.

- (22) Van Armenian (Acharyan 1952: 213)
mot Mayram dudu-ji-n
 near Mayram dudu-DAT-DEF
 ‘to Mayram dudu’s place’ (lit. ‘near Mayram dudu’)

- (23) Van Armenian (Acharyan 1952: 213)
metʰ pæk-i-n
 in garden-DAT-DEF
 ‘in the garden’

¹Acharyan’s examples (14) and (16) are in fact ambiguous between genitive and dative, which on most lexical nouns can only be distinguished by the fact that the dative may be accompanied by the definite article, while the genitive may not. However, the definite article does not appear on nouns modified by a possessive pronoun in Mush dialect (Acharyan 1911: 118), so this criterion does not apply to the examples in question. Based on the comment of Grigoryan (1957), we have glossed them as dative.

- (24) Dersim Armenian (Andranik 2024: 232)

gə-ts^hol-a *vər mēndz sar-i-n*
 PTCL-shine-PRS.3SG on big mountain-DAT-DEF
 ‘It shines on the big mountain.’

9.6 Discussion

The Neo-Aramaic dialects have largely remained unchanged with respect to their case marking typology (see also Haig et al. 2025). They did not develop new case declensions and the case-marking, wherever present, has remained largely prepositional. Several NENA dialects, however, developed dedicated object pronouns, which could be partly due to influence from Iranian (see §??). Moreover, the Kurdish-Gorani preposition *bo* ‘to, for, because of’ both directly and indirectly influenced the development of new dative case-markers (*bo-d*, *ba-* and *baqa*) in the Trans-Zab Jewish varieties of NENA. In the Iranian languages, the former case declensions have largely collapsed. There is considerable case syncretism in the Kurdish languages, since case morphology, where present, is mainly narrowed down to an opposition between ‘direct’ and ‘oblique’; the first is derived from an originally nominative case, and the latter is derived from a former genitive-dative case. Azeri and Armenian have preserved their original case morphology to a large extent. In Armenian, similarly to Iranian, there is a tendency towards syncretism of genitive and dative case, which are generally kept separate in Neo-Aramaic and Arabic. The opposition between an unmarked case and oblique case at times converges with NENA ‘direct’ and ‘oblique’ person markers used in verbal constructions, see Chapter ?? In Kurdish, however, case morphology is notably absent on the head noun of an *ezafe* construction. Unlike other related Neo-Iranian languages, such as Persian, Kurdish did not develop novel accusative case markers. In Neo-Aramaic, Eastern Armenian dialects, as well as Persian, a dative case marker was readapted to differentially mark salient objects, on which see further Chapter ?. In terms of adpositional marking, the Neo-Semitic languages in the region have never undergone a shift from prepositional to postpositional marking (see Haig et al. 2025 for a discussion). This can be contrasted with Armenian, which has largely shifted to postpositional marking, probably promoted by influence from Turkic. In Iranian, and also in Kurdish, the adpositional marking strategies have remained more mixed.

10 Demonstratives

10.1 Demonstratives in NENA

NENA demonstrative pronouns may be used independently or they may be used adnominally. Various configurations of systems of demonstratives are attested across the dialects. Dialects use demonstratives as deictic or anaphoric pronouns. Deictic pronouns point out a referent that is visible in the speech situation. Anaphoric pronouns mark a referent as recoverable from the preceding discourse or from the speech situation without pointing. An example of recoverability from the speech situation is the following:

- (1) C. Urmi (Khan 2016a: vol. 2, 14)
hal-li *'o-ctava*.
 give.IMP-OBL.1SG DEM.ANAPHORIC-book
 ‘Give me that book’ (said, without pointing, when the addressee is holding the book in their hand).’

Most dialects have distinct demonstratives for near and far deixis. In archaising dialects, such as those on the Mosul plain, the singular near deixis demonstrative contains the stem *ḏ*, which is the reflex of the stem of near deixis demonstrative pronouns **d-* in earlier Aramaic, e.g., Biblical Aramaic *d-anā* SG.M (Rosenthal 2006: 24), Geonic Babylonian Aramaic *d-ēn* (Bar-Asher Siegal 2016: 80). The far deixis singular demonstrative has a reflex of the morpheme **hū* (SG.M) / **hī* (SG.F), which is the historical form of the far deixis demonstrative and third person pronoun. These morphemes are augmented by additional deictic elements that have the historical form **hā*, e.g.,

- (2) C. Qaraqosh
- | | Near deixis | Far deixis |
|------|---------------------------------|----------------------------------|
| SG.M | <i>'aḏa</i> (< <i>*hā-d-ā</i>) | <i>'awa</i> (< <i>hā-hū-hā</i>) |
| SG.F | <i>'aḏi</i> (< <i>*hā-d-ī</i>) | <i>'aya</i> (< <i>hā-hī-hā</i>) |

Compare the earlier eastern variety of Aramaic Syriac: *hā-nā* (near deixis SG.M < **hā-dnā*), *haw* (far deixis SG.M < **hā-hū*) (Nöldeke 1904: 47).

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The plural form of demonstratives have the same stem (ʾan-) in the near deixis and the far deixis form, e.g.,

- (3) C. Qaraqosh
- | | | |
|------|-------------|------------|
| | Near deixis | Far deixis |
| PL.C | ʾana | ʾane |

The stem ʾan- of the plural demonstratives remains stable across the NENA dialects. There is, however, considerable diversity across the dialects in the form of the singular demonstratives.

The *d of the historical singular demonstrative morpheme has undergone lenition or other types of phonetic development in some dialects, e.g.,

- (4) C. Urmi
Near deixis (SG.M)
ʾaha (< *hā-dā)
- (5) C. Bədyal
Near deixis (SG.M)
ʾačča (< *hā-dā)

In many dialects in the northern sector of NENA, the stem of the singular far deixis form with the /w/ element has been extended to the singular near deixis form and replaced the historical stem of the near deixis form derived from *dā (SG.M) / *dī (SG.F), e.g.,

- (6) C. Barwar
- | | | |
|------|---------------------|-------------------------|
| | Near deixis | Far deixis (SG.M) |
| SG.M | ʾawwa (< *hā-hū-hā) | ʾāwaha (< *hā-hū-hā-hā) |
| SG.F | ʾayya (< *hā-hī-hā) | ʾāyaha (< *hā-hī-hā-hā) |

In a few dialects where this extension takes place the sequence /aw/ is contracted to /o/ in such near deixis forms, e.g.,

- (7) C. Derabun
- | | | |
|------|--------------------|-------------------------|
| | Near deixis | Far deixis |
| SG.M | ʾoha (< *hā-hū-hā) | ʾāwaha (< *hā-hū-hā-hā) |
| SG.F | ʾeha (< *hā-hī-hā) | ʾāyaha (< *hā-hī-hā-hā) |

Some dialects have a /i/ element in the stem of the singular near deixis demonstrative. This is found in the Jewish Trans-Zab dialects, e.g.,

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(8) J. Arbel

Near deixis (SG.C)	Far deixis (SG.C)
'iyya (< *hī (?) -hā)	'o (< *hāhū)

It is possible that the /i/ in the near demonstrative form 'iyya derives historically from the SG.F demonstrative stem *hī, which has been extended to both genders. Alternatively, it may be a deictic particle unrelated to this feminine stem. The /o/ morpheme of the far demonstrative form 'o clearly derives historically from the SG.M demonstrative stem *hū.

In some dialects a /y/ or /i/ element is combined with a near demonstrative that has a *d, e.g.,

(9) Near deixis (SG.C)

C. Alqosh	C. Ankawa
'ađ-i	'ad-i

This, likewise, is of common gender and may have arisen by the extension of the historical SG.F form to a common gender form (cf. the C. Barwar SG.F form 'ađi).

A number of dialects have the element /ay/ in the stem of the near demonstrative. The historical derivation of such demonstratives, however, is ambiguous. It may have derived from *hā + i or from a form *hādi in which the *d has been lenited, e.g.,

(10) Near deixis (SG.C)

C. Telkepe	C. Tisqopa	C. Sulemaniyya
'ayi	'ay	'aya

The /i/ element is used in some dialects on the Mosul plain as a focus augment to deixis demonstratives to express intensified deixis, e.g.,

(11) C. Qaraqosh

Near deixis (SG.M)	Far deixis (SG.M)
'i-'ađa	'i-'awa
'this very one'	'that very one'

This /i/ is also used in deictic adverbs in these dialects, e.g.,

(12) C. Qaraqosh

'axa	'i-'axa
'here'	'right here'

10 Demonstratives

Near and far deixis demonstratives can be used adnominally. In some dialects the form of the adnominal demonstrative is the same as that of the independent demonstrative, e.g.,

(13) C. Qaraqosh

	Independent	Adnominal
SG.M	'aḏa 'this (M)'	'aḏa gora 'this man'
SG.F	'aḏi 'this' (F)	'aḏi baxta 'this woman'
SG.M	'awa 'that' (M)	'awa gora 'that man'
SG.F	'aya 'that' (F)	'aya baxta 'that woman'
PL.C	'anə 'these'	'anə kθawə 'these books'
PL.C	'ane 'those'	'ane kθawə 'those books'

In some dialects, adnominal forms of demonstratives are shortened by removing the final vowel of the corresponding independent form, sometimes resulting in the contraction of a diphthong, e.g.,

(14) C. Sulemaniyya

	Independent	Adnominal
SG.C	'aya 'this (M)'	'ay gora 'this man'
SG.C	'aya 'this' (F)	'ay baxta 'this woman'
SG.M	'awa 'that' (M)	'o gora 'that man'
SG.F	'oya 'that' (F)	'o baxta 'that woman'
PL.C	'ani 'these'	'an ksawane 'these books'
PL.C	'oni 'those'	'on ksawane 'those books'

Some dialects have a distinct pronoun to express very far deixis. In some dialects a very far deixis pronoun is formed by a morphological augment on the far deixis form, e.g.,

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- (15) J. Betanure (Mutzafi 2008a: 42)
- | | |
|-------------------|------------------------|
| Far deixis (SG.M) | Very far deixis (SG.M) |
| 'ăwaha | 'awa'ha |

In some dialects the far deixis form is a modified form of the far deixis pronoun. The modification typically takes the form of the glottalisation of the stressed vowel, e.g.,

- (16) C. Barwar
- | | |
|-------------------|------------------------|
| Far deixis (SG.M) | Very far deixis (SG.M) |
| 'ăwaha | 'awa'ha |

This increases the muscular tension of the articulation, which iconically represents distance. An example of the use of a very far deixis pronoun is as follows:

- (17) C. Barwar (Khan 2008c: 469)
- | | | | |
|-------------------|-----------------|-------------------------------------|-------------------------------|
| <i>mâr-e</i> | <i>kê=le</i> | <i>béθa díya?</i> | <i>... mâr-a</i> |
| say.PST-OBL.3SG.M | where=COP.3SG.M | house ATTR.3SG.F | ... say-PST.OBL.3PL |
| 'ăwá'ha | <i>qáşra</i> | <i>ṭ-it-xzàyə-lle.</i> | |
| that.VERY.FAR | palace | SBRD-COP.2SG.M-see.PROG-OBL.3SG.M | |
- 'He said "Where is her house?" ... They said "That villa that you can see over there."'

In some dialects an original /h/ is strengthened to the pharyngeal fricative /ħ/ in very far deixis forms, e.g.,

- (18) C. Qarawilla
- | | |
|-------------------|------------------------|
| Far deixis (SG.M) | Very far deixis (SG.M) |
| 'ăwaha | 'ăwaħa |

The two features may be combined in the very far deixis form, e.g.,

- (19) C. Haşşan
- | | |
|-------------------|------------------------|
| Far deixis (SG.M) | Very far deixis (SG.M) |
| 'ăwaha | 'ăwa'ħa |

In some dialects the pharyngeal fricative is found also in far deixis pronouns, e.g.,

- (20) C. Peshabur
- | | |
|-------------------|------------------------|
| Far deixis (SG.M) | Very far deixis (SG.M) |
| ăwaħa | 'ăwa'ħa |

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In most dialects speakers can express far deixis by extension of the duration of the stressed vowel of the far deixis form (e.g., C. Shaqlawa *ʾawáha* > *ʾawáaha*). Only the glottalised forms (e.g., *ʾawaʾha*) are classified in the taxonomy below as distinct morphological forms.

All dialects use near and far deictic pronouns both exophorically, pointing out referents in the speech situation, and endophorically with anaphoric function within discourse. Many dialects have, in addition, a pronoun that is only used anaphorically. Those dialects that have an anaphoric pronoun differ as to whether the anaphoric pronoun is used both independently and adnominally or whether it is used only independently. When there is no dedicated anaphoric pronoun, the far deixis pronoun is most frequently used to express the anaphoric function. Within discourse there are differences between the anaphoric function of near, far and dedicated anaphoric pronouns. Broadly speaking, the distinction relates to differing degrees of discourse salience, in the following rising scale:

(21)

Type of pronoun	Salience
dedicated anaphoric pronoun	least salient
far deixis pronoun	
near deixis pronoun	most salient

Examples illustrating the endophoric uses of these various pronouns are as follows.

- (22) C. Qaraqosh (Khan 2002a: 628)
zál-lə *ʾàhu*[|] *ʾu-rkú-lə* *sústa*
go.PST-OBL.3SG.M PRO.ANAPH.SG.M and-mount.PST-OBL.3SG.M mare
dìd-əh.[|]
ATTR-3SG.M
'He went and mounted his mare.'
- (23) C. Qaraqosh (Khan 2002a: 650)
ʾáwa *kíwa* *kəm-táʾən-ne* *laššətt-əh.*[|]
DEIC.DIST.SG.M COP.PST.3SG.M PAR-raise.PRS-OBL.3SG.M corpse-3SG.M
'He had raised his corpse.'
- (24) C. Qaraqosh (Khan 2002a: 638)
w-áða *hár kil-kə-mşàyəh.*[|]
and-DEM.PROX.SG.M still COP.3SG.M-PAR-cry.PRS.3SG.M
'He is crying out'

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In (22), a dedicated anaphoric pronoun is used in a clause in narrative that continues the subject referent of the preceding clause and the event is not presented as prominent. Such a non-salient independent anaphoric pronoun is equivalent to a third person pronoun. The far deixis pronoun is used in (23) in a clause in which the subject has some degree of salience since it marks a change of subject referent from that of the preceding clause, but the clause is not a salient event in the discourse but is an explanatory statement in the background of the narrative. In (24) the near deixis pronoun is used to shift subject reference to a prominent participant in the narrative. The motivation for using the maximally salient pronoun is the prominence of the participant, the prominence of the activity in the narrative, and to draw attention to a shift in reference.

In some dialects a demonstrative pronoun of a particular type has a basic form and variant forms constructed by adding augmented syllables, especially in the independent forms. In C. Mawana, for example, the basic SG.M independent near deixis and far deixis forms are *wa* and *wə* respectively. The dialect also uses forms with the augments *-ha* and *-hane*:

- (25) Near deixis: *wa wăha wăhane*
Far deixis: *wə wəha wəhane*

In all dialects adnominal demonstratives are placed before the nominal, e.g.,

- (26) C. Barwar (Khan 2008c)
'awwa *naša* 'this man'
'ayya *baxta* 'this woman'
'anna *naše* 'these people'

A few dialects use a modified form of the anaphoric pronoun to mark a referent that the speaker assumes is accessible to the hearer not in the immediate discourse history but rather in the hearer's memory. In C. Barwar, for example, the anaphoric pronoun 'o (SG.M) is combined with the indefinite article *xa* (derived from the cardinal numeral 'one') to express this assumed accessibility in memory, e.g.,

- (27) C. Barwar (Khan 2008a: 486)
'ana 'o-xà qāša | t-ín 'əθya gu-Ṭiyàre, |
I DEM-INDF priest REL-COP.1SG.M come.RES.PTCP in-Ṭiyare
qəm-doq-át-li xoš-át-li gu-gòma. |
PAR-seize.PRS-2SG.M-OBL.1SG confine.PRS-2SG.M-OBL.1SG in-stable
'I am the priest who (you remember) came to Ṭiyare, whom you seized and confined in the stable.

10 Demonstratives

In most NENA dialects singular demonstratives are inflected for gender (masculine and feminine) whereas plural demonstratives are of common gender. In the Jewish Trans-Zab dialects, however, singular demonstratives are not inflected for gender (Mutzafi 2008c). Rather the historical masculine form has come to be used for common gender. This has arisen by convergence with the model of contact languages in the region, in particular Kurdish, in which singular demonstratives do not express gender distinctions.

In NENA dialects in the northern and western sectors, oblique forms of demonstrative pronouns have developed by the bonding of the particle **d-* to the front of the pronoun. This particle marks the dependence of a phrase to a head noun in an attributive (typically genitive) relationship, e.g.,

- (28) C. Barwar (Khan 2008a: 397)
čadra do-malka
tent ATTR.DEM.SG.M-king
'the tent of that king'

The formation and morphosyntax of these oblique demonstratives in NENA dialects are discussed in the chapter on the morphosyntax of nominal modification (§11).

In what follows the various demonstrative systems of the NENA dialects are classified according to the number of sets of demonstratives and their function. Only the SG.M non-oblique forms are presented. For a map of the distribution of these systems see Figure 10.1.

10.1.1 Two sets of Demonstratives (Near and Far Deixis) without a Dedicated Anaphoric Pronoun

A wide range of dialects, both Christian and Jewish, have only a near deixis and far deixis pronoun. The function of anaphoric pronoun is typically performed by the far deixis pronoun.

10.1.1.1 Christian Dialects

- (29) C. Armota/Koy Sanjaq (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	'aya	'aya, 'e
Far deixis (SG.M)	'awa	'awa
Anaphoric (SG.M)	'awa	'awa

10.1 Demonstratives in NENA

(30) C. Sulemaniyya (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	^ʔ <i>aya</i>	^ʔ <i>ay</i>
Far deixis (SG.M)	^ʔ <i>awa</i>	^ʔ <i>o</i>
Anaphoric (SG.M)	^ʔ <i>awa</i>	^ʔ <i>o</i>

(31) C. Bēšpən (Cudi, SE Turkey) (Sinha 2000)

	Independent	Adnominal
Near deixis (SG.M)	^ʔ <i>awaha</i>	^ʔ <i>awá</i>
Far deixis (SG.M)	^ʔ <i>awa</i>	^ʔ <i>aw</i>
Anaphoric (SG.M)	^ʔ <i>awa</i>	^ʔ <i>aw</i>

(32) C. Jilu (Fox 1997)

	Independent	Adnominal
Near deixis (SG.M)	^ʔ <i>aha</i>	^ʔ <i>aha</i>
Far deixis (SG.M)	^ʔ <i>aw</i>	^ʔ <i>aw</i>
Anaphoric (SG.M)	^ʔ <i>aw</i>	^ʔ <i>aw</i>

(33) C. Txuma (Jacobi 1973)

	Independent	Adnominal
Near deixis (SG.M)	^ʔ <i>awa</i> , ^ʔ <i>aha</i>	^ʔ <i>awa</i> , ^ʔ <i>aha</i>
Far deixis (SG.M)	^ʔ <i>aw</i>	^ʔ <i>aw</i>
Anaphoric (SG.M)	^ʔ <i>aw</i>	^ʔ <i>aw</i>

10.1.1.2 Jewish Trans-Zab Dialects

(34) J. Arbel (Khan 1999a)

	Independent	Adnominal
Near deixis (SG.M)	^ʔ <i>yya</i>	^ʔ <i>yya</i>
Far deixis (SG.M)	^ʔ <i>o</i>	^ʔ <i>o</i>
Anaphoric (SG.M)	^ʔ <i>o</i>	^ʔ <i>o</i>

(35) J. Urmi (Khan 2008b)

10 Demonstratives

	Independent	Adnominal
Near deixis (SG.M)	<i>ya</i>	<i>ya</i>
Far deixis (SG.M)	<i>’o</i>	<i>’o</i>
Anaphoric (SG.M)	<i>’o</i>	<i>’o</i>

(36) J. Sanandaj (Khan 2009)

	Independent	Adnominal
Near deixis (SG.M)	<i>’ay, ’ea, ’exa</i>	<i>’ay</i>
Far deixis (SG.M)	<i>’o, ’oa, ’oxa</i>	<i>’o</i>
Anaphoric (SG.M)	<i>’o, ’oa, ’oxa</i>	<i>’o</i>

(37) J. Barzan (Mutzafi 2002a)

	Independent	Adnominal
Near deixis (SG.M)	<i>’iya</i>	<i>’iya</i>
Far deixis (SG.M)	<i>’o, yowa</i>	<i>’o, yowa</i>
Anaphoric (SG.M)	<i>’o, yowa</i>	<i>’o, yowa</i>

(38) J. Aradhin (Mutzafi 2002b)

	Independent	Adnominal
Near deixis (SG.M)	<i>’iya</i>	<i>’iya</i>
Far deixis (SG.M)	<i>’o, yowa</i>	<i>’o, yowa</i>
Anaphoric (SG.M)	<i>’o, yowa</i>	<i>’o, yowa</i>

10.1.2 Three Sets of Demonstratives (Near, Far and Very Far Deixis) without a Dedicated Anaphoric Pronoun

In some dialects in the northern sector of NENA, there are three sets of distinct pronouns for near, far and very far deixis but no dedicated anaphoric pronoun. The function of anaphoric pronoun is typically performed by the far deixis pronoun.

10.1.2.1 Christian Dialects

(39) C. Baz Maha Xtaya (Mutzafi 2000)

10.1 Demonstratives in NENA

	Independent	Adnominal
Near deixis (SG.M)	<i>’aha</i>	<i>’aha</i>
Far deixis (SG.M)	<i>’aw</i>	<i>’aw</i>
Very far deixis (SG.M)	<i>’āwaha, ’awa’ha</i>	<i>’āwaha, ’awa’ha</i>
Anaphoric (SG.M)	<i>’aw</i>	<i>’aw</i>

(40) C. Jønnet (Noorlander, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>’oha</i>	<i>’oha</i>
Far deixis (SG.M)	<i>’owa</i>	<i>’oha</i>
Very far deixis (SG.M)	<i>woḥa</i>	<i>woḥa</i>
Anaphoric (SG.M)	<i>’owa</i>	<i>’o</i>

10.1.2.2 Jewish *Lišana Deni* Dialects(41) J. Betanure ([Mutzafi 2008a](#): 42–43)

	Independent	Adnominal
Near deixis (SG.M)	<i>’ayya, ’oha</i>	<i>’ayya</i>
Far deixis (SG.M)	<i>’awa, ’oha</i>	<i>’awa, ’o</i>
Very far deixis (SG.M)	<i>’āwaha</i>	<i>’āwaha</i>
Anaphoric (SG.M)	<i>’awa</i>	<i>’awa, ’o</i>

Note that the alternative deictic forms *’oha* and *’o* do not make a distinction between near and far deixis.

(42) J. Dohok ([Molin 2024](#))

	Independent	Adnominal
Near deixis (SG.M)	<i>’oha</i>	<i>’o</i>
Far deixis (SG.M)	<i>’awa</i>	<i>’awa, ’aw</i>
Very far deixis (SG.M)	<i>’āwaha</i>	<i>’āwaha</i>
Anaphoric (SG.M)	<i>’awa</i>	<i>’awa, ’aw</i>

(43) J. Challa ([Fassberg 2010](#))

	Independent	Adnominal
Near deixis (SG.M)	<i>’iya, ya</i>	<i>’iya, ya</i>
Far deixis (SG.M)	<i>’aya, ’e, ’a</i>	<i>’aya, ’e, ’a</i>
Very far deixis (SG.M)	<i>’āyaha</i>	<i>’āyaha</i>
Anaphoric (SG.M)	<i>’aya, ’e, ’a</i>	<i>’aya, ’e, ’a</i>

10 Demonstratives

10.1.3 Three sets of Independent Demonstratives (Near and Far Deixis, Anaphoric) and Two Sets of Adnominal Demonstratives (Near and Far Deixis)

Dialects on the Mosul plain in the western sector of NENA and in adjacent areas exhibit a system that has a dedicated anaphoric independent pronoun in addition to near and far deixis pronouns, but not an anaphoric adnominal pronoun. In such dialects the function of an adnominal anaphoric is typically performed by the far deixis adnominal pronoun. The independent far deixis pronoun, moreover, is frequently used as an anaphoric. When used in this way, it typically expresses more discourse salience than the anaphoric independent pronoun (for further details see [Khan 2002a](#): 252–270). As in all dialects, a high degree of discourse salience is expressed by using the near deixis form anaphorically.

(44) C. Qaraqosh ([Khan 2002a](#))

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔaḏa</i>	<i>ʔaḏa</i>
Far deixis (SG.M)	<i>ʔawa</i>	<i>ʔawa</i>
Anaphoric (SG.M)	<i>ʔahu</i>	<i>ʔawa</i>

(45) C. Telkepe ([Coghill 2018](#))

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔaḏi</i>	<i>ʔaθ</i>
Far deixis (SG.M)	<i>ʔawɒ</i>	<i>ʔawɒ, ʔo</i>
Anaphoric (SG.M)	<i>ʔawu</i>	<i>ʔawɒ, ʔo</i>

(46) C. Alqosh ([Coghill 2003](#))

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔaḏi</i>	<i>ʔaθ</i>
Far deixis (SG.M)	<i>ʔawa</i>	<i>ʔo</i>
Anaphoric (SG.M)	<i>ʔaw</i>	<i>ʔo</i>

(47) C. Karamlesh (Borghero, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔaḏa</i>	<i>ʔaḏa</i>
Far deixis (SG.M)	<i>ʔawa</i>	<i>ʔawa</i>
Anaphoric (SG.M)	<i>ʔawun</i>	<i>ʔawa</i>

10.1 Demonstratives in NENA

(48) C. Ankawa (Borghero, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔadi</i>	<i>ʔadi</i>
Far deixis (SG.M)	<i>ʔawa</i>	<i>ʔawa</i>
Anaphoric (SG.M)	<i>ʔawun</i>	<i>ʔawa</i>

(49) C. Bədyal (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔačča</i>	<i>ʔačča</i>
Far deixis (SG.M)	<i>ʔoxa</i>	<i>ʔo</i>
Anaphoric (SG.M)	<i>ʔawun</i>	<i>ʔo</i>

(50) C. Hertevin (Jastrow 1988b)

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔohá</i>	<i>ʔad</i>
Far deixis (SG.M)	<i>ʔawa</i>	<i>ʔo</i>
Anaphoric (SG.M)	<i>ʔahu</i>	<i>ʔo</i>

(51) C. Bohtan (Fox 2009b)

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔawa</i>	<i>ʔaw</i>
Far deixis (SG.M)	<i>ʔowa</i>	<i>ʔo</i>
Anaphoric (SG.M)	<i>ʔoyən</i>	<i>ʔo</i>

10.1.4 Three Sets of Demonstratives (Near and Far Deixis and Anaphoric)

Many dialects in the northern sector of NENA have three sets of demonstratives (near, far, anaphoric) that include both independent and adnominal forms of an anaphoric pronoun.

(52) C. Aradhin (Krotkoff 1982; Borghero, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔawwa</i>	<i>ʔawwa</i>
Far deixis (SG.M)	<i>ʔāwaʔha</i>	<i>ʔāwaʔha</i>
Anaphoric (SG.M)	<i>ʔawa</i>	<i>ʔaw</i>

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(53) C. Billin (Borghero, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	'awwa	'awwa
Far deixis (SG.M)	'ăwaha	'ăwaha
Anaphoric (SG.M)	'awa	'awa

(54) C. Umra d-Shish (Borghero, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	'awwa	'awwa
Far deixis (SG.M)	'ăwaha	'ăwaha
Anaphoric (SG.M)	'awa	'awa, 'o

(55) C. Umra (Hobrack 2000)

	Independent	Adnominal
Near deixis (SG.M)	'uha	'uha
Far deixis (SG.M)	'awa	'awa
Anaphoric (SG.M)	'awən	'aw

(56) C. Zakho (Mole 2000)

	Independent	Adnominal
Near deixis (SG.M)	'awwá	'awwá
Far deixis (SG.M)	'ăwaha	'ăwaha
Anaphoric (SG.M)	'awa	'aw

(57) C. Shaqlawa (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	'ăya	'e
Far deixis (SG.M)	'ăwaha	'ăwaha
Anaphoric (SG.M)	'awa	'aw

(58) C. Mawana (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	wa, wăha, wăhane	wa
Far deixis (SG.M)	wo, woha, wôhane	wo
Anaphoric (SG.M)	'aw, 'awən	'o

10.1 Demonstratives in NENA

(59) C. Tazacand (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>wa, wǎhane</i>	<i>wa</i>
Far deixis (SG.M)	<i>wo, wóhane</i>	<i>wo</i>
Anaphoric (SG.M)	<i>ʔaw, ʔawən</i>	<i>ʔo</i>

(60) C. Darband (Coghill, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔewá, ʔewǎhane</i>	<i>ʔewá</i>
Far deixis (SG.M)	<i>ʔǎwóhane</i>	<i>ʔǎwó</i>
Anaphoric (SG.M)	<i>ʔawən</i>	<i>ʔo</i>

(61) C. Darbandoke (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔǎwa</i>	<i>ʔǎwa</i>
Far deixis (SG.M)	<i>ʔǎwó</i>	<i>ʔǎwó</i>
Anaphoric (SG.M)	<i>ʔawən</i>	<i>ʔo</i>

(62) C. Dawidiya (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔǎwa</i>	<i>ʔǎwa</i>
Far deixis (SG.M)	<i>ʔǎwaha</i>	<i>ʔǎwaha</i>
Anaphoric (SG.M)	<i>ʔawa</i>	<i>ʔawa</i>

(63) C. Bebede (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔawwa</i>	<i>ʔawwa</i>
Far deixis (SG.M)	<i>ʔǎwaha</i>	<i>ʔǎwaha</i>
Anaphoric (SG.M)	<i>ʔaw</i>	<i>ʔo</i>

(64) C. Dere (Borghero, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔawwa</i>	<i>ʔawwa</i>
Far deixis (SG.M)	<i>ʔǎwaha</i>	<i>ʔǎwaha</i>
Anaphoric (SG.M)	<i>ʔawa</i>	<i>ʔaw</i>

10 Demonstratives

(65) C. Komane (Borghero, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	'awwa	'awwa
Far deixis (SG.M)	'āwaha	'āwaha
Anaphoric (SG.M)	'awa	'aw

(66) C. Deragne (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	'awwa	'awwa
Far deixis (SG.M)	'āwaha	'āwaha
Anaphoric (SG.M)	'awa	'aw

(67) C. Nerwa (Talay 2001)

	Independent	Adnominal
Near deixis (SG.M)	'awwa	'awwa
Far deixis (SG.M)	'awwaxa	'awwaxa
Anaphoric (SG.M)	'awu	'aw

(68) C. Harmashe (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	'awwa	'awwa
Far deixis (SG.M)	'awwa'ha	'awwa'ha
Anaphoric (SG.M)	'aw	'aw

(69) C. Shiyuz (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	'awwa	'oha
Far deixis (SG.M)	'awa	'awa
Anaphoric (SG.M)	'awa	'o

(70) C. Derabun (Borghero, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	'oha	'oha
Far deixis (SG.M)	'āwaha	'āwaha
Anaphoric (SG.M)	'awa	'o

10.1 Demonstratives in NENA

(71) C. Mangesh (Coghill, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>’oho</i>	<i>’oho</i>
Far deixis (SG.M)	<i>’ǎwaha</i>	<i>’ǎwaha</i>
Anaphoric (SG.M)	<i>’awa</i>	<i>’awa</i>

(72) C. Marga (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>’awwa</i>	<i>’awwa</i>
Far deixis (SG.M)	<i>’ǎwaha</i>	<i>’ǎwaha</i>
Anaphoric (SG.M)	<i>’awa</i>	<i>’aw</i>

(73) C. Meze (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>’awwa</i>	<i>’awwa</i>
Far deixis (SG.M)	<i>’ǎwaha</i>	<i>’ǎwaha</i>
Anaphoric (SG.M)	<i>’awa</i>	<i>’aw</i>

(74) C. Jilu (Khabur) (Talay 2008a)

	Independent	Adnominal
Near deixis (SG.M)	<i>’aha</i>	<i>’aha</i>
Far deixis (SG.M)	<i>’ohó</i>	<i>’ohó</i>
Anaphoric (SG.M)	<i>’aw</i>	<i>’aw</i>

(75) C. Garganaye (Borghero, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>’ǎwá</i>	<i>’ǎwá</i>
Far deixis (SG.M)	<i>’ǎwó</i>	<i>’ǎwó</i>
Anaphoric (SG.M)	<i>’awən</i>	<i>’o</i>

(76) C. Sat (Mutzafi 2008b)

	Independent	Adnominal
Near deixis (SG.M)	<i>da</i>	<i>da</i>
Far deixis (SG.M)	<i>wó</i>	<i>wó</i>
Anaphoric (SG.M)	<i>’awən</i>	<i>’o</i>

10 Demonstratives

(77) C. Harbole (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>'uwa</i>	<i>'uwa</i>
Far deixis (SG.M)	<i>wahá</i>	<i>wahá</i>
Anaphoric (SG.M)	<i>'awa</i>	<i>'aw</i>

(78) C. Salamas (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>'aha</i>	<i>'aha</i>
Far deixis (SG.M)	<i>'ová</i>	<i>'ová</i>
Anaphoric (SG.M)	<i>'avən</i>	<i>'o</i>

(79) C. Van (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>'awwa</i>	<i>'awwa</i>
Far deixis (SG.M)	<i>'awwá</i>	<i>'awwá</i>
Anaphoric (SG.M)	<i>'av</i>	<i>'o</i>

(80) C. Baz Aruntus (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>'aha</i>	<i>'aha</i>
Far deixis (SG.M)	<i>'a'ha, 'ǎwaha</i>	<i>'a'ha, 'ǎwaha</i>
Anaphoric (SG.M)	<i>'aw</i>	<i>'o</i>

(81) C. Baz Argab (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>'aha</i>	<i>'aha, 'a</i>
Far deixis (SG.M)	<i>'ayó, 'ayóha</i>	<i>'ayó, 'ayóha</i>
Anaphoric (SG.M)	<i>'aw, 'ahu</i>	<i>'aw</i>

(82) C. Baz Bi-Sillem (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>'aha</i>	<i>'aha, 'a</i>
Far deixis (SG.M)	<i>'ǎwá</i>	<i>'ǎwá</i>
Anaphoric (SG.M)	<i>'aw</i>	<i>'aw</i>

10.1 Demonstratives in NENA

(83) C. Xərpa (Coghill, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	^ʔ awwa	^ʔ awwa
Far deixis (SG.M)	^ʔ āwa ^ʔ a	^ʔ āwa ^ʔ a
Anaphoric (SG.M)	^ʔ awwən	^ʔ aw

(84) C. Shosh u-Sharmən (Coghill, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	^ʔ awwa	^ʔ awwa
Far deixis (SG.M)	^ʔ āwaha	^ʔ āwaha
Anaphoric (SG.M)	^ʔ awwən	^ʔ aw

10.1.5 Four Sets of Demonstratives (Near, Far, Very Far Deixis and Anaphoric)

Many dialects in the northern sector of NENA have four sets of demonstratives (near, far, very far, anaphoric) that include both independent and adnominal forms of an anaphoric pronoun.

10.1.5.1 Christian Dialects

(85) C. Barwar (Khan 2008a)

	Independent	Adnominal
Near deixis (SG.M)	^ʔ awwa	^ʔ awwa
Far deixis (SG.M)	^ʔ āwaha	^ʔ āwaha
Very far deixis (SG.M)	^ʔ āwa ^ʔ ha	^ʔ āwa ^ʔ ha
Anaphoric (SG.M)	^ʔ aw	^ʔ aw

(86) C. Tən (Coghill 2025b)

	Independent	Adnominal
Near deixis (SG.M)	^ʔ awwa	^ʔ awwa
Far deixis (SG.M)	^ʔ āwaha	^ʔ āwaha
Very far deixis (SG.M)	^ʔ āwa ^ʔ ha	^ʔ āwa ^ʔ ha
Anaphoric (SG.M)	^ʔ awa	^ʔ awa

(87) C. Rekan (Khan, fieldnotes)

10 Demonstratives

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔaha</i>	<i>ʔaha</i>
Far deixis (SG.M)	<i>ʔǎwá</i>	<i>ʔǎwá</i>
Very far deixis (SG.M)	<i>ʔǎwaʔha</i>	<i>ʔǎwaʔha</i>
Anaphoric (SG.M)	<i>ʔaw</i>	<i>ʔo</i>

(88) C. Baz (Shawwa) (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔaha</i>	<i>ʔaha, ʔa</i>
Far deixis (SG.M)	<i>ʔǎwá</i>	<i>ʔǎwá</i>
Very far deixis (SG.M)	<i>ʔǎwaʔha</i>	<i>ʔǎwaʔha</i>
Anaphoric (SG.M)	<i>ʔaw</i>	<i>ʔaw</i>

(89) C. Qarawilla (Khan, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔawwa</i>	<i>ʔawwa</i>
Far deixis (SG.M)	<i>ʔǎwaha</i>	<i>ʔǎwaha</i>
Very far deixis (SG.M)	<i>ʔǎwaḥa</i>	<i>ʔǎwaḥa</i>
Anaphoric (SG.M)	<i>ʔaw</i>	<i>ʔaw</i>

(90) C. Urmi (Khan 2016a)

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔaha</i>	<i>ʔaha</i>
Far deixis (SG.M)	<i>+ʔavva</i>	<i>+ʔavva</i>
Very far deixis (SG.M)	<i>+ʔavvó</i>	<i>+ʔavvó</i>
Anaphoric (SG.M)	<i>+ʔav, +ʔavən</i>	<i>ʔo</i>

(91) C. Zariwaw (Napiorkowska 2015)

	Independent	Adnominal
Near deixis (SG.M)	<i>ʔawwa</i>	<i>ʔawwa</i>
Far deixis (SG.M)	<i>ʔǎwó</i>	<i>ʔǎwó</i>
Very far deixis (SG.M)	<i>ʔǎwóha</i>	<i>ʔǎwóha</i>
Anaphoric (SG.M)	<i>ʔawən, ʔo</i>	<i>ʔo</i>

(92) C. Haşsan (Damsma, fieldnotes)

10.1 Demonstratives in NENA

	Independent	Adnominal
Near deixis (SG.M)	'oha	'oha
Far deixis (SG.M)	'āwaha	'āwaha
Very far deixis (SG.M)	'āwa'ha	'āwa'ha
Anaphoric (SG.M)	'awa	'o

(93) C. Peshhabur (Coghill 2013)

	Independent	Adnominal
Near deixis (SG.M)	'awwa	'awwa
Far deixis (SG.M)	'āwaha	'āwaha
Very far deixis (SG.M)	'āwa'ha	'āwa'ha
Anaphoric (SG.M)	'awa	'aw

(94) C. Nargəzine-Kharjawa (Coghill, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	'awwa	'awwa
Far deixis (SG.M)	'āwaha	'āwaha
Very far deixis (SG.M)	'āwa'ha	'āwa'ha
Anaphoric (SG.M)	'awwən	'aw

(95) J. Amedia (Greenblatt 2011)

	Independent	Adnominal
Near deixis (SG.M)	'ayya	'ayya
Far deixis (SG.M)	'āwaha	'āwaha
Very far deixis (SG.M)	'āwa'ha	'āwa'ha
Anaphoric (SG.M)	'awa, 'ahu, 'o	'awa, 'o

(96) J. Nerwa (Mutzafi, fieldnotes)

	Independent	Adnominal
Near deixis (SG.M)	'iyya	'iyya
Far deixis (SG.M)	'awa	'awa
Very far deixis (SG.M)	'āwaha	'āwaha
Anaphoric (SG.M)	'awa	'e

(97) J. Zakho

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	Independent	Adnominal
Near deixis (SG.M)	^ʔ o	^ʔ o
Far deixis (SG.M)	^ʔ oha	^ʔ oha
Very far deixis (SG.M)	^ʔ awaha	^ʔ awaha
Anaphoric (SG.M)	^ʔ aw	^ʔ aw

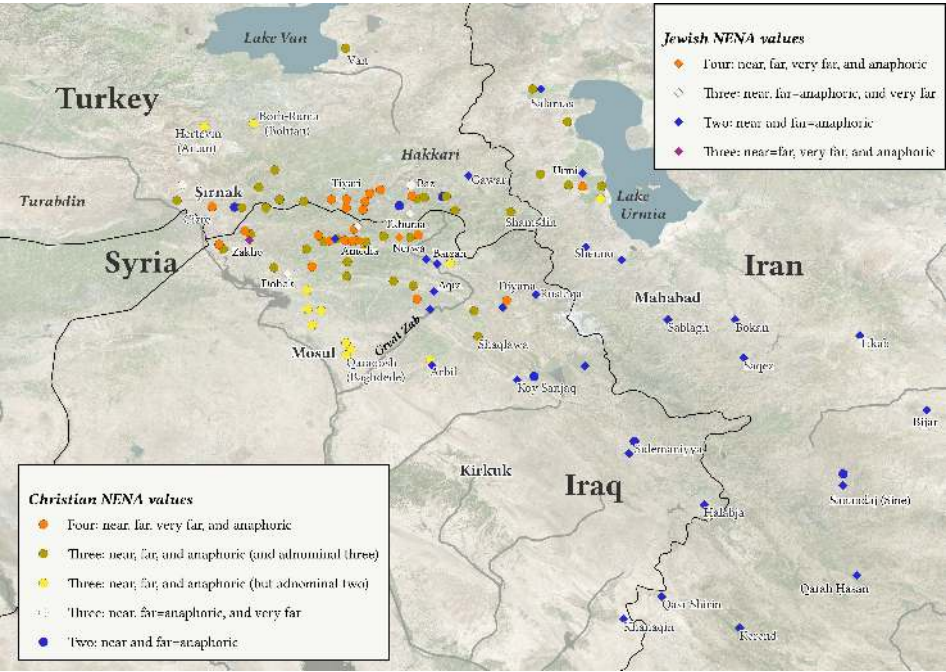


Figure 10.1: Distribution of demonstrative systems across NENA dialects

10.2 Demonstratives in Iranian

Within Iranian languages of the region, anaphoric pronouns tend to be derived from demonstrative pronouns. This is the case, for example, in major varieties of Kurdish. In central and southern varieties of CK, the anaphoric set in the singular lacks the demonstrative clitic =e. The demonstrative set should be interpreted as being a combination of the demonstrative determiners *em* (PIr **ima*-) and *ew* (PIr **awa*-) with the demonstrative clitic =e (of uncertain origin), e.g., *eme* ‘this’ *em ker*=e ‘this donkey’.

10.2 Demonstratives in Iranian

Table 10.1: Anaphoric and demonstrative pronouns in central and southern varieties of CK

Anaphoric pronouns			Demonstrative pronouns	
	PROXIMATE	REMOTE	PROXIMATE	REMOTE
SG	<i>em</i>	<i>ew</i>	<i>eme</i>	<i>ewe</i>
PL	<i>eman</i>	<i>ewan</i>	<i>emane</i>	<i>ewane</i>

In the northernmost Central Kurdish varieties, i.e., Shaqlawa, Erbil, Mukri, the proximate and remote forms have merged into the form *ew(e)* due to the phonological merger of /v/ with /w/, eliminating the deictic distinction. In contrast to the other CK varieties, the anaphoric and demonstrative set inflect for case (dir: -Ø; obl. -î) in the singular number.

Table 10.2: Anaphoric and demonstrative pronouns in northernmost varieties of CK

	Anaphoric			Demonstrative		
	DIR.SG	OBL.SG	PL	DIR.SG	OBL.SG	PL
PROXIMATE	<i>ew</i>	<i>(e)wî</i>	<i>(e)wan</i>	<i>ewe</i>	<i>(e)we-y</i>	<i>(e)wan</i>
REMOTE						

In the neighbouring Behdîni NK varieties (and generally in NK as a whole), the anaphoric and demonstrative pronouns tend to merge into one set. The pronoun set is unmarked for number and gender in the direct case, but these features are marked in the oblique case.

Table 10.3: Demonstrative pronouns in NK

PROXIMATE				REMOTE		
	M	F	PL	M	F	PL
DIR	<i>ev</i>			<i>ew</i>		
OBL	<i>(e)vî</i>	<i>(e)vê</i>	<i>van</i>	<i>(e)wî</i>	<i>(e)wê</i>	<i>(e)wan</i>

The Behdîni dialects of Duhok and further north in Hakkari have a three-way distinction in independent and adnominal demonstratives. The far deixis

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demonstratives may optionally be augmented by the particles *he*, *henê*, which are connected to the demonstrative through *ezafe* linkers.

- (98) Behdîni NK
ewê he ‘that (one) over there’
ew mêrikê he ‘that man yonder’
ew mêrikêt henê ‘those men yonder’

The masculine forms of demonstratives are presented in 10.4.

Table 10.4: Demonstrative pronouns in Behdîni, the masculine direct forms

	Anaphoric	Proximal	Distal
Independent	<i>ew/eve</i>	<i>eve</i>	<i>ew</i> <i>ewê he</i> <i>ewê henê</i>
Adnominal	<i>ew</i>	<i>ev ... e</i>	<i>ew</i> <i>ew ... ê he</i> <i>ew ... ê henê</i>

The anaphoric pronoun refers back to an element either already introduced in the preceding discourse context or a referent that is identifiable without being pointed out due to its salience in the speech context, as in (99):

- (99) Behdîni NK
ew ya min e
3SG.DIR EZ.SG.F 1SG.OBL COP.3SG
‘It is mine’ (said, for example, when referring to a book that the interlocutor is holding in their hand)

Sometimes a proximal demonstrative pronoun *eve* can also be used anaphorically when the referent is salient in the discourse, e.g.,

- (100) Behdîni NK
ev=e di-hat-in=e bajêr-ê Amêdiy-ê.
DEM.PROX=DEM IPFV-come.PST-3SG=to city-SG.M.OBL Amêdî-SG.F.OBL
‘These would come to Amadiya city’ (referring to the members of a number of tribes from surrounding regions that have been mentioned previously)

10.2 Demonstratives in Iranian

The proximal is generally marked with the marker *-e* (101):

(101) Behdîni NK

- a. *ev zelam=e mirov-ê me ye.*
 DEM.PROX man.SG.M.DIR=DEM person-EZ.SG.M 1PL.OBL COP.3SG
 ‘This man is our relative.’
- b. *eve ya min e.*
 DEM.PROX EZ.SG.F 1SG.OBL COP.3SG
 ‘This is mine.’

In the Duhok-Hakkari dialects, however, the proximate marker *-e* is lost in the oblique case of singular nouns (102) but is retained in this context in the Shamdinan dialect (103):

(102) Behdîni Duhok-Hakkari NK

- gund-ê vî zelam-î*
 village-EZ.SG.M DEM.PROX.SG.M.OBL man-M.SG.OBL
 ‘the village of this man’

(103) Behdîni Shamdinan NK

- gund-ê vî zelam=e-y*
 village-EZ.SG.M DEM.PROX.SG.M.OBL man=DEM-SG.M.OBL
 ‘the village of this man’

In the Behdîni dialects, far deixis can be expressed by *ew*, the same form that is used as an anaphoric pronoun, e.g.,

(104) Behdîni NK

- ew kê ye li wêrê?*
 DEM.DIST who COP.3SG in there
 ‘Who is that (person) over there?’

(105) Behdîni NK

- gund-ê wî zelam-î*
 village-EZ.SG.M DEM.DIST.SG.M.OBL man-M.SG.OBL
 ‘the village of that man (said when pointing to the man).’

The far deixis can also be expressed with the addition of a deictic particle *he/henê*:

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(106) Behdîni NK

eve dayik-a wî ye, ew-a
 DEM.PROX mother-EZ.SG.F DEM.DIST.SG.M.OBL COP.3SG DEM.PROX-EZ.SG.F
henê jî xwîşk-a wî ye.
 DIST ADD sister-EZ.SG.F DEM.DIST.SG.M.OBL COP.3SG
 ‘This is his mother, and that one there is his sister.’

(107) Behdîni NK

ev jin=e dayik-a wî
 DEM.PROX womanSG.F.DIR=DEM mother-EZ.SG.F DEM.DIST.SG.M.OBL
ye, ew jin-a henê dayik-a
 COP.3SG DEM.PROX womanSG.F.DIR-EZ.SG.F DIST mother-EZ.SG.F
wî ye.
 DEM.DIST.SG.M.OBL COP.3SG
 ‘This woman is his mother, that woman is his (another person’s) mother.’

However, the particle *he/ henê* does not express a greater distance from the speaker than *ew* by itself. It seems to endow the referent with greater salience than is the case with the morphologically simpler form *ew*. This is exploited in examples such as (106) and (107) to express a contrast.

Very far deixis is expressed by combining the distal demonstrative with the particle *he/henê* and either inserting a glottal stop between the two or by lengthening the stressed vowel demonstrative I (SG.M) and *ewa* (SG.F) and *ewêt* (PL).

(108) Behdîni NK

- a. *ka biwir li kêrê=ye?*
 Q.PTCL axe at where=COP.3SG
 ‘Where is the axe?’
- b. *ew-ê henê=ye li ber dîwar-î.*
 DEM.DIST-EZ.SG.M DIST=COP.3SG from front wall-SG.M.OBL
 ‘It is yonder by the wall.’

(109) Behdîni NK

ewaaa he di-bîn-î?
 DEM.DIST VIS IND-seePRS-2SG:A
 ‘Do you see that (woman) there?’

The conservative Hewramî varieties of Gorani have a complex demonstrative system. Unlike Kurdish, the deictic demonstrative pronouns are consistently dis-

10.2 Demonstratives in Iranian

tinguished morphologically from anaphoric pronouns. These three sets of pronouns inflect for case, gender, and number; see Table 10.5. The use of each of these sets of pronouns is conditioned by discourse factors (See [Mohammadirad 2026](#): Ch.5 for details).

Table 10.5: Anaphoric and demonstrative pronouns in Hewramî

Set 1: Anaphoric third person pronouns				
	PROXIMAL		REMOTE	
	DIR	OBL	DIR	OBL
M	<i>êd</i>	<i>êdî</i>	<i>ađ</i>	<i>ađî</i>
F	<i>êde</i>	<i>êdê</i>	<i>ade</i>	<i>adê</i>
PL	<i>êdê</i>	<i>êdîşa</i>	<i>ađê</i>	<i>ađîşa</i>
Set 2: Anaphoric demonstratives				
M	<i>îd</i>	<i>îdî</i>	<i>ew</i>	<i>ewî</i>
F	<i>îd(e)</i>	<i>îdê</i>	<i>ew(e)</i>	<i>ewê</i>
PL	<i>îdê</i>	<i>îdîşa</i>	<i>ewê, ewêşa</i>	<i>ewîşa</i>
Set 3: Demonstrative pronouns				
M	<i>îne</i>	<i>îney</i>	<i>ane</i>	<i>aney</i>
F	<i>înê, îni</i>	<i>înê</i>	<i>anê</i>	<i>anê</i>
PL	<i>înê, îni</i>	<i>îna, înişa</i>	<i>anê</i>	<i>ana, anişa</i>

A paradigm of demonstratives of similar shapes are found in Şebekî. Table 10.6. However, no study to date has explored the precise functional distribution of Shabaki demonstrative pronouns. [MacKenzie \(1956\)](#) lists the form *ewe*, but he does not make any comments about its function beyond being a third-person pronoun.

Table 10.6: Anaphoric and demonstrative pronouns in Şebekî

	M.SG.DIR	M.SG.OBL	F.SG.DIR	F.SG.OBL	PL
PROX.ANAPH	<i>êd</i>	<i>êdî</i>	<i>?êde</i>	<i>?êdê</i>	<i>êşan</i>
ANAPH.?	<i>ew</i>	<i>*ewî</i>	<i>*ewe</i>	<i>*ewê</i>	<i>ewşan</i>
PROX.DEM	<i>îne</i>	<i>îney</i>	<i>?îne</i>	<i>?înê</i>	<i>înegle</i>
DIST.DEM	<i>ûne</i>	<i>ûney</i>	<i>?ûne</i>	<i>?ûnê</i>	<i>ûnegle</i>

In Zazaki there are two sets of demonstratives, proximal and distal, which are inflected for gender, number, and case. The proximal set is expressed by means

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of a nasal prefix *n-* in Northern and Southern Zazaki, whereas in Central Zazaki a prosthetic vowel *e-* or *i-* is added as well.

In general, all demonstratives can be used either adnominally as determiners or independently as pronouns. The pronominal forms differ from the determiners only in the oblique case and show typical dialectal variation.

Table 10.7: Demonstrative pronouns in Zazaki

Proximal			Distal	
	DIRECT	OBLIQUE	DIRECT	OBLIQUE
M	<i>no / eno</i>	<i>ney/enî</i>	<i>o</i>	<i>ey</i>
F	<i>na / ena</i>	<i>naye / enay</i>	<i>a</i>	<i>aye</i>
PL	<i>nê / enî</i>	<i>nînan / enîn</i>	<i>ê</i>	<i>înan</i>

Table 10.8: Adnominal demonstrative in Zazaki

Proximal			Distal	
	DIRECT	OBLIQUE	DIRECT	OBLIQUE
M	<i>no / eno</i>	<i>nê / enî</i>	<i>o</i>	<i>ê</i>
F	<i>na / ena</i>	<i>na / ena</i>	<i>a</i>	<i>a</i>
PL	<i>nê / enî</i>	<i>nê / enî</i>	<i>ê</i>	<i>ê</i>

Both distal and proximal demonstratives can also be used anaphorically, i.e., to mark discourse-salient referents.

In addition, there is a third, less frequent set of demonstratives, used especially in the northern dialects, which serves to point to something close by, to express emphasis, or to mark topicalisation (cf. also Keskin 2008: 76, who labels this “second level of near deixis”). This third set of proximal demonstratives features a prosthetic vowel *a-* or *e-*. In the northern dialects there is likewise an additional distal form with the single invariable form *ha* (Table 10.9):

As seen in tables 10.7 and 10.8, the proximal demonstratives have an optional *e-* or *i-* prefix without any functional difference. This likely results from paradigm levelling with the third set of demonstratives of the northern dialects (Keskin 2008: 76, Paul 2024: 56,72).

Although Selcan (1998: 310) describes *ha* as a marker of very far deixis, it seems to acquire that meaning only in combination with another distal demonstrative or the copula.

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Table 10.9: Third set of demonstratives

	PROXIMAL	DISTAL
M	<i>ano / eno</i>	<i>ha</i>
F	<i>ana / ena</i>	<i>ha</i>
PL	<i>anê / enî</i>	<i>ha</i>

- (110) Northern Zazaki (fieldnotes)
ha nî-yo uca =de roniştî =yo?
DEM NEG-COP-3SG.M there =at sit.PTCP =COP-3SG.M
'Isn't that him sitting over there?'

The distal *ha* can also function as an independent discourse particle, or it may combine with a proximal demonstrative to express emphasis (see §??). In general, it is not used adnominally.

The near–far distinction also surfaces in the morphology of certain local and temporal adverbs, e.g., *nika* 'now' vs. *a-nika* 'right now' or *na-ca* 'here' vs. *ha-ca* 'there'.

10.3 Demonstratives in Arabic

In *Qeltu* Arabic dialects the demonstratives, which serve as both independent pronouns and adnominal modifiers, are attested in either two or three distinct sets. Demonstrative systems consisting of two sets include near and far deixis. Systems that consist of three sets include, according to the terminology of Jastrow (1978: 102), near, far and middle deixis. The middle deixis demonstratives serve to point to referents in the far but visible environment of the interlocutors (Jastrow 1978: 107). In contrast to the middle forms, the far deixis forms are used to point to referents in the far distance or anaphorically to refer to referents that are outside the field of vision but recoverable from the discourse history (Jastrow 1978: 107). When there are only two sets of demonstratives, the far form is presumably used both for far deixis (pointing at visible objects) and to express anaphora (recoverability from the discourse history).

The near deixis demonstratives in *Qeltu* dialects have the following typical forms or cognates of these (Jastrow 1978: 102-105):

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- (111) sg.m *hāḏa*
sg.f *hāḏi* (alternatively *hāy* or *hāye*)
pl *hawḏe*

The far deixis/anaphoric demonstratives have the following typical forms or cognates of these (Jastrow 1978: 105-107):¹

- (112) sg.c *hāk* or *hāke*
pl *hawk* or *hawke*

Bəḥzāni and Mērgi, on the Nineveh Plain, both have a gender distinction in the far deixis singular (Jastrow 1978: 106; Hassan 2023: 583).

The middle deixis demonstratives in systems with three sets are formed from the far/anaphoric demonstratives by the addition of a suffix *-ā* (< **hā*), e.g.,

- (113) Bəḥzāni (Jastrow 1978: 107)
sg.m. *hākā* < **hāk-hā* ‘that (one) there’

The demonstratives described above and deictic adverbs can be combined in many Qəltu dialects with an ‘intensivising’ prefix *ē-* or *ī-*, meaning ‘just, exactly’ (Jastrow 1978: 102), e.g.,

- (114) Kəndērīb *ēhāy* ‘precisely these’
Bəḥzāni *īhawni* ‘just here’

The following tables present the type of demonstrative systems attested in Arabic dialects in or around the NENA area, in so far as information on this feature is currently available.

Table 10.10: Arabic demonstratives within the NENA area

Mosul (M, C, J)	2 sets (Jastrow 1979: 44)
Bəḥzāni (C, Y, M)	2 sets (Jastrow 1979: 44; Jastrow 1978: 107)
Mērgi (C)	2 sets (Hassan 2023: 583)
Arbīl and ‘Aqra (J)	3 sets (Jastrow 1990a: 43–45)

¹Other forms are found in some Qəltu dialects: see Jastrow (1978) for a fuller picture.

Table 10.11: Arabic demonstratives on the periphery of the NENA area

Arabic demonstratives on the periphery of the NENA area	
Āzəx (C)	3 sets (Wittrich 2001: 37-39)
Siirt town (M)	2 sets (Biṭună 2016a: 137-138)
Tillo (M)	2 sets (Lahdo 2009: 79-80)

10.4 Demonstratives in Armenian

Classical Armenian had three series of demonstratives, which can be described as proximal, medial, and distal, although factors other than distance are also involved in the distinction between them. These are based on the root consonants *-s-* (proximal), *-d-* (medial), and *-n-* (distal). The same deictic distinction is found in the article system of Classical Armenian, which had a proximal form *-s*, medial *-d*, and distal *-n* (see §11). There are also series of adverbs and adjectives with the same distinction, which continue to be used in modern forms of the language.

Some dialects of Armenian have only two sets of demonstratives (Feature 68 of Jahukyan 1972). Tigranakert/Diyarbakir has lost the medial set, whose function is now included in that of the distal demonstratives (Haneyan 1978: 88), but is the only dialect out of those examined in Jahukyan (1972) to have done so. The loss of the proximal set has a much wider distribution, being found across a continuous area including the whole Urmia region, most of the dialects of the Van group (except for Van city itself), and some dialects belonging to the Mush group, as seen in Figure 10.2.

According to Asatryan (1962: 91), in Khoy/Urmia, the originally medial demonstrative *et* (< CA *ayd*) combines the functions of the proximal and medial demonstratives, so that *et tun-ə* (the suffix *-ə* on the noun is the definite article, which is obligatory with demonstratives in most dialects) can mean either ‘this house, near me’ or ‘that house, near you’.

According to Muradyan (1962: 118-119), Shatakh (Van group) seems to have lost the original CA proximal demonstrative *sa/ays*, like most other dialects of the area, but unlike these, has replaced it with a different form, so that we have proximal *næ*, medial *æt*, distal *æn*.

Modern Armenian demonstrative pronouns are mainly derived either from the Classical Armenian series *sa*, *da*, *na*, which were only used as independent pronouns, or *ays*, *ayd*, *ayn*, which could be used either as independent pronouns or as adnominal modifiers. In Classical Armenian, when used as modifiers, the

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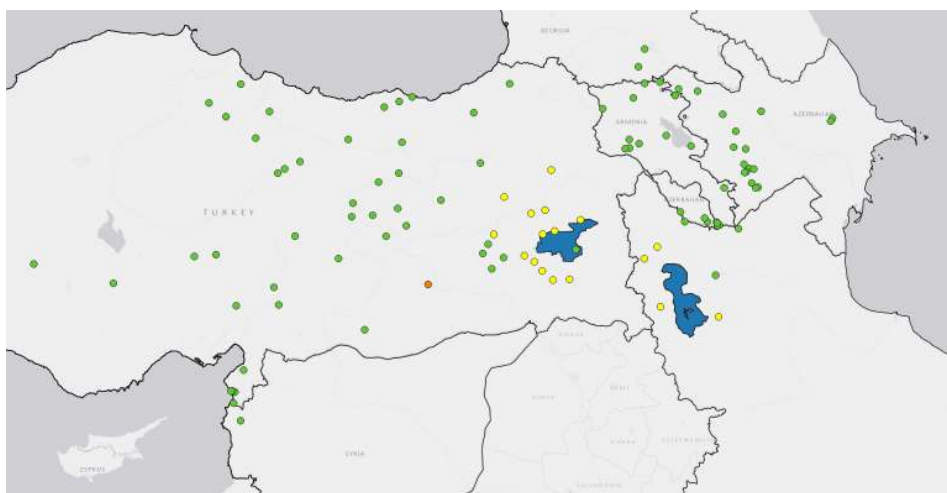


Figure 10.2: Armenian dialects with all three sets of demonstratives (green), lacking the proximal set (yellow), lacking the medial set (red) (according to [Jahukyan 1972](#)).

demonstratives *ays*, *ayd*, *ayn* could either precede (115) or follow (116) the modified noun.

- (115) Classical Armenian ([Abajyan et al. 2017: 375](#))

ayd giwt
DEM2 village
‘that (medial) village’

- (116) Classical Armenian ([Abajyan et al. 2017: 374](#))

giwt-d ayd
village-ART2 DEM2
‘that (medial) village’

As can be seen, when the demonstrative followed the noun, the noun usually took the definite article (116), while when the demonstrative preceded the noun, it did not (115). Classical Armenian possessed alternative forms of the demonstratives with the extension *-ik*, which resembles the diminutive morpheme *-ik*.

As a rule, the forms in *-ik* follow the modified noun ([Abajyan et al. 2017: 374](#)), as seen in (117).

- (117) Classical Armenian ([Abajyan et al. 2017: 374](#))

10.4 Demonstratives in Armenian

Table 10.12: Paradigm of the Classical Armenian proximal demonstrative *ays* with alternative forms in *-ik* (Abajyan et al. 2017: 374)

	Singular	Plural
NOM	<i>ays</i>	<i>ayskʻ, aysokʻik</i>
ACC	<i>(z)ays</i>	<i>(z)ays, (z)aysosik</i>
GEN	<i>aysr, aysorik</i>	<i>ayscʻ, aysocʻik</i>
DAT	<i>aysm, aysmik</i>	<i>ayscʻ, aysocʻik</i>
ABL	<i>yaysm, yaysmanē</i>	<i>yayscʻ, yayscʻanē, yaysocʻik</i>
INS	<i>aysow, aysowik</i>	<i>aysokʻiwkʻ, aysokʻimbkʻ</i>
LOC	<i>yaysm, yaysmik</i>	<i>yays, yaysosik</i>

giwt-kʻ-d aydorik
village-PL-ART2 DEM2.NOM.PL
‘those (medial) villages’

Demonstratives with the extension *-ik* and phonetic variants are still used in most Western Armenian dialects, and in Urmia, but not in most Eastern Armenian dialects (Jahukyan 1972), e.g., *esik, etik, enik* in Van (Acharyan 1952: 152), *etikʻ, enikʻ* in Urmia (Asatryan 1962: 89), which, like most Van subdialects, with the exception of Van itself, lacks the proximal set.

In modern Armenian, the demonstrative generally precedes the noun, but must be accompanied by the definite article on the noun, as seen in (118). Unlike in Classical Armenian, where they are declined for number and case, as seen in Table 10.12, in modern Armenian the forms derived from *ays*, *ayd*, *ayn* are indeclinable, as seen in (119).

(118) SEA
ays gyut-ə
DEM1 village-DEF
‘this village’

(119) SEA
ays gyut-er-i-n
DEM1 village-PL-DAT-DEF
‘to these villages’

The pragmatic usage of the demonstratives is as follows. The proximal demonstratives are generally used to refer to something close to the speaker. The medial

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demonstrative is most often used for things in the addressee's space or within a shared nearby context. However, in practice, many speakers also use it for something that is not immediately at hand for the speaker but is still within visible or accessible range, such as across the table or across the room. The distal demonstrative is associated with greater physical or metaphorical distance from both speaker and addressee. For referents that are out of visible range, this form must be used. It is also associated with formal registers. For example, if the speaker points to an object on the other side of the room, the medial demonstrative would be more natural in colloquial registers, but the distal form would be used in formal contexts.

In both SEA and SWA, the distal demonstrative pronoun *na/an* is considered to play the role of an anaphoric pronoun, signalling that a referent that is not in the speech situation is recoverable from the discourse. When used for non-subjects, this pronoun indicates non-coreference with the subject of the clause (Donabedian 2007), thus in (120), *nra* (genitive of *na*) cannot be interpreted as referring to the subject Armen.

(120) SEA

Armen-ə korcʻr-el=ē nra girkʰ-ə
 Armen-DEF lose-PST.PTCP=be.3SG.PRS 3SG.GEN book-DEF
 'Armen_i lost his_i book.'

Thus, *na* seems to have a “switch-subject” (see Kibrik 2011: 154) or “antireflexive” (see Kibrik 2011: 315) function, contrasting particularly with the “emphatic/reflexive” pronoun *inkʻə*, which is used for referents which are coreferent with the subject of the clause, or with prominent topics in general (see Donabedian 2007, Sigler 2001, Hodgson et al. 2026).

The medial demonstrative *da/ayd* is also used as an anaphoric pronoun, mainly for non-human referents, as seen in 121 (*eti* is a colloquial form of *ayd*), while *na* is used for human referents.

(121) Colloquial EA (EANC)

u eti axʃʰəka gər-ats yerʰkʰ-n=a təy-u masin
 and DEM2 girl.GEN write-RES.PTCP song-DEF=be.PRS.3SG boy-GEN about
 'And that is the song the girl wrote about the boy.'

10.5 Demonstratives in Turkic

Most Turkic languages feature a three-way deictic distinction (proximal, medial, distal), though many systems are more elaborate, incorporating additional pa-

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rameters like visibility or topicality. Demonstratives can serve as both modifiers and anaphoric pronouns.

Azeri has for the most part lost the three-way distinction at an early stage, a well-known isogloss between Oghuz varieties of the Anatolian type and those of the Azeri type. Analyses of Middle Azeri translations of Persian texts from the fifteenth and sixteenth centuries reveal that the system was already predominantly two-way and contrasted proximal and distal (Bellér-Hann 1995: 88; Stein 2014b: 220). This two-way distinction persists in most modern Azeri dialects, where the basic demonstratives are proximal *bu* and distal *o*. The medial demonstrative *şu* is attested to a very limited extent in the Şahsevən variety of Ardabil (Çam 2021: 210–211) and is encountered solely in poetic registers in Urmia Azeri (Doğan 2010: 128). In Iraqi Turkic, the medial demonstrative is confined to written language under the influence of the Istanbul Turkish norm, and is absent from spoken language (Bayatlı 1996: 387–388).

Azeri modifier demonstratives show no morphological agreement with their heads, e.g., *bu kitab* ‘this book’ vs. *bu kitablar* ‘these books’.

Like most Turkic languages, Azeri lacks dedicated third-person pronouns, instead employing anaphoric demonstratives. Distal *o* serves as the pragmatically neutral anaphoric pronoun, while proximal *bu* is associated with physical proximity, topicality or affection. When anaphoric demonstratives are inflected for case, they take an *n*-extension, historically an oblique stem marker, but often analysed synchronically in scholarly works as part of the case suffix, e.g., *bunun* ‘this.GEN’ (historically *bun-un*), *onda* ‘that.LOC’ (historically *on-da*). A notable phonological feature in most Azeri dialects is the regressive assimilation of initial /b/ in *bu* to /m/ in the nasal environment created by the *n*-extension, yielding, e.g., *munun* ‘this.GEN’, *muna* ‘this.DAT’, *munda* ‘this.LOC’ (Dybo & Levitskaja 2002: 147; Bulut 2019: 364), a tendency already observable in sixteenth-century texts (Stein 2014a: 22).

In most dialects, the “oblique” stems also host the plural suffix *-lar*, which derives plural anaphoric forms: *bunlar* ‘they (i.e., these ones)’, *onlar* ‘they (i.e., those ones)’, subject to a pre-nasal liquid assimilation: *bunnar* ~ *munnar*, *onnar*. In other dialects, including Urmia Azeri and Eastern Azeri, it is the “direct” stem that hosts the plural suffix (hence no nasalisation of *b* in the proximal): *bular*, *olar* (Doğan 2010: 138–139).

Azeri dialects display various ways of expressing stronger pronominal deixis. Modifier demonstratives include *həmin* and *həmən*, copied from Persian (< proximal *hamīn* and distal *hamān* respectively). While these originally conveyed proximal and distal distinctions, their semantic contrast has largely eroded, and they

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are now used almost interchangeably, with Urmia Azeri showing a preference for *həmən* (Doğan 2010: passim).

In addition, both modifier and anaphoric inherited demonstratives can be modified by further elements for emphatic deixis. One of them is *bax*, etymologically the 2sg imperative form of *bax*- ‘see!’ (which does not inflect for number even if the addressee is plural):

bu kitab ‘this book’ > *bax bu kitab* ‘this very book’, lit. ‘this book, see!’

o kitab ‘that book’ > *bax o kitab* ‘that very book’, lit. ‘that book, see!’

Another set of emphatic elements are the so-called ostensive predicators *budē* and *odē* (cf. Urmia Azeri *odey*, according to Doğan 2010: 140), which can roughly be compared to French predicators *voici* and *voilà* (Ragagnin 2022b: 68-70, transcription and glosses adapted):

- (122) a. *Budē telefon-un.*
 OST telephone-POSS.2SG
 ‘Here is your phone!’
 b. *Odē telefon-un.*
 OST telephone-POSS.2SG
 ‘There is your phone!’

The ostensive markers derive from the combination of a demonstrative and *dē*. Ragagnin (2022b: 68-70) further notes that in spoken Azeri, the demonstrative component can be omitted when the deixis is proximal:

- (123) *Dē telefon-un.*
 OST telephone-POSS.2SG
 ‘Here is your phone!’

The origin of the component *dē* remains a subject of debate. Ragagnin (2022b: 68-70) suggests it may stem from the merging of the additive particle *dA* with the exclamation particle *ey*. However, an alternative analysis, proposed here, suggests that *dē* originates in the third-person copula *-dlr* combined with the exclamation particle *ey*, especially given the existence of structurally parallel forms in other dialects, e.g., Qax Azeri *buduā* and *oduā* (Mämmädli 2019: 216), almost certainly originating from *bu-dur ha* <this-cop.3 excl> and *o-dur ha* <that-cop.3 excl> respectively. Another potential origin for *dē* may link it to dental-initial demonstratives attested sporadically across Turkic languages; e.g., Salar, an Oghuz outlier spoken in China, features the distal demonstratives *tū* and *cugü*,

while Bashkir, a Kipchak language of the Volga region in Russia, has *tigi*. Alternatively, *dē* may share its origin with the Azeri imperative particle *di* ‘come on, go on, go ahead’, as seen in expressions like *Di başla!* ‘Come on, begin!’.

All three ostensive markers—*budē*, *dē* and *odē*—can themselves modify demonstratives, whether modifier or anaphoric, to further emphasise deixis. The proximal ostensive *budē* usually combines with the proximal demonstrative *bu* to yield *budē bu* ‘this (one) right here’, while the distal ostensive *odē* combines with the distal demonstrative *o* to form *odē o* ‘that (one) over there’. Some speakers judge the “mismatched” construction *budē o* as acceptable as a medial emphatic demonstrative but its exact semantic distribution needs to be studied further. Meanwhile *dē* can combine with either *bu* and *o*, resulting in *dē bu* ‘this (one) right here’ and *dē o* ‘that (one) over there’.

- (124) Ardabil Azeri (Karini 2009: 284, transcription adapted, original gloss)
məsələm budēy bu qəlyənnahar vaxtt gəl-ər-di-k.
 for.example ost this breakfast time.POSS.3? come-EVT-PST-1PL
 ‘For instance, we used to arrive right at this breakfast time.’

10.6 Discussion

The NENA demonstrative pronouns exhibit innovative developments that match closely those of languages in contact.

The innovative extension of the /w/ element of the stem of the far deixis form to the near deixis form in NENA dialects of the northern sector (e.g., C. Barwar *’awwa* near deixis, *’āwaha* far deixis) results in morphological forms that match closely the demonstratives forms of Kurdish dialects of the region. In Northern Central Kurdish dialects both near deixis and far deixis demonstratives have /w/ in their stem (see table 10.2). In Northern Kurdish near deixis demonstratives have /v/ in their stem and far deixis demonstratives have /w/. While not identical, these sounds are closely related and NENA /w/ could have been perceptually matched with Kurdish /v/.

The development of three sets of demonstratives in NENA dialects in the northern sector of NENA matches the system of three sets (near deixis, far deixis and anaphoric) that is found in Behdîni Kurdish. Moreover, there is also a striking similarity in morphological shape of the forms in NENA and Kurdish. The morphological components of the NENA forms are of Aramaic heritage, but have been developed to resemble the Kurdish forms, e.g.,

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(125)

SG.M	Anaphoric	Near	Far
C. Barwar NENA:	<i>ʔaw</i>	<i>ʔawwa</i>	<i>ʔāwaha</i>
Behdîni Kurdish:	<i>ew</i>	<i>eve</i>	<i>ewê he</i>

A further feature of convergence between NENA and Behdîni is the formation of ‘very far deixis’ forms by glottalisation or lengthening of the vowel of the far deixis form:

(126)

	Far deixis	Very far deixis
Behdîni:	<i>ewê he</i>	<i>ewêʔ he, ewêêê he</i>
Barwar NENA:	<i>ʔāwaha</i>	<i>ʔāwaʔha, ʔāwaaaha</i>

An alternative augment of the Behdîni far deixis demonstrative is *-henê*, e.g., *ewê henê* (sg.c). An augment of the same shape is found in demonstratives in NENA dialects in the Targawar region on the periphery of the Behdîni Kurdish area, e.g.,

(127)

C. Mawana			
Near deixis:	<i>wa</i>	<i>wāha</i>	<i>wāhane</i>
Far deixis:	<i>wə</i>	<i>wəha</i>	<i>wəhane</i>

As can be seen, in C. Mawana NENA the augments *-ha* and *-hane* have been extended also to the near deixis forms. This seems to be an internal development in the NENA dialect.

In NENA the innovative extension of the stem of the Aramaic inherited far deixis form with a /w/ element to the near deixis form in imitation of Kurdish took place only in the dialects in the northern sector. In Christian dialects in the south-eastern sector (including those in the Arbil and Sulemaniyya provinces, and the Christian dialect of Sanandaj) and in Jewish Trans-Zab dialects, the near deixis demonstrative contains a /y/ or /i/ element in its stem.

The lack of extension of the /w/ element of the far deixis stem to the near deixis stem in these dialects may have been due to the fact that in most Central Kurdish dialects the near deixis demonstrative has an /m/ element rather than a /w/ element (see Table 10.1). The /y/ or /i/ element is a demonstrative particle that is attested as an ifying near deixis augment in NENA dialects on the Mosul plain (see 10.1) and also in Zazaki (in its variant form /e/, see 10.9) and Arabic *Qəltu* dialects (see 10.3). Its presence in the stem of near demonstrative forms in

NENA dialects in the south-eastern sector may have been reinforced by contact with Hewramî, in which the stem of near deixis demonstratives contain an /i/ element (see Table 10.5). This applies to all Jewish Trans-Zab dialects, which all have a /y/ or /i/ element in their singular near deixis forms. This includes Jewish Trans-Zab dialects in north-western Iran, such as J. Urmi, outside of the area of Hewramî and this, it seems, has resulted from migrations of Jewish Trans-Zab speakers from Hewramî area further south to north-western Iran.

The restriction to two sets of demonstratives in NENA dialects in the eastern sector matches Azeri Turkic, which has two sets of demonstratives (§10.5). The collapse of the gender distinction in the Jewish Trans-Zab dialects results, indeed, in the far deixis form being a phonetic match to Turkic genderless *o*. The deictic augment element *bax* ‘see!’ that is used in Azeri Turkic demonstrative, e.g., *bax bu kitab* ‘look this book’, could, furthermore, be identified as a match to the /y/ or *i* element in near demonstrative forms in NENA dialects of the eastern sector.

Some Armenian and Arabic varieties in the northern sector of the NENA area likewise exhibit three sets of demonstratives:

(128)

SG.M	Near	Middle	Far/anaphoric
Arabic (Bəḥzānī):	<i>hāza</i>	<i>hākā</i>	<i>hāk</i>
Armenian (SEA):	<i>sa</i>	<i>da</i>	<i>na</i>

The use of three sets of demonstratives, together with, in some cases, intensifying augments for near deixis and far deixis (cf. the suffixed *he* augment in Kurdish and the suffixed *hā* augment in Arabic), is, therefore, a feature of many languages of the northern sector of NENA. There is also, as we have seen, a coincidence in morphological shape of the forms of the NENA demonstratives with those of languages in contact. However, the pragmatic function of the various sets of demonstratives across the languages does not coincide exactly with NENA dialects. In the NENA dialects of the northern sector that have three sets of demonstratives, the form *ʔaw* is used only as an anaphoric pronoun, whereas in Behdînî Kurdish *ew* can be used as an anaphoric pronoun or as a far deixis demonstrative. The form *ewê* *he* expresses a more intense pointing and direction of attention to a distant referent than *ew*. In *Qəltu* Arabic dialects that have three sets of demonstratives, two degrees of distance are expressed by the middle deixis form (*hākā*) and by the far deixis form (*hāk*) respectively. The far deixis form *hāk* is also used as an anaphoric pronoun. This resembles the system of Behdînî, in which the morphologically less complex form *ew* can be a far deixis or anaphoric pronoun. It does not, however, express greater distance than the morphologically

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more complex form *ewê he*, in contrast to Arabic, in which the morphologically less complex form *hāk*, when used deictically, expresses greater distance than the morphologically more complex form *hākā*. The three sets of pronouns in Armenian dialects (near deixis *sa*, middle deixis *da*, far deixis *na*) would correspond functionally to the near, far and very far deixis forms respectively in NENA dialects in the northern sector. However, unlike these NENA dialects, which have a dedicated anaphoric pronoun, Armenian uses the far deixis form as anaphoric pronoun. So NENA has converged with the general structure of the demonstrative systems of three sets in the languages in contact in the northern sector, but the functional application of the demonstrative forms have undergone internal development within NENA.

In Kurdish the vocalic ending or augment of the independent forms of demonstrative pronouns is detached from the stem of forms in the adnominal forms and attached to the end of the modified noun:

(129) Behdîni

Independent	Adnominal
<i>eve</i> ‘this one’	<i>ev zêlam=e</i> ‘this man’
<i>ewê he</i> ‘that one’	<i>ew zêlamê he</i> ‘that man’

In NENA dialects, by contrast, this feature is not replicated. Only the form of the independent demonstrative is replicated and this is used also in an adnominal function, e.g.,

(130) C. Barwar

Independent	Adnominal
<i>’awwa</i> ‘this one’	<i>’awwa naša</i> ‘this man’
<i>’ăwaha</i> ‘that one’	<i>’ăwaha naša</i> ‘that man’

11 The morphosyntax of nominal modification

11.1 Linking dependent elements to a head noun in NENA

11.1.1 Noun-dependent noun

In NENA dialects, a noun may be syntactically annexed to a following noun. When two nouns are so linked, the second is an attribute of the first and there is, broadly speaking, a genitive relationship between them. The second noun, known as the dependent noun, modifies the head noun in some way. Such attributive annexation of nouns in a noun phrase was expressed in earlier forms of Aramaic by the particle *d* or *dī* placed before the dependent noun, e.g.

- (1) Syriac (Nöldeke 1904: 161)
malkā d-Bāḇel
 king ATTR-Babylon
 ‘the king of Babylon’

This particle *d/dī* is historically a demonstrative pronoun that formed the head of a phrase containing the dependent noun: ‘the king the one (of) Babylon’.

The particle also links some prepositions to their dependent noun and introduces subordinate clauses. In (1) and the ensuing discussion of the reflexes of the particle in Neo-Aramaic the particle is glossed as ATTR, which denotes its function as a linker introducing attributive dependents of various types.

11.1.2 Christian Dialects of the Mosul Plain

11.1.2.1 C. Qaraqosh

Some NENA dialects in the western sector of the dialect group on the Mosul plain have preserved the type of structure that is used in Syriac (1), such as C. Qaraqosh, e.g.,

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- (2) C. Qaraqosh (Khan 2002a: 207)
bšala d-našə
 cooking ATTR-people
 ‘the cooking of people’

In most cases in such dialects, however, the particle is suffixed to the head noun when the dependent noun begins with a consonant, e.g.,

- (3) C. Qaraqosh (Khan 2002a: 207)
ʔaxalə-d našə
 food-ATTR people
 ‘the food of people’

Note that the final vowel of the head noun is contracted to the central vowel /ə/.

In C. Qaraqosh, the particle that is suffixed to the head noun is sometimes devoiced to /t/. This occurs mainly when the first consonant of the second noun is voiceless, e.g.,

- (4) C. Qaraqosh (Khan 2002a: 207)
gəld-ət para
 skin-ATTR lamb
 ‘the skin of a lamb’

The /ə/ before the particle on the head noun is occasionally elided and the particle is syllabified with the following dependent noun:

- (5) C. Qaraqosh (Khan 2002a: 208)
yom d-gwara
 day ATTR-wedding
 ‘wedding day’

The nominal inflection of the head noun, therefore, becomes reduced through phonetic contract ($a > ə > \emptyset$). Also in some cases the morphological distinction between singular and plural of the head noun is lost:

- (6) C. Qaraqosh (Khan 2002a: 208)
šwawə-d babi
 neighbour-ATTR father.my
 ‘the neighbour of my father’ (< *šwawa d-babi*)

11.1 Linking dependent elements to a head noun in NENA

When the dependent noun begins with a laryngeal /ʔ/ followed by a vowel, the particle is regularly syllabified with this vowel:

- (7) C. Qaraqosh (Khan 2002a: 208)
 ʔitə d-Ašmoni
 church ATTR-Ašmoni
 ‘the church of Ašmoni’

Note that the underlying morphological form of this can be assumed to be ʔitəd ʔAšmoni with the particle attached to the head and the final -a vowel of the head contracted to /ə/. The same applies to constructions such as (5) in which the /ə/ of the head has been elided.

In fast speech, the particle *d* is sometimes assimilated to the initial consonant of the second noun, e.g.,

- (8) C. Qaraqosh (Khan 2002a: 208)
 ʔilanə z-zaʔθa
 tree ATTR-olive
 ‘olive tree’

The gemination is often lost, resulting in the complete loss of the attributive particle:

- (9) C. Qaraqosh (Khan 2002a: 208)
 ʔərə Bağdedə
 land Bağdedə
 ‘the land of Bağdedə’

In some tightly-knit conventionalised phrases, the head noun has no inflectional vowel and the attributive particle is elided, even before a word beginning with a vowel:

- (10) C. Qaraqosh (Khan 2002a: 208)
- a. *bəxət baba*
 wife father
 ‘wife (*baxta* of the father’ (= ‘stepmother’)
 - b. *syām-iḏa*
 laying-hand
 ‘laying (*syama*) of the hand’ (= ‘ordination’)

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The long vowel in forms such as *syām-iḏa* in (10b) can be explained by the hypothesis that the attributive particle is still present at an underlying phonological level and the vowel length is conditioned by the fact that the syllable is open syllable at this underlying level (see [Khan 2016b](#): vol. 1, 217).

The dependent noun may be a predicative attribute in a clause. In such cases the noun must be preceded by the attributive particle:

- (11) C. Qaraqosh ([Khan 2002a](#): 279)

ʾáhi ʾiyáwa d-ʾəzla.¹

it.F was ATTR-cloth

‘It was (made) of cloth.’

There is an independent attributive pronoun that has developed historically from the combination of the attributive particle *d* with other elements, e.g., *didux* ‘yours’, which is most likely derived historically from **di-lux* ‘the one belonging to you’. This attributive pronoun has the same generic attributive annexation function as the attributive particle *t/d*. A full paradigm is available for this attributive pronoun:

- (12) C. Qaraqosh

3SG.M *didəḥ*

3SG.F *didəḥ*

3PL *dədhən*

2SG.M *didux*

2SG.F *didax*

2PL *dədxun*

1SG *didi*

1SG *didan*

Examples:

- (13) C. Qaraqosh ([Khan 2002a](#): 83–84)

beθa didəḥ ‘his house’

beθa didux ‘your house’

The particle *d* is also used to link prepositions to their dependent nouns and introduce subordinate clauses, in particular relative clauses modifying a head noun. In these contexts the particle exhibits the phonotactics and contractions that are found when it takes a dependent noun as its complement:

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- (14) C. Qaraqosh (Khan 2002a: 239)
rišəd kalθa
 upon.ATTR bride
 ‘upon the bride’
- (15) C. Qaraqosh (Khan 2002a: 209)
yoma d-k-aθə mar-an b-gaw-əh
 day REL-PART-come.3SG.M lord-our in-inside-it.SG.M
 ‘the day on which our Lord is born’
- (16) C. Qaraqosh (Khan 2002a: 209)
šokə-d k-maθe-hə
 sprigs-ATTR PART-bring.3SG.M-OBL.3PL
 ‘the sprigs that they bring’

11.1.3 Christian dialects in the northern sector of NENA

In Christian dialects of the northern sector of the dialect area, the attributive annexation construction exhibits some further developments in its historical evolution.

11.1.3.1 C. Barwar

In the C. Barwar dialect the attributive particle is regularly suffixed to the head noun and this is by default devoiced to *-t*:

- (17) C. Barwar (Khan 2008a: 397)
dan-ət kawuθra
 time-ATTR lunch
 ‘time of lunch’

Even when the dependent noun begins with a voiced con-sonant, the suffixed particle remains voiceless:

- (18) C. Barwar (Khan 2008a: 397)
bab-ət brata
 father-ATTR girl
 ‘the father of the girl’

Unlike in dialects on the Mosul plain, when the dependent noun begins with a laryngeal /ʔ/ followed by a vowel, the laryngeal is generally retained and not elided:

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- (19) C. Barwar (Khan 2008a: 397)

ʾurx-ət ʾalaha
way-ATTR God
‘the way of God’

Similar contractions of the attributive particle on the head noun as have been described above in the C. Qaraqosh dialect, e.g.,

- (20) C. Barwar (Khan 2008a: 399)

- a. *xāṣ-t dawera*
back-ATTR mule
‘the back of a mule’ (< *xāṣət dawera*)
- b. *ʾənə miya*
spring.ATTR water
‘the spring of water’ (< *ʾənət miya*)
- c. *brōn malka*
son king
‘the son of a king’ (< *bronət malka*)

Occasionally the attributive particle is separated from the head noun and attached to the dependent noun. In such cases the particle has the form *ʾət*, which resembles the *-ət* ending of head nouns with the particle (e.g., *bəθət* ‘house of’):

- (21) C. Barwar (Khan 2008a: 398)

malka ʾət-Maġrəb
king ATTR-Maghreb
‘the king of the Maghreb’

When the head noun is a loanword without a final Aramaic nominal inflectional vowel *-a*, the attributive particle still has the form */ət/*, either suffixed to the head noun or separated from it. In (22b) the loanword *ḥabba* ends in the vowel *-a*, but this is its final vowel in the source language and not the Aramaic nominal inflectional vowel. It is connected to the *-ət* suffix by a glide */y/*:

- (22) C. Barwar

- a. *puwuš ʾət-ʾilana*
kindling ATTR-tree
‘kindling of a tree’ (Khan 2008a: 398)

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- b. *ḥabbay-ət xəttiθa*
 seed-ATTR grain
 ‘a seed (*ḥabba*) of grain’ (Khan 2008a: 397)

A phrase introduced by *’ət* may be used predicatively:

- (23) C. Barwar (Khan 2008a: 493)
zawna wiye=le ’ət-šlama.
 time be.PTCP=COP.3SG.M ATTR-peach
 ‘The time has become one of peace.’

When a demonstrative pronoun follows a head noun in an attributive annexation construction, the pronoun is prefixed by a voiced reflex of the particle *d*-, e.g.,

- (24) C. Barwar (Khan 2008a: 397)
čadra do-malka
 tent ATTR.DEM.SG.M-king
 ‘the tent of that king’

The demonstrative *do* here can be considered to be an oblique adnominal form of the demonstrative *’o*, which has formed by the bonding of the attributive particle *d* to it (Cohen 2010; Gutman 2018; 2021). In (24) the head noun does not have any attributive clitic. In many cases in C. Barwar, however, the head noun has the attributive suffix *-ət* in such constructions, which is sometimes contracted to *-ə*, e.g.,

- (25) C. Barwar (Khan 2008a: 397-399)
- a. *’ahwalt-ət do naša*
 condition-ATTR OBL.DEM.SG.M person
 ‘the condition of that person’
 - b. *rixə do-xamra*
 smell.ATTR OBL.DEM.SG.M-wine
 ‘the smell of that wine’

The *-ə* here can be considered to be a contracted form of *-ət*, so that the head noun is marked by the attributive particle and followed by an oblique pronoun (< *rixət do-xamra*). All sets of demonstrative pronouns have oblique forms (see Table 11.1).

Also the interrogative pronouns *’eni* ‘who? which?’ have oblique forms, e.g.,

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Table 11.1: Case distinction in demonstrative pronouns in C. Barwar

	SG.M	SG.F	PL
NEAR DEIXIS INDEPENDENT/ADNOMINAL			
Nominative	<i>’awwa</i>	<i>’ayya</i>	<i>’anna</i>
Oblique	<i>dawwa</i>	<i>dayya</i>	<i>danna</i>
FAR DEIXIS INDEPENDENT/ADNOMINAL			
Nominative	<i>’āwaha</i>	<i>’āyaha</i>	<i>’ānaha</i>
Oblique	<i>dāwaha</i>	<i>dāyaha</i>	<i>dānaha</i>
ANAPHORIC INDEPENDENT			
Nominative	<i>’aw</i>	<i>’ay</i>	<i>’ani</i>
Oblique	<i>daw</i>	<i>day</i>	<i>dani</i>
ANAPHORIC ADNOMINAL			
Nominative	<i>’aw/’o</i>	<i>’ay/’ε</i>	<i>’an</i>
Oblique	<i>daw/do</i>	<i>day/dε</i>	<i>dan</i>

(26) C. Barwar (Khan 2008a: 398)

- a. *bεθa dεni?*
house OBL.who
‘Whose house?’
- b. *bεθə dεni?*
house.ATTR OBL.who
‘Whose house?’

The formation of the oblique demonstrative and interrogative pronouns is not a purely phonetic process, since, as seen in (19), the attributive particle would not normally be bound with a following word beginning with a laryngeal and a vowel. One motivating factor may have been the analogy of the independent attributive pronoun:

- (27) *bεθa diya* ‘her house’
bεθa day ‘the house of that one (sg.f)’

The full paradigm of the attributive pronoun in C. Barwar is as follows:

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- (28)
- | | |
|-------|---------------|
| 3SG.M | <i>diye</i> |
| 3SG.F | <i>diya</i> |
| 3PL | <i>diyay</i> |
| 2SG.M | <i>diyux</i> |
| 2SG.F | <i>diyax</i> |
| 2PL | <i>diyexu</i> |
| 1SG | <i>diyi</i> |
| 1PL | <i>diyən</i> |

When the dependent noun is set up in a contrastive opposition with a dependent noun in a parallel clause, a demonstrative pronoun may be placed before the attributive particle to give it enhanced salience:

- (29) C. Barwar (Khan 2008a: 492)
- | | | | | | |
|-------------|----------------|-------------|----------------|-----------------|--------------|
| <i>bεθa</i> | <i>ʔo-t</i> | <i>xoni</i> | <i>goɾε=le</i> | <i>bas-bεθa</i> | <i>ʔo-t</i> |
| house | PRO.SG.M-GEN | brother.my | big=COP.3.SG.M | house | PRO.SG.M-GEN |
| <i>xaθi</i> | <i>zoɾε=le</i> | | | | |
| sister.my | big=COP.3.SG.M | | | | |
- ‘The house of my brother (literally: the house, the one of my brother) is big but the house of my sister is small.’

11.1.3.2 After Prepositions

An attributive particle is sometimes suffixed to a noun when it is the head of a phrase expressing a spatial or temporal relation to the dependent or to a preposition at the head of a prepositional phrase:

- (30) C. Barwar (Khan 2008a: 444)
- | | |
|----------------|---------------|
| <i>šawp-ət</i> | <i>bron-i</i> |
| in-place-ATTR | son-my |
- ‘in place of my son’

Many prepositions do not take the attributive suffix. When the complement of the preposition is a demonstrative pronoun, however, the demonstrative is generally in its oblique form beginning with *d-* after such prepositions.

- (31) C. Barwar (Khan 2008a: 440)
- | | |
|----|-------------------|
| a. | <i>mən-ʔilane</i> |
| | from-trees |
- ‘from the trees’

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- b. *mən-dε-ṭasa*
 from-OBL.DEM.SG.F bowl
 ‘from that bowl’

The attributive particle links a head noun to a relative clause:

- (32) C. Barwar (Khan 2008a: 952)
ʿε-brat-ət maluš-i-wa-la julle
 DEM.F.SG-girl-ATTR dress-3PL-PST-3SG.F clothes
 ‘the girl whom they dressed in clothes’

11.1.4 C. Urmi

The C. Urmi dialect in the north-east of the NENA area has a profile of attributive particle that is similar to that C. Barwar, with a few further developments.

The attributive ending *-ət* can be separated from the head noun:

- (33) C. Urmi (Khan 2016b: vol. 1, 536)
beta ʿət-malca
 house ATTR-king
 ‘the house of the king’

When the head noun is a loanword without an Aramaic nominal inflectional ending *-a*, the attributive particle still has the form *-ət*:

- (34) C. Urmi (Khan 2016b: vol. 1, 536)
+sarbazət malca
 soldier.ATTR king
 ‘the soldier (+sarbaz) of the king’

The head noun may be gapped, resulting in a nominal phrase consisting of the attributive element and the dependent noun. This is found where the gapped head is indefinite without a specific referent, e.g.,

- (35) C. Urmi (Khan 2016b: vol. 2, 77)
ʿət-leša jəpna ci-⁺kar-i-va-la.
 ATTR-dough trough IND-call-3PL-PST-OBL.3SG.F
 ‘(One) of dough (i.e., a basin for dough) was called a kneading trough.’

A phrase introduced by *ʿət* may act as an ascriptive predicate:

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- (36) C. Urmi (Khan 2016b: vol. 2, 78)

'aha beta 'ət-malcə=lə

this house ATTR-king=COP.3SG.M

'This house is one of the king (= is the king's).'

The head noun regularly has the *-ət* ending before oblique demonstrative pronouns and the inflected attributive pronoun:

- (37) C. Urmi (Khan 2016b: vol. 1, 540)

a. *bet-ət do naša*
house-ATTR OBL.DEM.SG.M person
'the house of that person'

b. *bet-ət diyu*
house-ATTR ATTR.3SG.M
'his house'

The free-standing attributive particle *'ət* has an oblique form *dət*, e.g.,

- (38) C. Urmi (Khan 2016b: vol. 1, 541)

'up 'ən 'iva max dət xinə
even if COP.PST.3SG.M like OBL.GEN others

'although it (an unspecified token of a class of bathrooms) was like one of the others (i.e., one of the bathrooms of other people)'

This indicates that the independent *'ət* particle in this context has a covert indefinite pronominal head ('one of').

A dependent phrase introduced by *'ət* may be fronted before the head phrase. In such cases, a pronominal copy of the dependent phrase is attached to the head in the form of a possessive suffix. The motivation for the fronting is to set up the referent of the dependent phrase in a contrastive opposition to other salient referents in the context:

- (39) C. Urmi (Khan 2016b: vol. 2, 75)

'ət-do brun-u sura mattəttun jull-u.

ATTR-OBL.DEM.SG.M son-his small put.2PL clothes-his

'Lay out the clothes of his youngest son.'

The fronted dependent phrase may be an interrogative, e.g.,

- (40) C. Urmi (Khan 2016b: vol. 2, 75)

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ʔat-dəm-našə=lə nob-u?
ATTR-OBL.which-person=COP.SG.M turn-his
'Which person's turn is it?'

As in C. Qaraqosh and C. Barwar, the attributive particle is also used to link a preposition to its complement and to link a head noun to a relative clause:

- (41) a. C. Urmi (Khan 2016b: vol. 1, 577)
 ʔaltix-ət beta
 under-ATTR house
 'under the house'
 b. C. Urmi (Khan 2016b: vol. 2, 451)
 ʔan-náš-ət ⁺bay-i ʔati
 DEM-people-ATTR want-3PL come.3PL
 'those people who want to come'

Many prepositions do not have a linking attributive particle but in all cases a following demonstrative is in its oblique form, e.g.,

- (42) C. Urmi (Khan 2016b: vol. 1, 575)
 ʔax do-naša
 like OBL.DEM.SG.M.-man
 'like that man'

11.1.5 Christian Dialects of the Eastern Sector of NENA

Some of the Christian dialects in eastern Iraq exhibit developments of the attributive annexation constructions distinct from those of the Mosul plain and northern sector.

11.1.5.1 C. Armoṭa

As in dialects considered so far, in C. Armoṭa, the attributive particle is used as a suffix on the head noun when followed by a dependent nominal phrase or relative clause and on prepositions:

- (43) C. Armoṭa (Khan, field notes)
 a. bet-ət xor-i
 house-ATTR friend-my
 'the house of my friend'

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- b. 'awa naš-ət gu-bete=le
 DEM man-ATTR in-house=COP.3SG.M
 'the man who is in the house'
- c. reš-ət mez
 on-GEN table
 'on the table'

When the dependent nominal begins with a demonstrative pronoun, the demonstrative may remain in its nominative form, e.g.,

- (44) C. Armoṭa (Khan, field notes)
 bet-ət 'awa naša
 house-ATTR DEM man
 'the house of that man'

Alternatively, the demonstrative may be preceded by the particle *di-* without any particle suffixed to the head noun:

- (45) C. Armoṭa (Khan, field notes)
 beta di-'awa naša
 house ATTR-DEM.SG.M man
 'the house of that man'

A sequence such as *di-'awa* is equivalent to an oblique demonstrative such as *dawa* in other dialects. The *di-* particle appears to have developed by analogy with the first syllable of the independent attributive pronoun *did-*, e.g.,

- (46) C. Armoṭa (Khan, field notes)
- a. beta didan
 house ATTR.PRO.1PL
 'our house'
- b. beta didax
 house ATTR.PRO.2SG.F
 'your (sg.f) house'

This supports the hypothesis that one of the factors that motivated the development of the oblique form of the demonstrative pronouns in other dialects was analogy with the inherited attributive pronoun.

The *di* particle may be used before dependent nominals, e.g.,

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- (47) C. Armoṭa (Khan, field notes)
'e-beta harzan di-xor-i
 DEM-house cheap ATTR-friend-my
 'the cheap house of my friend'

It can be combined with the attributive suffix *-t*, e.g.,

- (48) C. Armoṭa (Khan, field notes)
'e-beta harzan di-t xor-i
 DEM-house cheap ATTR-ATTR friend-my
 'the cheap house of my friend'

The *di-* particle is used in predicative constructions such as:

- (49) C. Armoṭa (Khan, field notes)
'aya beta di-māni=le?
 DEM house ATTR-who=COP.3SG.M
 'This house is whose? (= belongs to whom?)'

11.1.5.2 C. Sulemaniyya

The C. Sulemaniyya dialect has oblique demonstrative pronouns and these are preceded by the attributive particle, e.g.,

- (50) C. Sulemaniyya (Khan, field notes)
ksaw-ət dawa naša
 book-ATTR OBL.DEM.SG.M man
 'the book of that man'

The third-person oblique demonstratives take the place of the inherited third-person forms in the paradigm of the independent attributive pronoun:

- (51)
- | | |
|-------|----------------|
| 3SG.M | <i>dawa</i> |
| 3SG.F | <i>doya</i> |
| 3PL | <i>doni</i> |
| 2SG.M | <i>didux</i> |
| 2SG.F | <i>didax</i> |
| 2PL | <i>didoxən</i> |
| 1SG | <i>didi</i> |
| 1PL | <i>didan</i> |

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This is further evidence for the relationship between the paradigm of the independent attributive pronoun and oblique demonstrative forms.

Predicative genitive phrases are introduced by the particle *hən* (often shortened to *hi*) rather than the attributive particle *d/t*, e.g.,

- (52) C. Sulemaniyya (Khan, field notes)
hən/hi-Yosəp=ile
 ATTR-PN=COP.3SG.M
 ‘It is Yosəp’s/It belongs to Yosəp.’

This may be followed by an oblique demonstrative pronoun or the independent attributive pronoun:

- (53) C. Sulemaniyya (Khan, field notes)
- a. *hi-dawe=le*
 ATTR-OBL.DEM.3SG.M=COP.3SG.M
 ‘It is his.’
 - b. *hən-didi=le*
 ATTR-GEN.1SG=COP.3SG.M
 ‘It is mine.’

This *hən* particle resembles the shape of the particle *həl-*, which is attested in predicative genitive constructions in the J. Sanandaj dialect (Khan 2009: 568), and the particle *ʔəl-*, which is used in this context in the J. Sulemaniyya dialect (Khan 2004: 192). These all derive from the Aramaic preposition **l-* ‘to’.

The attributive particle is suffixed to prepositions, e.g.,

- (54) C. Sulemaniyya (Khan, field notes)
- a. *xes-ət mez*
 under-ATTR table
 ‘under the table’
 - b. *xes-ət day mez*
 under-ATTR OBL.DEM
 ‘under this table’

The attributive particle is generally not used, however, to link a head noun to a relative clause. Rather, the connection is either asyndetic or the head noun is linked by the Iranian relative particle *ka*:

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- (55) C. Sulemaniyya (Khan, field notes)
- a. 'o-*gora* 'ise-le 'ammi=le
DEM-man come.PST-OBL.3SG.M un-cle=COP.3SG.M
'The man who came is my paternal uncle.'
- b. 'o-*gora* ka maya šate=le xor-i=le.
DEM-man REL water drink.3SG.M=COP.3SG.M friend-my=COP.3SG.M
'The man who is drinking water is my friend.'

11.1.6 Jewish *Lišana Deni* Dialects

A distinctive feature of this cluster of dialects is the reduction of the stem of the attributive pronoun from *did-* to *d-* before plural suffixes, e.g.,

- (56) J. Dohok (Molin 2024: 70)

3SG.M	<i>did-e</i>
3SG.F	<i>did-a</i>
3PL	<i>d-oni</i>
2SG.M	<i>did-ox</i>
2SG.F	<i>did-ax</i>
2PL	<i>d-oxun</i>
1SG	<i>did-i</i>
1PL	<i>d-eni</i>

The attributive particle is used to link a head noun to a dependent noun, typically as a suffix in the form of *-at/-əd* on the head noun, but also sometimes *d-* attached to the dependent noun:

- (57) J. Dohok (Molin 2024: 77)

- a. *beθ-ət xur-i*
house-ATTR friend-my
'the house of my friend'
- b. *l'ine d-pare*
jugs ATTR-money
'jugs of money'

Oblique forms of demonstrative pronouns are used after a head noun with an attributive particle or after prepositions:

- (58) J. Betanure (Mutzafi 2008a: 168)

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- a. *praqyaθ-əd do-gora*
tonsils-ATTR OBL.DEM.SG.M-man
'the tonsils of that man'
- b. *gu-de dana*
in-OBL.DEM.SG.F time
'at that time'

The attributive particle is sometimes completely elided,

- (59) J. Duhok (Molin 2024: 77)
brat 'amoy-ox (< *brat-ət 'amoy-ox*)
daughter paternal.uncle-2SG.M
'your paternal cousin'

Head nouns may optionally be connected to the attributive pronoun with the attributive particle:

- (60) J. Betanure (Mutzafi 2008a: 42)
- a. *beθa did-i*
house GEN-1SG
'my house'
 - b. *beθ-ət did-i*
house-ATTR GEN-1SG
'my house'

The stem *did-* of the attributive pronoun paradigm is also used to link a head noun to a dependent noun, realised as *did* (Mutzafi 2008a: 92; Cohen 2008: 95) or *dət*:

- (61) J. Dohok (Molin 2024: 77)
kəsta dət pare
bag ATTR money
'a bag of money'

Occasionally attributive annexation is expressed by both *d-* or its allomorph and *did/dət*, e.g.,

- (62) J. Betanure (Mutzafi 2008a: 42)
ṁayə-d did de-kθeθa
water-ATTR ATTR OBL.DEM-chicken
'the broth (literally: water) of that chicken'

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The attributive particle links a head noun to a relative clause:

- (63) J. Betanure (Mutzafi 2008a: 41, 168)
səmmor-əd xze-lan
 squirrel-ATTR see.PST-OBL.1PL
 ‘the squirrel that we saw’

11.1.7 Jewish Trans-Zab dialects

Jewish Trans-Zab dialects generally place the stress in nouns on the final syllable. The attributive particle, however, does not take the stress, so it is best analysed as a clitic.

11.1.7.1 J. Arbel

In the J. Arbel dialect, the attributive clitic particle =*ət* connects a head noun to a dependent nominal phrase:

- (64) J. Arbel (Khan 1999a: 168)
son=ət bab-i
 grandfather=ATTR father-my
 ‘the grandfather of my father’

The *t* of the ending is occasionally elided:

- (65) J. Arbel (Khan 1999a: 168)
kolan=ə mšilmane
 street=ATTR Muslims
 ‘the street of the Muslims’

Full elision of the attributive suffix is attested in closely-knit conventionalised phrases, such as those headed by kinship terms:

- (66) J. Arbel (Khan 1999a: 169)
brōn-Maraj
 son-Maraj
 ‘the son of Maraj’

The attributive particle is sometimes attached to the dependent noun:

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- (67) J. Arbel (Khan 1999a: 169)

xalwa t-noš-aw
 milk ATTR-self-her
 ‘her own milk’

If the dependent nominal phrase begins with a demonstrative pronoun, this is generally not in an oblique form, though some sporadic cases of oblique demonstratives are attested, all near deixis demonstratives:

- (68) J. Arbel (Khan 1999a: 168, 315)

- a. *mewan=ət* *’o-bela*
 guests=ATTR DEM-house
 ‘the guests of that house’
- b. *bab=ət* *diyya* *fāqīr*
 father=ATTR OBL.DEM poor
 ‘the father of that poor man’

There is a full paradigm of the independent attributive pronoun. The head noun is generally connected to this by the attributive particle *-ət*:

- (69) J. Arbel (Khan 1999a: 219)

lišan=ət *did-an*
 language=ATTR GEN-1PL
 ‘our language’

In some cases, a dependent nominal is linked to the head noun by the phrase *’o=t*, i.e., a singular demonstrative combined with the attributive particle. This is also used when the head noun is plural:

- (70) J. Arbel (Khan 1999a: 224)

bšəlmane *’o=t* *’Arbel*
 Muslims DEM=ATTR PN
 ‘the Muslims of Arbel’

When the dependent component of an attributive annexation construction is used in an ascriptive predicate without a head noun, it is regularly introduced by the phrase *’o=t*, e.g.,

- (71) J. Arbel (Khan 1999a: 315)

kulla *’o=t* *bab=ət* *diyya* *faqīr=ile*
 all DEM=GEN father=ATTR OBL.DEM poor=COP.3SG.M
 ‘Everything belongs to the father of this poor man.’

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In the speech of an informant who spent his childhood in Batas, 50 kilometres north-east of Arbel, the attributive suffix on a head noun is sporadically combined with an additional particle *ʔi* linking it to a dependent noun. This is likely to be etymologically a demonstrative pronoun (cf. the /i/ element in the near deixis demonstrative *ʔiyya*). It is glossed here ATTR, e.g.,

- (72) J. Arbel (Khan 1999a: 169)
kulla mamlakat=ət ʔi-Kurdistan
 all towns-ATTR ATTR-Kurdistan
 ‘all the towns of Kurdistan’

The attributive particle links a head noun to a relative clause:

- (73) *ʔo-kep=ət pil-le reš-eu*
 DEM-stone=ATTR fall.PST-OBL.3SG.M upon-3SG.M
 ‘the stone that fell on him’

In many cases, a relative clause (both restrictive and non-restrictive) is linked to the head noun by the phrase *ʔo-t*, i.e., a singular demonstrative combined with the attributive particle. This is used also when the head noun is plural:

- (74) J. Arbel (Khan 1999a: 388)
- a. *ʔiyya Ḥatan k-imr-i-wa-le ʔo=t*
 DEM PN IND-say.PRS-PL-PST-OBL.3SG.M DEM=ATTR
me-le-lli
 bring.PST-OBL.3SG.M-OBL.1SG
 ‘the man called Ḥatan who brought me’
- b. *kre ʔo=t loš-i-lu*
 suits DEM=REL wearprs-textsc3pl-obl.3pl
 ‘the suits that they wear’

11.1.7.2 J. Ruwanduz

In the dialect of the Jews of Ruwanduz, in Iraq situated to the north-east of Arbel, attributive annexation is expressed both by the *=ət* clitic or, more frequently, by the *ʔi* particle:

- (75) J. Ruwanduz (Khan, field notes)

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- a. *tār=ət mala*
gate-ATTR village
'gate of the village'
- b. *kništa 'i hula'e*
syngagogue ATTR Jews
'the syngagogue of the Jews'

The two linking strategies may be combined:

- (76) J. Ruwanduz (Khan, field notes)
bab=ət 'i bārux-i
father=ATTR ATTR friend-my
'the father of my friend'

In the J. Ruwanduz dialect there is a paradigm of the independent attributive pronoun (*did-*), but it appears that oblique forms of demonstrative pronouns are not used:

- (77) J. Ruwanduz (Khan, field notes)
 - a. *ḥawš did-an*
courtyard GEN-1PL
'our house'
 - b. *xamr=ət did-u*
wine=ATTR GEN-3PL
'their wine'
 - c. *wil-li ta 'o-naša*
give.PST-OBL.1SG to DEM-man
'I gave to that man'

When the dependent nominal is predicative, it is introduced by *i*:

- (78) J. Ruwanduz (Khan, field notes)
ya mez 'i bārux-i=le
this table ATTR friend-my=COP.3SG.M
'This table is my friend's (belongs to my friend).'

The *'i* particle links a head noun to a relative clause:

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- (79) J. Ruwanduz (Khan, field notes)

ya gora 'i ga-šuqa gze-lox bāruḫ-i=le
 this man ATTR in-market see.PST-OBL.2SG.M friend-my=COP.3SG.M
 'The man whom you saw in the market is my friend.'

In a translation of the Hebrew Bible into the J. Ruwanduz dialect, the Hebrew relative particle *'ašer* is regularly translated by a particle with the form *'ay* (Rees 2008: 27–28). This is likely to be the archaic antecedent of the particle *'i* in the vernacular J. Ruwanduz dialect. The form *ay* has survived in the vernacular J. Urmi dialect (see §11.1.7.4).

Only the indefinite pronouns *manni* 'whoever' and *ma* 'whatever' are linked to a following relative clause by *-t*:

- (80) *manni=t g-be 'āde ba-'āde*
 who=ATTR IND-want.PRS.3SG.M come.PRS.3SG.M let-come.PRS.3SG.M
 'Whoever wants to come, let him come.'

- (81) *wəl ma=t g-be-t*
 give.IMP.SG what=ATTR IND-want.PRS-2SG.M
 'Give whatever you want.'

11.1.7.3 J. Rustaqa

In the dialect of the Jews Rustaqa, in Iraq lying to the north-east of Ruwanduz, only the particle *i* is used to connect a head noun to a dependent noun.¹ This is typically attached to the end of the head noun as a clitic:

- (82) J. Rustaqa (Khan 2002b: 408–9)
- a. *kništa=i mala*
 synagogue=ATTR village
 'the synagogue of the village'
 - b. *hule=i Rustaqa*
 Jews=ATTR Rustaqa
 'the Jews of Rustaqa'

The pratical *i* introduces a predicative genitive phrase:

¹In this dialect there is no insertion of a glottal stop ' before a word-initial vowel.

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- (83) J. Rustaqa (Khan 2002b: 408–9)
'anna bate kull-u i rešane=lu
 these houses all-3PL ATTR king=COP.3PL
 ‘all those houses belong to the king’

The particle *i* connects a head noun to a relative clause:

- (84) J. Rustaqa (Khan 2002b: 408–9)
'o gora=i tammal ədye-le dost-i=le
 DEM man=ATTR yesterday come.PST-OBL.3SG.M friend-my=COP.3SG.M
 ‘The man who came yesterday is my friend.’

The =*t* particle is only used to link a head to a relative clause when the head is one of the indefinite pronouns *manni* ‘whoever’ and *ma* ‘whatever’:

- (85) J. Rustaqa (Khan 2002b: 408–9)
- a. *manni=t ade bel-an paşəx*
 who=ATTR come.PRS.3SG.M house-our rejoice.PRS.3SG.M
 ‘Whoever comes to our house will be pleased.’
 - b. *ma=t kayf-ox made 'ol*
 what=ATTR pleasure-your bring.PRS.3SG.M do.IMP.SG
 ‘Do whatever you want.’

In such cases the =*t* clitic is sometimes followed by the *i* particle:

- (86) J. Rustaqa (Khan 2002b: 408–9)
- a. *manni=t=i abe maşe ade*
 who=REL=DEM want.PRS.3SG.M can.3SG.M come.PRS.3SG.M
 ‘Whoever wants can come’
 - b. *ma=t=i abe-t 'ol*
 what=REL=DEM want.PRS-2SG.M do.IMP.SG
 ‘Do what you want’

11.1.7.4 J. Urmi

In the J. Urmi dialect in north-western Iran the attributive particle *-ət* is cliticised to the head noun:

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(87) J. Urmi (Khan 2008b: 174)

tar=ət bela

door=ATTR house

‘the door of the house’

When the dependent noun of a head noun or preposition begins with a demonstrative pronoun or the interrogative particle *emni/emnu* ‘which’, these are in an oblique form beginning with *d-*:

(88) J. Urmi (Khan 2008b: 175, 192)

a. *⁺nišān do parda*

symbol OBL.DEM curtain

‘the symbolic meaning of that curtain’

b. *dad=ət do brona*

father=ATTR OBL.DEM son

‘the father of that son’

c. *bar do gora*

behind OBL.DEM man

‘behind that man’

An attributive relation between two nominals may also be expressed by the particle *ay*.² This is historically a demonstrative pronoun but is glossed here ATTR. It is normally used in combination with the attributive particle *ɾ* on the head nominal, e.g.,

(89) J. Urmi (Khan 2008b: 175)

lel=ət ay xlula

night-ATTR ATTR wedding

‘the night of the wedding’

The attributive particle is sometimes attached to the *ay* in its voiced form *d-*, viz. *day*. This resembles an oblique demonstrative form:

(90) J. Urmi (Khan 2008b: 175)

a. *lele day xlula*

night OBL.DEM wedding

‘the night of the wedding’

²In the J. Urmi dialect a glottal stop ʔ is not inserted before a word-initial vowel.

11.1 Linking dependent elements to a head noun in NENA

- b. *bel=ə day flankas*
 ouse=ATTR obl.dem so-and-so
 ‘the house of so-and-so’

If the dependent nominal begins with a demonstrative pronoun, the demonstrative is in its oblique form:

- (91) J. Urmi (Khan 2008b: 176)
⁺*qayd=ət ay d-o tka*
 custom=ATTR ATTR OBL-DEM place
 ‘the custom of that place’

Sporadically, the attributive particle *ṭ* is omitted, and the construction contains only the particle *ay*:

- (92) J. Urmi (Khan 2008b: 175)
qadi ay goyme
 judge ATTR Muslims
 ‘the judge of the Muslims’

When the dependent nominal of an attributive annexation construction is used in an ascriptive predicate without a head noun, it is regularly introduced by the particle *ay*:

- (93) J. Urmi (Khan 2008b: 233)
kalo ay noš-ew=ila
 bride ATTR self-3SG.M=COP.3SG.F
 ‘The bride is his.’

The *ay* particle is used to connect a nominal to a following nominal in apposition:

- (94) J. Urmi (Khan 2008b: 219)
brona ay ⁺hatān masq-i-wa-le reš-⁺gare
 boy ATTR groom take.upprs-textsc3pl-pst-obl.3sg.m on-roof
 ‘They would take up the boy, the groom, onto the roof.’

When a head nominal is qualified by a relative clause, the linker is generally the Iranian particle *ki*:

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- (95) J. Urmi (Khan 2008b: 353)
qus ki baxta=ila
 bird REL wife=COP.3SG.F
 ‘the bird who is the wife’

The attributive particle *-t* is restricted to a closed set of pronominal and adverbial heads:

- (96) J. Urmi (Khan 2008b: 357)
māni=t ade
 who-ATTR come.PRS.3SG.M
 ‘whoever comes’

11.1.7.5 J. Sulemaniyya

In the J. Sulemaniyya dialect, the head nominal is sometimes linked to a dependent nominal by the attributive particle *-ət*, e.g.,

- (97) J. Sulemaniyya (Khan 2004: 192)
tat=ət ɣatan
 father=ATTR groom
 ‘the father of the groom’

More frequently, however, the head noun and dependent noun are juxtaposed without a linking particle, e.g.,

- (98) J. Sulemaniyya (Khan 2004: 192)
šamma brona
 name boy
 ‘the name of the boy’

Demonstrative pronouns, however, are often in an oblique form beginning with *d-* after head nouns or prepositions:

- (99) J. Sulemaniyya (Khan 2004: 192)
- a. *šwawe do naše*
 neighbours OBL.DEM people
 ‘the neighbours of those people’
 - b. *bar do bela*
 behind OBL.DEM house
 ‘behind that house’

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Nominal phrases beginning with adnominal demonstratives may also be juxtaposed with the head noun without the *d* element:

- (100) J. Sulemaniyya (Khan 2004: 261)
baba 'o-brona
 father DEM.SG-boy
 ‘the father of that boy’

The oblique form is obligatory when it is an independent demonstrative:

- (101) *bela do*
 house OBL.DEM.SG.M
 ‘the house of that one (= his house)’

The oblique demonstratives are used in free variation with the inherited third-person forms in the paradigm of the independent attributive pronoun:

- (102)
- | | |
|-------|-----------------------|
| 3SG.M | <i>didew do, dido</i> |
| 3SG.F | <i>didaw do, dido</i> |
| 3PL | <i>didu doni</i> |
| 2SG.M | <i>didox</i> |
| 2SG.F | <i>didax</i> |
| 2PL | <i>didāxun</i> |
| 1SG | <i>didi</i> |
| 1PL | <i>didan</i> |

This alternation and also the hybrid form *dido* supports the hypothesis that the oblique demonstratives arose by analogy with the paradigm of the independent attributive pronoun.

Occasionally, a head noun is linked to a dependent noun by the particle *'i* (generally attached to the head noun as a clitic). This is attested after loanwords:

- (103) J. Sulemaniyya (Khan 2004: 192–93)
maktab=i hulaye
 school=ATTR Jews
 ‘school of the Jews’

When an attributive phrase is used predicatively, it is introduced by the particle **'əl* ‘to’ rather than an attributive particle:

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- (104) J. Sulemaniyya (Khan 2004: 192)
'ay torxa 'əl-do gora=ya
 this bag to-OBL.DEM man=COP.3SG.F
 'This bag belongs to that man.'

The attributive particle =*ət* has been completely lost in relative clauses with heads that are lexical nouns. Instead, a head noun is linked to the relative clause by the Iranian particle *ka/ga*:

- (105) J. Sulemaniyya (Khan 2004: 414)
baba 'o-brona ga-ləbl-a-le
 father DEM-boy REL-take.PST-3SG.F-OBL.3SG.M
 'the father of the boy who took her'

The attributive particle is restricted to pronominal heads:

- (106) J. Sulemaniyya (Khan 2004: 418-419)
- a. *'o=t bazərgān=ye-le*
 DEM.SG.M=REL clothier=COP.PST-OBL.3SG.M
 'whoever was a clothier'
 - b. *mani=t ḥatan ḥāwe-wa*
 who=ATTR groom be.PRS.3SG.M-PST
 'whoever was a groom'

11.1.7.6 J. Sanandaj

In the J. Sanandaj dialect in western Iran, an attributive relation between a head noun and a following dependent noun is generally expressed by juxtaposition without any linking particle:

- (107) J. Sanandaj (Khan 2009: 199)
- a. *bela bārux-i*
 house son-my
 'the house of my son'
 - b. *lele xlula*
 night wedding
 'the night of the wedding'

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On some occasions, the head noun is linked to the dependent noun by the particle *ʿe* (generally attached to the end of the head noun as a clitic). This is likely to be a demonstrative in origin and have the same etymology as the particles *ʿay* and *ʿi* in other Trans-Zab dialects:

- (108) J. Sanandaj (Khan 2009: 199)
bela=e bārux-i
 house=ATTR friend-my
 ‘the house of my friend’

A preposition may likewise be linked to the dependent nominal by *ʿe*:

- (109) J. Sanandaj (Khan 2009: 220)
dawr=e mez
 around=ATTR table
 ‘around the table’

The attributive particle *d* is used only when the dependent component of an annexation construction contains a demonstrative pronoun. The *d* is attached to the demonstrative rather than to the preceding head noun. Such cases can be considered to be oblique forms of demonstratives:

- (110) J. Sanandaj (Khan 2009: 200)
bela do naša
 house OBL.DEM.SG.M man
 ‘the house of that man’

This also applies to phrases with a preposition as head:

- (111) J. Sanandaj (Khan 2009: 219)
baqa do naša
 to OBL.DEM man
 ‘to that man’

The oblique demonstrative after a head noun may be preceded by the linking particle *ʿe*:

- (112) J. Sanandaj (Khan 2009: 200)
fəšar=e do mae
 pressure=ATTR OBL.DEM.PL water
 ‘the pressure of that water’

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Nominal phrases beginning with demonstratives may also be juxtaposed with the head noun without the *d* element:

- (113) J. Sanandaj (Khan 2009: 200)
bela 'o naša
 house DEM man
 'the house of that man'

When, however, the demonstrative is an independent pronoun, the use of the oblique demonstrative is obligatory:

- (114) J. Sanandaj (Khan 2009: 200)
bela do
 house OBL.DEM.SG.M
 'the house of that one (= his house)'

Constructions such as these stand in a paradigmatic relationship with phrases containing the independent attributive pronoun *did-* with 1st or 2nd person suffixes. Unlike in J. Sulemaniyya, the inherited third person forms of the paradigm have been completely supplanted by the oblique demonstratives:

- (115)
- | | |
|-------|----------------|
| 3SG.M | <i>do</i> |
| 3PL | <i>doni</i> |
| 2SG.M | <i>didox</i> |
| 2SG.F | <i>didax</i> |
| 2PL | <i>didăxun</i> |
| 1SG | <i>didi</i> |
| 1PL | <i>didan</i> |

When an attributive phrase is used predicatively, it is introduced by the particle *həl-* (< *'əl 'to') rather than an attributive particle *t/d*:

- (116) J. Sanandaj (Khan 2009: 568)
'ay bela həl-do=ye
 this house to-OBL.3SG.M=COP.3SG.M
 'This house is his (= belongs to him).'

In J. Sanandaj, head nouns are connected to a relative clause by the particles *ya*, *'e* or *ke*:

- (117) J. Sanandaj (Khan 2009: 377-381)

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- a. 'o-naše ya da'wat kol-i-wa-lu
DEM-people REL invitation do-.PRS.3PL-PST-OBL.3PL
'the people whom they invited'
- b. 'o-baxta=e ləxm-äke k-ol-a-wa-le-o
DEM-woman=ATTR bread-ART IND-do-3SG.F-PST-OBL.3SG.M-PART
'the woman who opened out the bread'
- c. 'ay-bšəlməne ke xalwa zabn-i ta-did-an
DEM-Mulsims REL milk sellPRS-textsc3pl to-ATTR-1PL
'those Muslims who sell milk to us'

The relative particle *ya* appears to be an Aramaic demonstrative in origin. The particle *ke* is a loan from Iranian. Unlike in J. Sulemaniyya, where the Aramaic attributive particle *-t* is preserved after generic pronominal heads (see 106a), in J. Sanandaj it has been lost in this context also. In J. Sanandaj such heads are expressed by the Iranian constructions *har-kas* 'whoever' and *har-či* 'whatever' connected asyndetically to what follows or by an Iranian relative particle:

(118) J. Sanandaj (Khan 2009: 382-383)

- a. *har-kas bi-zoa hāwe-le biš*
any-person more-more be.PRS.3SG.M-OBL.3SG.M more
'ayza=y.
good=COP.3SG.M
'Anybody who has more is (considered) better.'
- b. *har-či 'āt k-əmr-at*
any-INDF you.SG say.PRS-2SG.F
'every thing that you say'
- c. *har-či ke mən-xalwa=ye-le*
any-INDF REL from-milk=COP.PST-OBL.3SG.M
'anything that was (made) from milk'

11.1.7.7 J. Kerend

In the Jewish dialect of Kerend, the most southerly place where NENA was spoken in western Iran, head nouns are regularly juxtaposed with a dependent noun without a linking particle:

(119) J. Kerend (Khan, field notes)

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bela bǎrux-i
house friend-my
'the house of my friend'

Unlike other Jewish Trans-Zab dialects, the independent attributive pronominal paradigm has been replaced by juxtaposition of default independent pronouns, though oblique pronouns remain an alternative in the third person:

(120) J. Kerend (Khan, field notes)

- a. 3SG.M
bela 'o ~bela do
house DEM.SG ~house OBL.DEM.SG
'his/her house'
- b. 3PL
bela 'oni ~bela d-oni
house DEM.PL ~house OBL.DEM.PL
'their house'
- c. 2SG
bela 'at
house you.SG
'your (SG) house'
- d. 2PL
bela 'axtu
house you.PL
'your (PL) house'
- e. 1SG
bela 'ana
house I
'my house'
- f. 1PL
bela 'axni
house we
'our house'

Complements of prepositions are, likewise, followed by independent pronouns rather than the attributive pronoun:

(121) J. Kerend (Khan field notes)

11.1 Linking dependent elements to a head noun in NENA

- a. *ba 'o* ~*ba do*
to DEM.SG ~to OBL.DEM.SG
'to him'
- b. *ba 'ana*
to 1SG
'to me'

When an attributive phrase is used predicatively, it is introduced by the particles 'əl- 'to', *xeta* or 'e. The latter particle can be identified with the linking particles 'e, 'ay and 'i in other Trans-Zab dialects, which are demonstratives in origin:

(122) J. Kerend (Khan field notes)

- a. 'ay *bela* 'əl-'o=*ye*
this house to-DEM.SG.=COP.3SG.M
'This house is his.'
- b. 'ay *bela* *xeta*-'o=*ye*
this house belongs.to-DEM.SG.=COP.3SG.M
'This house belongs to him.'
- c. 'ay *bela* 'e-'ana=*ye*
this house ATTR-1SG=COP.3SG.M
'this house is mine.'

11.1.8 Adjectival modifiers

In NENA dialects adjectival modifiers are normally placed after the head noun without any linking element, e.g.,

- (123) C. Barwar
bεθa gora
house big
'big house'

In some dialects an adjective may be placed before the head noun when it is an evaluative, emotion-driven judgement rather than objective description. This is shown in (124a) and (124b):

- (124) C. Barwar (Khan 2008c: 520-521)

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- a. *spay naša* ‘A good man (subjective judgement)’
good man
- b. *susa spay*
horse good
‘a good horse (objective description)’

In dialects in the north-eastern sector of NENA, such as C. Urmi, preposing of adjectives is more frequent and is used when the adjective is evaluative but also when it is in contrastive focus or when the speaker wishes to express an intensification of the property, e.g.,

(125) C. Urmi (Khan 2016a: vol. 2, 39-40)

- a. *cumə +jilasə*
black cherries
‘black cherries (contrastive with cherries of other colours)’
- b. *xa xəšcana lelə*
on dark night
‘A (particularly) dark night’

Sporadically one finds in NENA dialects constructions in which a head noun is connected to a following adjectival modifier by an annexation linker, e.g.,

(126) C. Urmi (Khan 2016a: vol. 2, 45)

- tre məššənnər-ət* *’amricayə*
two missionaries-ATTR American
‘two American missionaries’

(127) J. Arbel (Khan 1999a: 168)

- dukk=ət* *rumanta*
place=ATTR high
‘a high place’

In (128) the attributive clitic before the adjective has been elided:

(128) J. Arbel

- nāš* *xriwe* (< *našət xriwe*)
people bad
‘bad people’

11.1 Linking dependent elements to a head noun in NENA

In the J. Arbel dialect, the linking *ʔo=t* phrase is sometimes used before an adjective or an independent attributive pronoun:

(129) J. Arbel (Khan 1999a: 219, 229)

- a. *šakar ʔo=t dāqīqa*
sugar DEM=REL fine
‘fine sugar’
- b. *gure ʔo=t did-an*
men DEM=REL GEN-1PL
‘our husbands’

In dialects that do not use annexation linkers to connect a noun to a following adjective, a preposed evaluative adjective may be connected to following noun by a linker, e.g.,

(130) C. Barwar (Khan 2008c: 521)

- ʔo-māṣken-ət xon-i*
DEM-poor-ATTR brother-my
‘That poor brother of mine.’

When the head of an annexation construction is a complex noun phrase consisting of a noun and its modifier, the attributive particle is added at the end of the noun phrase. In such constructions the particle has the prosodic status of a clitic in both the northern and southern sectors of NENA:

(131) C. Urmi (Khan 2016a: vol. 2, 73)

- beta támməz=ət xat-i*
house clean=ATTR sister-my
‘The clean house of my sister’

When an adjectival attribute is set up in contrastive opposition with another adjective, the adjective may be introduced by a demonstrative pronoun:

(132) C. Barwar (Khan 2008a: 515)

- brat-ət malka ʔε-gorta ... brat-ət malka*
daughter-ATTR king DEM.SG.F-big ... daughter-ATTR king
ʔε-zorta
DEM.SG.F-small
‘the elder daughter of the king ... (literally: the daughter of the king the big one) ... the younger daughter of the king’

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When such noun phrases are dependent phrases in an annexation construction, the demonstrative pronoun is in the oblique form:

- (133) C. Barwar (Khan 2008a: 517)
bεθ-ət xona do-gorā
 house-ATTR brother OBL.DEM.SG.M-big
 ‘the house of the big brother’

11.2 Ezafe and linking dependent elements to a head noun in Kurdish

Attributive constructions in most Kurdish varieties (Kurmanji, Sorani, Zazaki, Hewramî) require the use of *ezafe* morphemes. In the more southern varieties, including CK dialects of Sanandaj and Iranian varieties spoken further south (Laki, Southern Kurdish), these constructions are formed by a simple juxtaposition of the head noun and the dependents. In some of these dialects, however, the marking of the *ezafe* is optional.

The following is a simplified comparative list:

- | | | | |
|-------|-------------|------------------------|-----------------------|
| (134) | | | ‘the son of the king’ |
| | Kobani NK | <i>kuř-i paşê</i> | (son-EZ.M king.OBL.M) |
| | Akre NK | <i>kuř-ê paşa-y</i> | (son-EZ.M king-OBL.M) |
| | Erbil CK | <i>kuř-i paşa</i> | (son-EZ king) |
| | Mukri CK | <i>kuř-i (de) paşa</i> | (son-EZ PRT king.OBL) |
| | Sanandaj CK | <i>kuř paşa</i> | (son king) |
| | Ilam SK | <i>kuř(-i) paşa</i> | (son-EZ king) |
| | Hewramî | <i>kuř-û paðşa-y</i> | (son-EZ.M king-OBL) |
| | Laki | <i>kuř(-i) paşa</i> | (son-EZ king) |

The *ezafe* morphemes are affixes in NK as they take the word-final stress. However, in CK and SK (as in the rest of Iranian) they are clitics since they are unstressed and the default word-final stress of Kurdish is shifted to the penultimate syllable when *ezafe* morphemes are attached:

- (135) NK *miróv* ‘man’ vs. *mirov-ě pîr* ‘old man’
 CK *mirów* ‘man’ vs. *mirów=i pîr* ‘old man’

In addition to the possessor and adjectives, *ezafe* morphemes also link any other dependent element that follows the head noun, such as adjectives, prepositional phrases, and relative clauses. These additional functions are especially relevant in NK.

11.2 Ezafe and linking dependent elements to a head noun in Kurdish

11.2.1 Northern Kurdish

11.2.1.1 Attributive constructions

In NK, the *ezafe* suffixes attach to the head noun, and they agree with the gender and number of the head noun, followed by an attributive element or a noun or pronoun in the oblique case, often in the function of possessor. The *ezafe* suffixes are given in Table 11.2, where the alternatives are dialectal forms (see §11.2.1.2 for ‘secondary *ezafe*’ forms).

Table 11.2: *Ezafe* forms in basic nominal attribution

	SG		PL
	M	F	PL
Primary (definite)	-ê / -î	-a / -ê	-ê / -ên / -êt
Secondary(indefinite)	-ê / -î	-a / -e	-ên / -(in)e

The phrases in (136) illustrate the primary *ezafe* forms in context with nominal and pronominal possessors in the Behdînî dialect of NK:

(136) Behdînî NK (fieldnotes)

- a. *kur-ê paşa-y*
son-EZ.M king-OBL.M
‘the son of the king’
- b. *qesr-a paşa-y*
palace-EZ.F king-OBL.M
‘the palace of the king’
- c. *kiç-êt paşa-y*
girl-EZ.PL king-OBL.M
‘the daughters of the king’
- d. *mal-a min*
house-EZ.F 1SG.OBL
‘my house’

Most of the NK dialects (including Behdînî, Hakkari, Urmi) follow this pattern, with further southern and western dialects preferring the -ê and -ên alternatives for plural. Thus, the number and gender are systematically distinguished except

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in the central and southern dialects of Mardin-Sinjar, where the masculine singular and plural are syncretised, so that *kur-ê min* can be ‘my son’ or ‘my sons’. As stated above, the primary *ezafe* forms in NK, unlike the rest of Iranian, are stressed.

11.2.1.2 Secondary *ezafe* with indefinite nouns

An indefinite head noun is expressed with the indefiniteness marker *-(y)ek*. When this marker is used, the Duhok-Shirnak dialects retain the same *ezafe* forms as with the definite nouns, while varieties further western, including the Shamdinan-Akre and Urmi dialects, have the singular forms with *-î* (sg.m) and *-e* (sg.f):

(137) Duhok-Shirnak (fieldnotes)

- a. *kur-ek-ê min*
son-INDF-EZ.M 1SG.OBL
‘a son of mine’
- b. *keç-ek-a min*
girl-INDF-EZ.F 1SG.OBL
‘a daughter of mine’

(138) Akre-Urmi (fieldnotes)

- a. *kur-ek-î min*
son-INDF-EZ.M 1SG.OBL
‘a son of mine’
- b. *kiç-ek-e min*
girl-INDF-EZ.F 1SG.OBL
‘a daughter of mine’

With plurals, in most dialects the same plural *ezafe* suffix (*-ê*, *-ên*, *-êt*) is used, as in:

(139) NK various dialects (fieldnotes)

- hindek mirov-ên/-êt me*
some human-EZ.PL 1PL.OBL
‘some relatives of ours’

However, the southern Mardin-Qamishli dialects have developed a specific plural indefinite morpheme *-in*, and the linking suffix used to connect a plural

11.2 Ezafe and linking dependent elements to a head noun in Kurdish

indefinite head noun to its dependent (possessor or adjective) is an invariant *-e*, as in:

- (140) *xanî-n-e* *paşê*
house-INDF.PL-LNK king.OBL
‘some houses of the king’

11.2.1.3 Demonstrative or pronominal *ezafe*

In a complex head noun phrase with multiple dependent elements, the second dependent element is linked to the dependent element with *ezafe*. In this case, however, the *ezafe* morphemes are prosodically either enclitic or independent words. The *ezafe* in this function is called variously ‘independent’, ‘demonstrative’ or ‘pronominal’ *ezafe*. The forms of these are given in Table 11.3.

Table 11.3: Pronominal/demonstrative *ezafe* forms

	SG		PL
	M	F	PL
Enclitic	=ê	=a	=ê / =ên / =êt
Independent	(y)ê	(y)a	(y)ê / (y)ên / yêt

In Behdîni (and most of the other central dialects), these *ezafe* forms are identical to their primary forms. They differ, however, in terms of their prosodic status.

- (141) NK various dialects (fieldnotes)
- a. *mêz-a* *me=ya* *kevn*
table-EZ.F 1PL.OBL=EZ.F old
‘our old table’
 - b. *qelem-ê* *Hesen* *yê* *reş*
pen-EZ.M Hasan.OBL EZ.M BLACK
‘Hasan’s black pen’
 - c. *pêlav-êt* *we* *yêt* *nû*
shoe-EZ.PL your EZ.PL new
‘your (pl.) new shoes’

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- (142) Behdîni (fieldnotes)
bira-yê min yê mezin
 brother-EZ.M 1SG.OBL EZ.M big
 ‘my big brother’

11.2.1.4 Pronominal *ezafe*

These demonstrative *ezafe* forms are identical to a third category of *ezafe* forms, known as ‘pronominal’ *ezafe*. These independent *ezafe* forms are used to form a noun phrase with the *ezafe* expressing the pronominal head of the phrase:

- (143) Behdîni NK (MacKenzie 1961a)
- a. *ya min*
 EZ.SG.F.PRO 1SG.OBL
 ‘mine (lit. the one that is mine)’
 - b. *yê te*
 EZ.SG.M.PRO 2SG.OBL
 ‘yours (lit. the one that is yours)’
 - c. *yê/yên/yêt wan*
 EZ.PL.PRO 3PL.OBL
 ‘theirs (lit. the ones that are theirs)’
 - d. *yêt xoş ev-e=ne*
 EZ.PL.PRO good DEM-PROX=COP.3PL
 ‘the good ones are these’

11.2.2 Central Kurdish

The *ezafe* in Central Kurdish behaves quite differently from comparable constructions in Northern Kurdish. There is a division between attributive constructions and possessive constructions in the way that they interact with definiteness.

11.2.2.1 Possessive constructions

The *ezafe* forming genitive constructions in Sorani is an invariable unstressed clitic suffix =î:

- (144) *kuř-î paşa* ‘the son of the king’
kiç-î paşa ‘the daughters of the king’
qesr-î paşa ‘the palace of the king’

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Note that the plural phrase has a more generic and unspecific reading. For a specific reading, the use of the plural definite marker is required:

- (145) *kiç-ek-an-î paşa*
 daughter-DEF-PL-EZ king
 ‘the daughters of the king’

There is a slight dialectal variation. For instance, southern Central Kurdish varieties use simple juxtaposition in possessive constructions (see 146).

- (146) Central Kurdish Sanandaj (Mohammadirad 2022: G, 0714)
kenişk pawşa
 daughter king
 ‘the king’s daughter’

11.2.2.2 Attributive constructions

The *ezafe* forming attributive constructions in Sorani is an invariable unstressed clitic suffix =î when indefinite or generic:

- (147) *kuř-î drêj* ‘tall son(s)’
kiç-êk-î cwan ‘a beautiful daughter’

However, when the head noun is definite, the *ezafe* -e is used, and all definite morphology occurs on the following adjective (see Karim 2022, 2023).

- (148) a. *kuř-e drêj-ek-an*
 boy-EZ.DEF tall-DEF-PL
 ‘the tall sons’
 b. *ew kiç-e cwan=e*
 DEM.DIST girl-EZ.DEF beautiful=DEF
 ‘that beautiful daughter’

11.2.3 Further functions of the *ezafe* functions

As stated above, *ezafe* in NK and CK introduces the post-head dependents of a noun. These can be a possessor (149a), adjective (149b, 149c), prepositional phrase (149d, 149e).

- (149) Behdîni NK (MacKenzie 1961a: 160, 204)

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- a. *kur-ê min*
son-EZ.M 1SG.OBL
'my son'
- b. *qeza-yek-a kevn*
district-INDF-EZ.F old
'an ancient district'
- c. *heywan-ê kîvî*
animal-EZ.PL wild
'wild animals' (p.204)
- d. *ew hakim-ê di sindoq-ê-da*
DEM governor-EZ.M in chest-OBL.F-POSTP
'the governor (who is) in the chest'
- e. *ew tişt-ê bin-ê behrê-da*
DEM thing-EZ.PL bottom-EZ.M sea-OBL.F-POSTP
'those things (which are) at the bottom of the sea'

In NK dialects (with the exception of some Behdîni dialects), *ezafe* also connects adjectives and nouns in compound verb constructions to their complements. This usage of *ezafe* is called 'generic' *ezafe* by MacKenzie (1961a, hence glossed as EZ.GEN):

(150) Behdîni NK (MacKenzie 1961a: 160, 161)

- a. *teslîm-î ...wezîr-î bi-k-e*
surrender-EZ.GEN vizier-OBL.M SBJV-do.PRS-IMP
'give it to the vizier'
- b. *nêzîk-î hakim-î bû*
near-EZ.GEN judge-OBL.M be.PST.3SG
'(he) was near the judge'

The dependent of the head noun introduced by *ezafe* can also be a relative clause, especially in NK. In the Behdîni dialects of NK, the *ezafe* can introduce a relative clause by itself, without any other subordinator:

(151) Behdîni (Amadiya, Akre) NK (MacKenzie 1961a: 203)

- a. *tişt-ê min da=ye=v hinge*
thing-EZ.M 1SG.OBL give.PST=DRCT=ADP you.OBL
'the thing that I gave you'

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- b. *ew kesê ewwîlî b-ê-t*
 DEM person-EZ.M first SBJV-come.PRS-3SG
 ‘that person who comes first’

Crucially, as MacKenzie (1961a: 204) notes, when the relative clause is separated from its antecedent (i.e., head noun), it is introduced by a demonstrative/pronominal *ezafe*:

- (152) Behdînî NK (Akre) (MacKenzie 1961a: 204)
ez=ê hat-î-m=e wê desrok-a zêr-a, ya
 1SG=EZ.M come.PST-PTCP-1SG=DRCT that kerchief-EZ.F gold-PL EZ.F:DEM
subey zû filan kes-î îna-y
 morning early so.and.so person-OBL.M bring.PST-PTCP
 ‘I have come for that kerchief (full) of gold that so-and-so brought early this morning.’

The particle *ku/ke* can also introduce a relative clause, with or without the *ezafe*. The use of *ezafe* necessarily renders the relative clause ‘restrictive’:

- (153) Behdînî NK (Akre) (MacKenzie 1961a: 204)
ser av-a ko t-çî-t l dirêjî-ya gelî
 over water-EZ.F SUB IND-go.PRS-3SG in length-EZ.F valley
 ‘over the river that goes along the valley’
- (154) Mardin Tor NK (Sahin 2018: 103)
mal-a ku pez-ê wa zahf ba-n-e
 family-EZ.F SUB sheep-EZ their much be.PST.IRR-3PL-CON
 ‘the families who had a lot of wheat’

CK also uses *ezafe* for relative clauses, but most often together with the subordinator:

- (155) CK Sulemaniyya (Zhang 2025)
ew kiç-e-y ke dayk=î mamosta=ye
 DEM girl-DEM-EZ SUB mother=3SG teacher=COP.3SG
 ‘The girl whose mother is a teacher’

Ezafe relative clauses without a subordinator are rarer in CK and often appear with a pronominal head (156, 157):

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- (156) CK Sulemaniyya (MacKenzie 1961a: 132)
ew kitêb-e-y da=m-xist-û-we
 DEM book-DEM-EZ PVB=1SG-drop.PST-PTCP-COP.3SG
het=î-gir-e
 PVB=3SG-take.PST-IMP
 ‘Pick up that book which I have thrown down.’
- (157) CK Sulemaniyya (fieldnotes)
ewe-y hat bira-y min-e
 DEM-EZ come.PST brother-EZ 1SG-COP.3SG
 ‘the one who came is my brother’

Note that these *ezafe* constructions in relative clauses are not possible in CK dialects spoken further south in the Sanandaj region.

In Central Kurdish, there is a particle *hî*. This has been described as a demonstrative *ezafe* in previous studies (e.g. Öpengin 2016). This particle is used in predicative possession (see 158).

- (158) CK Kerkuk (fieldnotes) *ew ktêb=e hî Şwan=e*
 DEM.DIST book=DEM EZ Shuan=COP.3SG
 ‘that book is Shuan’s.’

Note that this construction also occurs in Goranî varieties, e.g., Hewramî (see 159).

- (159) Hewramî Text (Mohammadirad in review) *hin-û*
 EZ.PRON-EZ.GEN
dil-û qirôt-û nere-(e)key
 inside-EZ.GEN hollow.M-EZ.GEN oak_tree.M-DEF.OBL.SG.M
 ‘The one raised in the tree hollow’

Finally, note that, especially in the Behdînî dialects but also in some other NK and CK dialects, the *ezafe* is used in the formation of a number of predicative constructions, such as (i) syntactic possession, (ii) attributive predication, (iii) locative predication, (vi) perfect, and (v) progressive. These are treated separately in the chapters dedicated to these topics.

11.2.4 Evolution of *ezafe* in Kurdish

The various *ezafe* elements in the Iranian languages are traced back to the Old Iranian forms *(t)ya* and *(h)ya*. These were in Old Iranian used as relativiser, demonstrative, and as Haig (2011b) claims ‘nominaliser’, as in, for example, ‘the strong

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one'. The standard widespread account (Karimi 2007) is based on the evolution from a relative pronoun. In such constructions, often the overt copula of such relative clauses would be missing, and this would result in a construction identical to a noun with simplex adjective or possessor. In languages such as Persian the *ezafe* lost its subordinator/relativiser function and was reduced to adnominal linking, while in Kurdish it retained its subordinator/relativiser function and at the same time evolved into adnominal linking elements.

The demonstrative *ezafe* used in complex noun phrases with multiple dependents (141) as well as the pronominal *ezafe* (143) are more readily connected to their historical relativiser/demonstrative function. The phrase *pêlavêt we=yêt nû* 'your new shoes' above (141c) must have developed from readings such as 'your shoes, those that are new', from a putative construction as *pêlav-êt we, ew yêt nû-ne* with a demonstrative head noun and a relative clause complement that contains a copular predicate.

11.2.5 The plural *ezafe* in Kurmanji

The *ezafe* in Iranian is most commonly an invariant form that does not distinguish number; the restriction of that distinction to a small subset of Northern Kurdish leads to the conclusion that this is an innovation therein. In many Kurmanji dialects, in addition to gender, number is also distinguished in *ezafe* forms. The distinct plural forms *-ên/-êd ~-êt* (and their independent demonstrative forms *yên/yêt*) thus require an explanation as to how they have evolved.

One likely scenario is that the Kurmanji plural must have developed from the combination of the historical Iranian *ezafe* *-ê* and a particular particle *di* (i.e., *ê di*). The particle became encliticised onto the *ezafe* of the preceding head noun, and the fusion resulted in *-êd* (and further devoicing in some dialects into *-êt*). The variant *-ên* is easier to account for: the *n* formative is a feature of plural marking throughout the nominal and verbal systems, i.e., *-n* (1/2/3)PL, *-an* -OBL.PL, and *-no* -VOC.PL.

The more conservative Behdînî and Yezidi dialects retained the source form *-êd/-êt*, while the neighbouring Kurmanji dialects of Shirkak evolved it further into the *-ên* form. However, the majority of NK dialects of central-southern-western areas did not develop any distinct plural *ezafe*, relying on the *-ê* form. In fact, MacKenzie (1961a: 159) notes that in all dialects the form *-ê* may appear in the plural.

What exactly is the particle *di* has not yet been clarified. MacKenzie (1961a: 159) notes it as a separate 'plural' article and considers it (in MacKenzie 1995: 413) a borrowing from the Aramaic linking particle *dā-*.

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There is only one area in Kurdish where the particle appears on its own without the *-ê* form of *ezafe*: in Behdînî NK preceding attributive predicates that are impermanent features of their subject:

(160) Behdînî (Duhok) NK (fieldnotes)

- a. *sêv çawan-in?*
apple how-COP.3PL
'How are the apples?'
- b. *di xoş-in*
PRT nice-COP.3PL
'(They) are nice.'
- c. *ev sêv-e di xoş-in*
these apple-DEM.PROX PRT nice-COP.3PL
'These apples are nice.'

That it must have been grammaticalized from the combination of the *-ê ezafe* form with this pre-attributive *di* particle is supported also by the fact that in classical Kurdish poetry the combination occurs often and always in the form of separate elements, viz. *-ê di*, and only later in the 18th century texts the fusion into *-êt* is found (Öpengin 2023: 377):

(161) Literary Kurmanji (16th century)

- roj-ê di sipî şev-ê di tîrî*
day-EZ.M PRT white night-EZ.M PRT dark
'light days and dark nights'

The same separate pair of elements is also found in the more archaic northern CK dialects of Mukri and Bingird, especially in their formulaic registers such as ballads:

(162) Mukri CK

- a. *wezîr-î de min*
vizier-EZ PRT my
'my viziers'
- b. *gul-î de sosin-ê*
rose-EZ PRT lily-OBL
'lily flowers'

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Finally, in the Baku enclave dialect of NK, the particle is retained with its vowel, used with the *ezafe* *-ê* form as *-ê-de*, while in the Yezidi dialects of Sinjar and Yerevan the form is fused as *-êd*. But in both of these dialects the particle is used unsystematically, probably as a relic:

- (163) NK Baku dialect (Stilo 2021: 328)

je diz-an yek-î be haval-e-de xwe go...
 from thief-PL.OBL one-OBL.M to friend-EZ-PRT self say.PST
 ‘One of the burglars said to his friends.’

- (164) NK Yerevan dialect (Cindî 2005: 222)

nimûn-êd zargotinê
 example-EZ.PL folklore=OBL.F
 ‘examples (i.e., genres) of folklore’

Geographically, the plural *di* particle, and its use with the *-ê ezafe* form, is restricted to core Kurdish areas as well as two enclaves, which are also very likely related to the core areas by way of migration and retention. This limited diffusion of the particle and the incorporation of the particle by the plural *ezafe* suggest that it is the result of a recent development in Kurdish, as argued in Paul (forthcoming).

11.2.5.1 Syntax of the noun phrase

In a complex noun phrase with both a possessor and an attributive element, the order is always head-initial, but the relative order of the possessor and attributive element differs among Kurdish dialects.

Northern Kurdish, especially with a pronominal possessor, is strictly possessor first and adjective second:

- (165) Behdînî NK (fieldnotes)

- a. *xanî-yêt min=êt xoş*
 house-EZ.PL 1SG.OBL=EZ.PL nice
 ‘my pleasant houses’
- b. *kur-ê paşa-yê yê mezin*
 son-EZ.M king-OBL.M EZ.M big
 ‘the king’s elder son’

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This order is sometimes compromised, especially in the writing of new generations, but the exact deviations are still to be explored.

Central Kurdish, on the other hand, has the adjective first and then the possessor:

(166) CK literary language (fieldnotes)

- a. *kur-î gewre-y paşa-ke=man*
son-EZ big-EZ king-DEF=1PL
'our king's elder son'
- b. *firokexane-y nawneteweyî-y Hewlêr*
airport-EZ international-EZ Erbil
'International Erbil Airport'

It is more natural, however, in a construction with both attributive and genitival modifiers to use the definite *ezafe* described in §11.2.2.2. Recall that in these constructions a separate *ezafe* marker *-e* is used and all other morphology that typically lands on the head noun, i.e., number and definiteness, occurs on the adjective. Note that these constructions are inseparable to the point that these forms are often referred to as compounds (Thackston 2006: e.g., following). The Adjective < Possessor order is the only possible order in these constructions.

(167) CK literary language (fieldnotes)

- xanûw-e xoş-êke-y ewan*
house-EZ nice-DEF-EZ their
'their nice house'

11.2.6 *Ezafe* in Zazaki

The Zazaki *ezafe* system is fairly complex, since it distinguishes gender, number, case, definiteness, and also whether the head is a noun or an adjective. In addition, many dialects, especially in the central dialect area, show erosion, where suffixation is replaced by vowel alternation (raising and/or fronting), or alternatively by palatalization.

The *ezafe* suffixes attach to the head noun (Table 11.4). Depending on whether the head is followed by a noun/pronoun or an adjective, different suffixes are used, although the difference is limited to the singular masculine. If the *ezafe* is followed by a noun (referred to as genitive or nominal *ezafe*, cf. Paul 2009: 563–564), the dependent noun takes the oblique case. In all cases, the *ezafe* suffix agrees with its head in gender, number, and case (168). In adjectival *ezafe* constructions (169) the adjective agrees with its head.

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Table 11.4: *Ezafe* suffixes in Zazaki

DIRECT		OBLIQUE	
	SG	PL	
M	-o (+ ADJECTIVE)		
		-ê	-ê -anê
	-ê (+ NOUN)		
F	-a	-ê	-a -anê

(168) Genitive *ezafe* in Northern Zazaki (fieldnotes)

- a. *meyman-ê veyve-y*
guest-EZ.M wedding-OBL.M
'wedding guest (male)'
- b. *meyman-a veyve-y*
guest-EZ.F wedding-OBL.M
'wedding guest (female)'
- c. *meyman-ê veyve-y*
guest-EZ.PL wedding-OBL.M
'wedding guests'

(169) Adjectival *ezafe* in Northern Zazaki (fieldnotes)

- a. *meyman-o rind-Ø*
guest-EZ.M good-M
'good guest (male)'
- b. *meyman-a rind-e*
guest-EZ.F good-F
'good guest (female)'
- c. *meyman-ê rind-î*
guest-EZ.PL good-PL
'good guests'

In parts of Central Zazaki, especially in the Çewlîg (Bingöl) and Palu regions, the *ezafe* suffixes differ: the masculine and plural forms are *-î*, while the feminine is *-ê*. However, the *ezafe* suffix is often omitted, especially in genitive *ezafe* constructions. In such cases, vowel alternation in the environment, most often

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palatalisation, substitutes for the suffix. See (170) and (171) for a comparison with the word *her* ‘donkey’.

- (170) Central Zazaki (Paul 2024: 199)

hêr-Ø mi
donkey-Ø my
‘my donkey’

- (171) Northern Zazaki (Selcan 1998: 663)

her-ê mi
donkey-EZ.M my
‘my donkey’

11.2.6.0.13 Second set of oblique *ezafe*

In Southern and parts of Central and Eastern Zazaki, there exists a second set of oblique *ezafe* suffixes, formed by inserting *d-* before the regular *ezafe* suffix (called Oblique II in Paul 2024: 31–36). It has been speculated that this is in origin the Aramaic linking particle *d-* (MacKenzie 1995: 412–413 but see also Karim 2021: 249–264 for an alternative explanation). This secondary oblique *ezafe* is used only when the head noun is (a) linked to a preceding head via another *ezafe* (i.e., in complex noun phrases), (b) the complement of adpositions, (c) a post-predicate goal or recipient, or (d) part of certain adverbial phrases. These *d-*form obliques are absent from most of Central Zazaki and from all of Northern Zazaki, but they are attested in Eastern varieties spoken in Batman and Bitlis.

- (172) Southern Zazaki (Paul 2024: 34)

este-yê gan-dê mi
bone-EZ.PL soul-EZ.M.OBL my
‘the bones of my body’

11.2.6.0.14 Indefinite *ezafe*

For indefinite noun phrases, indefinite *ezafe* suffixes are used, which exhibit dialectal variation. In Northern and Southern Zazaki, the *d-*form *ezafe* follows the indefinite suffix (*-ê* in Northern Zazaki, *-ê(n)* in Southern Zazaki), before the regular *ezafe* suffix attaches (see 173). In Central Zazaki, however, no *d-*form exists for the indefinite. Instead, the indefinite suffix *-êk*, likely borrowed from Kurmanji, is followed directly by the *ezafe* suffix (174). The indefinite *ezafe* is restricted to the singular (Paul 2024: 43–45).

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(173) Indefinite *ezafe* in Northern Zazaki (fieldnotes)

- a. *meyman-ê-do rind-Ø*
guest-INDF-EZ.M good-M
'a good guest (male)'
- b. *meyman-ê-da rind-e*
guest-INDF-EZ.F good-F
'a good guest (female)'

(174) Indefinite *ezafe* in Central Zazaki (fieldnotes)

- a. *misafir-êk-o hol*
guest-INDF-EZ.M good
'a good guest (male)'
- b. *misafir-êk-a hol*
guest-INDF-EZ.F good
'a good guest (female)'

11.2.6.0.15 Multiple *ezafe* constructions

If the head is followed by a combination of dependent nouns and adjectives linked via *ezafe*, the genitive *ezafe* typically precedes adjectives. With pronouns, this order is obligatory. In continuous chains of possessors and possessee, each *ezafe* suffix attaches to its head, agreeing in gender, number, and case (175a). In discontinuous chains, e.g., where nominal and adjectival constructions co-occur, the *ezafe* suffix detaches and surfaces as an independent word. Even then, it must agree with its head, though no longer directly attached (175b, 175c).

(175) Multiple *ezafe* Constructions

- a. (fieldnotes)
ko-yê dew-a muxtar-ê ma
mountain-EZ.PL village-EZ.F mukhtar-EZ.M 1PL
'the mountains of our mukhtar's village'
- b. (fieldnotes)
bira-yê Xidir-î yo pîl-Ø
brother-EZ.M Xidir-OBL.M EZ.M older-M
'Xidir's older brother'

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c. (Selcan 1998: 269)

ma-a tu ya kor-a kokim-e
 mother-EZ.F 2SG.OBL EZ.F blind-EZ.F old-F
 ‘your blind, old mother’

11.2.6.0.16 Absolutive/Pronominal *ezafe*

As in Kurmanji, the *ezafe* may also appear independently, called absolutive, pronominal, or anaphoric *ezafe*. This type of *ezafe* has a single, invariant form *yê* or *ê*—depending on dialect—regardless of phrase type, gender, number, or case. In this use, the *ezafe* itself functions as the head of the phrase.

(176) Pronominal *ezafe* (fieldnotes)

- a. *na kitav yê min=o*
 DEM book.M EZ 1SG.OBL=COP.3SG.M
 ‘This book is mine.’
- b. *yê min=o*
 EZ 1SG.OBL=COP.3SG.M
 ‘It is mine.’

11.2.6.0.17 Genitive constructions without *ezafe*

With body parts and a small set of clothing items (e.g., bags, pockets) denoting inalienable possession, genitive constructions may be reversed: the possessor precedes the head, takes the oblique case (if available), and no *ezafe* suffix appears. This reverse genitive is restricted to two constituents and is most common in adpositional phrases or with recipients and goals.

(177) Genitive constructions without *ezafe*

- a. (Selcan 1998: 265)
kitab lazek-î dest=de bî-Ø
 book boy-OBL.M hand=in be.PST-3SG.M
 ‘The book was in the boy’s hand.’
- b. (Paul 2024: 47)
hak-ê ken-o xo tûnik-î
 egg-INDF do.PRS-3SG.M REFL bag-OBL.M
 ‘He puts an egg into his bag.’

11.3 Attributive linkers in Arabic

Qaṭu Arabic dialects, like other sedentary Arabic dialects (Näim 2006: 672), possess an independent attributive particle,³ although the form it takes varies widely. The particle may serve to connect the head noun with another noun, e.g.,

- (178) C. Mērgi (Hassan 2023: 588)
ktīg muṣaddāt lāt az-zarāʿa
 many tool.PL POSS.PL DEF-agriculture
 ‘a lot of agricultural tools’

Alternatively, the attributive particle may take pronominal suffixes and serve as an independent possessive pronoun, e.g.,

- (179) C. Mērgi (Hassan 2023: 596)
aj-jaḥš līt-u
 DEF-donkey POSS.SG-3SG.M
 ‘his donkey’

When it connects the head noun with another noun, it is an alternative to the construct state (*iḍāfa*) construction: compare (180a) and (180b))

- (180) Mosul (Procházka 2019: 253-254)
 a. *daftar xadma māl zōj-i*
 record.CSTR service POSS husband-1sg
 ‘my husband’s service record’
 b. *sīyār-t al-pōlīs*
 car-F.CSTR DEF-police
 ‘the police car’

When it takes pronominal suffixes, it may be an alternative to possessive suffixes attached directly to the head noun, e.g.,

- (181) C. Mērgi (Hassan 2023: 585)
ʔabū-hu
 father-3SG.M
 ‘his father’

³The survey in Jastrow (1978: 125-126) assumes—rather than explicitly states—the universal existence of such a particle in the dialects and concentrates on presenting the diversity of forms.

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In both functions, it may be used to avoid repetition of the head noun in predicative constructions (Jastrow 1978: 126), e.g.,

- (182) a. J. ‘Aqra (Jastrow 1990a: 48)
hadā l-bayt līt māni=we?
this DEF-house POSS.SG who=PRS.COP.3SG.M
‘Whose is this house?’
b. J. Arbīl (Jastrow 1990a: 48)
hawdā l-ʕazāl mān-na=me
these DEF-thing.PL POSS-1PL=PRS.COP.3PL
‘These things are ours’

Not all descriptions mention both functions (i.e., with both noun and pronominal suffixes as possessors), so it cannot be ascertained whether or not the particles always have both. In the tables below, both the independent form (used with following noun) and attached form (used with pronominal suffixes) are given, where both could be found in the source (in the description or attached texts).⁴

Table 11.5: Arabic attributive particles within the NENA area

Mosul (M, C, J)	<i>māl ~ mālāt; māl- ~ mālāt-</i> (Procházka 2019: 253–254), <i>gayy-</i> (Jastrow 1979: 62, n. 67)
Bəḥzāni (C, Y, M)	sg <i>līt</i> , pl <i>lāt</i> (Jastrow 1978: 125)
Mērgi (C)	sg <i>līt</i> , pl <i>lāt</i> ; sg <i>līt-</i> (Hassan 2023) ⁵
Səndōṛ (J)	<i>lāt</i> (Jastrow 1991b: 775)
Arbīl (J)	invar. <i>māl</i> ; <i>māl-</i> (Jastrow 1990a: 47)
‘Aqra (J)	sg <i>līt ~ lāt</i> , pl <i>lāt</i> ; <i>līt-</i> (Jastrow 1990a: 47)

⁴An exhaustive search was, however, not made of the texts available.

⁵The source does not explicitly say that the first is singular and the second plural, but this appears to be the case judging by the examples of both in the texts, with one exception on p.593 where *līt* is used with a plural noun: *mumaṭəlayyāt hawnēk līt ən-nəzūh*(exemplary.school.PL there POSS DEF-displacement) ‘Exemplary Schools, there, for the displaced people’.

⁶Jastrow gives the form for the Siirt dialects in general as *lē* (Jastrow 1978: 125), but does not specify the precise form for the town of Siirt.

11.4 Nominal Modification in Armenian

Table 11.6: Arabic attributive particles on the periphery of the NENA area

Āzəx (C)	invar. <i>ləl</i> ; <i>ləl</i> - (Jastrow 1978: 125; Wittrich 2001: 125)
Siirt town (M)	invar. <i>lə</i> ; <i>lə</i> - (Biṭună 2016a: 140) ⁶
Tillo (M)	invar. <i>lay</i> (Lahdo 2009)

11.4 Nominal Modification in Armenian

11.4.1 Position of the possessor/adjective in Armenian

In modern Armenian, possessors and adjectives almost always precede the noun:

- (183) SEA
Armen-*i* *tun-ə*
Armen-ATTR house-DEF
‘Armen’s house’
- (184) SEA
im *tun-ə*
1SG.GEN house-DEF
‘my house’
- (185) SEA
mec tun
big house
‘(a) big house’

It is possible under certain circumstances for an adjective (as seen in (186)) or possessor (as seen in (187)) to follow the noun. As discussed in Hodgson (2019), this generally occurs for reasons connected to information structure, when the modifier is categorised as [-focus]. It may also occur in expressions taken from or imitating Classical Armenian, where the order of noun and modifier was more flexible (see eg., Donabédian 2000).

- (186) SEA (Hodgson 2019: 154)

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Hayastan-a leŕ-ner-ov šrĵapatvac ē, inč'pes mi hat
 armenia-DEF mountain-pl-INS surrounded be.3SG.PRS, as one piece
šlt'a-yov gelec'ik
 chain-INS beautiful
 'Armenia is surrounded by mountains, as if by a beautiful chain.'

- (187) SEA (Hodgson 2019: 154)
kaŕavarut'yun-a mer
 government-DEF 1PL.GEN
 'our government'

11.4.2 Morphology of noun modification in Armenian

When a noun is modified by an adjective, there is no linking morpheme present. The adjective is invariant, showing no agreement morphology, as seen in (188).

- (188) SEA
mec tun
 big house
 '(a) big house'

Nouns modified by a possessor obligatorily take the definite article, as seen in (183), (184), and (187). This is the case even if the noun is semantically indefinite (189), as discussed by Sigler (1996) (her discussion concerns SWA, but the facts are similar across most Armenian dialects, including SEA).

- (189) SWA (Sigler 1996: 113)
ekelec'i-i mə tanik'-a
 church-ATTR one roof-DEF
 'a church's roof'

However, when a noun is modified by a noun in the genitive that serves as a modifier, but is not interpreted as a possessor, the definite article is not obligatory, as seen in (190).

- (190) SEA (Hodgson 2022b: 139)
sa Hayastan-i ciran=ē
 DEM1 Armenia-GEN apricot=be.3SG.PRS
 'These are/this is (an) Armenian apricot(s).'

11.4 Nominal Modification in Armenian

If the possessor is 1st or 2nd person, instead of the ordinary definite article, the 1st person or 2nd person “possessive article” may be used, as in (191) and (192).

- (191) SEA
im tun-s
 1SG.GEN house-ART1
 ‘my house’

- (192) SEA
k'o tun-d
 2SG.GEN house-ART2
 ‘your house’

In some dialects, including SWA, this is obligatory, i.e., with a 1st or 2nd person possessor, the 1st or 2nd person “possessive article” must be used, unlike in SEA, where the construction in (184) with the ordinary definite article is preferred when the possessor is a 1st or 2nd person pronoun.

The “possessive articles” may also be used without a pronoun to express 1st or 2nd person possessors, as seen in (193) and (194), and indeed a noun with the regular definite article can also be interpreted as being possessed by a 3rd person referent, as in (195).

- (193) SEA
tun-s
 house-ART1
 ‘my house’

- (194) SEA
tun-d
 house-ART2
 ‘your house’

- (195) SEA
tun-ə
 house-DEF
 ‘the house’ OR ‘his/her house’

The “possessive articles” are derived from the proximal and medial articles of Classical Armenian, while the distal article has become the ordinary definite

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article of modern Armenian.⁷

The Classical Armenian definite article had a three-way deictic distinction, parallel to that of the demonstratives, perhaps representing “an intermediate stage in the development from demonstratives to definite articles” (Vaux 1994–1995), as seen in Table 11.7.

Table 11.7: The Classical Armenian definite articles

<i>ayr-s</i>	<i>ayr-d</i>	<i>ayr-n</i>
man-ART1	man-ART2	man-ART3
‘the/this man	‘the/this man	‘the/this man
(close to me)’	(close to you)’	(close to neither)’

11.5 Discussion

The subject of linkers in Neo-Aramiac and Iranian languages has been studied extensively. Of particular note is Gutman (2018), a detailed study of these constructions in Neo-aramaic (based on the state of knowledge at that time) and neighbouring Kurdish varieties. Other works of note include NENA: Cohen (2015, 2010), Gutman (2021) and Iranian: Karim (2023, 2022), Karimi (2007), Ghomeshi (1997), Haig (2011a), Sahin (2018), Samiian (1994), Parsafar (2010), Larson & Samiian (2020), Kahnemuyipour (2014), Samvelian (2007). Many of these works focus on theoretical issues related to the morphosyntax of linking constructions. This chapter does not engage with many of these, but rather focuses on the comparison of variation based on a large quantity of new data from NENA and Iranian. One topic that has received much attention is the *ezafe* (linking morpheme)’s status as a suffix, head-marker or dependent-marker. One particularly interesting proposal is Larson & Samiian’s (2020) approach that the *ezafe* is a case marking suffix with two governors. According to this view, the case of a noun phrase is governed by an adnominal modifier and a verb, for example. The verb would assign a direct or oblique case relation to a subject or object noun phrase, and within the noun phrase the adnominal modifier would assign *ezafe* case. We remain agnostic about this discussion here, and we acknowledge that there is no solution that rings equally true for all languages with morphological linking elements.

⁷In Classical Armenian, and in some modern dialects, including Mush, this element was pronounced *-(ə)n*, but in the modern standard languages, it is pronounced *-ə*, with deletion of the final *-n*, except when it follows or precedes a vowel, in which case the *-n* is preserved.

11.5 Discussion

In the NENA dialects, the historical Aramaic attributive particle *d* has converged phonotactically and functionally with the Iranian *ezafe* particle. In NENA dialects in the northern sector, the particle is typically attached as a suffix to the head noun when it is annexed in a genitive relation to a following dependent noun, e.g.,

- (196) C. Barwar
bεθ-ət malka
 house-ATTR king
 ‘the house of the king’

This matches the phonotactics of the Northern Kurdish *ezafe* particle, which also has the status of a suffix (11.2) (Cohen 2015).

Unlike Northern Kurdish, however, the NENA attributive particle that is suffixed to the head noun does not inflect for gender and number. It is possible that NENA *-ət* suffix was derived historically from a genitive pronominal suffix + attributive particle **d*, but, if so, one would have to assume that the suffix had become frozen in the 3SG.M form and did not vary for gender or number, e.g., **e* (SG.M) + **d* > *-ət* (see Mengozzi 2005; Gutman 2021). Furthermore, there is no distinction between the form of the NENA attributive particle on definite and indefinite nouns, as there is in Northern Kurdish (§11.2.1.2).

In NENA dialects in the Central Kurdish region, the attributive particle is best analysed as a clitic. This is seen, for example, in the J. Arbel dialect, which has a general word-final stress but does not place the stress on the particle, e.g.,

- (197) J. Arbel (Khan 1999a: 168)
báb=ət bratá
 father=ATTR girl
 ‘the father of the girl’

Both the generalised word-final stress and the unstressed clitic status of the attributive particle match Central Kurdish, in which the *ezafe* has the status of an unstressed clitic (11.2).

The NENA attributive particle may be attached to an adjective expressing a spatial or temporal relation of the dependent. This matches the use of *ezafe* in such contexts in Kurdish, e.g.,

- (198) a. NENA C. Barwar
qurb-ət maθa
 near-ATTR village
 ‘near to the village’

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- b. NK Behdîni
nêzîk-î ħakimî
 near-EZ.GEN judge-OBL.M
 ‘near the judge’

In NENA dialects in the northern sector, the *-ət* suffix can be detached from the head and placed before the dependent noun (21) or placed before a predicative attribute (23). This may have been facilitated by the pronominal use of *ezafe* as an independent particle in northern Kurdish (§11.2.1.4).

In NENA, head nouns are generally not connected to an adjectival modifier by a linker particle, just as there was no linker historically in earlier Aramaic in such constructions. By contrast, in Kurdish, a head noun is regularly connected to an adjectival modifier by *ezafe*. There are, however, some sporadic examples of linker particles before adjectival modifiers in NENA dialects (see (126) and (127)) and this is presumably due to convergence with Kurdish.

Constructions in NENA dialects with a demonstrative pronoun and linker particle before an adjectival modifier, such as (129), or with a demonstrative before the adjective without linker, such as (132), are likely to have developed by convergence with Northern Kurdish constructions with a pronominal *ezafe* before the adjective, such as (142). These are repeated in (199):

- (199) a. NENA J. Arbel
šakar ’o=t dāqîqa
 sugar DEM=REL fine
 ‘fine sugar’
- b. NENA C. Barwar
brat-ət malk ’ε goṛta
 king-ATTR DEM.SG.F big
 ‘the elder daughter of the king’
- c. NK, Behdîni
bira-yê min yê mezin
 brother-EZ.M 1SG.OBL EZ.M big
 ‘my big brother’

In Northern Kurdish, the pronominal *ezafe* is the norm after complex noun phrases. In NENA dialects, on the other hand, such constructions are used to give focus to the adjective, typically contrastive focus. Non-contrastive adjectives are placed immediately after the head noun, e.g.,

- (200) C. Barwar (Khan 2008c: 488)
 ʔumra xaθ-ət Mar Sawa
 church new-ATTR Saint Sawa
 ‘the new church of Saint Sawa

The NENA dialects, therefore, have used the Kurdish model of pronominal *ezafe* after complex noun phrases to express an internal functional distinction between contrastive and non-contrastive adjectives. Alternatively, it is possible that at an earlier historical period, the construction with demonstrative *ezafe* may have been used contrastively in Kurdish as well, but has now been extended to non-contrastive use. The placement of adjectival modifiers before the head noun in NENA to express evaluative attributes or focused attributes is a sporadic feature that is found in dialects of the northern sector (124, 125). It is most common in dialects of the north-eastern sector, such as C. Urmi. This may have been facilitated in NENA in that region due to its contact with Armenian and Turkic, in both of which the placement of an adjective before the noun is the norm (for Armenian see §11.4.1). Again, therefore, NENA may have replicated the construction in contact languages to serve a specific functional purpose within NENA, thus enriching its strategies of expression of pragmatic distinctions.

In the Jewish Trans-Zab NENA dialects in the south-eastern sector, the NENA linking particle is omitted in genitive annexation phrases (98, 107a). This replicates the pattern of southern Central Kurdish dialects of the region, such as the Kurdish of Sanandaj, where the *ezafe* is typically omitted (Khan & Mohammadi-rad 2024b: 108). In J. Sulemaniyya, there are vestiges of the NENA linker particle, but these are mainly restricted to relative clauses with pronominal heads (106a). When the head of a relative clause is a full noun in J. Sulemaniyya an Iranian relative particle *ka/ga* is used. This distribution matches that of the *ezafe* particle in relative clauses in the Central Kurdish dialect of Sulemaniyya, in which it is mainly restricted to clauses with pronominal heads (157).

Although the NENA annexation is generally omitted in J. Sulemaniyya and J. Sanandaj, the oblique form of the demonstrative (*do* sg., *doni* pl) still preserves a historical vestige of the particle (99a, 110). Such oblique demonstratives, however, have lost their historical compositionality. They appear to have developed by NENA-internal analogy with the paradigm of the attributive pronoun reinforced by the presence of oblique pronouns in Iranian. In the J. Sulemaniyya and J. Sanandaj NENA dialects, contact with Gorani is likely to have been a factor, since Gorani of the region preserved oblique third person pronouns, whereas these have been lost in southern Central Kurdish dialects. The oblique demonstratives have undergone attrition in the NENA dialect J. Kerend further south

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(120) outside the region of Gorani influence. In the Christian dialects C. Sulemaniyya (§11.1.5.2) and C. Sanandaj (Panoussi 1990) the NENA attributive particle has been preserved, reflecting the greater conservativeness of these dialects than the Jewish dialects of the area.

In several Jewish Trans-Zab dialects, a particle resembling the shape of Iranian *ezafe* particles is used in place of, or in combination with, the regular NENA attributive particle. These include the forms *i* (75b, 82a, 103), *ay* (94) and *e* (108). These are unlikely to be loans of Iranian *ezafe* morphology. Rather, they are original NENA demonstrative elements, which can be identified as components of demonstratives in the paradigms of the demonstrative pronouns in these dialects. They have developed by matching the form and syntax of the Iranian *ezafe*, reflecting an identification of the *ezafe* as a demonstrative element.

In NENA dialects in the south-eastern sector, a predicative possessor is introduced by the preposition *ʔal* ‘to’ (or phonetic variants thereof) rather than the attributive particle *t/d* (104, 116, 122a). In C. Sulemaniyya, the phonetic variant of this is *hən* or *hi* (52). This is phonetically similar to a particle that is used for the same purpose before predicative possessors in Central Kurdish of the region. This strategy of expressing predicative possession, therefore, may have been induced by contact with Central Kurdish.

In Northern Kurdish, a *d* element, often devoiced to *t* or shifting to *n*, appears in the plural *ezafe* (§11.2.5). This element is found in the singular and plural of some *ezafe* constructions in Zazaki (§11.2.6.0.13, 11.2.6.0.14). This resembles the phonetic shape of the NENA attributive particle, and Iranists have speculated that it is a borrowing from Aramaic. It seems, however, unlikely that Northern Kurdish and Zazaki would borrow a component of Aramaic inflectional morphology, while the influence of Aramaic syntax and lexicon on these Iranian varieties is otherwise minimal. It is more likely to be an internal development of Iranian. That said, it is a recognised phenomenon of NENA that internal developments have often been reinforced by contact with Iranian, so, theoretically, the same may have happened, *vice versa*, in Iranian.

The genitive linking particles in Arabic *Qəltu* dialects are independent particles in all contexts (§11.3), e.g., (178), repeated here:

- (201) C. Mērgi (Hassan 2023: 588)
kṭīg muʕəddāt lāt əz-zəṛāʕa
 many tool.PL POSS.PL DEF-agriculture
 ‘a lot of agricultural tools’

Their closest analogue in NENA is the combination of the stem of the attributive pronoun *did-* and the annexation linker particle in constructions such as (61),

repeated here:

- (202) J. Dohok (Molin 2024: 77)
kəsta dət pare
 bag ATTR money
 ‘a bag of money’

The Arabic constructions are also similar to the independent use of the Iranian *ezafe* (see §11.2.1.4).

In summary, the developments surveyed in this chapter point to a sustained process of convergence between NENA and the surrounding Iranian languages, especially Kurdish, in the domain of nominal linkage and modification. The outcomes of this convergence are not simple borrowings of morphological material, but rather the replication and adaptation of patterns that were integrated into existing Aramaic structures. Across the NENA dialect continuum, contact-induced innovations interact with inherited Aramaic features in different ways, resulting in a wide range of local configurations. The diversity of linker constructions found in NENA thus provides a particularly clear illustration of how language contact can reshape grammatical systems while preserving continuity with earlier stages of the language

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North-Eastern Neo-Aramaic in the Context of the Languages of Western Asia

North-Eastern Neo-Aramaic in the Context of the Languages of Western Asia offers a comprehensive investigation of the development of North-Eastern Neo-Aramaic (NENA) through sustained contact with the languages of one of the world's most linguistically diverse regions. Bringing together evidence from NENA and its principal contact languages—including Kurdish, Gorani, Zazaki, Armenian, Arabic and Turkic—the volume explores how centuries of multilingual interaction have shaped the phonology, morphology and syntax of these endangered Aramaic varieties. The book brings together leading experts in the respective languages and offers a detailed description of most of the internal diversity documented to date.

Drawing on extensive documentation of NENA dialects and detailed comparative analysis, the book examines a wide range of linguistic phenomena, from sound systems, nominal modification and verbal morphology to alignment, word order and argument marking. Rather than treating shared structures simply as products of contact, it investigates the historical, geographical and social conditions under which linguistic convergence has taken place. Particular attention is paid to the effects of migration, patterns of settlement and the boundaries of social and communal networks, which can produce striking differences even between communities occupying the same geographical area.

The volume also makes a broader contribution to the study of language contact. Its concluding synthesis develops a fine-grained account of contact-induced change, distinguishing processes such as matter borrowing, pattern replication, simplification, enrichment and grammaticalisation, while showing how replication may be partial, functionally transformed or shaped by multiple causes. It demonstrates that linguistic change cannot be understood through geographical proximity alone: social networks, historical layering, internal developments and speakers' interpretations of structures in other languages all play crucial roles.

Combining rich empirical documentation with comparative and theoretical perspectives, this book will be an essential resource for scholars of Aramaic and the languages of Western Asia, as well as for researchers interested in historical linguistics, linguistic typology, multilingualism and the dynamics of language contact.